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Tobias Glas

Asset Pricing and Investment Styles in Digital Assets

A Comparison with Traditional Asset Classes

Springer

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Preface

This book was originally meant to become a cumulative PhD thesis consisting of five research papers. Four of the five draft papers were written in cooperation with one or several co-authors. The remaining one [1] was produced completely by me and was published in the *Journal of Alternative Investments*. I wrote another paper in cooperation with Prof. Dr. Thorsten Poddig [2], which was published in the *Quarterly Journal of Economic Research* (*DIW Vierteljahresheft*). Based on the remaining three drafts, I developed this book. The following paragraphs detail the work and input of all co-authors per chapter to acknowledge their help and contributions, for which I am very thankful. In particular, I would like to express gratitude to Prof. Dr. Thorsten Poddig, PD Dr. Christian Fieberg, and Prof. Dr. Armin Varmaz for their invaluable comments and considerations.

Chapters 1–5 are completely written by me. Chapter 6 is motivated by the paper "Kryptowährungen in der Asset-Allokation," which is based on data that I collected from www.coinmarketap.com. Most of the software used (for data collection, data description, and tables) were programmed by me, and some others (portfolio optimization) were taken from the book *Computational Finance* written by Prof. Dr. Thorsten Poddig, Prof. Dr. Armin Varmaz, and PD Dr. Christian Fieberg. The code for the spanning tests was taken from https://sites.google.com/site/mcliffweb/programs. I developed the argument motivated by the early hedge fund literature. The chapter has been revised several times. I wrote all the drafts. Prof. Dr. Poddig made helpful comments and valuable recommendations each time, which were incorporated into the respective new draft. Prof. Dr. Armin Varmaz contributed by making several remarks before the final publication of the paper.

Chapter 7, which was initially meant to become a German paper called "Zur Effektivität von technischen Indikatoren in Kryptowährungen," is based on data I collected from www.coinmarketap.com. All the software and text were written by me. I developed the argument on my own, based on a controversial opinion regarding technical indicators in the academic literature. The chapter has been revised several times. All drafts were written by me; Prof. Dr. Poddig contributed by suggesting a different structure and slightly different methodology, thereby enhancing the readability. I also received feedback from an anonymous referee

associated with the German magazine *Corporate Finance*, and I incorporated the points of criticism.

Chapter 8 was motivated by Glas [1] and extends the analysis of the published paper in several areas. All the associated software and text were written by me.

Chapter 9 is based on data that I collected by hand from several online sources and Thomson Reuters Datastream. All software and text were written by me (except for the principal component analysis function, which was programmed by Prof. Dr. Thorsten Poddig and adjusts the covariance matrix through spectral decomposition). I developed the argument mostly by myself, inspired by two papers focusing on the extraction of risk factors in foreign exchange rates and hedge funds, respectively. Prof. Dr. Armin Varmaz contributed by suggesting an additional section (clusters and idiosyncratic risk), which is relevant for the subsequent analyses. The chapter was rewritten multiple times, all drafts being written solely by me. Prof. Dr. Poddig contributed by recommending specific methodologies and in the interpretation of results. He also ran an empirical analysis which is used as basis for the characteristics vs. covariances section (see Appendix I).

Chapter 10 is the logical consequence of the previous chapters and uses the data mentioned before. The argument was developed and text written solely by me, and I used code for the parametric portfolio policy written by Prof. Dr. Thorsten Poddig and PD Dr. Christian Fieberg.

In order to soften possible claims of weak results due to bad data, I describe robustness tests in Chap. 11. There, I report the outcomes of the same analyses from the previous chapters with a higher-quality dataset from another data source. Due to the relatively short initial dataset, claims of spurious results could be made. Therefore, I obtained a second, longer, and broader, dataset from www.coingecko.com, which I used to support the findings of this thesis. Thus, the programming of the different tests has already been described. The text was all written by me.

The summary and conclusion in Chap. 12, as well as the appendices, were written and produced solely by me. Prof. Dr. Thorsten Poddig also commented on the drafts of this book, which helped to increase its readability and stringency.

Bremen, Germany Tobias Glas

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1. Glas TN (2019) Investments in cryptocurrencies: handle with care! J Altern Invest 22(1):96–113. https://doi.org/10.3905/jai.2019.22.1.096
2. Glas TN, Poddig T (2018) Kryptowährungen in der asset-allokation: Eine empirische untersuchung auf basis eines beispielhaften deutschen multi-asset-portfolios. Vierteljahresh Wirtsch 87(3):107–128. https://doi.org/10.3790/vjh.87.3.107

Acknowledgments

Working and studying in the Chair of Finance at the University of Bremen has taught me several things: (i) I learned to use programming languages like Matlab or Python in more depth; (ii) being able to cope with criticism is a valuable skill—not only in academia; (iii) time management, self-organization, determination, and persistence are important factors for success. So keep them up!

Without the support and understanding of Prof. Dr. Thorsten Poddig, this work would not have been possible. His intellectual capacity and academic experience contributed a lot, especially when working with (more or less) unexplored topics like digital assets, for which I am very grateful. He did not mind reading and commenting on different topics in stressful or unsuitable situations, which cannot be taken for granted.

I also want to thank Prof. Dr. Armin Varmaz for reviewing and grading the thesis on which this work is based. Mr. Varmaz and I both share a passion for cryptocurrencies, which is why I am all the more happy to have had him as a reviewer. Likewise, I want to express gratitude to PD Dr. Christian Fieberg for co-authoring and supporting me with my first ever academic paper.

Without my colleagues Richard Glende, Dr. Sergej Uschakow, Dr. Corinna Frädrich, Dr. Richard Mertens, Nico Schäfer, Ulrich Hammerich, Michael Falge, Tobias Tubbenhauer, Daniel Metko, Paige Michael-Shetley, Steffen Günther, and Carlos Osorio, my time at university would have been only half as fulfilling. Special thanks are owed to Ms. Sebbes for running and organizing everything so well.

Last but not least, special thanks is also owed to my parents and family for always supporting me in my academic and professional career. The biggest thanks, however, go to my girlfriend and significant other Helene. I must pay tribute to her for coping and sticking with me during hard and demanding times, but also during happy and great times.

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Acronyms

ADA	Cardano
ALGO	Algorithm employed for a digital asset
APP	Application
APT	Arbitrage pricing theory
BB	Bollinger bands
BTC	Bitcoin
BU	Bekaert and Urias
CAPM	Capital asset pricing model
CDS	Credit default swap
CER	Certainty equivalent return
CILQ	Crypto illiquidity (factor)
CMA	Conservative minus aggressive
CMKT	Crypto market (factor)
CoCo	Contingent convertible bond
COV	Covariance
CPM	Coins found per minute
CRIX	Crypto index
CSMB	Crypto size (factor)
CSTR	Crypto short-term reversal (factor)
CTA	Commodity trading adviser
DMI	Directional movement index
DOGE	Dogecoin
ERC	Equal risk contribution
ERC-20	Ethereum request for comment
ETH	Ether
FF	Fama–French
FFK	Ferson, Forster, and Keim
FM	Fama–MacBeth
FX	Foreign exchange
GHS	Gigahashes per second
GOC	Ichimoku

GSCI	Goldman Sachs commodity index
HK	Huberman and Kandel
HML	High minus low
ICAPM	Intertemporal capital asset pricing model
ICO	Initial coin offering
IOS	Investment opportunity set
LTC	Litecoin
MACD	Moving average convergence divergence
MSCI	Morgan Stanley capital international
NAA	Number of active addresses
NEM	New economy movement
Nonce	Number-used-once
OLS	Ordinary least squares
P2P	Peer to peer
PC	Principal component
PCA	Principal component analysis
PE	Private equity
PLS	Partial least squares
POS	Proof-of-stake
POW	Proof-of-work
REIT	Real-estate investment trust
RMW	Robust minus weak
RSI	Relative strength index
S&P	Standard and Poors
SHA	Secure hash algorithm
SMB	Small minus big
STO	Stochastic
TRX	Tron
USDT	Tether
VAR	Variance
VAT	Volume at time
WML	Winners minus losers
WTI	West Texas intermediate
XMR	Monero
XRP	Ripple

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Chapter 1
The Emergence of a New Asset Class: Digital Assets



The title of this volume, *Asset Pricing and Investment Styles in Digital Assets*, is an imperfect attempt to summarize the content of this study. While most of my work focuses on a relatively young and emerging asset class, namely cryptocurrencies or digital assets, the remainder draws connections to other traditional and alternative asset classes. I not only show the outcomes of several asset pricing analyses, but beforehand I also dive deeper into data and other connected issues. Moreover, I briefly describe the broader conception of the term *blockchain* and some single and well-known digital assets.

Cryptocurrencies are a relatively young and emerging phenomenon within the financial world. The use cases of such virtual tokens are manifold, ranging from technical to legal to financial applications. In the years 2016 and 2017, for example, start-ups discovered initial coin offerings (ICOs) to raise funds, circumventing traditional venture capitalists. Since the number of possible applications is vast for cryptocurrencies, I prefer the term *digital assets* as a broader conception of the whole asset class (as already used by several institutions like Fidelity Investments or Goldman Sachs, as well as various media outlets).

Several groups of digital assets can be distinguished:

- "Traditional" cryptocurrencies like bitcoin (BTC) or ethereum (ETH) that are mediums of (value) exchange and are connected to their own blockchain.
- Other virtual coins, like ERC-20 tokens, are situated on such a blockchain and give the issuer of those tokens the possibility to operate on the underlying blockchain. ERC-20 tokens, however, are mainly used for capital raising purposes and could therefore also be regarded as some kind of "equity" token.
- Some virtual coins serve as the so-called utility tokens that allow their holder to access or use a specific service or platform connected to the token.
- Furthermore, it is possible to link the value of a digital asset to a specific underlying index like, e.g., the Standard and Poors (S&P) 500 or the USD. Hence, this group could be called "commodity tokens" or "stable coins."

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This classification methodology is intensively discussed within the crypto community and not meant to be conclusive. As can be seen, cryptocurrency (in the sense of virtual currencies with their own blockchain) is no longer a homogeneous term, which is why I stick to the notion of digital assets in this monograph.

Due to the comparably short data history (e.g., when compared to equities) and small size (i.e., total market capitalization) of digital assets, research in this asset class is still in its early stages. There might also be other reasons for diminished research interest in this topic. When talking to people about bitcoin, ethereum, and other cryptocurrencies,[1] two groups emerge. The first consists of people who believe in the idea of bitcoin (which is decentralization) and its future. The second group, on the other hand, do not take this asset class seriously and think it is about to disappear sooner rather than later. Perhaps there exists a third group who just try to make money out of this (especially hedge funds, which started trading digital assets in 2017). Next to these groups, the popularity and/or acceptance of digital assets moves in cycles. During a time of (relatively) high prices, more people are interested. During distress or low prices, nobody cares. Of course, this relationship could also be the other way round. Figure 1.1 plots the interest in web searches[2] for the term "bitcoin" versus the price of bitcoin[3] (the y-axes are log scales).

The first wave of interest emerged in the years 2013/2014, with relatively low interest and prices in the subsequent years. The second wave started in 2016 and peaked at the end of 2017 with record highs in the price of bitcoin. Prices and interest then tumbled again into 2018. This leads to several questions. Can digital assets, with the flagship asset bitcoin, really be considered a serious asset class? That is not easy to answer since most virtual coins and tokens are tradeable and quoted on different exchanges, but regulation may still not produce meaningful laws or restrictions. Regardless of this, there is a more important question to ask. Since it is possible to trade digital assets, does it really make sense to trade or to invest in these assets?

This is also the central problem of this book. It is not only of interest for private and institutional investors, but also for the research community. Finance research experienced similar issues (short or bad data, errors in variables, theory not in line with empirical results, ...) decades ago, and it appears that history is repeating itself. That opens the opportunity to learn from previous research and apply the findings to recent problems. All of this is described in more detail in the different chapters of this work.

To answer the central question(s) above, it is sometimes helpful to break the problem into several pieces. After some context, which is provided in Chaps. 1

[1] Cryptocurrency is a more specific term for digital assets with their own blockchain. Chapter 5 gives a more detailed overview of the technical aspects and terminology.

[2] See https://trends.google.com/trends/explore?date=today%205-y&q=bitcoin and https://trends.google.com/trends/explore?date=2018-09-30%202020-05-02&q=bitcoin; data obtained on October 8, 2018 and May 3, 2020.

[3] See https://coinmarketcap.com/currencies/bitcoin/historical-data/; data obtained on May 3, 2020.

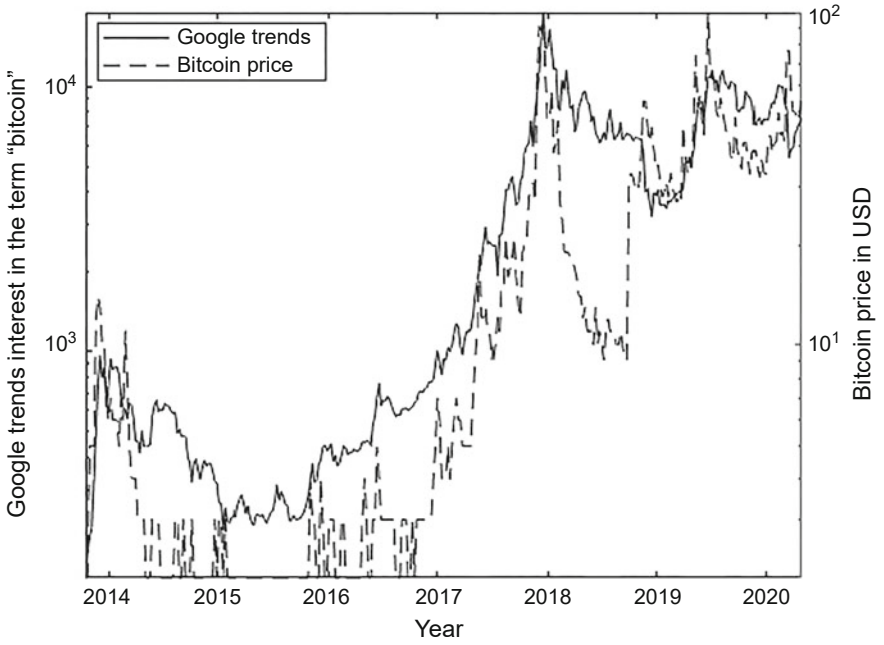


Fig. 1.1 Google trends (left-hand scale) and the price of bitcoin (right-hand scale); both series are on a log scale

to 5, we need to consider whether digital assets add value when compared to traditional asset classes (see Chap. 6). A similar issue was of importance for the hedge fund literature around the turn of the millennium.[4] To investigate possible benefits, when adding a (new) asset (class) to a reference portfolio of other asset(s) (classes), Huberman and Kandel (1987) proposed a "mean–variance spanning" test which is based on the seminal work of Markowitz (1952). The empirical results however, point to only a limited advantage of digital assets when added to a portfolio consisting of traditional asset classes. Therefore, it is important to focus on digital assets exclusively (and no allocation to other (traditional) asset classes), and to investigate other approaches described in the finance literature. Should it be possible to explore or replicate similar effects and comparable findings (like, e.g., investment styles, risk factors, etc.) from other asset classes, digital assets could be considered a serious asset class, and thus profitable investment strategies could be set up.

Due to the long history and wide range of the finance literature, which mainly started following the work of Markowitz (1952), it is not possible to investigate here every stream of the literature with regard to digital assets. However, I aim

[4] See, among others, Agarwal and Naik (2000), Edwards and Liew (1999), Schneeweis and Martin (2001), and Karavas (2000).

to include the most recent or controversial ones. One of the most controversial discussions definitely surrounds the applicability of technical indicators in almost any asset class. Technical indicators find their origin in the theory of Charles Dow, the founder of Dow Jones & Company and the *Wall Street Journal*. Early in the twentieth century, Dow proclaimed that stocks move in trends, as opposed to Bachelier (1900), who posited a random walk of stock prices. Based on this proposition more sophisticated approaches emerged, namely technical analysis. Technical analysis can be split into chart analysis and technical indicators. Chart analysis tries to predict future moves of a financial asset based on (past) chart patterns. That approach, however, is difficult to quantify and consequently others started to develop indicators, which measure the extent to which a stock has moved (e.g., the relative strength index (RSI) or moving average convergence divergence (MACD)). Such indicators can be easily tested and divide the research community: One half believes in the effectiveness of technical indicators and the other half does not. The general opinion is probably tilted toward the refusing side, since it cannot clearly be shown that technical indicators actually work. The same finding also applies to digital assets. In Chap. 7 I show that only a reversal strategy based on Bollinger bands produces meaningful excess returns when compared to a simple buy-and-hold benchmark.

This positive outcome, however, is very similar to another effect described in the literature. (Long-term) reversal was introduced early by De Bondt and Thaler (1985), and Jegadeesh and Titman (1993) also found a short-term reversal effect in US stocks. This effect has already led to another very intensively discussed stream of the literature: asset pricing, risk factors, and characteristics. The pioneers in this field were definitely the two proponents of neo-classical capital market theory, Eugene Fama and Kenneth French. Fama and French (1993) developed a model to explain the cross-section of stock returns. Motivated by their work, hundreds of researchers worldwide found effects (i.e., characteristics) that do not fit into their model.[5] These results are also of interest for this book. Chapter 8 finds that several characteristics appear to drive the returns of digital assets, at least to some extent. These findings also extend the work of Glas (2019). He formulated a first empirically motivated four-factor model for digital assets. Looking at the same issue more deeply, I find evidence for two to three risk factors in Chap. 9. On the other hand, these risk factors are not priced clearly, which leaves the results of Glas (2019) in doubt. It emerges in the same chapter that the development of an asset pricing model specific to digital assets is not yet feasible. However, I find first indications for priced characteristics but not covariances (see Sect. 2.3 for more details).

Digital asset research has thus far produced relatively disappointing results. The main message of these chapters is that almost no findings from traditional asset classes can be applied to digital assets. This also leaves the question unanswered of whether digital assets can be seen as their own asset class. Chapter 10, however,

[5] More information on the history of asset pricing and connected problems can be found in the next chapter.

shows that it does indeed make sense to invest in such virtual coins. Incorporating the main findings of this book into the portfolio construction process yields significant excess returns when compared to a naive benchmark. Moreover, it appears that investment strategies based on characteristics like volume, size, or nominal price can produce significant returns but are priced neither clearly nor significantly. In Chap. 11 I also test extensively for the robustness of the obtained results. It appears that the results do not change for different datasets or periods of time. Consequently, the findings of this book can be deemed robust.

Putting all of these outcomes into context, I contribute to the literature in several ways. I am one of the first to investigate digital assets in more detail. In general, I show that well-known capital market mechanics as observed in traditional asset classes do not apply directly to digital assets (yet). As such, standard asset pricing models from these asset classes do a poor job when tested with digital asset data. Moreover, I find indications for two to three risk factors in the returns of digital assets. These risk factors, on the other hand, are not priced clearly, and therefore the development of an asset pricing model specific to this young asset class is not (yet) feasible. Additionally, I find first indications of priced characteristics, namely volume, size, and nominal price, in digital assets, which could lay the groundwork for future asset pricing studies and models. These characteristics can also be used to construct promising investment styles. However, it needs to be noted that careless investments into this asset class are not recommended since digital assets still suffer from various issues (hacked exchanges, fraud and scam, regulatory uncertainty, . . .). Moreover, I warn against simple usage of digital asset data. Data panels are simply too unbalanced and too short to support meaningful and unequivocal findings.

This book is structured as follows. Chapter 2 describes commonly used methods and asset pricing tests, which are important in understanding the investment and asset pricing literature more deeply. Chapter 3 introduces characteristics and their corresponding investment styles in diverse asset classes. In Chap. 4, I summarize the current state of the academic digital asset literature. Chapter 5 illustrates technical aspects of blockchain and tokens, and includes some stylized facts to allow a deeper understanding of this emerging asset class. Chapter 6 then investigates the allocation of digital assets to a portfolio comprised of traditional asset classes. Since the inclusion of digital assets does not yield much benefit, I investigate the effectiveness of time-series measures like technical indicators in Chap. 7. After some disappointing results, I focus on the cross-section of digital asset returns in Chap. 8 and test several characteristics that are known from other asset classes. Chapter 9 extends this analysis and arrives at three apparently priced characteristics in digital asset returns. In Chap. 10 I bundle the findings from the previous chapters and show how characteristics can be incorporated into the digital asset portfolio construction process. Due to various data issues, I run the main analysis of each chapter again in Chap. 11, using alternative, higher-quality, datasets. Chapter 12 provides some conclusions.

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Chapter 2
An Asset Pricer's Toolkit



Since one of the main motivations of this book is to explore the outcomes of asset pricing methods and different investment styles in digital assets, the following chapters start with more general information, eventually arriving at very narrow and special issues and questions. Therefore, I describe asset pricing and investment styles in general in the beginning, before arriving at the asset class under consideration: digital assets. I aim to answer various questions and describe the results in the subsequent chapters, all dealing with asset pricing and investment styles in connection with digital assets.

The methods used in today's asset pricing and investment styles literature mostly developed during the twentieth century. These quantitative approaches can also be applied to nearly any asset class. Initially, most of the methods were intended to be used for US stock data. Then, the scope was extended to international stock markets and other asset classes like foreign exchange (FX), commodities, or bonds. The latest asset class to which the same methods can be applied are digital assets. Here and in other asset classes, (almost) no theoretical adjustments need to be made to the original concept. This chapter is therefore not necessarily related to digital assets specifically, but provides a methodological background for the analyses used in the subsequent chapters.[1] Since the capital asset pricing model (CAPM) and other models find their roots in the mean–variance optimization introduced by Markowitz (1952, 1959), mean–variance optimization is described in this chapter as well.

[1] For readers not interested in the methodological framework, this chapter and Chap. 3 can be skipped.

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2.1 A Brief Overview of the Evolution of Asset Pricing

The birth of equity and other financial markets ranges back several centuries. The development of the first academic asset pricing model, however, happened about 300 years after the first (stock) exchange(s) were founded in the Benelux countries. Back in the day, there were no such vehicles as stocks or bonds as we know them today. At that time it was common to trade private and public debt issuances. In the 1600s, European shipping companies started doing business with East India and Asia, and thus had a huge financing need. Therefore, these limited liability companies gathered money in the form of stocks. Generally, one stock was valid only for one voyage. If the ship managed to come back to Europe, investors received a small share of the proceeds from the trade: the interest or dividend. If the ship did not come back, the invested money was lost. Later, companies began to merge their operations and stocks were then valid for all voyages of a company and not only for a single expedition. This was the point when the first stocks similar to today's form were born.[2]

It took some more years until the first real stock exchange was formed in London in 1773. However, one of the first major research pieces on equity markets dates back to 1900. In his seminal work, Bachelier (1900) proposed that stock prices follow a random walk, which means any forecast or rule-based investment strategy is deemed to fail. Bachelier's ground-breaking usage of the Wiener process and his later development of diffusion processes laid the foundation for today's option pricing models (Black and Scholes 1973; Merton 1973b). The cornerstone of modern asset pricing theory, however, can be found in the work of Markowitz (1952). He developed an optimization problem that leads to the construction of efficient portfolios. This means that a portfolio on the efficient frontier cannot be dominated by another portfolio in terms of return and a given level of risk, or in terms of risk and a given return. Tobin (1958) added cash or the risk-free rate to the Markowitz model, and showed that the composition of the risky part of a portfolio is independent of an investor's individual risk preference. This concept (which is also called two-fund separation) was further developed by Sharpe (1964), Lintner (1965), and Mossin (1966), who arrived at the capital asset pricing model. The striking point of the CAPM is its proposition of a linear relationship between risk and return. Thereby, it aims to explain the (expected) return of any given asset by its dependence on the market portfolio (which is also called β). An alternative to the CAPM can be found in arbitrage pricing theory (APT). The beta on the market portfolio, which is a measure of systematic risk in the CAPM, is less important for APT. Ross (1976) developed a more general framework, which allows several risk factors at the same time. It also does not define the nature of the risk factors, which is why the factors can vary for different analyses. APT, on the other hand, also assumes systematic dependencies on risk factors in the form of priced factor sensitivities. This linear

[2] See, among others, Petram (2011), Stringham (2015), Boettke (2015), and Goetzmann et al. (2005).

relationship is captured by the slope coefficient of a simple linear regression (mostly ordinary least squares (OLS)) and cannot be diversified. Thus, the lambda of the slope coefficient can be interpreted as risk premium (Huberman and Wang 2008). Another point that APT and CAPM have in common is that both hold in equilibrium situations only. In an empirical test of APT, Roll and Ross (1980) found evidence for three to even four priced risk factors in US equity data. Consequently, the CAPM and APT are the first asset pricing models in the modern finance literature.

The rationale behind the CAPM and other asset pricing models is of specific interest in this chapter.

2.1.1 *Efficient Portfolios*

Markowitz (1952) assumed a risk-averse investor who only cares for the mean and variance of their one-period investment. In his seminal work, Markowitz (1952) described a portfolio optimization problem that minimizes the variance of a portfolio σ_P^2 for a given level of (expected) return μ. More formally, this problem can be read as

$$\min_{w} \sigma_P^2 = \mathbf{w}'\mathbf{V}\mathbf{w} \tag{2.1}$$

subject to the constraints

$$\boldsymbol{\mu}'\mathbf{w} = \mu_P^*, \tag{2.2}$$

$$w_i \geq 0 \qquad \text{for all } i = 1, \ldots, N, \tag{2.3}$$

$$\sum_{i=1}^{N} w_i = 1 \qquad \text{for all } i = 1, \ldots, N, \tag{2.4}$$

where $\mathbf{w}$ is an $N \times 1$ vector of weights, $\mathbf{V}$ is the variance-covariance matrix, and μ_P^* the target return. The efficient frontier is given by every combination of σ_P^2 and μ_P^*, which cannot be dominated by another combination in terms of risk for a given level of return, or in terms of return for a given level of risk. The combination of σ_P^2 and μ_P^* that leads to the lowest variance is called the *minimum-variance portfolio*. The combination resulting in the highest possible return is termed the *maximum-return portfolio*.

2.1.2 *CAPM, APT, and Jensen's Alpha*

Based on Markowitz's initial hypothesis, Sharpe (1964) and Lintner (1965) added more assumptions. One is the agreement about the "true" distribution of returns

or homogeneous expectations. This means that the return distribution is assumed to stay constant over time and markets are frictionless. Another assumption is that investors can borrow and lend funds at the same risk-free interest rate r_f, and that r_f is independent of the specific investor or the face amount of the investment. With all investors agreeing about the return distribution, opportunities for arbitrage, etc. do not exist, and thus any investor sees exactly the same investment opportunity set. As such, by caring only for the mean and variance of a portfolio, every investor invests funds only in a portfolio on the efficient frontier.[3] When adding the possibility of risk-free borrowing and lending, any portfolio becomes a combination of a risky and a risk-free investment. A mean–variance-efficient portfolio thus consists of a tangency portfolio T and a risk-free investment—the so-called Tobin two-fund separation theorem (Tobin 1958). Due to complete agreement, the tangency portfolio is held by every investor and reduces to a market-capitalization-weighted market portfolio M. More formally, the CAPM can be described as

$$E(r_i) = r_f + \frac{\text{cov}(r_i, r_m)}{\sigma^2(r_m)}(E(r_m) - E(r_f)), \tag{2.5}$$

where r_i is the return of asset i, and r_m is the return of the market portfolio M.

Ross (1976) generalized the idea of the CAPM by dropping its assumptions. He assumed that random returns r of i assets can be captured by a simple factor δ:[4]

$$\tilde{r_i} = E(r_i) + \beta_i \tilde{\delta} + \tilde{\epsilon_i}, \tag{2.6}$$

with $\tilde{\delta}$ being a mean-zero common factor and $\tilde{\epsilon_i}$ the mean-zero error term with the vector $\tilde{\epsilon}$; $\tilde{\epsilon}$ is assumed to be satisfactorily independent of the model, such that the law of large numbers holds. From here, a highly diversified arbitrage portfolio η is introduced, which follows the assumption of $\sum_i w_i = 0$ (with w as the weight of an asset i). Equation 2.6 then reduces to

$$\eta\tilde{r} = \eta E(r) + (\eta\beta)\tilde{\delta} + \eta\tilde{\epsilon} \approx \eta E(r) + (\eta\beta)\tilde{\delta}. \tag{2.7}$$

If the arbitrage portfolio η exhibits no systematic risk ($\eta\beta = 0$), $E(r)$ becomes

$$\eta\tilde{r} = \eta E(r). \tag{2.8}$$

To avoid opportunities for arbitrage, $\eta E(r) = 0$ needs to hold. As this restriction must hold for any η ($\eta\epsilon = \eta\beta = 0$), the expected return $E(r)$ is in the span of ϵ and β. This fact can also be described as

$$E(r_i) = \rho + \lambda\beta_i \tag{2.9}$$

[3] Portfolios on the efficient frontier cannot be dominated by another portfolio in terms of risk (and a given level of return) or in terms of return (and a given level of risk).

[4] The following equations are reproduced in a slightly different form from Ross (1976).

with the constants λ and ρ. If the existence of an r_f is true, $\rho = r_f$. Otherwise, ρ is equal to the return of a portfolio with $\beta = 0$. Since APT allows any risk factor, it can also facilitate the CAPM:

$$E(r_i) = \rho + \beta_i(E(r_m) - \rho). \tag{2.10}$$

Having this relationship in mind, it is very easy to measure the success (i.e., excess return or α) of a given investment style/strategy/portfolio with respect to some factor model (in this specific case the CAPM; for more details see Jensen (1968)):

$$r_i - r_f = \alpha_i + \beta_i(r_m - r_f) + \epsilon_i. \tag{2.11}$$

In the case that the model is correct, α_i should equal zero, or at least not be statistically different from zero.

2.1.3 Capital Market Anomalies

Especially after the introduction of the CAPM (and APT), empirical asset pricing uncovered many anomalies that contradict the CAPM. An anomaly or factor in this context can be seen as a systematic (price) behavior that is not explained by the CAPM or other asset pricing models (like the Fama–French models described below). Such an anomaly is usually defined as an investment strategy based on a specific characteristic like, e.g., past return volatility. Harvey et al. (2015) arrived at 316 different factors or characteristics, which have been described in the asset pricing literature between 1962 and 2012. Most of those factors exhibit a statistical significance at least at the 5% level. Not surprisingly, Cochrane (2011) called this a "factor zoo," which needs to be tamed to arrive at "true" risk factors.[5]

A characteristic can itself come in different forms. An accounting-related number or ratio (like book-to-market ratio), past return information (like cumulative returns or return volatility), or some other kind of information (like environmental, social, and/or governance-related data) are just a few examples of possible characteristics. Subrahmanyam (2010) provided an overview of the predictive variables identified by academic research, arriving at four larger clusters:

- Predictors *not* based on ex ante reasoning (e.g., firm size or book-to-market ratio).
- Risk–return model variants (e.g., idiosyncratic risk or macro factors).
- Behavioral biases (e.g., inverse relation between long-run earnings growth forecast and future returns).
- Market frictions (e.g., (il)liquidity or non-synchronicity).

[5] Feng et al. (2020) and Green et al. (2017) were among the first to tame the zoo.

2.1.4 Fama and French Models in Depth

Inspired by the findings of the empirical asset pricing literature, Fama and French (1993) added two more factors to the CAPM: high minus low (HML; value) and small minus big (SMB; size). They connected both factors to the underlying firm-specific properties earnings and profitability, respectively (Fama and French 1995). The value factor dates back to the work of Graham and Dodd (1934) and can be measured by several accounting ratios like, e.g., the price–earnings ratio (Basu 1977) or the book-to-market ratio (Fama and French 1992). Size, on the other hand, is commonly known as a firm's market capitalization (Banz 1981). Size and value effects were extensively documented in the earlier asset pricing literature, but Fama and French (1992, 1995) connected both factors to an underlying economic rationale. They argued that firms with high book-to-market ratios tend to earn less than firms with low book-to-market ratios, which is due to a higher required equity return for high book-to-market firms. This effect also holds four years before and about five years after the measurement of book-to-market equity, according to their analysis. Next to earnings, the profitability of a firm is an important requirement for future success. After controlling for the value effect, Fama and French (1992) showed that small firms still experience lower earnings on assets than big firms. However, Fama and French (1995) found no reasonable explanation for that outcome. This fact might only be an indication to explain the negative relation between firm size and stock returns, and could therefore be seen as a (common) risk factor. The explanatory power of the Fama–French three-factor model surpasses the CAPM, but still does a poor job when other anomalies like net stock issues, accruals, or momentum (Fama and French 2008) are tested. By adding value and size to the market factor, Fama and French (1993) incorporated the findings of the empirical asset pricing literature in a smart move into the neo-classical theory:

$$r_{it} - r_{ft} = \alpha_i + \beta_i (r_{mt} - r_{ft}) + s_i SMB_t + h_i HML_t + \epsilon_{it}. \tag{2.12}$$

Here, *SMB* mimics the size factor and *HML* is a value-factor-mimicking portfolio. About two decades of tests and reviews later, Fama and French (2015) added two more factors to their three-factor model. After the introduction of their first model, it emerged that the Fama and French (1993) model was still unable to explain return variations connected to profitability and investment. Following a closer look at the dividend discount model—see also Gordon (1959) and Eq. 2.13—they argued that profitability and investment are the natural choice:

$$P_t = \sum_{i=1}^{\infty} E(d_{t+i})/(1+r)^i, \tag{2.13}$$

with the stock price of a firm P_t, the expected dividend per share $E(d)$ at time $t+i$, and the risk-adjusted rate of return r. Based on Eq. 2.13, Fama and French (2015) showed that two stocks with the same expected dividend in t but different

stock prices need to have different internal rates of returns. Hence, the firm with the higher stock price has a lower long-term expected return. Taking into account the findings of Miller and Modigliani (1961), the market value MV of a firm can be measured by

$$MV_t = \sum_{i=1}^{\infty} E(Y_{t+i} - \Delta B_{t+i})/(1+r)^i \tag{2.14}$$

with the expected earnings Y and the change in total book equity ΔB. Dividing Eq. 2.14 by B_t yields

$$\frac{MV_t}{B_t} = \frac{1}{B_t} \sum_{i=1}^{\infty} E(Y_{t+i} - \Delta B_{t+i})/(1+r)^i. \tag{2.15}$$

From here, Fama and French (2015) argued that three relations become apparent. First, when fixing everything in Eqs. 2.13–2.15 but MV_t and r in Eq. 2.15, then a higher book-to-market ratio $\frac{B_t}{MV_t}$ results in a higher expected return. Second, when fixing everything but expected earnings and expected return in Eq. 2.15, the positive relation between expected earnings and expected stock return becomes obvious. Finally, the link between investment and expected return can be seen when fixing B_t, MV_t, and expected earnings. A higher Δ of B results in a lower expected return. Incorporating all these findings into another model yields the Fama and French (2015) five-factor model:

$$r_{it} - r_{ft} = \alpha_i + \beta_i(r_{mt} - r_{ft}) + s_i SMB_t + h_i HML_t + m_i RMW_t + c_i CMA_t + \epsilon_{it}, \tag{2.16}$$

where *RMW* is the return spread between portfolios of robust profitability and weak profitability (robust minus weak), and *CMA* is the difference between conservatively (low) and aggressively (high) investing firms (conservative minus aggressive). CAPM, APT, and both Fama–French models are specific to the equities asset class only. When applying these models to other asset classes, or even geographical regions, pricing errors tend to increase (Fama and French 1998; Hou et al. 2011). Of course, additional models have been developed over time.

Nonetheless, the first Fama–French model in particular is considered the state-of-the-art model in academia, and rival asset pricing models with different focuses have emerged over time. For example, Carhart (1997) added momentum, which was found by Jegadeesh and Titman (1993), to the Fama–French three-factor model. Another neo-classical model based on the q theory of investment was introduced by Hou et al. (2014). They incorporated the triple-sorted factors market, size, profitability, and investment into their model. The q-factor model does a much better job, by explaining about half of the 80 capital market anomalies investigated, than the Fama–French models. Another model for stock market anomalies was developed by Stambaugh and Yuan (2016), who described a four-factor model containing two

"mispricing" factors in addition to the market and size factors. All those models based their findings on priced regression coefficients. Such an approach, however, is not shared by every researcher.[6]

A major issue of these models, on the other hand, is their empirical character. This means that after research has uncovered many different anomalies; the same anomalies are used to explain the cross-section of stock returns. Such an approach obviously leads to much disagreement. The "correct" way of formulating an asset pricing model would be to base any factor on a predefined rationale and test it afterwards. Subsequently, claims of data mining, survivorship bias, or simply luck had to be resolved after the introduction of the Fama–French three-factor model. Black (1993) and MacKinlay (1995) were the first to point at data mining issues. This claim was rejected by Barber and Lyon (1997), who found a value premium in financial firms (which were excluded by Fama and French). They also presented evidence that survivorship bias does not affect the size and value premiums, which is also in line with Chan et al. (1995). A third problem mentioned by Kothari et al. (1995) appears to be beta estimation based on monthly data. Investors usually have a longer investment horizon of around one year, which is why annual data should be used for beta estimations. Fama and French (1996) disagreed with this proposition and showed the same inferences for a beta premium based on monthly data as on annual data.

2.1.5 Portfolio Sorts

The construction of risk factors is mostly achieved by constructing different portfolios, also called portfolio sorts. Beforehand, it needs to be mentioned that portfolio sorts qualify for both a time-series and a cross-sectional classification (Cochrane 2011). Sorting stocks into portfolios is a standard procedure in asset pricing nowadays and was introduced in the early 1970s.[7] There are two main reasons to follow such an approach. First, the estimation of "true" (regression) betas (e.g., of single stocks) is not possible, due to the so-called errors in variables problem. The beta normally cannot be predicted, and thus only observed or realized betas can be used. Also, errors in the data can lead to imprecise estimates. Second, the regression residuals tend to have common sources of variation (i.e., they are positively correlated), because of industry or other specific covariation. This leads to a downward bias in the standard errors of the regression slopes within an ordinary least squares regression framework (Fama and French 2004).

The formation of portfolios mostly solves the errors in variables problem, but also reduces the number of assets in a dataset to the number of portfolios formed. Therefore, it is important to sort the stocks in the investment opportunity set in

[6] See Sect. 2.3 for more details.

[7] See, e.g., Black et al. (1972), Blume and Friend (1973), and Basu (1977).

a certain direction. This means, when constructing portfolios of high-beta to low-beta stocks, significant differences between the portfolios of varying betas should emerge. Hence, portfolio sorts often involve the identification of terciles, quintiles, or deciles of a specific characteristic (like beta, size, or value) to form the portfolios. Based on the hedge portfolio returns (i.e., top percentile average returns minus bottom percentile average returns), dependencies between the characteristic itself and expected returns can be observed.

In general, Cochrane (2011) described portfolio sorts as univariate non-parametric cross-sectional regressions with non-overlapping histogram weights. Thus, portfolio sorts are capable of dealing with non-linearity. This claim was illustrated in more detail by Berk (2000), who additionally argued that the variance of the data panel is reduced when applying portfolio sorts. This then yields a misstated explanatory power (i.e., R^2) of a factor model when testing the portfolio returns.

There are several ways to conduct portfolio sorts. The most simplistic is described above. Assets are sorted into different portfolios based on a common characteristic. This means that when sorting assets into quintiles, five portfolios are considered. Assets with the most distinct characteristic are placed in the top quintile, whereas assets with the least distinct characteristic are sorted in the bottom quintile. The same applies to the portfolios/quintiles in the middle. Assets within each quintile can be equally weighted or market cap weighted; other weighting approaches can also be considered. To investigate the specific effect or characteristic, the return spread of the top and the bottom portfolios is of interest. Thus, the returns are calculated in $t+1$ based on the weighted portfolio constituents in t. Averaging the returns over the whole sample period and calculating the return difference of top–bottom portfolios results in the average return of the investigated characteristic. This approach can therefore also be called a single sort, or in some other cases also investment style.

In addition to single sorts, it is possible to sort for two or even three characteristics. We therefore need to distinguish between dependent and independent sorts. Independent double sorts sort for two characteristics *independently*. This means that conclusions are drawn from the intersections of the independent sorts. Fama and French also applied independent double sorts to construct factor-mimicking portfolios. Fama and French (1993) formed six portfolios based on all stocks from the three US American stock exchanges Amex, NYSE, and NASDAQ. They double sorted the factor-mimicking portfolios of value and size since Fama and French (1992) found a (weak) linkage between these effects. Both, however, are also standalone effects and need to be included in the asset pricing model. By controlling for the other effect respectively, multicollinearity is avoided and both single effects can be included in the model. Thus, the *SMB* factor can be obtained via

$$SMB = \tfrac{1}{3}(\text{small value} + \text{small neutral} + \text{small growth}) - \tfrac{1}{3}(\text{big value} + \text{big neutral} + \text{big growth}). \quad (2.17)$$

The *HML* factor is constructed similarly:

$$HML = \tfrac{1}{2}(\text{small value} + \text{big value}) - \tfrac{1}{2}(\text{small growth} + \text{big growth}). \tag{2.18}$$

The reason for sorting *SMB* into three value groups but *HML* only in two size groups follows the results of Fama and French (1992), who found a more pronounced value than size effect, and hence size should be controlled for through three value groups. A robustness test, however, was not part of their work.

Conditional or dependent double sorts, on the other hand, represent the return spread of a given characteristic *dependent* on the first sort (Fama and French 2008). Based on the example above, illustrating dependent double sorts more formally yields

$$dHML = \text{small value} - \text{small growth}. \tag{2.19}$$

In this case, *dHML* is the return spread of value and growth stocks conditional on firm size (here, small stocks). The same analysis could also be conducted for medium-sized or large companies and can be applied to any other characteristic. The same methodology can include three or even more characteristics. Such an approach, however, would decrease the number of assets within each portfolio tremendously, and thus limits the information content obtained from such an analysis.

Applying double or even triple sorts still cannot avoid one severe issue. The asset pricing literature uncovered several hundred characteristics, which appear to explain average stock returns. For analyses involving more variables, other approaches than portfolio sorts must be used.

2.2 Another Perspective: Cross-Sectional Regressions

Therefore, researchers switched from time-series to cross-sectional regressions. Cross-sectional regressions can take into account several variables or characteristics at once. I present two approaches below. The first was developed by Fama and MacBeth (1973). Their approach works relatively well for up to five different characteristics. To consider more characteristics, Cochrane (2011) demanded a panel regression approach in future asset pricing studies.

2.2.1 Fama–MacBeth Regressions

In their work, Fama and MacBeth (1973) aimed to investigate the relationship between average returns and risk based on a two-parameter portfolio model. This model goes back to the works of Tobin (1958), Markowitz (1959), and Fama (1965),

and assumes perfect markets, normally distributed stock returns, and risk-averse investors who maximize their expected utility. Thus, the efficient portfolio for any investor may not experience lower risk for a given or higher expected return. Based on this proposition, Fama and MacBeth (1973) proposed a cross-sectional regression approach to show that US American stocks are indeed "efficient."

Their approach is relatively straightforward and consists of two steps that aim to determine if the betas with regard to a common factor are priced significantly in the underlying dataset. Therefore, the first step is to estimate the rolling betas for any given stock or asset with respect to a common factor or several factors. The estimation window would normally be from 36 to 120 observations (i.e., months). The respective factor model could be one of the models described above, for example. This first step, however, is only of importance for studies in line with the covariance approach. For the characteristics approach, betas do not need to be estimated and the characteristic(s) itself (themselves) can be applied to the second step.[8] In the second step, Fama and MacBeth (1973) switched from the time dimension to the cross-section. Thus, at every point t, the betas or characteristics are used as the independent variable and the stock returns as the dependent variable. Hence, every point t yields a coefficient with the respective t- and p-value, standard error, and all other test statistics. To arrive at one final result, these values are simply averaged over every t. Formally, the Fama and MacBeth (1973) regression can be described by

$$r_{it+1} = \lambda_{i0} + \lambda_1 d_{1it} + \cdots + \lambda_k d_{kit} + \epsilon_{it}, \tag{2.20}$$

with r_{it+1} being the return of asset i in $t + 1$, d_{kit} being the beta or characteristic under consideration and ϵ the error term. The t-statistic is calculated as

$$t_{\lambda_k} = \sqrt{N}\frac{\bar{\lambda_k}}{\sigma_{\lambda_k}}, \tag{2.21}$$

with $\bar{\lambda}$ the average of the lambdas; $\sigma_{\lambda_k}/\sqrt{N}$ is the standard error of the average lambdas.

2.2.2 OLS Panel Regressions

The Fama and MacBeth (1973) regression has two serious drawbacks: It does not take into account the variation over time of one specific variable, and it cannot cope with higher dimensionality, i.e., more than around five characteristics. Cochrane (2011) described the overarching problem very simply. Researchers mostly aim to

[8] More details follow in Sect. 2.3.

describe expected returns $E(r)$ as a function of different characteristics X,

$$E(r_{t+1} \mid X_t), \tag{2.22}$$

with X being a vector containing all (*relevant*) characteristics like momentum, size, value, and others (see Green et al. (2017) for the first identification of independent characteristics in US stocks). To find the most important variables of the known ones, Cochrane (2011) proposed a simple panel-data regression approach:

$$r_{it+1} = \alpha_i + X_{it}\beta_i' + \epsilon_{it}. \tag{2.23}$$

This simple and standard panel-data regression framework allows several different models and a large number of characteristics. Here, i again corresponds to assets and t denotes time. The stacked variables X and the error term ϵ are assumed to be independent. A simpler notation is[9]

$$Y = X\beta + \epsilon, \tag{2.24}$$

where the coefficient β can be obtained via the standard OLS formula

$$\hat{\beta} = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}Y. \tag{2.25}$$

The asymptotic variance of β is given by

$$AVar[\hat{\beta}_{\mathrm{OLS}} - \beta] = \frac{1}{N}(T\sigma_X^2\sigma_\epsilon^2)(T\sigma_X^2)^{-2} = \frac{\sigma_\epsilon^2}{\sigma_X^2 NT}. \tag{2.26}$$

The most general model might be a pooled panel-data regression. It allows for heterogeneity over time and in the cross-section. However, it can lead to a systematic bias in the estimated parameters. Therefore, two other methods can be considered. The first is the random effects model. It assumes no correlation between (unobserved) firm specific effects and the explanatory variables. Since that assumption does not hold for nearly any asset pricing study, this model is not frequently used in the related literature (Petersen 2008). Another possibility is to run a fixed effects model. A fixed effect describes a firm related effect γ that exists in the data, i.e., the residuals ϵ contain an idiosyncratic effect ζ as well as the firm effect:

$$\epsilon_{it} = \gamma_i + \zeta_{it}. \tag{2.27}$$

[9] The equations below are reproduced in a slightly adjusted form from Petersen (2008) and Hildebrandt (2009).

Petersen (2008) also assumed such a firm specific effect μ in X:

$$X_{it} = \mu_i + \nu_{it}. \tag{2.28}$$

Here, γ, ζ, μ, and ν are independent of each other with zero mean and finite variance. Petersen (2008) also showed that the independent variable(s) and the residuals are uncorrelated across firms, but are correlated over time for the same firm. Having this relationship in mind, the fixed effects standard error is given by

$$\begin{aligned} AVar_{\text{fixed}}[\hat{\beta}_{\text{OLS}} - \beta] &= \frac{1}{N}\big(T\sigma_X^2\sigma_\epsilon^2 + T(T-1)\rho_X\sigma_X^2\rho_\epsilon\sigma_\epsilon^2\big)(T\sigma_X^2)^{-2} \\ &= \frac{\sigma_\epsilon^2}{\sigma_X^2 NT}(1 + (T-1)\rho_X\rho_\epsilon). \end{aligned} \tag{2.29}$$

Within-cluster residual correlations, Petersen (2008) recommended the use of cluster-robust standard errors. In that case, the standard error is adjusted by estimating the covariance between the within-cluster correlations:

$$AVar_{\text{cluster}}(\beta) = \frac{N(NT-1)\sum_{i=1}^{N}\big(\sum_{t=1}^{T} X_{it}\epsilon_{it}\big)^2}{(NT-k)(N-1)\big(\sum_{i=1}^{N}\sum_{t=1}^{T} X_{it}^2\big)^2}. \tag{2.30}$$

Fama–MacBeth and panel-data regressions can also be applied the other way round to identify significant variables out of many potential ones. Starting with univariate regressions, a full enumeration, i.e., every possible combination of variables or characteristics $\left(\frac{n!}{k!(n-k)!}\right)$ can lead to similar results. In another approach—a forward algorithm—univariate regressions are estimated in the first place. After that, a combination of the identified significant variable and any other remaining variable is investigated. Thus, the second step is a multivariate regression consisting of two independent variables (a significant one from the first step and every combination with the remaining variables). In the third step, combinations with two significant variables from the second step are included in a multivariate regression model, with a third variable being added out of the remaining ones. Every other variable is again being added to the two significant ones. This approach is repeated as long as no new variable can be added. Sometimes the combination of independent variables is of interest to explain the cross-section of returns. A full enumeration or a forward algorithm are suitable approaches to achieve this.

In the subsequent chapters I also apply the models described above to digital assets. While the theory holds for this asset class too, the results are rather disappointing. Standard asset pricing models cannot explain the returns of virtual currencies, and characteristics are only priced to a small extent—see Glas (2019) and Chap. 9. As shown by Harvey et al. (2015), many anomalies, characteristics, or even factors have been described in the literature. Since a test of all of these studies on digital assets is (i) not reasonable for digital assets and (ii) would go beyond the

scope of this book, I aim to test only the most robust characteristics in the form of investment styles and other tests.

2.3 Covariances Versus Characteristics

In early tests of the CAPM,[10] it emerged that the relationship between return and beta is too flat and that the intercept is higher than the risk-free rate. This leads to two diverging explanations of why the proposition of the CAPM cannot be observed in empirical data. On the one hand, calls for more complex asset pricing models were made, which resulted in the intertemporal CAPM (ICAPM), developed by Merton (1973a). In the ICAPM, an agent not only cares for their end-of-period payoff but also for opportunities of consumption and covariation with macroeconomic state variables. Other explanations can be found in De Bondt and Thaler (1987), Lakonishok et al. (1994), and Haugen (1995). These behavioralists explained the too-flat relationship by investor over- and underreaction. This means that investors assume a further extension of good times (i.e., positive past stock returns) or bad times (i.e., negative past stock returns), respectively. That results in stocks that are not fairly priced, and eventually corrects itself. If that is the case, value stocks appear to exhibit high past returns and growth stocks appear to have experienced lower past returns.[11]

Daniel and Titman (1997) explored this issue more deeply and found a priced risk premium for the characteristic itself but not for the slope coefficient. This claim would lead to the rejection of the Fama–French three-factor model since stock returns could be better explained by the characteristic (characteristics model) than by the covariance structure (risk model). This finding was confirmed for other geographical regions like Japan (Daniel et al. 2001) and Germany (Fieberg et al. 2016b). For France, the risk model seems to work better (Lajili-Jarjir 2007). Davis et al. (2000) replicated the work of Daniel and Titman (1997) with a longer sample period and arrived at the exactly opposite result of a better risk model when compared to the characteristics model. The choice between the two approaches, on the other hand, is not always trivial. While the characteristics model allows behavioral finance explanations, the risk model requires a lot of empirical work. In a horse race based on the work of Brandt et al. (2009), Fieberg et al. (2016a) showed better performance of the characteristics model versus the risk model in an investment context. In a mean–variance framework, using covariances rather than characteristics reduces portfolio volatility more (Kim 2015). The characteristics model is also used to evaluate the performance of mutual funds. Daniel et al. (1997)

[10] See Douglas (1968), Black et al. (1972), Merton and Scholes (1972), and Blume and Friend (1973).

[11] The CAPM assumes an inverse relationship with high returns for growth stocks and low returns for value stocks.

showed some success by mutual fund managers selecting the best stocks within a characteristic cluster, but found no timing ability. Up to now, there has been no consensus concerning which of the models is more accurate. Chapter 9 of this work provides indications of priced characteristics in digital asset returns and thus supports the characteristics model.

Other chapters also deal with this issue, at least indirectly. While Chap. 8 shows that investment styles can hardly be found in digital assets, Chap. 9 arrives at first indications of priced characteristics within this emerging asset class, but still no priced covariances. Chapter 10 utilizes these findings in an investment context.

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Chapter 3
Investment Styles in General



The above-mentioned models and methods have been used extensively in the asset pricing and investment styles literature. By doing so, research has found that some characteristics achieve similar good results (i.e., positive excess returns, low correlations to long-only strategies, etc.) across different asset classes. If those characteristics are robust over asset classes like equities, fixed income, commodities, and others, it could be assumed they can be applied to digital assets in the same manner. Such an application is investigated more deeply from Chap. 6 onward, but beforehand I aim to portray the most important characteristics for each asset class. Since it is not yet clear if digital assets can be considered as their own asset class, I describe characteristics in popular asset classes below to draw connections in subsequent chapters.

3.1 Equities

While there is evidence of anomalies evaporating after publication in the USA only (Jacobs and Müller 2020; McLean and Pontiff 2016), some (risk) factors can be proxied by several characteristics. Asness et al. (2015) argued that only the three factors (proxied by the characteristics) momentum, value, and defensive (and to some extent illiquidity) are robust over different asset classes and periods of time, and are liquid enough for large institutional portfolios.[1] The term "factor" in this context is commonly used in the asset pricing literature, whereas "style" is mostly connected with the investment literature.[2] An investment style can usually be described by any rule-based investment strategy (which often employs portfolio

[1] Asness et al. (2015) also mentioned carry and size as robust investment styles for asset classes like fixed income, foreign exchange, or commodities.

[2] The difference between these terms is described in Sects. 2.1 and 2.2.

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sorts based on characteristics) to come up with investment decisions. McLean and Pontiff (2016) defined four categories of possible characteristics, which are reproduced in Table 3.1. The category "event" is not investable on an ongoing basis, since these strategies are always dependent on the event taking place. This means an IPO for example normally happens once in the lifetime of a company and not thereafter. This is why Asness et al. (2015) focused on easy-to-implement scalable styles, which are mainly included in the two categories market and valuation. The styles momentum, defensive, value, and illiquidity can also be proxied by several characteristics and are just umbrella terms for a specific effect.

Momentum assumes a higher expected return for firms that performed better in the past relative to the remainder of the investment universe. It is commonly calculated as the 2–12 months return, as described by Jegadeesh and Titman (1993). Here, the most recent month is skipped due to a short-term reversal effect. Other proxies are 2–6 months returns, long-term momentum, industry momentum (Moskowitz and Grinblatt 1999), characteristics momentum (Lewellen 2002), Sharpe ratio, and to some extent technical indicators (George and Hwang 2004).

Defensive is also classified as a market anomaly by McLean and Pontiff (2016). It stands in contrast to the CAPM, which assumes a linear relationship between the systematic risk of an asset and its return. Defensive, on the other hand, arrives at the conclusion of the same or even higher returns for low-risk assets. It is often described by the beta of an asset over different time horizons with respect to some benchmark (Black et al. 1972). The volatility realized over different periods of time is another proxy for defensive (Blitz and van Vliet 2007). A lookback period of 12, 24, 36, or up to 60 months is mostly taken into consideration when talking about this investment style.

Next to momentum and defensive, the longest-known style might be value. It had already emerged with the work of Graham and Dodd (1934), who came up with different valuation methodologies for companies/stocks to see if it is favorable to buy or sell a specific firm. Thus, value tries to identify over- and undervalued stocks with regard to a given investment opportunity set. Value is also part of the Fama–French three-factor model and has been extensively researched. Thus, the ways of calculating value are manifold. Most of the variables defined in the valuation column of Table 3.1 can be used. The best-known proxy, however, might be the book-to-market ratio (Fama and French 1992). Other possible variables are dividend yield (Naranjo et al. 1998), cash flow/market value (Lakonishok et al. 1994), earnings-to-price (Basu 1977), and sometimes also long-term reversal (De Bondt and Thaler 1985).

Illiquidity, on the other hand, is a compensation for holding relatively illiquid stocks or assets compared to very liquid ones. Illiquidity was first proposed by Amihud and Mendelson (1986), who found a relationship between the return of a stock and its liquidity. It was then described more formally by Amihud (2002), whose newly developed measure is another prominent proxy for illiquidity. The calculation of this style is not trivial and sometimes requires daily instead of monthly data. It can be calculated as the sum of the absolute price change per dollar of trading volume over the last 12 months, standardized by the number of observations with

Table 3.1 Investment style categories defined by McLean and Pontiff (2016)

Event	Market	Valuation	Fundamental
Change in asset turnover	52-week high	Advertising/MV	Accruals
Change in profit margin	Age-momentum	Analyst value	Age
Change in recommendation	Amihud's measure	Book-to-market	Asset growth
Change in forecast & accrual	Beta	Cash flow/MV	Asset turnover
Debt issuance	Bid–ask spread	Dividends	Cash flow variance
Dividend initiation	Co-skewness	Earnings-to-price	Earnings consistency
Dividend omission	Idiosyncratic risk	Enterprise component of B/P	Forecast dispersion
Dividends	Industry momentum	Enterprise multiple	G index
Down forecast	Lagged momentum	Leverage component of B/P	Gross profitability
Exchange switch	Long-term reversal	Marketing/MV	G-score
Growth in inventory	Max	Org. capital	G-score 2
Growth in LTNOA	Momentum	R&D/MV	Herfindahl
IPO	Momentum and long-term reversal	Sales/price	Investment
IPO + age	Momentum ratings		Leverage
IPO no R&D	Momentum reversal		M&B accruals
Mergers	Price		NOA
Post earnings drift	Seasonality		Operating leverage
R&D increases	Short interest		O-score
Ratings downgrades	Short-term reversal		Pension funding
Repurchases	Size		Percent operating accrual
Revenue surprises	Volume		Percent total accrual

(continued)

Table 3.1 (continued)

Event	Market	Valuation	Fundamental
SEOs	Volume trend		Profit margin
Share issuance (1 year)	Volume variance		Profitability
Share issuance (5 year)	Volume momentum		ROE
Spin-offs	Volume/MV		Sales growth
Sustainable growth			Tax
Total external finance			Z-score
Up forecast			
Δcapex − Δindustry capex			
Δnon-current op. assets			
Δsales − Δinventory			
Δsales − ΔSG&A			
Δworking capital			

This table is reproduced from the appendix of McLean and Pontiff (2016)

available data. Another way to measure illiquidity is simply to look at turnover data (Datar et al. 1998).

These measures and proxies alone are not yet a final investment style, though. Such a style can be implemented in different ways. The literature commonly involves two types of portfolios, one with the best or most distinct characteristic and the other with the least distinct characteristic. Therefore, the long portfolio includes the stocks with the desired characteristic. To arrive at a market- or dollar-neutral portfolio that solely measures the effect of interest, a second portfolio is needed. This short portfolio short sells the stocks with the least distinct characteristic. The classification of "(non)-distinct" stocks can be found via portfolio sorts. Stocks are sorted into different percentiles based on their specific characteristic. The top and the bottom percentiles then serve as long and short portfolios, respectively. Alternatively, more sophisticated approaches like characteristics-based optimization (Brandt et al. 2009; Fieberg et al. 2016a) can be taken into consideration.

Since investment styles usually consider a single characteristic, it might be beneficial to consider more than one characteristic at the same time. Thus, the opinion emerged that it is advantageous to integrate investment styles (i.e., an integration of (several) styles directly into the allocation process, like characteristic-based optimization) compared to a mix of different investment styles (a simple combination of single styles). Integration was proposed by Clarke et al. (2016), Bender and Wang (2016), and Fitzgibbons et al. (2017). Israel et al. (2017) even saw the "correct" or best implementation of investment styles as a new source of excess returns (i.e., alpha). Leippold and Rueegg (2017), however, called the dominance of the integrated approach a "statistical fluke." Fernandez-Perez et al. (2019) also arrived at the conclusion that an integration is not superior to the $\frac{1}{N}$ rule.

So far, almost every study mentioned above has focused on equity market data only. Investment styles, however, also generate sufficient results in other asset classes. This is explicitly true for bonds, commodities, FX, and even investment strategies, and might be an indication that these styles could work in digital assets, too.

3.2 Commodities

Commodity trading advisers (CTAs), who are by nature trend-following investors, rely heavily on the momentum effect in commodities and FX. Such an effect was shown to exist by Miffre and Rallis (2007) and Shen et al. (2007). A short-term reversal effect as in equities, however, generates negative returns. Chaves and Viswanathan (2016), on the other hand, showed that momentum strategies work especially well in future but not spot markets, and mean reversion yields positive results in spot, but not future markets. Since market dynamics vary between different asset classes, other fundamentals tend to influence overall commodity market movements. Therefore, Fuertes et al. (2015) included term structure and

idiosyncratic volatility next to momentum in a triple-screen investment strategy, which outperforms the S&P GSCI, a well-known commodity index.

Term structure mostly describes the price difference of two future contracts of the same commodity with differing maturities. This difference or delta can be decomposed into the carry and the expected price appreciation. In general, the term structure can have three forms: It is either flat, in contango, or in backwardation. Contango means a higher price for long-term maturities than for short-term maturities. Backwardation describes the opposite relationship. The term structure of commodity markets can even be used to predict the returns of equity markets or the business cycle (Fernandez-Perez et al. 2016a). When not accounting for contango and backwardation, Fernandez-Perez et al. (2016b) found a strong past idiosyncratic volatility effect as an explanatory variable for commodity returns. When controlling for the term structure in asset pricing models, this effect disappears. Basu and Miffre (2013) constructed a successful investment strategy based on the hedging pressure (as measured by open interest) in commodity markets. Fernandez-Perez et al. (2018) also constructed a hedge portfolio based on skewness, which outperformed different commodity asset pricing models. Another study (Lubnau and Todorova 2015) combined mean-reverting calendar spreads with Bollinger bands, a technical indicator, to challenge the proposition of weak-form efficient energy markets. Another technical indicator, namely a simple moving average, was shown to be useful when constructing market-timing strategies in commodity markets (Han et al. 2016). A phenomenon that might one day be of interest for digital assets is the so-called financialization of commodity markets. With ever-increasing inflows in the asset class commodities, (stronger) co-movements of commodities and other asset classes like equities or bonds have developed.[3]

3.3 FX

An asset class that is often perceived to be close to digital assets or crypto*currencies* is foreign exchange rates. FX returns can also largely be explained by term structure and carry (Fama 1984; Lustig et al. 2011). Carry is the dominant characteristic when talking about FX markets. It describes the difference between two interest rates of different countries or regions. Hence, investors exploit this spread by borrowing funds in the currency with lower interest rates and investing the proceeds in the currency with higher interest rates. Assuming a constant spot price should result in a risk-less profit. The returns of the carry factor, however, are lower during times of global recession. This finding implies that the USD is also a measure of worldwide risk, since it is the main funding currency for some developed as well as emerging market countries (Koijen et al. 2018). There could also be a parallel with

[3] See, among others, Adams and Glück (2015), Fernandez-Perez et al. (2016a), and Prokopczuk et al. (2017).

bitcoin, which is the most prominent and largest digital currency (as measured by market capitalization) in the digital asset space. Such a relationship, however, is not investigated in this book.

Other well-known characteristics like momentum[4] or value[5] yield similar results in FX markets when compared to other traditional asset classes. Technical analysis is also often applied to currencies (Menkhoff and Taylor 2007) and appears to be profitable as well. Those profits, however, disappear after traders learn about their success.[6] Another more "exotic" characteristic in FX markets is seasonality, where assets are chosen based on historical same-calendar-month returns. Seasonality appears to exist in FX, too, but is harder to exploit than in other asset classes.[7]

3.4 Fixed Income

Investment styles in one of the largest and most important asset classes worldwide have not been researched as deeply as in the asset classes described above. Research in fixed income, which includes diverse financial vehicles like corporate and governmental bonds, convertible and contingent convertible (CoCo) bonds, or credit default swaps (CDS) only began in more depth recently. The concept of term structure and carry can be applied to fixed income markets in a similar manner. In this asset class, carry can be seen as the yield on a bond. Term structure is also called "roll down" of a bond, which is the change in price by rolling down (or up) the yield curve until maturity (Bieri and Chincarini 2005). Most investment styles like momentum, value, or defensive are not special to equities and can be applied to fixed income as well (Asness et al. 2013, 2015).

3.4.1 Corporate and Government Bonds

Corporate bonds are debt obligations issued by a company. Government bonds, on the other hand, are issued by the authorities of a country. Investors in general demand a risk premium for holding risks associated with economic growth and default rates (Asvanunt and Richardson 2016), which is also called the credit risk premium. Israel et al. (2018) and Brooks et al. (2018) documented the existence of four investment styles—carry, defensive, momentum, and value—in a cross-section

[4] See, e.g., Okunev and White (2003), Menkhoff et al. (2012), and Moskowitz et al. (2012).

[5] See Kroencke et al. (2013) and Asness et al. (2013).

[6] See Pukthuanthong-Le et al. (2007), Neely et al. (2009), and others.

[7] See, among others, Tse (2017) for FX seasonality; Lucey (2010) for commodities seasonality; and Blume and Stambaugh (1983), Alves (2014), and Keloharju et al. (2016) for seasonality in stock markets.

of corporate bond returns. Man-Kit (2014) similarly found evidence of a low-risk (i.e., defensive) anomaly, and Houweling and van Zundert (2017) documented size, defensive, value, and momentum effects in corporate bonds. The value effect was documented by Correia et al. (2012) as well, and momentum in corporate bonds is further supported by Jostova et al. (2013). An illiquidity premium seems not to exist (Richardson and Palhares 2019). Others (Chordia et al. 2017; Haesen et al. 2017) focus on the application of equity-related styles, which contain information for bond markets. Also, the covariances versus characteristics story is of interest for Gebhardt et al. (2005), who arrived at better results for the risk model.

Government bonds are even less well researched than corporate bonds. The only study (to my knowledge) investigating solely government bonds is Brooks and Moskowitz (2017). They found that the value characteristic of government bonds can explain a risk factor connected to the level of the yield curve. Momentum and carry, on the other hand, explain the slope and the curvature of the yield curve. Asness et al. (2013) included government bonds in a multi-asset evaluation, as did Asness et al. (2015), who included global government bond futures in their study.

3.4.2 CDS

To my knowledge, other niche asset classes like convertible bonds have not yet been subject to studies explicitly involving investment styles. CDS markets, which provide insurance in the case of the default of a bond issuer, have already been investigated.

The most striking finding for CDS markets is the divergence from the yield of the underlying corporate bond (which is also referred to as the basis). Normally, the price (i.e., yield or spread) of a CDS should be equal to (or higher than) the yield of the underlying bond (Duffie 1999). Such a deviation can be explained by the lead of CDS markets over credit spreads in the price discovery process (Alexopoulou et al. 2009; Blanco et al. 2005; Zhu 2006). Nashikkar et al. (2011) found liquidity from both bond and CDS markets to be an explanatory variable for the basis. Bai and Collin-Dufresne (2018) added funding costs, counterparty risk, and collateral quality to the explanatory variables for the CDS-bond basis. Vice versa, the basis can be used to construct a basis factor that cannot be explained by other bond-specific risk factors (Kim et al. 2017). Next to the basis, only momentum has been investigated, by a non-peer-reviewed working paper. Lee et al. (2021) found a significant momentum effect in CDS markets based on equity momentum signals and CDS-only signals. To my knowledge, investment styles like value, defensive, or size have not yet been part of an academic study focusing on CDS markets.

3.5 Investment Strategies

Based on all the above-mentioned investment styles, a second form of investment style can be introduced. Here, the investment universe is not comprised of assets from a specific asset class but of investment strategies/investment styles from one or several asset classes.

3.5.1 Characteristics Momentum

Lewellen (2002) was the first to document characteristics momentum in US stock returns. Characteristics momentum is very similar to the momentum effect as known from equity markets. Here, Lewellen (2002) used an underlying investment opportunity set of investment strategies based on different characteristics. He calculated the past returns of these characteristics/strategies, which were used to form the characteristics momentum investment style. The main difference is therefore in the data used, as no single assets are considered, but investment strategies based on single assets. Therefore, a more general term for such a second-layer investment strategy could be "style of styles."

Barberis and Shleifer (2003), shortly after, introduced a theoretical framework for such a style of styles. Since then, several others (but still a comparably low number of researchers) have contributed to this stream of the literature. Since style of styles is not limited to US characteristics only, I include most of the existing studies regarding style of styles in Table 3.2. As can be seen there, most studies focus on US stock data and investigate style momentum only. Most of these studies arrived at the conclusion that style momentum is able to form a successful second-layer investment strategy. In many studies, this effect can be explained by the classical momentum effect.

Another (unpublished) working paper, which I wrote in collaboration with Prof. Dr. Thorsten Poddig and PD Dr. Christian Fieberg, aims at exactly this stream of the literature (Glas et al. 2019). Since the scope of the studies described above is somewhat limited, I aimed to broaden the style of styles literature by investigating two more second-layer investment styles. Glas et al. (2019) investigated momentum, value, defensive, and other style of styles proxies based on a large European data sample. They investigated 110 European characteristic portfolios and arrived at results that are in line with the current literature. Only style momentum is a pronounced style of styles in European equities. Defensive appears to exist, too, but in a weaker form. On the other hand, they only found a little evidence for a style value effect.

Table 3.2 Literature on style of styles

Authors	Used data	Style characteristics
Lewellen (2002)	US stocks	Momentum
Barberis and Shleifer (2003)	—	Theoretical framework
Chen (2003)	US stocks	Momentum
Chen and Bondt (2004)	S&P500 stocks	Momentum
Teo and Woo (2004)	S&P500 stocks, mutual funds	Momentum, reversal
Aarts and Lehnert (2005)	UK stocks	Momentum
Peltomäki and Peni (2009)	US stocks	Momentum
Clare et al. (2010)	UK stocks	Momentum
Peltomäki and Peni (2010)	Hedge funds, US stocks	Momentum
Chen et al. (2012)	US stocks	Momentum
Kim (2012)	Multi asset	Momentum
Chao et al. (2012)	International stocks	Momentum
Zaremba and Szyszka (2016)	Polish stocks	Momentum
Zaremba and Shemer (2017)	International stocks	Momentum
Zaremba (2018)	International stocks	Momentum
Gupta and Kelly (2019)	International stocks	Momentum
Glas et al. (2019)	European stocks	Momentum, value, defensive

This table contains most if not all the studies concerning style of styles at the time of writing

3.5.2 Characteristics Mix and Integration

Next to characteristics momentum, a mix or an integration can also be considered when taking into account several characteristics at the same time. It is thereby possible to incorporate the characteristic itself or the covariances. Although Daniel and Titman (1997) found priced characteristics but not covariances in average stock returns, Davis et al. (2000), on the other hand, claimed that the sample size used by Daniel and Titman (1997) was too short for meaningful results. Meanwhile, several other studies found evidence for the characteristics model in other countries, and for other characteristics (Fieberg et al. 2016b; Grundy and Martin 2001; Hirshleifer et al. 2012; Hou et al. 2011).

The discussion of covariances versus characteristics also carries some serious implications for practitioners. Lin and Zhang (2013) and Fieberg et al. (2016a) showed that incorporating characteristics (as opposed to covariances) into the investment process leads to more favorable results. This can be achieved by several means. The easiest way to incorporate characteristics into an investor's portfolio is to build factor portfolios.[8] A factor portfolio is long only on assets with the strongest desired characteristic. For example, a combination of standalone long-

[8] Factor portfolios in this context should not be mistaken for factor-mimicking portfolios. A "factor portfolio" is a more practical term for a portfolio tilted toward a desired characteristic, and does not involve shorting of assets, as opposed to the classical style construction approach.

only portfolios consisting of the strongest turnover (here low turnover), (low) prices, and e.g. (low) market capitalizations would be termed a mix approach. Since one asset could be part of several (long-only) characteristic portfolios and therefore have an increased weight in the overall portfolio, the integrated approach emerged. Integration, however, is even more important for long–short portfolios, since the same asset could be in the long part of one characteristic and the short part of another characteristic, and subsequently offset its exposures.

Integration can be achieved in several different ways. Fitzgibbons et al. (2017) used several styles to arrive at an expected return forecast for each stock. Then, a portfolio was formed based on the expected returns. Clarke et al. (2016) applied factor betas to determine the portfolio weights for each asset. Bender and Wang (2016) constructed an integrated portfolio by determining an average rank over several characteristics. This average rank was then multiplied by an initial portfolio weight. In the last step, the portfolio weights were rescaled to add up to 100%. This approach carries two benefits. First, the mix and the integrated portfolio construction process are directly comparable. Second, the integrated approach is straightforward and easy to comprehend. Bender and Wang (2016) also showed that an optimization framework involving characteristics cannot easily be compared to the former two approaches, since portfolio optimization would involve a (non-linear) quadratic objective function. Such a parametric optimization considering several factors was developed by Brandt et al. (2009). This is another possibility for incorporating characteristics into a portfolio with respect to a specific benchmark. However, the optimization framework is no more directly comparable to the mix or integrated approach.

Chapter 2 and this chapter have shown that arbitrage pricing theory can be applied to several asset classes, and subsequently several different asset pricing models emerged. These asset pricing models often try to connect the findings of the empirical asset pricing literature with neo-classical capital market theory. Thus, different characteristics, which assume proxy-specific risk factors, have found their way into standard asset pricing models. It has also been shown that these characteristics (e.g., value, defensive, and momentum) are robust across several asset classes. This leads me back to both the research questions initially formulated. Do the same findings apply to a young and emerging asset class, like digital assets, too? That would be a first indication that digital assets could indeed be termed a "serious" asset class. And second, is it possible to construct profitable investment strategies/styles that would justify investments into that asset class? Before I report the results of my research, which aims to answer these questions, it is important to understand digital assets more deeply. Therefore, the next chapter briefly revisits the state of the digital assets literature. Chapter 5 then explains technical details associated with blockchain and virtual currencies, and describes digital asset data in the form of some stylized facts.

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Chapter 4
Related Literature



4.1 Digital Assets in General

Due to the short (data) history of digital assets, research in this asset class is still in its early stages. Meanwhile,[1] many different peer-reviewed works have been published, which mostly investigate various different topics in only a handful of virtual assets. The full cross-section of digital assets is only investigated sparsely. Nevertheless, I would like to provide some context regarding this young stream of the finance literature, especially with regard to asset pricing and investment styles.

One of the few studies to explore the complete investment universe of digital assets is El Bahrawy et al. (2017). They showed several interesting facts connected to this emerging asset class. First, the number of actively traded virtual coins is relatively stable over time. Second, the relative share of the total market capitalization is more or less stable. Furthermore, they found a monthly "birth" rate of new digital assets of 1.16% on average, compared to a slightly lower monthly death rate of 1.04% on average. Dyhrberg (2016) explored similarities between bitcoin, gold, and the USD, and subsequently classified bitcoin as a hybrid version, thematically located in the middle between a pure means of exchange and a pure store of value. Other works have aimed to predict the price of bitcoin via an economic policy uncertainty index or global geopolitical risks, respectively (Aysan et al. 2019; Demir et al. 2018). Colianni et al. (2015) analyzed postings on the social media platform Twitter to successfully predict the price of bitcoin. A similar approach was used by Chuen et al. (2018), who arrived at similar findings. Wang and Vergne (2017) suggested the "innovation potential" of a specific virtual coin to be a significant determinant of future returns. Media coverage ("buzz"), on the other hand, seems to be negatively related with returns. The one-day-ahead value-at-risk of bitcoin was also of interest for Ardia et al. (2018). Other studies have

[1] At the time of writing: February 2020.

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focused on general market dynamics: Caporale et al. (2018) found persistence in the cryptocurrency market by employing different long-memory approaches. They also found a positive correlation between past and future prices, which is similar to the results described in Sect. 5.3.3 and subsequent chapters. Corbet et al. (2018) found evidence for a bubble behavior in the prices of bitcoin and ethereum. Informational efficiency in bitcoin prices was investigated by Feng et al. (2018) and Zargar and Kumar (2019). Feng et al. (2018) showed that informed trading indeed happens ahead of important digital-asset-related events. Zargar and Kumar (2019), on the other hand, found increased market inefficiency with higher data frequency.

4.2 Asset Pricing in Digital Assets

As described in the previous chapters and above, many different factors can describe the return behavior of a specific asset class. This is why I have needed to run as broad an analysis as possible for the mostly uncharted territory of digital assets. Studies regarding asset pricing in that asset class are, however, still rather limited. There are (to my knowledge) only a few (peer-reviewed) works that aim at explaining the cross-section of digital assets. Hayes (2017) developed a three-factor model based on "fundamental" variables, using computational power, number of coins per minute, and the type of algorithm employed to explain the price of bitcoin. Wheatley et al. (2018) valued bitcoin and predicted bubbles by Metcalfe's law, approximating the value of a network by the number of active wallet addresses. Both studies, though, regressed the factors on prices and not on returns.[2] Glas (2019) was one of the first to formulate a four-factor model of tradeable assets/investment strategies to explain the cross-section of digital asset returns. He included market, size, illiquidity, and short-term reversal in his model. Slightly differing results were reported by Li and Yi (2019), who found that high momentum, defensive, and large capitalized digital assets were outperformed in a set of highly liquid and large virtual tokens. Liu et al. (2020) explored three factors in a set of 78 digital assets, namely market, size, and momentum. A set of about 1700 digital assets was utilized by Shen et al. (2019), who also arrived at the conclusion that market, size, and reversal are suitable to be used in a cryptocurrency-specific factor model. While most of the studies mentioned used lower-frequency data, Kakushadze (2018) utilized daily data and identified three significant short-term factors: size, momentum, and intra-day volatility. (Trading) volume, on the other hand, lost explanatory power over a short horizon. Kakushadze (2018) also found a strong short-term mean-reversion effect, which has also been documented by other studies (see above).

To summarize, many of the studies mentioned arrived at similar findings. Market beta, size, momentum, and short-term reversal are better documented than trading volume (i.e., illiquidity). Fundamental factors associated with technical aspects like

[2] More information regarding this issue can be found in Sect. 9.2.

hash ratio, mining difficulty, and so on have not been investigated with the same intensity. However, all these studies are just the beginning of the asset pricing literature in digital assets. With better data availability even more factors might be found, which then need to be put into context to avoid a factor zoo as seen in the asset class equities.

Investment styles and characteristics in asset classes other than digital assets are well documented in general (Asness et al. 2013, 2015; Harvey et al. 2015). In the cross-section of digital assets, however, characteristics like value, momentum, or defensive appear to be less robust and less profitable. Glas (2019) and some chapters of this book find other characteristics that are able to describe the cross-section of digital asset returns. The strongest of them might be nominal price and trading volume, as well as (short-term) reversal and size. In contrast to the neo-classical asset pricing theorem, which assumes priced betas (Fama and French 2015; Ross 1976; Sharpe 1964), Chap. 9 of this work does not arrive at any significantly priced beta for digital assets.

Hence, Chap. 10 aims at a set of questions. First, I investigate whether the inclusion of characteristics in the portfolio construction process for digital assets yields any benefit. Second, I test four different portfolio construction approaches in total (one approach that serves as benchmark and three other approaches based on characteristics). Thereby, I can determine the most promising method for a digital asset investor. Third, I contribute to the mix versus integrated debate by comparing both portfolio construction processes in the young and emerging asset class of digital assets.

In general terms, this book aims at a set of goals (in addition to the two overarching problems described in the introduction). In Chap. 5 I show that digital assets share almost no commonalities with existing traditional asset classes. Therefore, it needs to be seen (in Chap. 6) whether it is beneficial to add such virtual tokens to other traditional asset classes. Since that appears not to be the case, I look more closely at this young asset class individually. One approach heavily discussed in the literature and intensively used by practitioners is technical analysis. Technical analysis is a time-series tool to predict the future movement of an asset by its past performance. While the literature is still discussing such tools for traditional asset classes, evidence for digital assets is rather clear. It does not work well in this emerging asset class, except for one indicator (see Chap. 7). That indicator, however, can be explained by an effect known from the characteristics literature. As described in Chap. 2, characteristics are generally formed by considering the whole cross-section of a dataset or even an asset class. Therefore, I investigate several characteristics in Chap. 8, which are further evaluated in Chap. 9. It thereby emerges that some characteristics indeed seem to explain the cross-section of digital asset returns. In a final step (see Chap. 10), I aim to incorporate these findings in the portfolio construction process for digital assets, which yields promising results. The outcomes of all these chapters are checked for robustness in Chap. 11 with two higher-quality datasets.

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Chapter 5
Digital Assets



Before going into further detail, it is necessary to understand that digital assets do not carry any intrinsic value like stocks or bonds. Digital assets are more comparable to gold, for example. Gold has no intrinsic value either and its use cases are limited. However, gold serves as a haven asset in financial markets and has been attracting mankind for ages. Digital assets, on the other hand, do not even exist in physical form. Coins like bitcoin or ethereum can only be created (i.e., "mined") and stored in virtual form as computer code.

5.1 Technical Aspects of Blockchain

Bitcoin finds its origin in the work of Nakamoto (2008). Disappointed by the financial system and its institutions, Nakamoto (2008) aimed to find an alternative way to process financial transactions. These transactions should be peer to peer (P2P), anonymous, and not be sent to a central institution like a (central/commercial) bank. To keep track of all the processed transactions, some way of storing the transaction information is needed. This is where a digital ledger comes into play, which also called a "blockchain." Thus, the blockchain has two important tasks: (i) store historical transaction information and (ii) verify new transactions. All transactions are stored in blocks with a specific size. Except for some special instances, verified blocks cannot be changed.

The blockchain as a whole is run by a P2P network of computers that are also referred to as "nodes." The verification of transactions is also termed "mining." Mining requires a lot of computational power (in the form of CPUs and GPUs) to solve complex mathematical problems. Therefore, the sending process of a transaction follows five steps:

1. Setting the receiving address and the amount to be sent from the wallet. Afterwards, the transaction can be sent.

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2. The transaction information is obtained by a P2P network and gets distributed to all available nodes.
3. Additionally, other transactions are gathered and combined into a new block (which may not exceed the maximum block size).
4. A complex mathematical problem must be solved via a trial-and-error method. The solution to the problem is called the "number-used-once" (nonce).
5. The first miner who obtains the nonce adds the new block to the existing chain of blocks. Each block contains a hash value (a digital fingerprint), the hash value from the previous block, the nonce, a timestamp, and the original transaction information.

A schematic description of steps 3 to 5 is displayed in Fig. 5.1. All incoming transactions are combined into one new hash value for the new block. Additionally, each block contains the hash value of the previous block.

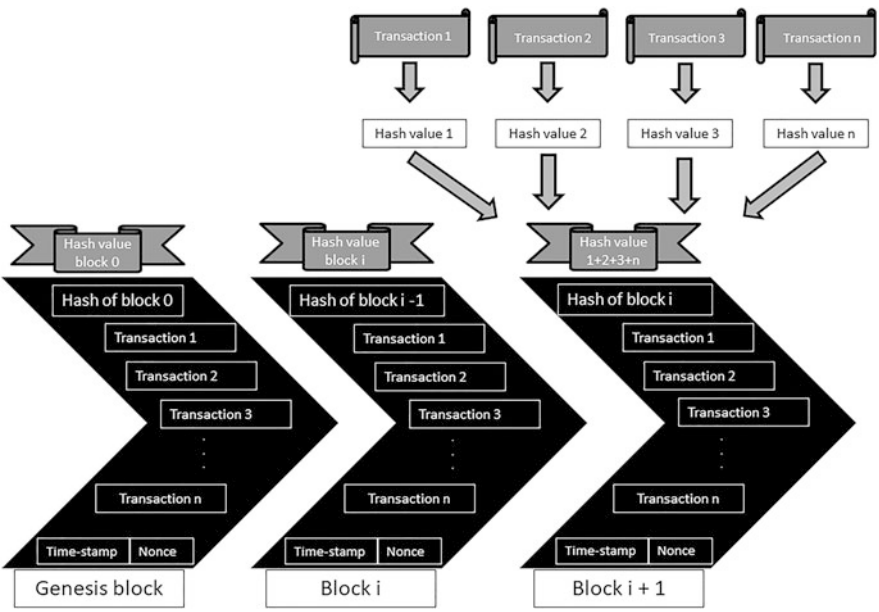


Fig. 5.1 A depiction of blockchain. Usually, every blockchain starts with the first ever created block—the genesis block. This first block gathers transaction information for n transactions until the maximum block size is reached. Based on all the transactions contained in the block, the hash value of block 0 can be obtained. After a miner solves a complex mathematical problem and finds the nonce, the block can be approved by other miners and the transactions within the block actually take place. In the meantime, new transactions have been sent by (other) users of the blockchain. Block i again gathers the new transaction information and miners compete to solve the nonce. Additionally, block i contains the hash value of block $(i - 1)$, which serves for concatenation and verification purposes. This process is repeated as long as there are no more new transactions or no more persons or mining pools willing to mine new blocks

Nakamoto (2008) wanted the miner who solves the nonce to be rewarded for their efforts. By mining the block, the miner is rewarded with a certain amount of the corresponding cryptocurrency. Additionally, the sender of a transaction can determine their own transaction fee. The higher the fee, the faster the transaction will be processed. The reward for one mined bitcoin block started at 50 bitcoins per block. This number is halved every three to four years. Furthermore, the complexity of solving the nonce increases with time. This means that more energy (i.e., power) needs to be consumed to mine a block. With less reward per block, the price of a coin and/or the transaction fee would need to rise to compensate miners for running the whole system.[1] After all 21,000,000 possible bitcoins are mined, miners will be compensated solely through transaction fees (Böhme et al. 2015).

As described above, a blockchain is not dependent on a central institution. Depending on the task of the blockchain, centralization could be favorable (Kakushadze and Liew 2018). Other use cases for blockchains are also possible. Kakushadze and Russo (2018) discussed several other applications like data malls, coin economies, data provenance, keyless payments, or proxy voting. Also, more variations of blockchain have emerged, although it might not always be worthwhile to use blockchain for certain purposes.

In particular, ICOs aim to become disruptive businesses by applying blockchain technology to almost any issue. An ICO is a project, start-up, or early stage company, often with (at least some) connection to blockchain, which circumvents traditional venture capitalists for funding reasons. ICOs are even comparable to crowdfunding. An ICO provides information about itself in the form of a white paper and tries to attract mostly private investors. Since this form of financing is not yet fully regulated, ICOs gather funds from private as well as institutional investors without the requirement of providing sensitive information like balance sheet data. The technical realization of an ICO is rather straightforward. Cryptocurrencies like ethereum, waves, neo, nem, lisk, and others offer the possibility to start an ICO on their respective blockchain. This reduces programming costs and enhances reliability since those blockchains are relatively widespread and governed by an active community. Thus, ICOs normally do not need their own blockchain, but they are situated on an already existing one. ICOs on the ethereum blockchain, for example, are called ERC-20 tokens, which emerged as the go-to ICO form.[2] Therefore, ICOs should not be mistaken for actual cryptocurrencies, which have their own blockchain and serve as the *currency* of the respective blockchain. It is perhaps more appropriate to think of ICOs as stocks of a specific company. The virtual tokens obtained from an ICO can be sent/received to/from an existing exchange (i.e., the respective blockchain in this case), but they do not contribute at all to the underlying blockchain.

[1] Additional bitcoin blockchain-related information can be found at https://www.blockchain.com/en/charts, accessed October 22, 2018.

[2] A guide and explanation to issuing an ERC-20 token can be found at https://www.ethereum.org/token, accessed October 23, 2018.

Between 2013 and the middle of 2018, ICOs raised c.20 billion USD worldwide.[3] Adhami et al. (2018) arrived at a success rate of 81% for ICOs in their data sample and documented a significant underpricing effect (i.e., the returns of the first trading day after listing on a crypto exchange are significantly positive on average). Amsden and Schweizer (2018) connected the success of an ICO to the (planned) tradeability of a token, the team size, and the CEO's social and professional network. A (short-term) underpricing effect was also documented by Varmaz and Varmaz (2018), Momtaz (2020), and Benedetti and Kostovetsky (2021). Papers focusing on the long-term success of ICOs, however, have not yet been published to my knowledge.

5.2 Prominent Digital Assets

This first description of digital assets and blockchain might still feel somehow abstract and inconclusive. To provide some more insights, I present additional details on some of the most important virtual coins currently around.[4]

5.2.1 Brief Overview

The digital assets described below rank among the most prominent at the time of writing. "Most prominent" in this context means a set of different subjective factors, which are hard to quantify. One example is the media attention or recent history for each digital asset, as well as its tradeability on various cryptocurrency exchanges. More objective factors are market capitalization and turnover. Of course, the list below is subject to controversy and changes over time. Thus, it should be seen as a snapshot of the most relevant digital assets as of December 2018 based on the experience and impressions of the author. As can be seen, the different technical details and purposes of each virtual currency can vary tremendously; which specific blockchain dominates the others and emerges as the "winner" remains to be seen.

The trend of mergers and consolidation observed in other industries like the banking sector, agriculture, or technology also applies here. The high number of different blockchains and projects seems not to be healthy, which is why many projects already "died," or further reduction in the number of blockchains might be imminent. Table 5.1 summarizes the most important digital assets and adds some more technical details.

[3] See "All-time cumulative ICO funding," https://www.coindesk.com/ico-tracker/, accessed October 23, 2018.

[4] The following cryptocurrencies are chosen subjectively since no common measure of importance exists. Turnover or market capitalization as proxy for importance would yield a similar list, depending on when it is compiled.

Table 5.1 Summary of Sect. 5.2

Digital asset	Hash algorithm	Consensus scheme	Max. no. of coins	Block time	Block reward
BTC	SHA-256	Proof-of-work	21M	10.00 min	12.50
USDT	Merkle tree	Omni	2580M	—	—
ETH	EtHash	Proof-of-work	Unlimited	15 s	3
XRP	RPCA	—	100,000M	—	—
LTC	Scrypt	Proof-of-work	82M	2.50 min	25
EOS	SHA-256	Proof-of-stake	1006M	~1 s	—
MIOTA	Tangle	—	2780M	—	—
DASH	X11	Proof-of-work	22M	2.50 min	1.67
XEM	SHA3-256	Proof-of-importance	—	50 s	—
DOGE	Scrypt	Proof-of-work	100,000M	1.00 min	10,000
WAVES	KECCAK	Proof-of-stake	100M	1.00 min	≤5% on held balance
TRX	SHA2 & SHA3	Proof-of-stake	99,218M	3 s	32
ADA	BLAKE2b-256	Proof-of-stake	45,000M	20 s	—
XMR	CryptoNightV8	Proof-of-work	18.4×10^{18}	2.00 min	4.87

For each digital asset, most of the technical details are reported. Data was collected from https://coinwarz.com/, https://coinmarketcap.com/, and the respective white papers, as well as the respective block explorers; accessed on December 12, 2018. "M" denotes million. Block time describes the average time it takes until a block is mined. Block reward (in number of coins) is the number of coins received by the miner for adding a block. Possibly all values in this table can/will change over time and are a snapshot at the time of writing. "—" means there is no clear statement to make or information is not available. Note: TRON (TRX) uses a *delegated* proof-of-stake mechanism

Next to the ticker name of each digital asset, Table 5.1 lists the respective hash algorithm, consensus scheme, maximum number of mineable coins, the average time it takes to mine a block, and the average reward of coins per block. The hash algorithm and the respective consensus scheme are closely interrelated. Since no central institution like a bank to approve new transactions exists, all available nodes (or at least the majority) connected to a blockchain need to find a way of (i) agreeing about the latest valid version of a blockchain[5] and (ii) adding new transactions to the network without the need to trust a central institution or even other participants in the network (Mauri et al. 2018).[6] Therefore, bitcoin uses a proof-of-work (POW) mechanism. POW is a more general term for finding the nonce based on a specific hash algorithm. POW is an energy-intensive process, which has been criticized in the recent past. Consequently, other virtual tokens like EOS apply another, more energy efficient, consensus scheme.[7]

The hash algorithm in this context is a method to condense large amounts of data (Härdle et al. 2020). (Keyed) hash algorithms are widely used online for encryption of passwords or other data. Such encryption normally involves a public and a private key to decrypt the encrypted data at some later point. The (unkeyed) secure hash algorithm (SHA) serves another purpose when used for blockchains. As Fig. 5.1 shows, the hash value of each block is based on the transaction information inside the block. As such, the SHA-256 hash algorithm, for instance, yields a chain of numbers and letters 256 bits in length, which ensures the data integrity of all the transaction information in a storage-efficient way. Since this method involves no encryption in the classical sense, the original information can easily be obtained the other way round. Moreover, most hash algorithms are designed to be collision resistant, which means there cannot be two different inputs yielding the same output (Al Ahmad and Alshaikhli 2013; Paar and Pelzl 2009; Sobti and Geetha 2012).

Below I briefly describe and summarize the different digital assets listed in Table 5.1 according to the consensus scheme used. The technological basis might be the same, but the use cases differ significantly. Since those descriptions are not necessarily relevant for understanding the remainder of this work, but they aim to give the interested reader an overview of the vast application possibilities of digital assets, the next three subsections can be skipped without losing information essential for further reading.

[5] This is especially important since a blockchain grows with every newly mined block. Thus, miners usually need to download the whole blockchain to a local hard drive and update it regularly. Widely used blockchains have sizes of several gigabytes and, in the case of, e.g., a lost internet connection, the miner works with an older version of the blockchain. If this miner wants to add a new block, however, based on an older version of the blockchain, all nodes need to find a way to agree on the current "correct" state of the blockchain, which is usually ensured by a consensus scheme.

[6] This is important to avoid the addition of wrong, manipulated, or corrupted transactions.

[7] More information on each virtual coin can be found in Table 5.1.

5.2.2 *Proof-of-Work Assets*

Bitcoin (BTC) is often perceived to be the first ever cryptographic currency. Although this is only half the truth, bitcoin is indeed the first widely used and accepted cryptocurrency. Before bitcoin, Chaum (1983) was one of the first to formulate the idea of anonymous cryptographic money. His idea was developed further two years later (Chaum 1985) and finally resulted in the creation of a company called DigiCash, which uses public and private keys to make payments anonymous and untraceable for third parties. DigiCash could not manage to attract enough clients and hence defaulted some years later.[8] After a wave of other unsuccessful ventures, web-based payments took over in the 1990s.[9] During that time, companies like PayPal, WebMoney, and e-gold emerged. Further investigations into cryptographic payments were made at the same time by Law et al. (1997) and others. The first formulation of bitcoin, however, dates back to Nakamoto (2008). "Satoshi Nakamoto" is a pseudonym for an as yet unidentified person or group. The first ever bitcoin block, the genesis block, was created on January 3, 2009, and included 50 BTC in total.[10] The genesis address receiving this amount still holds c.67 BTC today, which is the equivalent of about 767,000 USD.[11]

The community that runs the bitcoin network is ever curious about new ideas and innovations. Thus, it regularly agrees on updates of the initial blockchain or later updates, which result in forks of the blockchain and subsequently give birth to a new cryptographic currency. Past and planned forks are numerous, with a total so far of 34 forks.[12] Bitcoin itself is nothing other than a medium of private and anonymous exchange. It is also possible to buy BTC with fiat money, i.e., a state-backed currency like USD, EUR, or others.

Next to bitcoin, ethereum might be similarly important in the digital assets universe. Actually, ethereum is the name of a decentralized platform (i.e., blockchain) having the ability to run various applications (apps). These apps, also called smart contracts, are comparable to any other smartphone application. However, ethereum does not involve third parties and is not subject to downtimes or other drawbacks

[8] See "Requiem for a Bright Idea," Julie Pitta, https://www.forbes.com/forbes/1999/1101/6411390a.html#5fa0a2de715f, accessed on December 6, 2018.

[9] See "A Quick History of Cryptocurrencies BBTC—Before Bitcoin," Ken Griffith, https://bitcoinmagazine.com/articles/quick-history-cryptocurrencies-bbtc-bitcoin-1397682630/, accessed on December 6, 2018.

[10] See "Block #0," https://www.blockchain.com/de/btc/block/000000000019d6689c085ae165831e934ff763ae46a2a6c172b3f1b60a8ce26f, accessed on December 6, 2018.

[11] As of August 28, 2020. See "Genesis of Bitcoin," https://www.blockchain.com/de/btc/address/1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa, accessed on December 6, 2018.

[12] See "A List of Upcoming Bitcoin Forks and Past Forks" for more information on bitcoin forks, https://cryptocurrencyfacts.com/a-list-of-upcoming-bitcoin-forks-and-past-forks/, accessed on December 6, 2018.

of conventional programming methods.[13] The "fuel" or the driving force of the ethereum blockchain is ether (ETH). It is a means of value exchange, which ensures the blockchain and the network stay healthy.[14] ETH is used by developers and by clients accessing apps on the ethereum blockchain. Perhaps one of the most frequent smart contracts on the ethereum blockchain are ICOs.

An ICO is basically a fundraising mechanism without the involvement of third parties like banks, custodians, etc. The (mostly) public code of the smart contract is used to exchange funds between an investor and the fundraising company or project. This means that an investor sends a certain amount of ETH to the smart contract address, and the smart contract automatically credits the investor with some of the company's or project's newly created tokens. Some time after the issuance, the tokens can usually be traded on the secondary market.

An older virtual currency based on the same technical prerequisites is litecoin (LTC). Its technological foundation is close to that of bitcoin. Litecoin, however, employs the hash algorithm scrypt as opposed to SHA-256 used by bitcoin. The hash algorithm transforms the original transaction information into the hash value. Due to its complexity, SHA-256 is more accurate than scrypt but results in slower block times. The block time of one bitcoin block is 10 min, whereas one litecoin block is mined after 2.5 min. Also, a miner's current reward per block is higher for litecoin (25 LTC per block) compared to bitcoin (12.5 BTC per block).[15]

DASH extends the idea of an initial blockchain by introducing "masternodes." A masternode is responsible for processing (private) transactions, comparable to a miner, but it also runs special tasks such as instant transactions. It also controls the quality of blocks calculated by miners. Mining rewards are shared between miners (45%), masternodes (45%, although masternodes do not actually mine), and the DASH network (10%, also called budget). Masternodes also serve as a governing institution. In this role, masternodes maintain and vote on proposals of the DASH core team for further development of the DASH blockchain. Such a voting system is clearly superior to classical blockchains. Whereas blockchains like bitcoin need to gather an absolute majority of mining power for a specific proposal to be implemented (which can take years), DASH is able to let masternodes vote and act accordingly. DASH as a means of payment can also be used for fast, private, and secure transactions.[16]

[13] See "Build Unstoppable Applications," https://www.ethereum.org/, accessed on December 7, 2018.

[14] See "What Is Ether?", https://www.ethereum.org/ether, accessed on December 7, 2018.

[15] See "Bitcoin Cryptocurrency Summary," https://www.coinwarz.com/cryptocurrency/coins/bitcoin, and "Litecoin Cryptocurrency Summary," https://www.coinwarz.com/cryptocurrency/coins/litecoin, accessed on December 7, 2018.

[16] See "Understanding Dash Governance," https://docs.dash.org/en/stable/governance/understanding.html, accessed on December 11, 2018.

The virtual coins described so far try to convince with technological advances, a wide variety of use cases, or simply higher speed and lower transaction costs. Dogecoin (DOGE), which is one of the earliest digital assets, does not aim at such enhancements. Originally, it was meant as a parody of bitcoin and features a Shiba Inu[17] meme. Its technology is very close to that of litecoin. As such, it simply serves as an "internet currency."[18] The community behind dogecoin, however, is known for its socially responsible awareness. In 2013, it raised about 30,000 USD to send the Jamaican bobsled team to the 2014 Winter Olympics in Sochi.[19]

The last cryptocurrency in this list is monero (XMR). Its main specialization is anonymity. Other blockchains like bitcoin or ethereum make the sending and receiving addresses, as well as the amount sent, publicly available. By doing so, it could theoretically be possible to link a person's identity to a specific address on the blockchain. That does not apply to monero. It anonymizes that information, which makes transactions untraceable.[20] As such, monero might suffer from money-laundering issues or other illegal activities.

5.2.3 Proof-of-Stake Assets

An alternative to the energy-intensive POW mechanism is proof-of-stake (POS; King and Nadal 2012). POS attributes mining power according to the stake held in a respective blockchain. This means that a miner holding 1% of all the available coins connected to a blockchain is allowed to mine 1% of the transaction volume (using the specific hash algorithm). This approach reduces energy consumption significantly, since there is no more competition among miners,[21] and it decreases the likelihood of a 51% attack. For POW assets, such an attack would be the manipulation of an existing blockchain by a miner with 51% of the total mining power (i.e., the majority). In that case, a miner with the majority of the mining power could theoretically change past blocks and add new ones at the miner's sole discretion. For POS assets, a miner would have to hold 51% of all available coins, which is relatively unlikely for high market capitalization digital assets but more likely for small capitalized assets.

[17] Shiba Inu is a Japanese dog breed. Thus, dogecoin holders refer to themselves as "shibes."

[18] See "What Is Dogecoin," https://dogecoin.com/, accessed on December 11, 2018.

[19] See "Jamaican Bobsledders Ride Dogecoin Into Olympics," Kavitha A. Davidson, https://www.bloomberg.com/opinion/articles/2014-02-04/jamaican-bobsledders-ride-dogecoin-into-olympics, accessed on December 11, 2018.

[20] See "What Is Monero (XMR)?", https://ww.getmonero.org/get-started/what-is-monero/, accessed on December 12, 2018.

[21] For POW assets, miners compete to add blocks by solving the nonce faster than others. That is achieved by having the most computational power relative to competitors, which can be translated into more/better hardware and higher power consumption.

Nowadays, more and more digital assets are using POS. One of them, EOS, is one of the largest ICOs so far and raised 4bn USD over the term of one year of fundraising.[22] The start-up behind EOS's technology, block.one, developed a software platform competing with ethereum. The EOSIO software enables the creation of vertically and horizontally scaled decentralized applications through its own blockchain structure.[23] This results in a variety of potential use cases and also increases speed and simultaneously decreases fees compared to conventional platforms. Like ETH, EOS is a form of payment associated with the EOSIO platform. This means that any application running on EOSIO needs to rely on EOS when making or receiving payments. Consequently, EOS is still in the group of the largest worldwide digital tokens.

Waves, on the other hand, aims to enhance a very specific aspect of the ethereum blockchain. Ethereum allows the issuing of ERC-20 tokens via smart contracts. Waves provides an even simpler way to create its own digital tokens. This can be accomplished by typing the desired values into a graphical user interface, which does not require any programming (skills). Another benefit for ICOs on waves is its acceptance of other digital coins like BTC, ETH, or USD, which are automatically converted to WAVES tokens. This makes the ICO space even more accessible to non-tech-savvy investors. Currently, about 21,400 WAVES tokens have been created on the waves blockchain.[24]

Tron (TRX) is not about enhancement, but more about re-establishing a past state. In the early days of the internet, the World Wide Web was organized very much as a decentralized network. That changed over the years with companies like Alphabet, Facebook, Microsoft, and others providing much of the accessible content and (storage of) data. Users on social media platforms even have to entrust the rights to personal pictures or videos to the platform. Tron aims to reverse the centralization of the internet by establishing a decentralized web (other projects like Internxt or Skycoin aim at related purposes). Tron thereby focuses on the establishment of decentralized (social) media platforms, which ensure that the rights to the content uploaded by a user remain with the user.[25]

Cardano (ADA) follows a scientific approach and focuses on technological aspects. Starting without a clear purpose, cardano defined several points of interest, most of which are about designing, programming, maintaining, and updating a blockchain. Therefore, the team working on cardano is comprised of academics and engineers.[26]

[22] See "Initial Coin Offerings Have Already Topped 2017's Record Pace," Olga Kharif, https://www.bloomberg.com/news/articles/2018-06-01/initial-coin-offerings-have-already-topped-2017-s-record-pace, accessed on December 9, 2018.

[23] See "What Is EOSIO Software?", https://eos.io/faq, accessed on December 9, 2018.

[24] See "Investors," https://wavesplatform.com/, and "Create a Multiple-Currency Wallet," https://wavesplatform.com/product, accessed on December 11, 2018.

[25] See "What Is TRON?", https://s3.amazonaws.com/tron.network/What+is+TRON.pdf, accessed on December 12, 2018.

[26] See "What Is Cardano?", https://www.cardano.org/en/what-is-cardano/, and "Philosophy," https://www.cardano.org/en/philosophy/, accessed on December 12, 2018.

5.2.4 Other Consensus Schemes

Purchasing digital assets with fiat money normally destroys anonymity. Therefore, the company running the crypto exchange Bitfinex created a cryptocurrency (tether) with its value linked to USD. Tether (USDT) is situated on top of open blockchain technologies,[27] and every tether in circulation is said to be backed by 1 USD. Thus, tether is an asset-backed cryptocurrency with a value equal to 1 USD.[28] When trading digital assets, it is possible to obtain tether with fiat money (which is normally not anonymous) and convert the tether holdings to basically any other virtual coin. By converting the trade back to tether, it can be used to transfer the funds anonymously between exchanges without being prone to exchange rate risk, since 1 tether should always be worth 1 USD. Hence, tether is not only used by (private) investors but also by crypto exchanges or merchants. For exchanges and merchants, tether is an important tool to (seemingly) convert digital asset holdings to USD, since a digital asset itself bears a high exchange rate risk. This relationship, on the other hand, leads to serious market manipulations. Griffin and Shams (2020) investigated the role of tether and found that it played a major part in the drastic rise of bitcoin and other digital payment alternatives from March 2017 to March 2018.

A cryptographic currency specifically meant to be an alternative way of processing payments is ripple (XRP). Ripple aims to transform the global payments system by reliability, enhanced transaction speed, lower cost, and easy access.[29] The blockchain RippleNet is a direct link between banks and merchants and does not rely on a central institution like central or other banks. Ripple praises itself as having settlement times of roughly 4 s (compared to 1–5 days for conventional payment systems). Also, RippleNet is capable of processing about 1500 transactions per second.[30]

Iota fully focuses on the "internet of things" and Web 3.0. Some time in the future, when fridges automatically order milk from Amazon or when large industrial machines order commodities and goods from the warehouse for further processing, some form of payment will be needed. In the case of Amazon, the credit or debit card would obviously be the first choice. For intra-company payments, fiat money might not lead to the desired results. Thus, IOTA provides a form of fee-less inter-machine payments, for micro-payments and data integrity. Also, the structure of IOTA's distributed ledger is not a chain of blocks (see, e.g., Fig. 5.1), but it is more comparable to the structure of a network. The computational work to verify transactions is also tremendously lower than for bitcoin. A user of IOTA cannot

[27] See "We've Combined the Best of Both Worlds," https://tether.to, accessed on August 28, 2020.

[28] See "Tether: Fiat Currencies on the Bitcoin Blockchain," https://tether.to/wp-content/uploads/2016/06/TetherWhitePaper.pdf, accessed on December 6, 2018.

[29] See "Meet RippleNet," https://ripple.com/, accessed on December 7, 2018.

[30] See "Benefits," https://ripple.com/xrp/, accessed on December 7, 2018.

mine the connected virtual currency MIOTA but needs to verify the two previous transactions in order to make their own transaction.[31]

The NEM blockchain is quite similar to ethereum and EOS. It differs from the other two variants by giving the user the possibility of "customizing" the blockchain to their own needs. Thus, the NEM blockchain is able to host various "mini-blockchains," which operate on the underlying NEM technology. This system is also called a "smart asset system." The technological foundation also provides predefined features (or code), which reduce the need to write smart contracts or other off-blockchain functions. The customized blockchain can then be used for mobile and web applications, and also to store data or other functionalities.[32]

NEM's consensus scheme differs from the likes of bitcoin, ethereum, or EOS as well. A consensus system is the actual link between a node and the blockchain. Since usually many nodes add (transaction) information to the blockchain simultaneously, some mechanism needs to be applied in order to establish a commonly accepted status of the blockchain for each node. This also ensures that no information is added twice. While bitcoin and ethereum rely on a computationally intensive proof-of-work mechanism, NEM applies a proof-of-importance approach. The proof-of-work mechanism adds blocks through solving complex mathematical issues. The "proof" in that case is the energy consumed in order to solve the nonce. Consequently, miners with the most computational power will add blocks more often and therefore dominate the blockchain. Proof-of-stake incorporates the time a node has been working on the blockchain and also the stake (i.e., the number of coins associated to a specific blockchain—for NEM, the coin is called "XEM") held in the account by the node. From these variables, a weighted random number is generated to determine the node, which is allowed to add the next block.[33] Proof-of-importance follows a similar approach by allowing mining only for nodes, which hold a balance of at least 10,000 XEM. The importance is then determined from the past 30 days of XEM transactions, with a higher amount of transferred XEM leading to a higher importance ranking.[34]

5.3 The Digital Asset Universe

While "true" virtual coins emerged from the start of 2009, most of the ICOs took place between 2016 and 2018. This leads to a very short data history for those tokens and ultimately yields a very unbalanced dataset when working with digital asset

[31] See "An Open-Source Distributed Ledger" and "Meet the Tangle," https://www.iota.org/get-started/what-is-iota, accessed on December 11, 2018.

[32] See "NEM Technology," https://nem.io/technology/, accessed on December 11, 2018.

[33] See "What Is Proof-of-Stake?", https://github.com/ethereum/wiki/wiki/Proof-of-Stake-FAQs, accessed on December 11, 2018.

[34] See "The Outlink Matrix," https://nem.io/wp-content/themes/nem/files/NEM_techRef.pdf, accessed on December 11, 2018.

data. That is why I would like to give a first insight into digital asset data in the form of stylized facts.

5.3.1 Data Issues

Before anything else, it is important to gather the respective data. In recent years, www.coinmarketcap.com has emerged as the leading data provider for digital assets. It lists c.1600 different digital assets[35] on its website with the corresponding open, high, low, close, market capitalization, and turnover data. It also excludes digital assets that are inactive; digital assets that do not change in price for seven days are deemed "inactive" and removed from the website. This procedure has two major implications.

First, by excluding inactive digital assets and removing them from the website, the investment universe is not appropriately described any more. By removing (mostly the loser) digital assets, which at some point became inactive, a survivorship bias develops in the data. "Dead" digital assets with (mostly) negative returns are removed, and "surviving" digital assets remain to constitute the available data.[36]

Second, this issue served as an inspiration to start collecting data on our own.[37] Regrettably, www.coinmarketcap.com changed its terms and conditions and prohibits the collection of data from their website, effective from July 28, 2018. Thus, the dataset under investigation in this book ranges from the end of April 2013 to the end of June 2018 ($t = 63$ months). Figure 5.2 gives a first impression of the data. It plots the cumulative log returns of an equally weighted and a market-capitalization-weighted index based on all available assets of the dataset.

Both return series share the same start and end dates, but the total number of assets (see Fig. 5.3) grows from just 6 (in April 2013) to 1580 (in June 2018). The growing number of assets leads to an unbalanced dataset with more assets between 2017 and 2018 than in the beginning. Also, the dominance of bitcoin is an issue when calculating market-cap-weighted returns. The average monthly weight of bitcoin over the whole sample is 77.46% and it therefore significantly influences

[35] As of June 30, 2018.

[36] The website https://www.deadcoins.com/ tracks the number of dead coins. As of December 4, 2018, 934 virtual coins had already passed away, compared to an active digital asset universe of about 2000 at the same time.

[37] This means that the chair of finance at the university of Bremen adopted the methodology of www.coinmarketcap.com and currently retrieves data from about 100 digital asset exchanges that are aggregated into a dataset comparable to that of www.coinmarketcap.com. At this point I would also like to thank Martin Schorfmann and Carolina Keil for their determination and persistence concerning this project. Both of them provided well-considered programming and are the reason for the project's success. I also thank Steffen Günther for continuing the project after I left university. Unfortunately, this project was still in its final phase at the time of writing, which is why I have to rely on coinmarketcap data for this study.

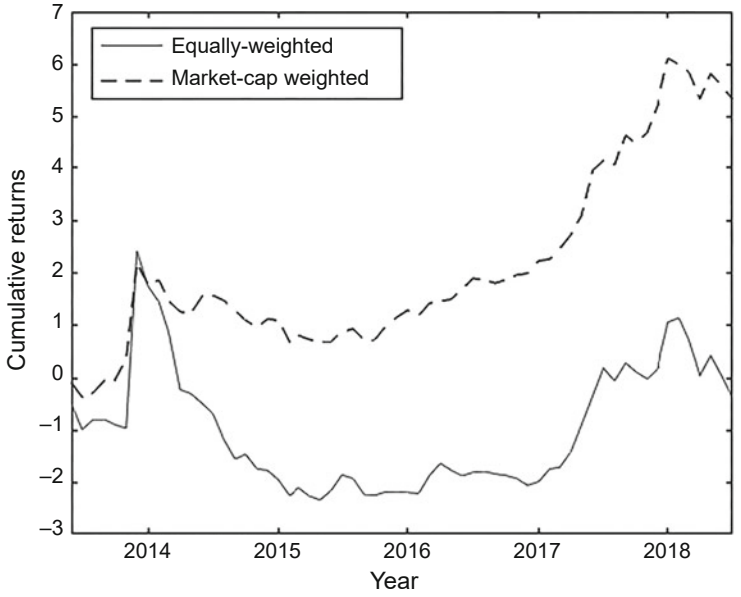


Fig. 5.2 Equal versus market-cap weights

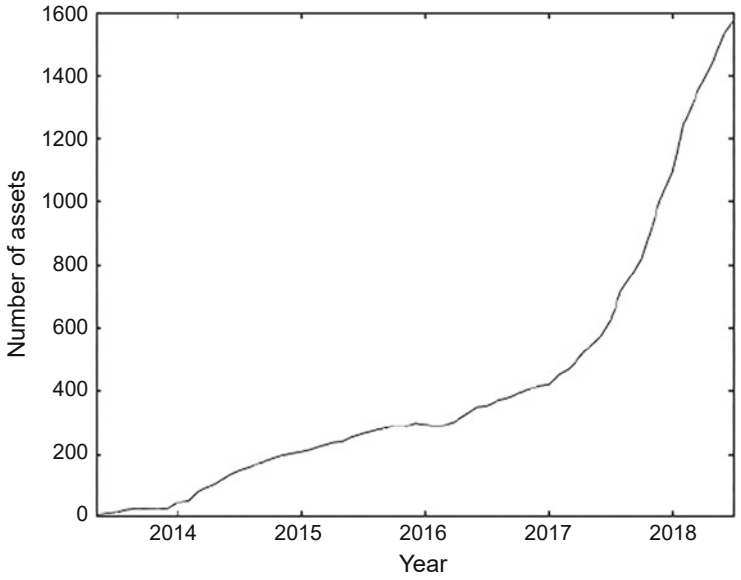


Fig. 5.3 Number of assets over time

Table 5.2 Monthly returns of the equally weighted versus cap-weighted dataset

Dataset	Dec 17	Jan 18	Feb 18	Mar 18	Apr 18	May 18	Jun 18	$\overline{r}$
Equal weights	88.17	8.65	−42.34	−67.40	38.08	−39.29	−42.02	−8.02
Cap weights	52.10	−27.87	−19.08	−55.24	43.23	−26.83	−27.37	−8.72
Δ	36.07	36.52	−23.26	−12.16	−5.15	−12.46	−14.65	0.70

This table contains the returns of an equally weighted and market-cap-weighted market index over the last months of the dataset from December 2017 to June 2018. All values are percentage values. Δ describes the difference between both return series; $\overline{r}$ denotes the average return of the reported returns over all seven months

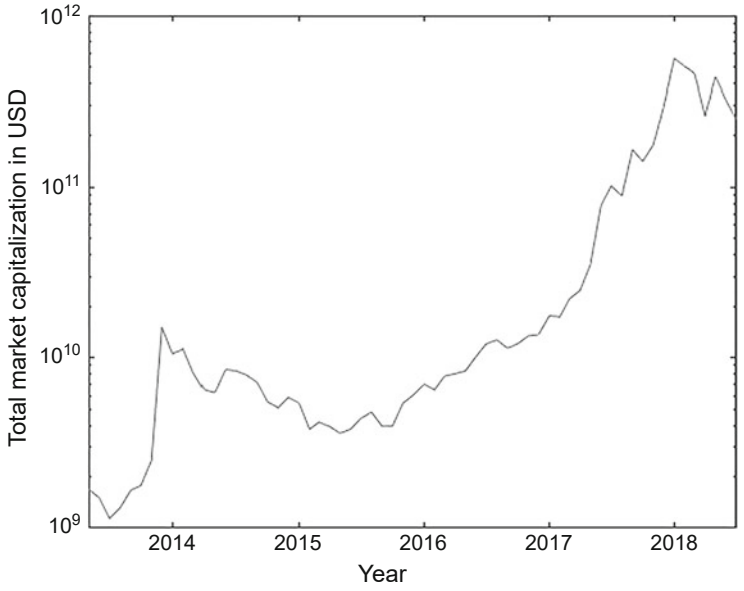


Fig. 5.4 Total market cap over time

the outcome of any analysis based on market-cap weights. Since Fig. 5.2 is quite imprecise, I also report the monthly returns of both return series from December 2017 to June 2018 in Table 5.2 for a more detailed but still illustrative impression.

December 2017 was the month when the market capitalization of all available digital assets hit an all-time high at c.559 billion USD (in this data sample). It decreased to c.248 billion USD in June 2018 (see Fig. 5.4). The trading volume even increased over the same period of time (see Fig. 5.5).[38]

For the time period under investigation, the equally weighted dataset outperformed the returns of a market-cap-weighted digital asset index by 0.70% on

[38] Figures 5.4 and 5.5 are on a log scale.

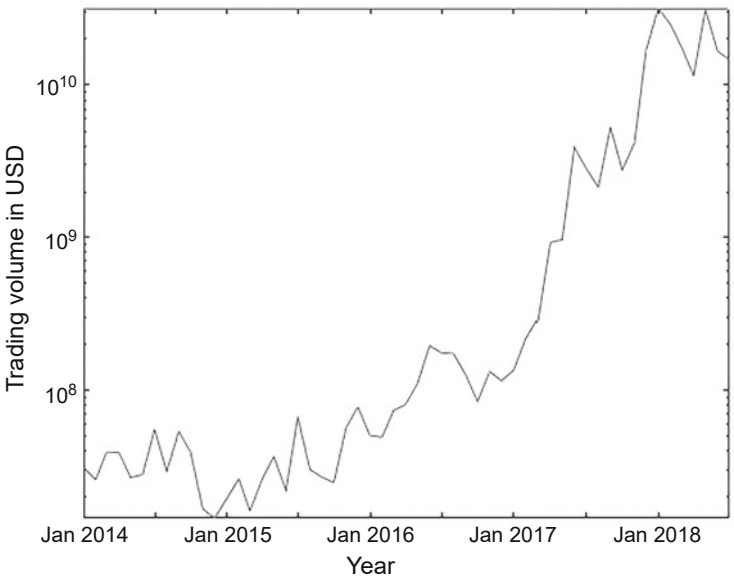


Fig. 5.5 Trading volume over time

average (see Table 5.2).[39] That, however, does not hold true for the whole sample period. A difference of 0.70% seems like a relatively small number, due to the magnitude of the monthly returns, but has at least one serious implication. With the possibility of a daily-changing dataset on www.coinmarketcap.com comes a risk of non-reproducibility for academic studies working with such a dataset. At worst, results could not be verified by other researchers and thus the outcomes of any academic work would be valid for only several months or even less. This, of course, strengthens the need for a survivorship-bias-free dataset, which could also be shared with other researchers.

[39] Alternatively, other index construction methods could be imagined. Major index providers like S&P, Nasdaq, or Deutsche Börse introduced different rules, like liquidity and/or market capitalization ranking criteria, to construct equity market indices. However, constructing a digital asset index would require reasonable methodological thinking and back-testing. That is why I prefer an equally weighted index of all available digital assets to a market-cap-weighted index (due to the issues described above) in the subsequent analyses. The usage of at least one of these approaches is also standard in the empirical asset pricing literature, which is why I rely heavily on an equally weighted index in this book. Trimborn and Härdle (2018), for instance, developed an index for cryptocurrencies, which includes 30 different digital assets. That index, however, also assigns the highest weight to bitcoin: 70.33% on November 9, 2019 (see "CRypto IndeX," https://thecrix.de/, accessed on November 9, 2019).

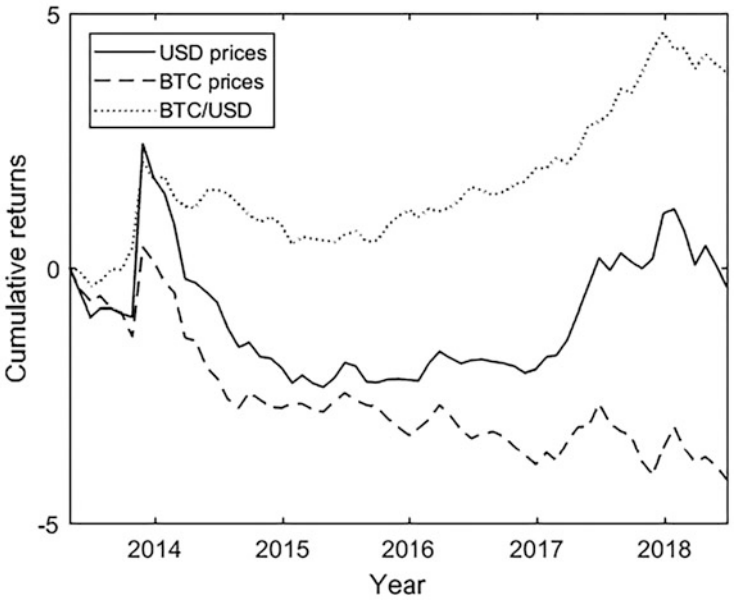


Fig. 5.6 BTC versus USD prices

5.3.2 USD-Denominated Prices Versus BTC Prices

The methodology of www.coinmarketcap.com also involves two highly volatile digital assets. The first pair is some given cryptocurrency quoted against BTC, which is the standard quotation on most digital asset exchanges. The price series of this pair is then converted to USD by taking the original data series times the BTC/USD price series. This leads to an unfavorable upward shift of an equally weighted market index (see Fig. 5.6).[40]

Converting the USD prices to BTC prices reduces the dependence on the BTC/USD exchange rate fluctuations and thus describes the data from an "internal digital asset point of view" (see Table 5.3). This means that when using USD-denominated price series, an investor looks at digital assets from an external point of view. Since such an investor would purchase digital assets with their own "home currency" (e.g., USD), it is totally reasonable to look at the digital asset universe in USD terms. However, when the view shifts from an external to an internal point of view, quoting digital asset prices in USD might be distorting. In that case, it might make more sense to look at digital assets measured in BTC.

[40] BTC prices are represented by the bottom line, USD prices by the line in the middle, and BTC/USD by the top line. The returns are log returns.

Table 5.3 BTC/USD influence regression

	α (%)	β	t-Value (β)	p-Value	R^2 (%)	ρ (%)
BTC prices	−9.55	0.4693	(3.53)	0.00	17.17	41.35
USD prices	−9.47	1.4418	(10.94)	0.00	66.62	81.59

This table reports the regression results of a simple univariate regression with the returns of an equally weighted market index based on all digital assets in the dataset as dependent variable. I ran this regression twice, once for returns based on BTC prices and again for returns based on USD prices. The returns of the BTC/USD exchange rate serve as the independent variable. ρ denotes correlation

Since the standard quotation on cryptocurrency exchanges is mostly in BTC terms, the influence of external factors like the BTC/USD exchange rate is neglected. Thus, the desired point of view is important to determine which quotation to use. Nonetheless, as shown in Appendices B and H, the results of most analyses in this book are not subject to the chosen quotation and can therefore be deemed robust. Also, most of the subsequent analyses involve portfolio sorts, which make this issue irrelevant. In USD terms, that means the long and short portfolios would also be long and short in USD and hence cancel each other out. That is why I refer to the respective point of view for each chapter specifically. Some chapters involve comparisons with other USD-denominated asset classes, so I use digital asset price series in USD terms. Other chapters solely look at the digital asset universe, for which BTC-denominated price series are more relevant in order to avoid the influence of the BTC/USD exchange rate as shown in Fig. 5.6.

Table 5.3 again contextualizes that issue by running a simple univariate regression model with an equally weighted market index (measured in BTC or USD terms) as dependent variable and BTC/USD as the independent variable. Naturally, the same index measured in BTC terms depicts a lower dependence on the BTC/USD exchange rate than an index measured in USD terms. Therefore, it should be reasonable enough to use BTC prices for an internal perspective and USD prices for an external perspective.

5.3.3 Some Stylized Facts

Converting Fig. 5.2 to numbers makes the digital asset landscape even more accessible. Table 5.4 reports the mean daily and mean monthly returns based on the equally weighted market index in USD (no data corrections have been made). It additionally shows the standard deviation of the daily and monthly return series, the average turnover, and the mean market capitalization. Table 5.4 also includes the skewness and kurtosis of the average returns, as well as the lowest and highest values for each category (on a single digital asset basis) and the 25th, 50th, and 75th percentiles.

Table 5.4 Descriptive statistics of the digital asset dataset

	Mean	Min	p25	p50	p75	Max
Daily data						
Return (%)	−0.08	−10.84	−0.92	−0.24	0.17	10.56
SD (%)	18.88	0.58	11.15	15.23	23.32	100.28
Skewness	0.71	−27.37	0.06	0.49	1.04	21.37
Kurtosis	16.39	−0.03	5.67	9.01	15.63	874.28
Average turnover	3.53M	—	9642.21	88,846.25	0.82M	23.84M
Average market cap	112.80M	—	0.66M	5.11M	22.80M	326.14M
Monthly data						
Return (%)	−0.62	−221.86	−27.84	−6.23	5.50	108.21
SD (%)	54.76	0.14	49.34	64.49	81.23	336.03
Skewness	3.77	−5.86	0.01	0.56	1.11	5.71
Kurtosis	24.86	−2.81	2.52	3.62	5.21	39.05

Mean return is the average log return of an equally weighted market index in USD. SD is the standard deviation of the equally weighted market index. The same applies to the third and fourth moments, skewness and kurtosis. Average turnover is the average turnover over all digital assets in USD. Market cap is the mean market cap over all digital assets in USD. Min, max, and the 25th, 50th, and 75th percentiles aim to describe the distribution of the respective return moments on a single digital asset basis. Therefore, the mean returns of all available digital assets serve as a cross-section from which the min, max, and percentile values are derived. I do not report average turnover and market cap for monthly data, since both represent end-of-month data drawn from the daily dataset. Both values could, however, be approximated by multiplying the daily values by 30 days

The minimum and maximum values in particular depict a highly heterogeneous digital asset dataset. First, I calculated the average returns of each available digital asset using its respective time series. The 50th percentile of the cross-section of average daily return data is then very close to 0. This means that the returns of digital assets average themselves out. A similar analysis was also conducted by Glas (2019), which yielded comparable results based on a shorter period of time under investigation. The same approach is used for the other return moments, the skewnesses of single digital assets are predominantly positive, and average returns tend to be leptokurtic.

Another note of importance is predictability. Predictability of asset returns in general is a huge topic in the financial literature. Therefore, researchers like Fama (1991) or Pagan (1996) investigated a possible autocorrelation in stock returns. If autocorrelation exists, an investor would be able to profit from the past information. Both found that stock returns can partly be predicted by the past autocorrelation of stock returns, which amounts to 0.10 on average. I measured autocorrelation in two ways. First, I simply took the daily log returns of an equally weighted index of digital assets to investigate autocorrelation (afterwards I used single digital assets). In an alternative approach, I aim to investigate a possible volatility clustering in digital assets. Therefore, I used squared daily returns that are commonly assumed to be a proxy for volatility (Goyal and Santa-Clara 2003). The autocorrelation ρ is

calculated in line with Box et al. (2015) as

$$\rho = \frac{\sigma_k}{\sigma_0}, \tag{5.1}$$

where

$$\sigma_k = \frac{1}{T}\sum_{t=1}^{T-k}(r_t - \bar{r})(r_{t+k} - \bar{r}), \tag{5.2}$$

with $k = 0, \ldots, K$ being the number of lags, r the return at time t, and σ_0 the sample variance of the return series. Figure 5.7 indicates a significant autocorrelation in digital asset returns (here I investigated an equally weighted market index in BTC) for up to six lags (which is six days). Its magnitude is also comparable to that known from equity markets. Figure 5.8 investigates volatility clustering in the (squared) returns of digital assets. The outcome of this analysis is even more striking than for the simple returns. Volatility can be predicted by up to 20 lags (or even longer). This

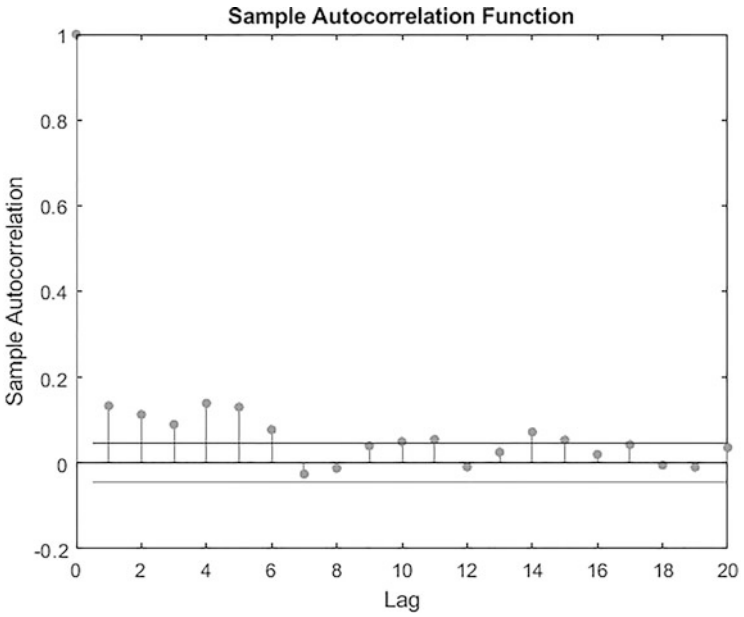


Fig. 5.7 Autocorrelation returns

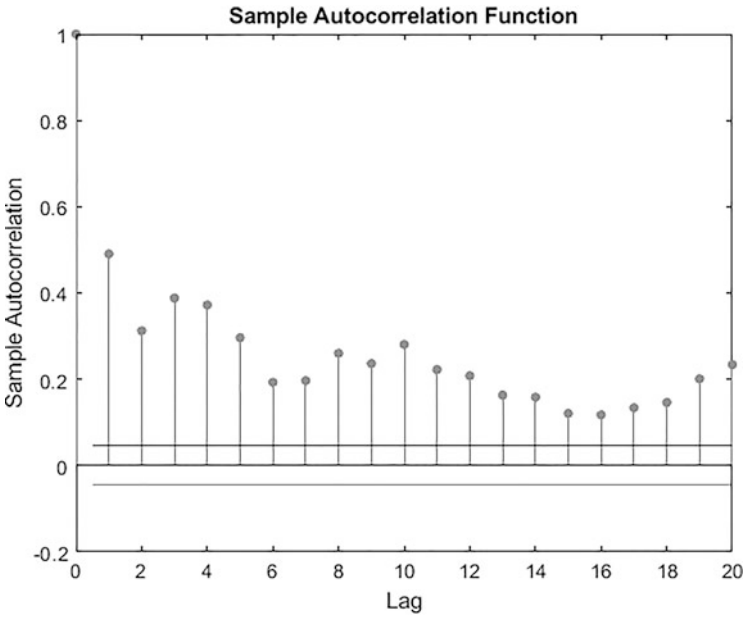


Fig. 5.8 Volatility clustering

means a high (low) past volatility indicates a high (low) future volatility. All in all, this result does not deviate much from the findings in other asset classes.[41]

However, using an equally weighted index might be misleading since it aggregates the information and does not look at each digital asset itself. Therefore, I ran a second analysis for every single virtual coin with at least 20 daily observations (to make the results comparable). After determining the autocorrelation for all single assets, I aggregated their autocorrelations to arrive at one final result. So, in this second approach, the analysis was done the other way round. First, I determined the autocorrelations per asset and then I equally weighted the autocorrelations. By doing so, the autocorrelation as shown in Fig. 5.7 disappears (see Fig. 5.9). While the positive autocorrelation in returns disappears, the autocorrelation in squared returns remains intact. Following the same approach for squared returns still yields a positive autocorrelation, although to a lesser extent (see Fig. 5.10). Thus, return prediction for individual digital coins based on autocorrelation might not be possible; for return volatility, on the other hand, it still seems feasible.

Some other, subjectively chosen, stylized fact is inspired by the work of French (1980). He investigated two different return-generating processes, the calendar time hypothesis and the trading time hypothesis. The first assumes a continuous return-

[41] The same analysis based on monthly returns does not yield comparable results. Monthly returns do not appear to exhibit any form of return autocorrelation. This is also in line with Fig. 5.7, which indicates an autocorrelation only for up to six days.

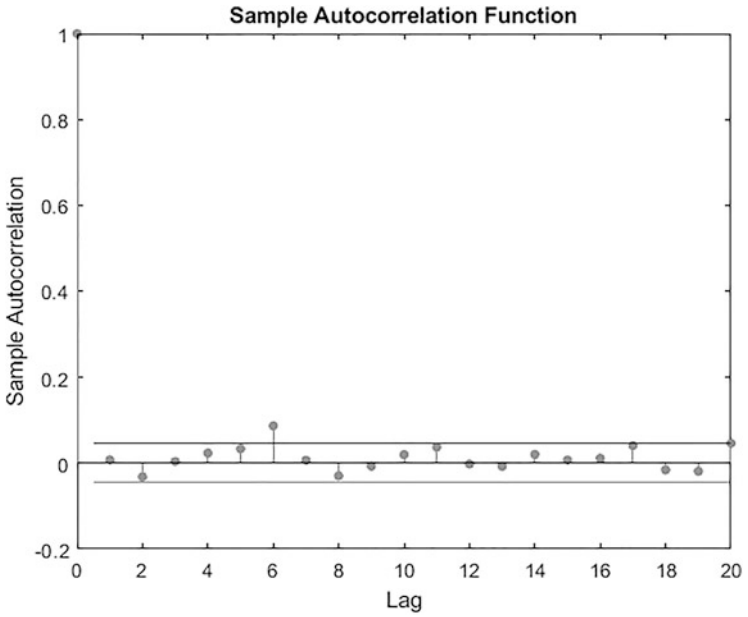


Fig. 5.9 Autocorrelation returns (2)

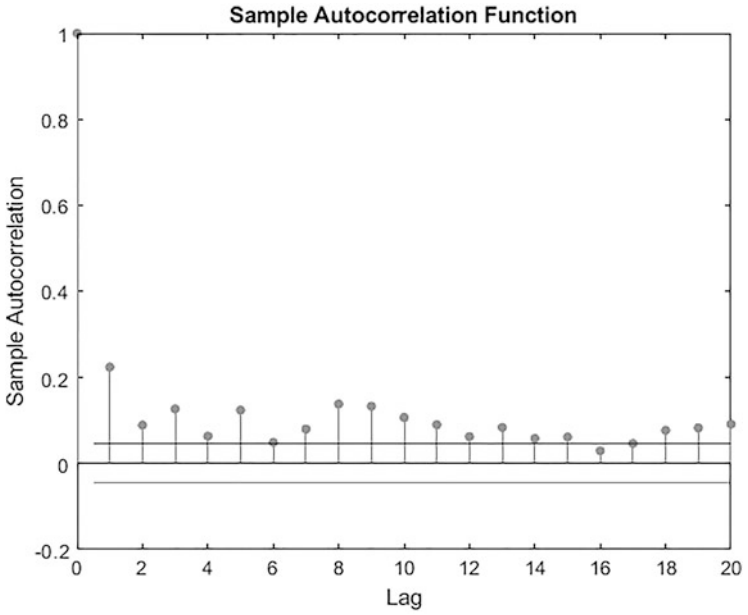


Fig. 5.10 Volatility clustering (2)

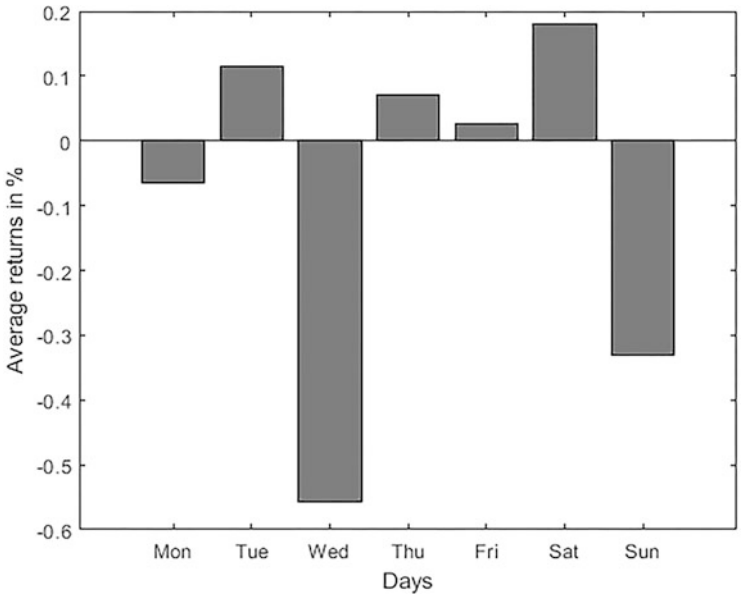


Fig. 5.11 Average returns per day

generating process with returns on Mondays being three times larger than the returns on the other days. Under this assumption, returns are generated only during (active) trading days, and thus returns should be the same for each day. Figures 5.11 and 5.12, respectively, show the average returns of an equally weighted digital asset index for each day and for each month.

As indicated by Fig. 5.11, weekday returns are positive on four out of seven days. Trading of digital assets happens 24/7, every day of the year, and thus returns can be investigated for every day of the week. The other three days with negative average returns are comparable or even higher in magnitude. In particular, Sunday and Wednesday appear to be weak days for digital assets. The lowest returns are achieved on Wednesdays with an average return of about -0.55%. The analysis for average returns per month is limited in its informational content (see Fig. 5.12). The dataset spans only about five years, and therefore average calendar month returns are calculated over 5–6 observations only. Regardless of that issue, November happens to be the month with the strongest returns on average. April and December also achieve positive returns, but all other months yield negative average returns. This leads to findings similar to French (1980). He also rejected both models for US stock returns.

Now that technical aspects of blockchain have been described, hints at data issues were made and some stylized facts gave a first impression of the data I am dealing with, I would like to get into more detail concerning other issues and chapters in this book. These issues obviously dig deeper into the data which is why the above section only includes three stylized facts with more to follow below.

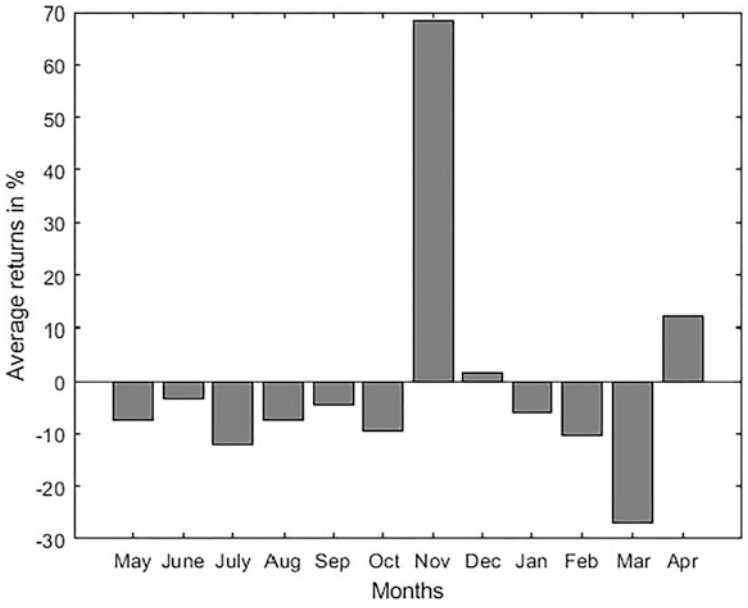


Fig. 5.12 Average returns per month

5.4 Chapter-Specific Descriptive Statistics

Inspired by (the robustness of) investment styles and their application as factors in asset pricing models, investment styles, or investment strategies, are also an area of interest for this book. Asset pricing factors are constructed in a similar way to investment styles. Thus, the difference between the two is not always clear. While factor construction is usually based on very large datasets, investment styles can be constructed using less data. As already mentioned, not every style is equal to a standalone risk factor. Several styles can proxy one and the same risk factor. This is also a criticism of Cochrane (2011), who asked for the "true" risk factors that lie underneath all the different anomalies, investment styles, and characteristics. While this is an important question for the asset pricing literature, this book focuses more on investment-specific implications. However, links to the asset pricing literature are drawn as well. Thus, the aim is to test, find, and understand investment styles and investment strategies in digital assets, and to investigate their intersections with other asset classes where possible. The academic literature in this context has not yet answered all questions connected to investment styles or characteristics (and perhaps never will), which leaves enough space for the subsequent analyses.

Furthermore, other approaches like portfolio optimization or even technical indicators are of great interest for practitioners and academics. These tools were described in the literature a long time ago and have led to interesting results (except for technical indicators, which produce rather controversial results). Therefore, an

application of "traditional" methods to an alternative and young asset class like digital assets seems especially exciting.

Chapter 6 therefore investigates whether an inclusion of digital assets in a multi-asset portfolio results in an upward shift of the efficient frontier. This claim is rejected, and thus I need to focus on different applications in digital assets only. Chapter 7 shows that the usage of technical indicators in digital assets is not recommended. Chapter 8 switches to a cross-section of digital asset returns and describes different investment styles and their intersections. This analysis is extended in Chap. 9, which explores the characteristics versus covariance discussion. The findings from all these chapters are put into an investment framework in Chap. 10 and verified again in Chap. 11. Chapter 12 provides some conclusions.

5.4.1 Digital Assets and Other Asset Classes

The digital asset data used in this book is described in more detail below. A list of all digital coins included in the dataset can be found in Appendix A. For the analysis in Chap. 6, I used monthly data, which is denoted in USD. Since digital asset research follows no commonly accepted database, no proposals for data adjustments (see Ince and Porter (2006) for suitable corrections when working with Datastream data) have been made. Table 5.5 contains return information for an equally weighted index (which is termed *index*) based on all available digital coins in the sample (see Figs. 5.2 and 5.3). I also include the digital coins described in Sect. 5.2 for additional insights. These single digital assets are more or less subjectively chosen (as described above) and constitute a set of the most prominent and important digital assets at the time of writing.

As noted in Table 5.5, nearly all of the single digital assets achieve high buy-and-hold returns as well as positive mean monthly returns. Only the equally weighted index and tether (USDT) produce negative results. This is due to the specific visualizations of the single virtual coins, which are the oldest and/or most prominent ones. Other less prominent digital assets achieve negative returns, which drag down the index. In particular, tether should have a value of 1 USD at any time, since it is said to be backed by the same dollar amount. However, the price of a tether fluctuates around 1 USD, which results in a slightly negative average return in the dataset used. I did not consider transaction costs in this analysis.[42] Average returns, on the other hand, are only significantly different from zero for ETH, DASH, and XEM at the 10% level. Standard deviations are clearly higher than for traditional assets (see Tables 5.6 and 5.7) and skewness is almost consistently positive. The returns of most of the assets in Table 5.5 are leptokurtic. I did not consider a market-cap-weighted index since such an index would be dominated by BTC. Vice versa,

[42] Transaction costs are not of interest at this point. Trading costs are taken into account in other chapters.

Table 5.5 Descriptive statistics for digital assets

Asset	r_{bh} (%)	Return (%)	*t*-Statistics	*p*-Value	SD (%)	Return/risk	Skewness	Kurtosis
BTC	383.02	6.18	1.57	0.12	30.96	0.20	2.43	15.23
USDT	−0.08	−0.00	−0.01	1.00	2.10	−0.00	2.50	21.75
ETH	581.32	17.10	2.13	0.04	46.71	0.37	0.32	2.71
XRP	441.62	7.61	1.00	0.32	58.09	0.13	1.83	6.81
LTC	294.04	4.74	0.77	0.44	48.28	0.10	3.46	21.00
EOS	154.17	14.02	0.66	0.52	70.56	0.20	0.80	1.84
MIOTA	92.23	7.69	0.38	0.71	70.93	0.11	0.85	1.84
DASH	607.04	11.67	1.77	0.08	47.51	0.25	1.25	6.73
XEM	695.30	18.30	1.78	0.08	63.21	0.29	0.98	3.47
DOGE	177.86	3.29	0.50	0.62	48.58	0.07	1.23	4.62
WAVES	304.17	12.67	1.16	0.26	53.31	0.24	0.98	3.94
TRX	283.28	31.48	0.86	0.42	110.36	0.29	2.17	7.93
ADA	150.92	18.87	0.55	0.60	97.82	0.19	0.87	1.83
XMR	421.26	8.60	1.31	0.20	46.01	0.19	0.90	4.32
Index	−38.33	−0.62	−0.09	0.93	54.76	−0.01	3.87	26.83

r_{bh} is the cumulative log buy-and-hold return starting from the first data point (end of April 2013) to the last available data point (end of June 2018). This does not necessarily mean that all digital assets share the same starting and ending points. Return is the average monthly log return percentage. The *t*-statistics describes the corresponding *t*-value of the mean return with the null hypothesis of an average return equal to 0% and the respective *p*-value. Standard deviation (SD) depicts the volatility of the average monthly returns (in %). Return/risk is the ratio of mean return and standard deviation. The third and fourth return moments are denoted by skewness and kurtosis. *Index* is an equally weighted portfolio of all digital assets in the dataset (see Appendix A). The index is rearranged monthly

Table 5.6 Descriptive statistics reference asset classes

Asset	r_{bh} (%)	Return (%)	*t*-Statistics	*p*-Value	SD (%)	Return/risk	Skewness	Kurtosis
S&P500	65.84	1.05	3.00	0.00	2.76	0.38	−0.16	3.35
Stoxx600	34.63	0.55	1.18	0.24	3.70	0.15	−0.05	2.28
Nikkei225	51.45	0.82	1.88	0.06	3.45	0.24	−0.14	3.69
MSCI Asia Pacific	34.34	0.55	1.25	0.22	3.47	0.16	−0.19	3.49
Equities	46.56	0.74	1.98	0.05	2.96	0.25	−0.23	3.78
US 10yr	3.60	0.06	0.28	0.78	1.65	0.03	0.10	3.71
German 10yr	8.80	0.14	0.48	0.63	2.29	0.06	−0.43	2.63
Japan 10yr	−6.82	−0.11	−0.28	0.78	3.09	−0.04	0.18	4.61
Asia Pacific 10yr	12.78	0.20	1.46	0.15	1.10	0.18	−0.55	3.53
Bonds	4.59	0.07	0.36	0.72	1.62	0.05	−0.53	4.24
Brent oil	−35.85	−0.57	−0.46	0.65	9.76	−0.06	−0.24	3.73
WTI oil	−27.10	−0.43	−0.40	0.69	8.56	−0.05	−0.32	3.92
Gold	−24.06	−0.38	−0.67	0.50	4.49	−0.09	−0.05	3.10
Silver	−56.52	−0.90	−1.04	0.30	6.87	−0.13	0.48	3.32
Copper	−12.87	−0.20	−0.31	0.76	5.22	−0.04	0.28	4.71
Aluminum	11.44	0.18	0.28	0.78	5.09	0.04	0.17	2.88
Corn	−36.84	−0.58	−0.85	0.40	5.46	−0.11	0.65	5.35
Wheat	−32.10	−0.51	−0.50	0.62	8.12	−0.06	0.22	3.85
Feeder cattle	−0.86	−0.01	−0.02	0.98	5.52	−0.00	−0.44	4.19
Lean hogs	−42.11	−0.67	−0.51	0.61	10.49	−0.06	−0.14	3.29
Commodities	−25.69	−0.41	−0.96	0.34	3.37	−0.12	−0.52	3.35

r_{bh} is the cumulative log buy-and-hold return starting from the first data point (end of April 2013) to the last available data point (end of June 2018). All assets share the same start and end dates. Return is the average monthly log return percentage. The *t*-statistics describes the corresponding *t*-value of the mean returns with the null of an average return equal to 0% and the respective *p*-value. Standard deviation (SD) depicts the volatility of the average monthly returns (in %). Return/risk is the ratio of mean return and standard deviation. The third and fourth return moments are denoted by skewness and kurtosis. The asset classes highlighted in italics describe an equally weighted portfolio of all assets that are part of the respective asset class. The portfolios are rearranged monthly

Table 5.7 Descriptive statistics reference asset classes (cont.)

Asset	r_{bh} (%)	Return (%)	*t*-Statistics	*p*-Value	SD (%)	Return/risk	Skewness	Kurtosis
EUR/USD	−9.52	−0.15	−0.52	0.60	2.29	−0.07	−0.30	3.35
USD/JPY	16.39	0.26	0.75	0.45	2.74	0.09	−0.14	4.50
GBP/USD	−13.99	−0.22	−0.70	0.49	2.53	−0.09	−0.42	3.91
USD/CAD	25.84	0.41	1.28	0.21	2.55	0.16	0.36	4.05
USD/SEK	31.91	0.51	1.53	0.13	2.62	0.19	−0.20	2.50
USD/CHF	−4.75	−0.08	−0.26	0.80	2.34	−0.03	0.57	3.74
FX	7.65	0.12	1.33	0.19	0.73	0.17	−0.50	4.20
Private equity	54.44	0.86	1.92	0.06	3.58	0.24	−0.16	3.23
Infrastructure	−17.05	−0.27	−0.41	0.69	5.30	−0.05	−0.58	4.08
Hedge funds	1.55	0.02	0.12	0.90	1.61	0.02	−0.62	3.55
Real estate	41.12	0.65	1.36	0.18	3.81	0.17	0.00	2.67
Alternatives	20.02	0.32	0.92	0.36	2.75	0.12	−0.53	4.15
Alternative energy	15.37	0.24	1.01	0.32	1.91	0.13	−0.49	2.95
Special situations	9.05	0.14	0.58	0.56	1.96	0.07	−1.25	5.03
Fundamental growth	13.29	0.21	0.73	0.47	2.29	0.09	−0.42	3.55
Fundamental value	15.91	0.25	1.31	0.20	1.53	0.16	−0.18	3.44

Commodity agriculture	0.94	0.01	0.11	0.91	1.08	0.01	−0.27	3.38
Commodity energy	16.68	0.26	0.78	0.44	2.69	0.10	0.10	4.22
Commodity metals	−36.80	−0.58	−0.93	0.36	5.01	−0.12	−0.08	3.85
Energy infrastructure	0.65	0.01	0.02	0.98	3.93	0.00	−1.22	6.36
Multi-strategy	1.18	0.02	0.17	0.87	0.89	0.02	−0.95	4.78
Real estate	19.83	0.31	1.95	0.06	1.28	0.25	−0.20	2.74
Systematic diversified	1.48	0.02	0.09	0.93	2.15	0.01	−0.50	4.70
Hedge funds	5.23	0.08	0.52	0.61	1.27	0.07	−0.42	2.91

r_{bh} is the cumulative log buy-and-hold return starting from the first data point (end of April 2013) to the last available data point (end of June 2018). All assets share the same start and end dates. Return is the average monthly log return percentage. The *t*-statistics describes the corresponding *t*-value of the mean returns with the null of an average return equal to 0% and the respective *p*-value. Standard deviation (SD) depicts the volatility of the average monthly returns (in %). Return/risk is the ratio of mean return and standard deviation. The third and fourth return moments are denoted by skewness and kurtosis. The asset classes highlighted in italics describe an equally weighted portfolio of all assets that are part of the respective asset class. The portfolios are rearranged monthly.

BTC could be seen as a proxy for a market-cap index. The reference asset classes, on the other hand, differ widely from this description and are reported below.

5.4.2 *Reference Asset Classes*

The reference asset classes of interest for the analysis in Chap. 6 switch from a country-centric to an asset class-based focus. While Chuen et al. (2018) and Glas and Poddig (2018) investigated the inclusion of digital assets for a US and German investor, I focus on the addition to single traditional asset classes themselves. That means I tested the outcomes of adding digital assets to six different traditional asset classes like equities, bonds, FX, and others. Therefore, I obtained reference asset class data covering the same period of time as the digital asset data.

Therefore, Tables 5.6 and 5.7 include six different asset classes in total. Each asset class consists of several single assets, which are combined into one final proxy for the respective asset class (highlighted in italics). I consider the traditional asset classes equities, bonds, commodities, and FX, as well as alternative investments in general and hedge funds in particular. For equities and bonds, the data is inspired by Fama and French (2012). They argued that the regions North America, Europa, Japan, and Asia Pacific (excluding Japan) cover most of the worldwide equity (and bond) markets. Thus, for equities I proxy these four regions through the inclusion of the S&P500 index, the Stoxx 600, Nikkei 225, and the MSCI Asia Pacific index. The same applies to bonds, where I assume the respective generic 10-year government bonds to be proxies for their regions.

The composition of the asset class commodities is inspired by the S&P Goldman Sachs commodity index (GSCI). It includes different assets from several groups of commodities. From each category, I include the first two single assets with the highest weight per category in 2018. For energy, I include Brent crude and WTI oil; for precious metals, the representatives are gold and silver. Industrial metals are comprised of copper and aluminum; agriculture consists of corn and wheat, and livestock is proxied by feeder cattle and lean hogs.[43] The asset class FX follows a similar methodology. For FX, I include all currencies that are included in the Dollar index.[44] Hence, EUR/USD, USD/JPY, GBP/USD, USD/CAD, USD/SEK, and USD/CHF describe the asset class FX in this analysis.

Furthermore, alternative investments are part of the following analysis. Digital assets themselves could be termed an alternative asset class, which is why it is especially interesting to explore whether the inclusion of digital assets leads to a beneficial shift in the efficient frontier when added to "older" alternative

[43] See https://www.spindices.com/documents/methodologies/methodology-sp-gsci.pdf for more information (accessed on December 5, 2018).

[44] See https://www.theice.com/publicdocs/futures_us/ICE_Dollar_Index_FAQ.pdf for more information (accessed on December 5, 2018).

Table 5.8 Cross-asset correlations

ρ	Digital assets	Equities	Bonds	Commodities	FX	Alternatives	Hedge funds
Digital assets	1	0.20	−0.04	0.04	0.11	0.05	0.10
Equities		1	0.07	0.34	0.20	0.81	0.67
Bonds			1	0.15	−0.43	0.18	0.23
Commodities				1	−0.21	0.29	0.45
FX					1	0.06	−0.10
Alternatives						1	0.79
Hedge funds							1

The table includes the correlation matrix for all asset classes under consideration

investments. The constituents of the asset class alternative investments are chosen subjectively and focus on the US market, since it was not possible to obtain meaningful global or regional variants. Thus, I use indices for US private equity (PE), infrastructure, hedge funds, and real estate as a proxy for alternative investments in general. Last, but not least, I use Hedge Fund Research indices, which are available in Datastream when searching for the term "hedge funds," to illustrate the US hedge fund universe. The single indices can be found in Table 5.7. I do not include a hedge fund index for North America or for hedge funds that focus on different regions to limit this specific asset class to the US only, since other country- or region-specific strategies are not available on Datastream.

For all assets, I downloaded monthly total return data from Thomson Reuters Datastream. The period of time was the same as for digital assets. The descriptive statistics for all reference assets are reported in Tables 5.6 and 5.7, analogous to Table 5.5. Comparing Table 5.5 with Tables 5.6 and 5.7 already yields some structural differences. Whereas the magnitude of the returns and volatilities is clearly lower for all reference assets, the assets in Tables 5.6 and 5.7 exhibit mostly negative skewness and lower kurtosis. These differences can also be observed in Table 5.8, which includes the cross-asset correlation coefficients. From there it emerges that digital assets are mostly uncorrelated to the reference asset classes. Summarizing Tables 5.6, 5.7, and 5.8 draws a relatively clear picture. Digital assets are a distinct asset class with no connection (i.e., low correlation) to other asset classes and fundamentally differing return behavior. However, the volatility of digital assets is a multiple of the volatilities observed in the other asset classes.

5.4.3 "Fundamental" Data

For the different analyses in Chap. 9, I applied several datasets from multiple sources. Table 5.9 includes all the relevant data sources and outlines descriptive statistics for the respective panels. Most important, and the basis of this analysis, is the data concerning traded digital assets. Here, I relied on the same digital asset data as described above. However, all values from Panel A are quoted in BTC, not

Table 5.9 Data composition for Chap. 9

No.	Variable	Source	Start	Mean	*t*-Stat	SD (%)	No. of obs.
Panel A: Digital assets							
	All	[1]	April 2013	−6.65	−1.49	35.06	63
	Mineable	[1]	April 2013	−6.39	−1.39	36.15	63
	Not mineable	[1]	April 2013	−5.23	−1.42	28.74	63
Panel B: BTC fundamental data							
1	Average block size	[2]	April 2013	0.57	14.32*	22.97	63
2	Transactions per day	[2]	April 2013	164,749.30	13.71*	21.51	63
3	Mempool size	[2]	April 2016	24,280,399.51	4.13*	156.49	27
4	Bitcoins in circulation	[2]	April 2013	14,620,313.49	63.75*	0.31	63
5	Blockchain size	[2]	April 2013	66,782.71	10.25*	1.63	63
6	Transactions per block	[2]	April 2013	1062.69	13.66*	22.69	63
7	Orphaned blocks	[2]	April 2013	0.41	3.92*	—	63
8	Median confirmation time	[2]	April 2013	9.40	23.05*	29.85	63
9	Hash rate	[2]	April 2013	4,498,508.58	4.05*	24.16	63
10	Difficulty	[2]	April 2013	581,464,638,188.91	4.07*	21.82	63
11	Mining revenue	[2]	April 2013	4,510,743.45	5.24*	36.16	63
12	Total transaction fees (BTC)	[2]	April 2013	84.23	5.42*	49.12	63
13	Cost % of transaction volume	[2]	April 2013	2.00	10.91*	36.35	63
14	Cost per transaction	[2]	April 2013	26.34	7.52*	30.52	63
15	Unique addresses	[2]	April 2013	323,414.14	13.4*	18.97	63

16	Transactions rate	[2]	April 2016	3.24	15.3*	37.67	27
17	Mempool transaction count	[2]	April 2016	19,488.93	3.96*	140.21	27
18	Mempool size growth	[2]	April 2016	1383.08	13.39*	45.86	27
19	Transaction volume ex popular addresses	[2]	April 2013	155,335.03	12.93*	21.43	63
20	Transaction volume ex long chains	[2]	April 2013	103,941.89	12.03*	18.65	63
21	Total output value	[2]	April 2013	1,646,596.89	9.05*	53.60	63
22	Estimated transaction value	[2]	April 2013	214,931.26	17.93*	34.28	63
23	Bitcoin wallet users	[2]	April 2013	7,996,149.52	8.38*	5.45	63
24	Number of unspent transactions output	[2]	April 2013	31,372,758.37	12.96*	5.36	63
Panel C: ETH fundamental data							
25	Total ETH supply	[3]	July 2015	87,169,020.14	58.64*	0.24	36
26	Total number of transactions	[3]	July 2015	242,028.50	27.32*	—	36
27	Block gas usage	[3]	July 2015	2,260,259.28	4.82*	39.54	36
28	Total gas usage	[3]	July 2015	12,194,590,624.36	4.66*	40.09	36
29	Average block usage	[3]	July 2015	32.11	5.4*	49.74	36
30	Hashrate	[3]	July 2015	30,942.89	1.19	224.72	36
31	Mining reward	[3]	July 2015	26,781.69	30.98*	13.28	36
32	Block mining reward	[3]	July 2015	24,755.42	26.10*	12.72	36

(continued)

Table 5.9 (continued)

No.	Variable	Source	Start	Mean	*t*-Stat	SD (%)	No. of obs.
33	Uncle mining reward	[3]	July 2015	1737.33	15.72*	33.44	36
34	Fee mining reward	[3]	July 2015	288.97	4.02*	61.19	36
35	Distinct miners	[3]	July 2015	77.25	6.65*	20.45	36
36	Mining revenue	[3]	July 2015	77.92	10.47*	39.35	36
37	Blocks per day	[3]	July 2015	5523.67	40.55*	18.06	36
38	Uncles per day	[3]	July 2015	590.83	9.81*	37.93	36
39	Block time	[3]	July 2015	15.64	27.96*	18.61	36
40	Difficulty	[3]	July 2015	919,689,797,743,802.90	4.48*	45.53	36
41	Block size	[3]	July 2015	8215.22	5.21*	26.05	36
42	Block gas limit	[3]	July 2015	4,948,129.36	13.94*	111.18	36
43	New accounts	[3]	July 2015	38,670.06	4.24*	116.25	36
44	Total accounts	[3]	July 2015	7,772,960.86	3.73*	52.03	36

This table describes the data used in the course of this work. I report the exact data source and the start of the time periods with available data, as well as the mean, the *t*-statistics of the mean, the standard deviation (in %) and the number of observations (months with available data). * denotes significance at the 5% level. For Panel A, all values are based on the returns of an equally weighted "market" index constructed from all available assets. All the statistics described represent the raw value of each variable itself and are neither standardized nor adjusted

Sources: [1] www.coinmarketcap.com; [2] www.blockchain.info; [3] www.etherchain.org

USD, terms. I additionally retrieved information for each digital asset as to whether it is mineable or not (i.e., if it is connected to its own blockchain). Data with respect to the four categories described in Chap. 1 (if a coin is a commodity, equity, or other token) could not be obtained. I used this data to conduct an initial partial least squares (PLS) and principal component analysis (PCA), and to calculate several investment styles based on characteristics, which are described in more detail in Chaps. 8 and 9. In Chap. 9, I also exclude digital assets with less than six (monthly) observations to make the PCA and PLS more reliable.[45]

Moreover, 44 "fundamental" variables (Panels B and C) associated with the two most prominent blockchains, bitcoin and ethereum, are added. This choice is pretty trivial. When quoting digital assets on crypto exchanges, two alternatives generally exist: either quote a digital coin against a fiat currency or quote against another digital currency. Since quotation against fiat money is exclusively done for the largest (in terms of market capitalization and thus turnover) digital assets, smaller virtual coins are quoted against other larger digital assets, in this case BTC and ETH. This means that both digital assets are heavily relied on and are part of multiple trades. Thus, obtaining fundamental data for these two digital coins should be suitable to cover a large share of the available information content, which is also an extension of the current literature (Hayes 2017; Wheatley et al. 2018).

5.5 Clustering of Digital Assets

For the subsequent analyses, it is still important to determine if the data can actually be used for asset pricing tests. As described in Chap. 2, researchers started sorting assets into portfolios in the early 1970s. Since single assets like stocks or bonds contain idiosyncratic variation, missing data, and other data issues, condensing the data into portfolios solves most of those problems. Also, digital assets, with very unbalanced datasets, may contain a large degree of idiosyncratic variation,[46] which would aggravate any asset pricing-related study. Therefore, the identification of different clusters or portfolios could solve several issues at once. First, by forming portfolios comprised of similar assets the dataset would become more balanced. Second, if different clusters exist in the data, these could be used as possible test assets in asset pricing analyses and diminish idiosyncratic variations. Third, a side product of such a clustering could be the identification of possible theoretical groups of digital assets like utility tokens, traditional cryptocurrencies, equity tokens, or commodity tokens (as stated in the introduction).

Therefore, I applied different approaches to cluster the (daily) return data. The first is classical hierarchical clustering based on Euclidean distances, and the second is clustering based on correlations between two respective digital assets.

[45] Details can be found in Chap. 9.

[46] Identifying common sources of risk/variation is one part of this book.

Hierarchical clustering is just one way of clustering data. Other methods could be machine learning algorithms like k-means or fuzzy c-means. Those approaches, however, rely on very high data availability, which is not a given for digital asset data, and consequently are not part of this work. As such, hierarchical clustering is able to work with missing data and consequently is the main clustering method applied in this section. For robustness, I utilized two distance measures (Euclidean distance and correlation) and several different clustering algorithms to ensure the results obtained are not subject to some special instances or data mining. If all of them point at the same results, different clusters can be assumed. If not, the data then just seems to be too heterogeneous or unbalanced for an appropriate cluster analysis.

The first distance measure, the Euclidean distance d_{ab} between the return vectors r_a and r_b, is calculated as (D'Agostino and Dardanoni 2008)

$$d_{ab} = \frac{n_{\text{all}}}{n}[(r_a - r_b)'(r_a - r_b)]. \tag{5.3}$$

Since the digital asset dataset is very unbalanced, I standardized the Euclidean distance by the number of total days, n_{all}, divided by days with overlapping observations, n. The second distance measure is simple (pairwise) correlation. The pairwise correlation ρ is the correlation between two vectors neglecting any row with missing data (Asuero et al. 2006):

$$\rho(A, B) = \frac{1}{N-1}\sum_{i=1}^{N}\left(\frac{A_i - \mu_A}{\sigma_A}\right)\left(\frac{B_i - \mu_B}{\sigma_B}\right), \tag{5.4}$$

with μ and σ being the mean and the standard deviation of a vector, respectively. An alternative form of the correlation can be written as

$$\rho(A, B) = \frac{\text{cov}(A, B)}{\sigma_A \sigma_B}. \tag{5.5}$$

Due to the unbalanced character of the data and the problem of how to measure distance,[47] I utilized four different kinds of linkage approaches to cluster the data. The first one is the most common and easiest way to cluster data. Two points with the smallest Euclidean distance are deemed to be similar and therefore put into the same assumed cluster. However, such a single linkage (linkage based on the

[47] Clustering based on Euclidean distances can be achieved in several ways (Johnson 1967). One obvious way is to look for the smallest distance between two points (Sakshi Patel and Jatain 2015). That approach, however, neglects the information available for all other points. Therefore, it is also possible to determine the link between two points (i.e., linkage) as the average distance to all other points. Another linkage method could be the sum of squared distances to all other points, or the smallest average distance to other points. Since the question of how to link distances is not trivial, I applied four different methods of linkage for robustness reasons.

distance between two points) tends to be subject to an issue called chaining.[48] Chaining describes the issue of exclusively looking at the closest pair. Since all other distances are not taken into account, the smallest pair will be put into the same cluster, regardless of the absolute value of the Euclidean distance. Consequently, linkage based on the smallest Euclidean distance is not able to cope with noise in the data, since noise might distort the results.

Next to the shortest distance/highest correlation between two assets (i.e., smallest Euclidean distance), I used centroid distance, weighted average distance, and the inner squared distance, which is comparable to a minimum variance algorithm. Centroid distance[49] assumes a "center"/"centroid" for each cluster. The centroid is the average of all points within one cluster, and therefore the distance between clusters can also be described as the distance between the centroids (Abdi 2009):

$$d_{ab} = ||\bar{x}_a - \bar{x}_b||_2, \tag{5.6}$$

with

$$\bar{x}_a = \frac{1}{n_a} \sum_{i=1}^{n_a} x_{ai}, \tag{5.7}$$

where $||\bar{x}_a - \bar{x}_b||_2$ is the Euclidean distance between $\bar{x}_a$ and $\bar{x}_b$ as the centroids of clusters a and b, and n_a is the number of elements in cluster a. Centroid distance is therefore able to cope with noise in the data, but it could theoretically result in inverse dendrograms, which cannot be interpreted properly.[50] Therefore, an alternative approach is weighted average distance. It describes the mean distance between all pairs in any two clusters p and q weighted by the number of assets within each cluster:

$$d_{ab} = \frac{d(p,b) + d(q,b)}{2}. \tag{5.8}$$

Clusters based on an average distance are less sensitive to noise but cannot yield inverse dendrograms. However, like centroid linkage, they might produce different results when using a monotone increasing transformation (i.e., distance^2).[51] Alter-

[48] See Mirkin (2005), Bacher et al. (2010), and "Clustering 2: Hierarchical clustering," https://www.stat.cmu.edu/~ryantibs/datamining/lectures/05-clus2.pdf, accessed on November 3, 2019.

[49] The following equations are reproduced in a slightly adjusted form from the Matlab documentation; see "More about (linkages)," https://www.mathworks.com/help/stats/linkage.html, accessed on June 26, 2019.

[50] See Mirkin (2005), Bacher et al. (2010), and "Clustering 3: Hierarchical clustering (continued); choosing the number of clusters," https://www.stat.cmu.edu/~ryantibs/datamining/lectures/06-clus3.pdf, accessed on November 3, 2019.

[51] See Mirkin (2005), Bacher et al. (2010), and "Clustering 2: Hierarchical clustering," https://www.stat.cmu.edu/~ryantibs/datamining/lectures/05-clus2.pdf, accessed on November 3, 2019.

natively, inner squared distance ("Ward's method") links distance as an incremental sum of squares and is a combination of the average and the centroid approach.[52] Sum of squares here is the sum of the squared distance between all assets within the cluster and the centroid of the cluster. The incremental increase of the within-cluster sum of squares thus serves as a distance measure, which needs to be minimized assuming an asset would join two clusters:

$$d_{ab} = \sqrt{\frac{2n_a n_b}{(n_a + n_b)}} ||\bar{x}_a - \bar{x}_b||_2, \tag{5.9}$$

where n_a and n_b are the numbers of elements in clusters a and b. Due to the different starting and ending points of the single digital asset data, I needed to calculate pairwise covariance coefficients first. Since this matrix is no longer necessarily positive semidefinite, I applied spectral decomposition to arrive at a positive-semidefinite correlation matrix.[53]

All four approaches, however, yield no clear results for either Euclidean distances or correlations. It appears that the results are relatively heterogeneous, so no clusters can be determined. Figures 5.13, 5.14, 5.15, and 5.16 illustrate this finding for Euclidean distances. The same findings apply to dendrograms based on correlations (see Figs. 5.17, 5.18, 5.19, and 5.20).

All eight dendrograms depict different results. While the Euclidean weighted average distance and centroid distance seem relatively similar, the dendrograms for the shortest Euclidean distance and the inner squared distance show other outcomes. Therefore, I cannot arrive at one or several *clear* cluster(s), which does not allow a combination or condensation of the dataset. The same is true for clustering methods based on correlations. All four dendrograms mainly depict different results, and the centroid approach even results in an inverse dendrogram (that is also due to the distance measure, correlation, which is not meant to be used for centroid linkage).

As a robustness check, however, I divided the data into three different samples for the subsequent analyses. One sample included all available digital assets (which is in line with this cluster analysis), another sample only included digital assets with their own blockchain (similar to traditional cryptocurrencies, with the purpose of functioning as a medium of exchange), and a third containing assets without their own blockchain (this would include utility, equity, and commodity tokens). If the subsequent results differ significantly from each other, this might indicate the existence of more than one cluster. If not, the finding of no clear cluster(s) is strengthened. Next to the above analysis, clustering could also be used to look for digital assets that share similar return and risk characteristics. Clustering such assets, which would not necessarily consist of the same kind (e.g., solely utility tokens),

[52] It also uses the same objective function as k-means clustering (Pang-Ning et al. 2019).

[53] For more information regarding spectral decomposition, see Bai and Silverstein (2010) and Wansbeek and Kapteyn (1982) or refer to Sect. 9.3.

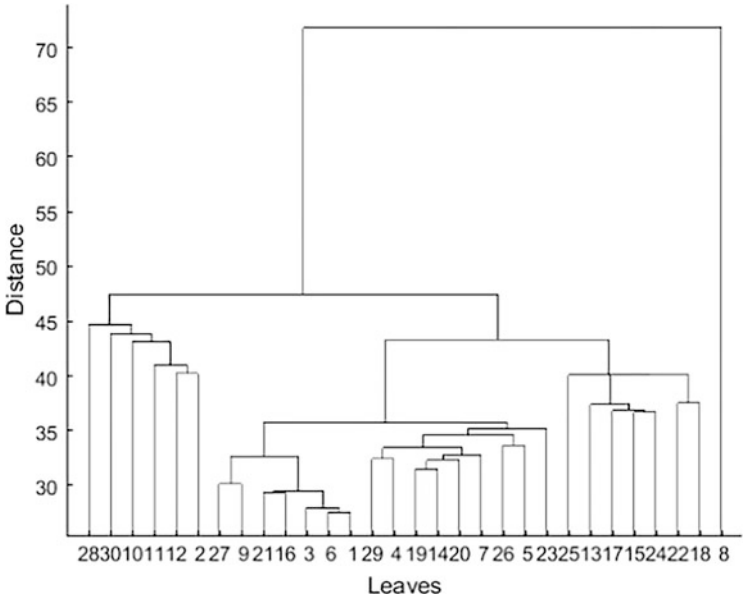


Fig. 5.13 Dendrogram shortest distance

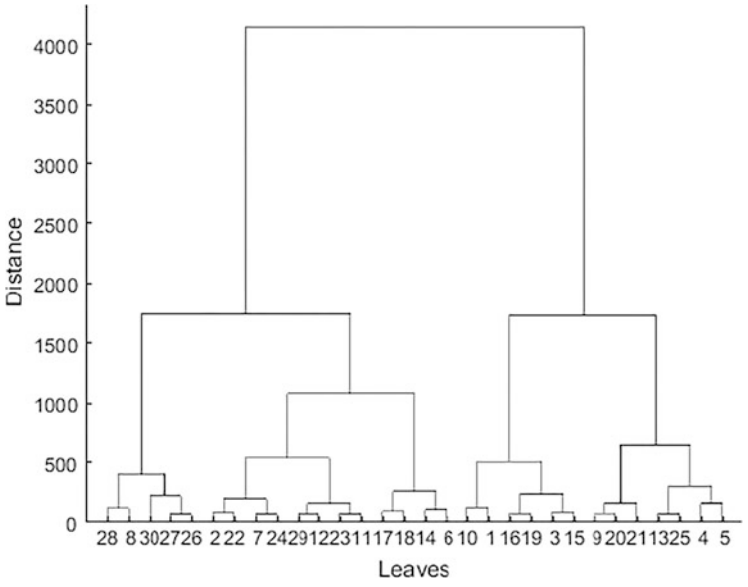


Fig. 5.14 Dendrogram squared distance

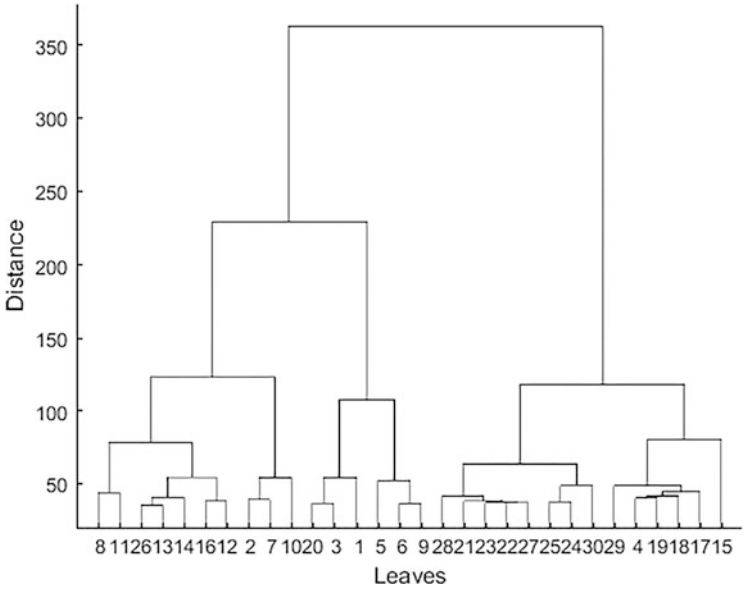


Fig. 5.15 Dendrogram weighted distance

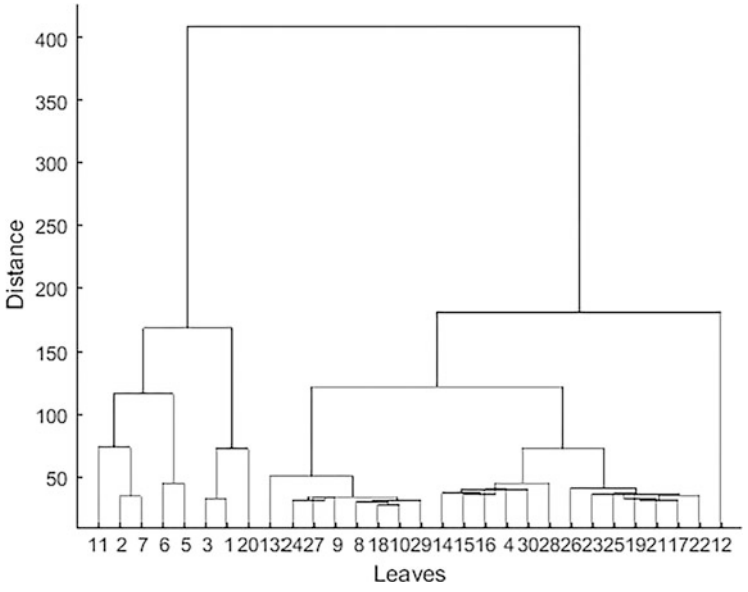


Fig. 5.16 Dendrogram centroid distance

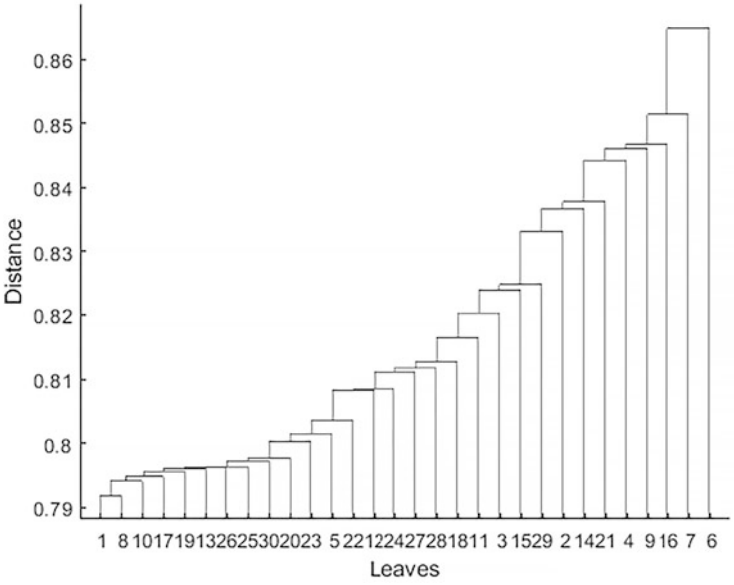


Fig. 5.17 Dendrogram shortest distance (2)

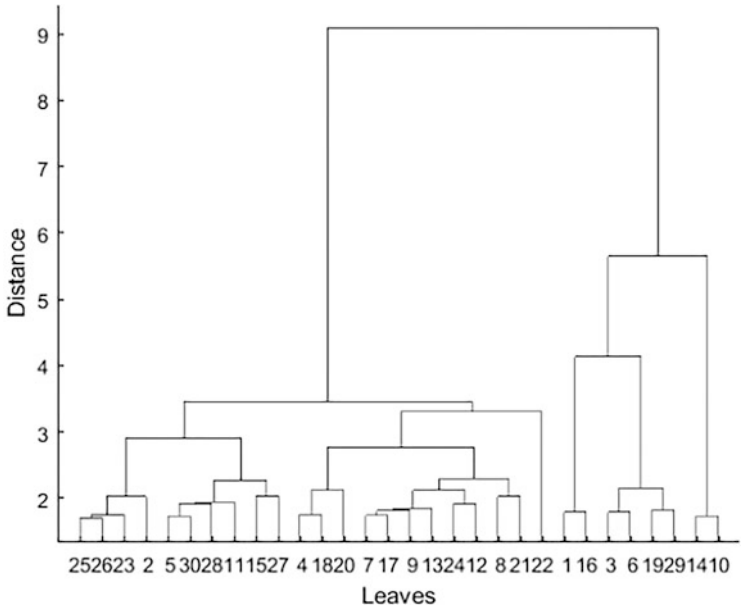


Fig. 5.18 Dendrogram squared distance (2)

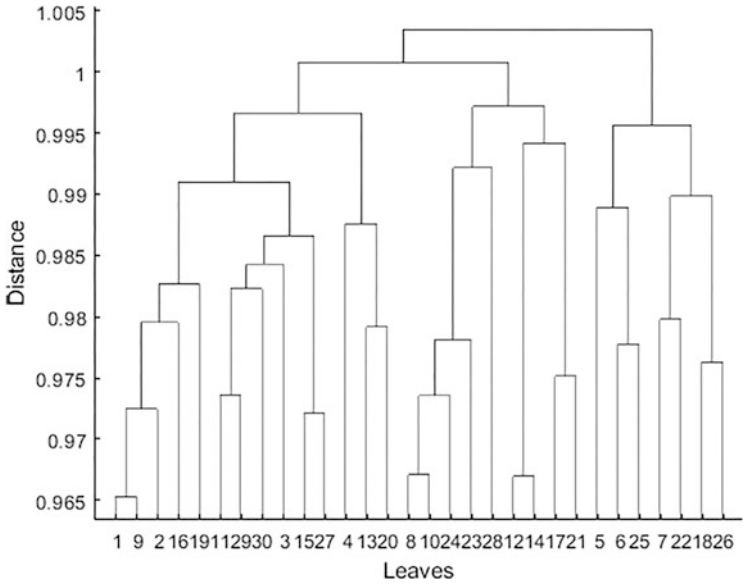


Fig. 5.19 Dendrogram weighted distance (2)

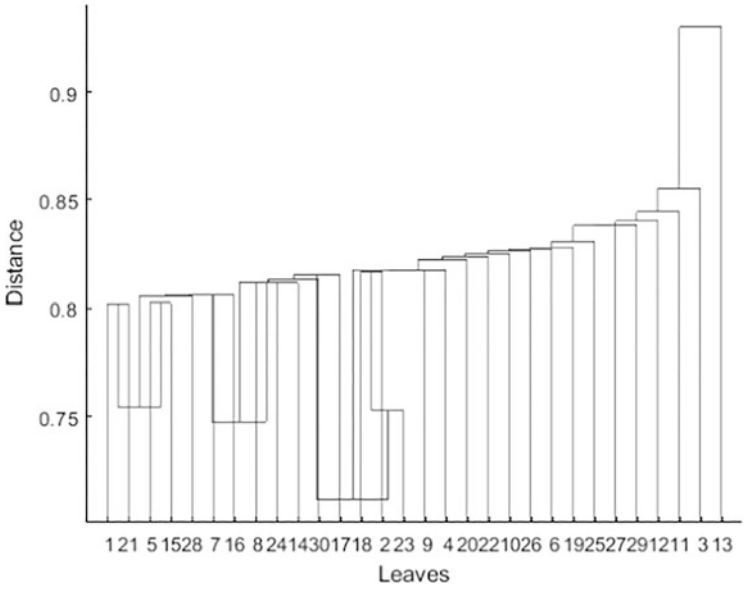


Fig. 5.20 Dendrogram centroid distance (2)

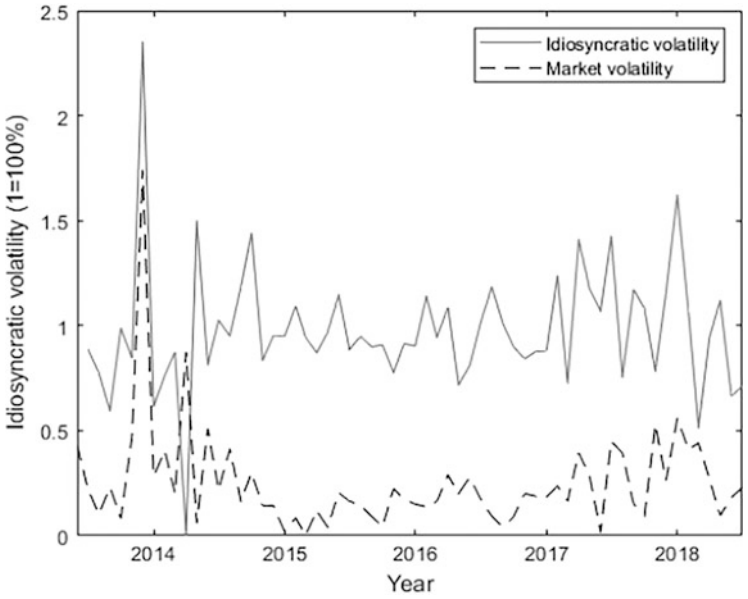


Fig. 5.21 *IR* original data

could result in the construction of proper test assets. That, however, is not part of this work,[54] which is why I leave that question open for further research.

The absence of different clusters could also be traced back to a high (daily) aggregate idiosyncratic variance, which is downsampled to monthly values for Figs. 5.21 and 5.22. Thus, I calculated the aggregate idiosyncratic variance V of asset i at time t in line with Goyal and Santa-Clara (2003) and Angelidis and Tessaromatis (2008) as

$$V_{i,t} = \sum_{d=1}^{D_t} r_{i,d}^2 + 2 \sum_{d=2}^{D_t} r_{i,d} r_{i,d-1}, \tag{5.10}$$

with D_t the number of observations in month t and $r_{i,d}$ the return of asset i on day d. If $V_{i,t} < 0$ holds, I dropped the latter part of Eq. 5.10 in accordance with Angelidis and Tessaromatis (2008). To distinguish between market and idiosyncratic risk, the aggregate idiosyncratic variance IR at time t is calculated as

$$IR_t = V_t - MV_t, \tag{5.11}$$

[54] Due to the weak clustering outcomes.

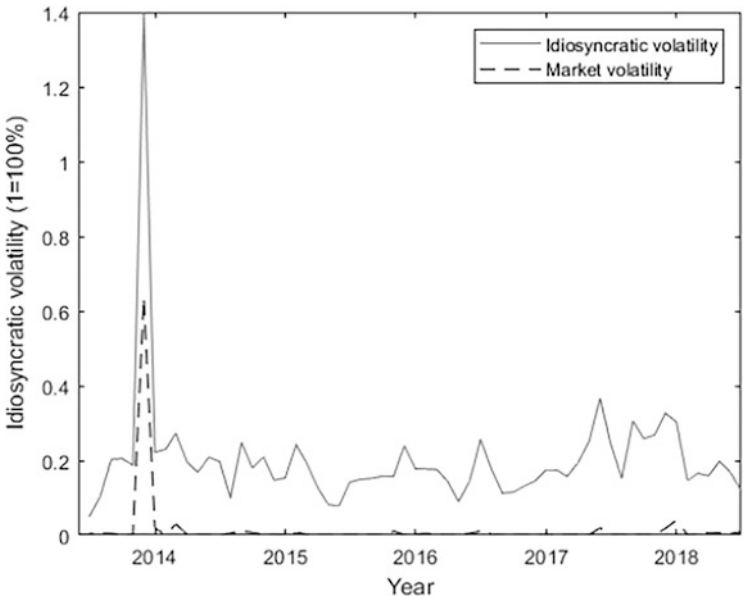


Fig. 5.22 *IR* condensed data

where V_t is the equally weighted idiosyncratic variance of all digital assets and MV_t is the variance of the returns of a monthly, equally weighted market index based on all available digital assets.[55] Figure 5.21 depicts the aggregate idiosyncratic volatility over time.[56] Such a high aggregate idiosyncratic volatility might consequently lead to no comprehensible cluster in the returns of digital assets.

In an attempt to reduce the idiosyncratic variance of the data for other analyses, I tried to condense the asset universe using an alternative approach to classical clustering. A condensation should not only reduce the idiosyncratic volatility but also enhance data availability. Thus, I calculated the Euclidean distances over the full data sample and equally weighted the returns of the asset pair with the smallest distance. Afterwards, I deleted the two original data series and added one combined return vector. This was done until I arrived at only half of the initial asset universe. I also did not consider Euclidean distances with regard to the newly created return

[55] Also measured by r^2 to be in line with Eq. 5.10.

[56] For comprehensibility, I convert the aggregate idiosyncratic variance to aggregate idiosyncratic volatility.

vectors to make sure the initial assets were combined at least once.[57] When running the cluster and all subsequent analyses on the condensed data, I did not arrive at significantly different results than when using the original data. Although the idiosyncratic and market volatility were reduced dramatically, the spread between both remained relatively high (see Fig. 5.22). That is why I report all results based on the original data described in the previous parts of this chapter.

Following the introduction, the methods of today's asset pricing literature, a literature review, and a first overview of digital assets and the respective digital asset data, I now continue to the empirical parts of this book. Since digital assets are a relatively young phenomenon, many unexplored topics and questions exist, which cannot be answered in one single study. Therefore, I focus on some "greater" topics, as described in Chap. 1.

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[57] One reason for this is the different starting and ending points of the single digital assets. For pairs with differing numbers of observations, the newly created series exhibits a correlation coefficient of 1 to the original data series until the similar asset kicks in. This results in an almost identical return series for the new and one of the initial assets. Thus, the Euclidean distance becomes very small and no other asset pairs can be found. Since this issue is not easy to solve due to very heterogeneous data availability and all previous results pointing at the same direction, this issue might become a topic of future research.

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Chapter 6
Digital Assets in a Multi-asset Context



6.1 Hedge Funds and Digital Assets

The previous chapter showed that digital assets are a relatively heterogeneous asset class with widely dispersed returns. This could lead to increased diversification when allocating digital assets to a portfolio of traditional assets. A similar thought was investigated around the turn of the millennium. During that time, hedge fund research kicked off and researchers investigated the effects of an allocation of hedge funds to other asset classes. The methodological basis, mean–variance spanning tests, was described earlier.[1] These spanning tests (the test for inclusion of a test asset in a portfolio of reference assets) are heavily relied on in the hedge fund literature.[2] Several studies employing this approach arrived at an upward shift of the efficient frontier when adding hedge funds to a given portfolio of reference assets. Similar analyses were conducted by Chuen et al. (2018) and Glas and Poddig (2018) with respect to the asset class digital assets. Chuen et al. (2018) arrived at beneficial results for the inclusion of some digital assets, but Glas and Poddig (2018) found encouraging results for bitcoin only. This chapter thus aims to extend the work of these two studies. I not only investigate one portfolio comprised of several asset classes, but I consider several reference portfolios of only one asset class and a portfolio that combines all asset classes under consideration.

Chuen et al. (2018) investigated the inclusion of digital assets (proxied by the CRIX index) in a portfolio comprised of asset classes of interest for a US-based investor. Their reference portfolio included the S&P500, gold, oil, the GSCI, real-estate investment trusts (REITs), and PE. Glas and Poddig (2018) ran a German-centric analysis and included the DAX index, German 10-year bonds, the

[1] More information can be found in Sect. 6.4.

[2] See, among others, Edwards and Liew (1999), Agarwal and Naik (2000), Karavas (2000), and Schneeweis and Martin (2001).

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GSCI, German REITs, and an FX basket comprised of four currencies denominated in Euro. Both studies found little evidence for the inclusion of digital assets in a traditional multi-asset framework.

This chapter extends both works in several ways. I used several reference assets and portfolios to verify the results for other asset classes, since the nature of digital assets is not yet clear. Some refer to digital assets as currencies, commodities, or some other possible type. Therefore, I tested the inclusion of this young asset class in almost every traditional asset class. By doing so, I aim to find similarities or differences between digital assets and all the other asset classes. In this case, both the reference asset classes and digital assets are denominated in USD, to make the results comparable. I also made use of the full sample of digital assets, which are equally weighted to a new test asset and rearranged monthly. Market-cap weights lead to an unfavorable dominance of bitcoin and a few other virtual coins. Investing equally in all available assets limits the influence of the largest digital coins and describes the investment universe more fairly. I ran mean–variance and risk-parity optimizations as well, in line with Glas and Poddig (2018). For the mean–variance framework I additionally report the results of the spanning tests.

The next section introduces the reference asset results of mean–variance portfolio optimization, which was developed by Markowitz (1952), in more detail. Section 6.3 shows the results when adding a test asset (an equally weighted index of digital assets) to different portfolios of reference assets. These results are further evaluated via mean–variance spanning tests in Sect. 6.4. Section 6.5 applies an equal risk contribution approach for robustness reasons, and Sect. 6.6 draws a first interim conclusion.

6.2 Mean–Variance Portfolio Optimization

As described in Sect. 2.1.1, the Markowitz (1952) framework is used to test the effects of adding digital assets to several portfolios of reference assets. Tables 5.6 and 5.7 depict the six different (traditional) reference asset classes. I additionally formed a seventh reference portfolio based on the equally weighted indices, which are highlighted in italics. Thus, I tested the six asset classes equities, bonds, commodities, FX, alternatives, and hedge funds for which the reference assets are given by the single assets associated with the respective asset class. The seventh portfolio references a multi-asset combination of the six single asset classes. That means I ran seven different analyses (i.e., mean–variance spanning tests). First, I added digital assets to each of the six single asset classes ranging from equities over FX to hedge funds; second, I constructed a multi-asset portfolio from these six asset classes and test the addition of digital assets to this example multi-asset portfolio.

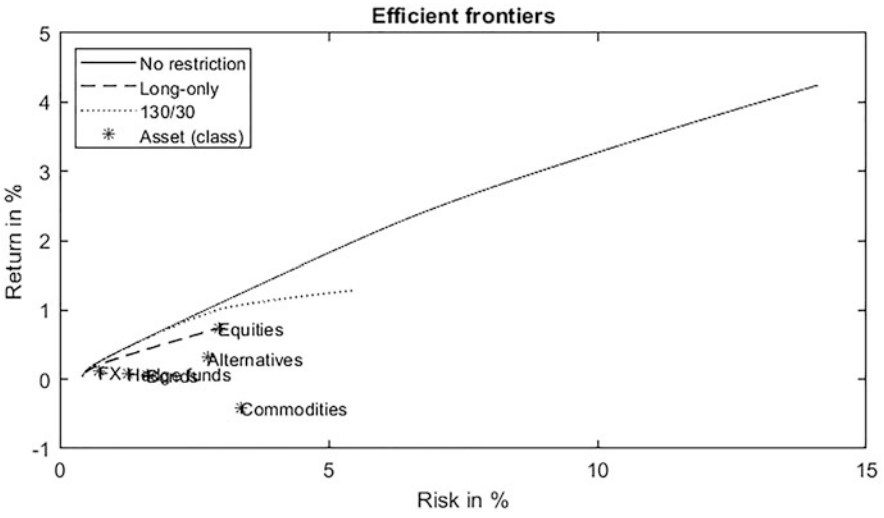


Fig. 6.1 Multi-asset efficient frontiers

For illustrative purposes I plot three different efficient frontiers in Fig. 6.1. This figure shows the optimization of the seventh reference portfolio (multi-asset). All other asset classes can be found in Appendix C. I ran the same analysis as Glas and Poddig (2018): I calculated a portfolio without any restrictions (see Eqs. 2.1–2.4),[3] a long-only portfolio, and a 130/30 portfolio where 130% of the portfolio value can be held in long positions and (−)30% in short positions. The transition map (see Fig. 6.2) depicts the relative weights per asset class for a given level of risk. In line with Chuen et al. (2018) and Glas and Poddig (2018) I ran a static in-sample analysis only. This is due to the very short data history of digital assets. Setting an initial optimization of, e.g., 12 months would reduce the dataset by 20% and make the results even less reliable.

Equities is the asset class with the highest relative weight in all three optimization frameworks (see Fig. 6.2). Equities is also the only asset class that lies on the efficient frontier (see Fig. 6.1). The allocation of capital to FX and bonds is higher for low risk levels but decreases with higher portfolio risk. Alternative investments and commodities play no important role and obtain only small weights. In the 130/30 case, the weight of alternative investments increases with higher portfolio risk.

[3] In line with Glas and Poddig (2018), I limited the weights of the unrestricted optimization to −100%.

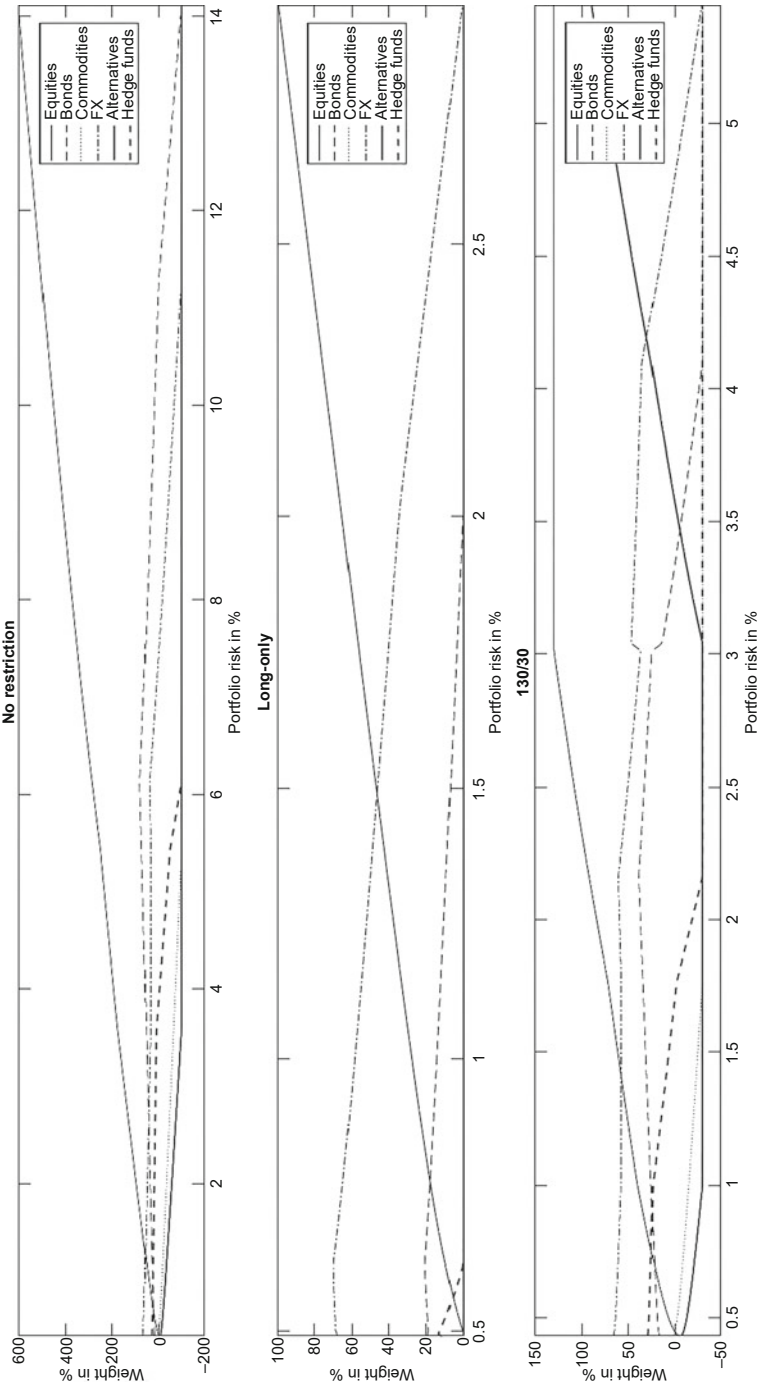


Fig. 6.2 Transition map multi-asset

6.3 The Inclusion of Digital Assets

Based on this portfolio of reference assets it is noteworthy to see whether the inclusion of the asset class digital assets leads to a more favorable return/risk structure, i.e., an upward shift of the efficient frontier. Thus, K denotes a portfolio comprised of the asset classes equities, bonds, commodities, FX, alternative investments, and hedge funds, and N represents the asset class digital assets. Hence, $N + K$ is the combination of six reference asset classes and the test asset digital assets. Figures 6.3, 6.4, and 6.5 compare the efficient frontiers of three portfolio optimization frameworks based on the reference asset classes K and a combination with N ($N + K$). Figure 6.3 presents the outcome of the almost unrestricted case.

When adding digital assets to K, the efficient frontier is extended and shifted slightly upwards. That is mainly achieved by a short position in digital assets. The inclusion of digital assets moves the efficient frontier to the upside, starting from a risk level of c.10%. The same applies to Fig. 6.5. Due to the restrictions, the $N + K$ efficient frontier is shorter than in the unrestricted case. The effect of the inclusion of digital assets, however, can already be seen at a risk level of about 7%–8%. Both variations lead to higher returns for a given level of risk when compared to K only (but not for the long-only case). For long-only, the portfolio is almost completely comprised of equities and/or FX (see Fig. 6.4). Digital assets do not find their way into the asset allocation, which results in weights of nearly 0%. Therefore, the efficient frontier of $N + K$ is close to or even the same as the efficient frontier of K.

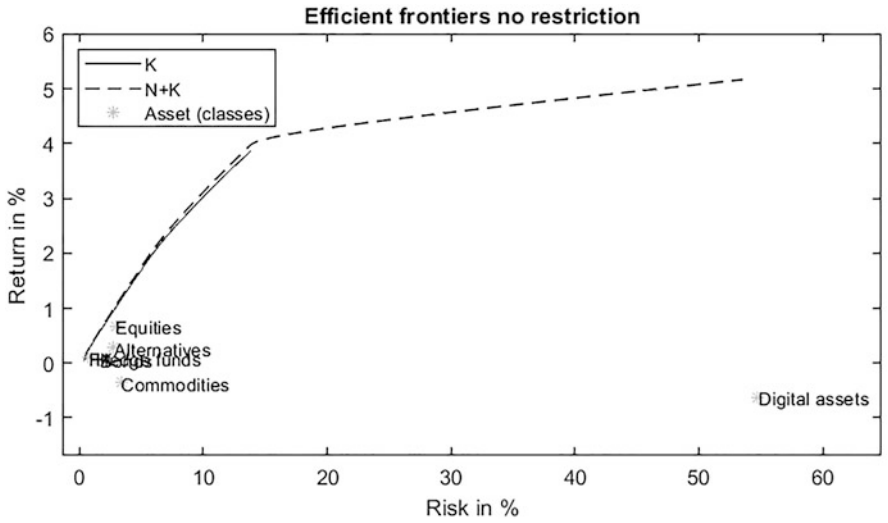


Fig. 6.3 Efficient frontiers, no restriction

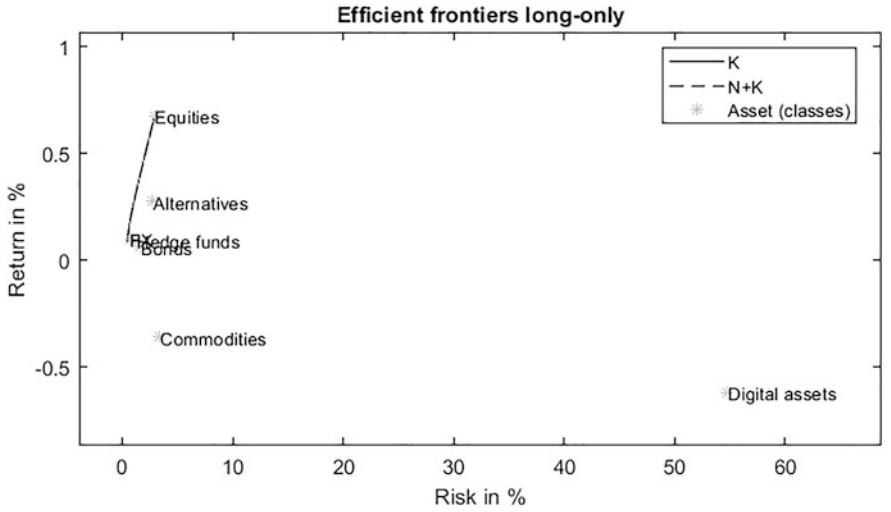


Fig. 6.4 Efficient frontiers, long-only

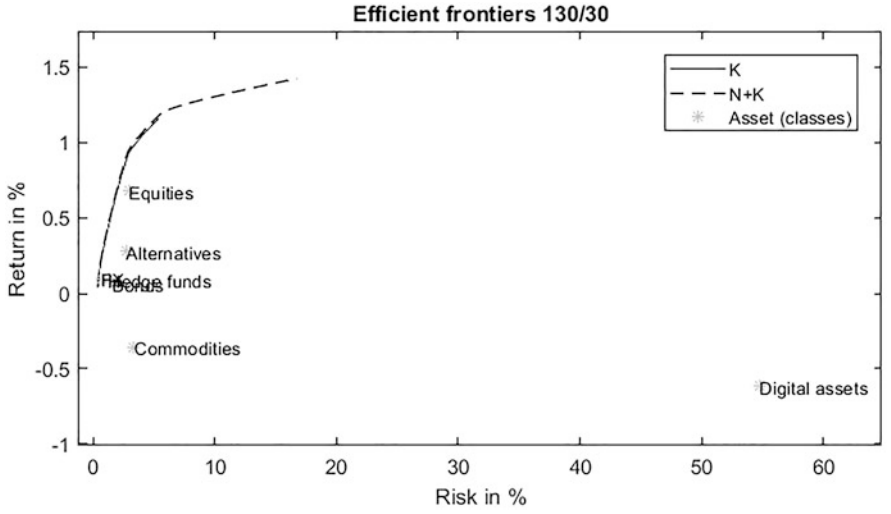


Fig. 6.5 Efficient frontiers, 130/30

The extension and the upward shift of the efficient frontiers in the unrestricted and 130/30 cases are again mainly due to an increasing short position in digital assets. An equally weighted index of digital assets yields a cumulative return of −38.33% (see Table 5.5). Shorting the index consequently results in higher returns at a given level of risk and extends the efficient frontier. At lower risk levels, digital assets carry only a very small weight with regard to the overall portfolio. The unrestricted portfolio invests the major part in equities. The 130/30 portfolio mainly allocates funds to equities and alternative investments. The remaining asset classes are considered at low risk levels or receive only small weights (see Fig. 6.6). Also, implementation is an issue at this point. Just a few digital assets can be shorted on unregulated exchanges. Regulated exchanges, which are of interest for institutional investors and banks, provide the possibility of shorting via bitcoin futures only.

The same analysis with respect to other asset classes leads to comparable outcomes. For almost every one of the asset classes considered (equities, bonds, commodities, FX, alternative investments, and hedge funds), the inclusion of digital assets yields an extended efficient frontier (see Appendix D). From the first visual inspection, I get the impression that digital assets shift the efficient frontier to the upside only for two asset classes. That is again only the case because of short positions in digital assets. These are commodities and alternative investments. This is interesting since digital assets themselves could be termed an alternative investment. Some also refer to digital assets as part of the asset class commodities. For a more similar asset class such as FX, the inclusion of digital assets only extends the efficient frontiers but does not shift them to the upside. Interestingly, digital assets are almost not considered for the long-only portfolio for any of the investigated reference asset classes.

Thus, it seems like digital assets are beneficial for the three reference portfolios: multi-asset, commodities, and alternative investments. To verify this claim, I further evaluated the inclusion of digital assets through mean–variance spanning tests.

6.4 Mean–Variance Spanning Tests

Mean–variance spanning tests are used to investigate the shift of the efficient frontier when adding one or several test assets to a given portfolio of reference assets K. The newly added test assets N should move the efficient frontier of a combined portfolio $(N + K)$ significantly upwards. N and K can each be comprised of one or several distinct assets. I aimed to evaluate the inclusion of digital assets for six single (reference) asset classes: equities, bonds, commodities, FX, alternative investments, and hedge funds. Additionally, I ran a multi-asset analysis that uses the equally weighted returns of each asset class for a multi-asset evaluation, similar to Chuen et al. (2018) and Glas and Poddig (2018). Like the work of Glas and Poddig (2018),

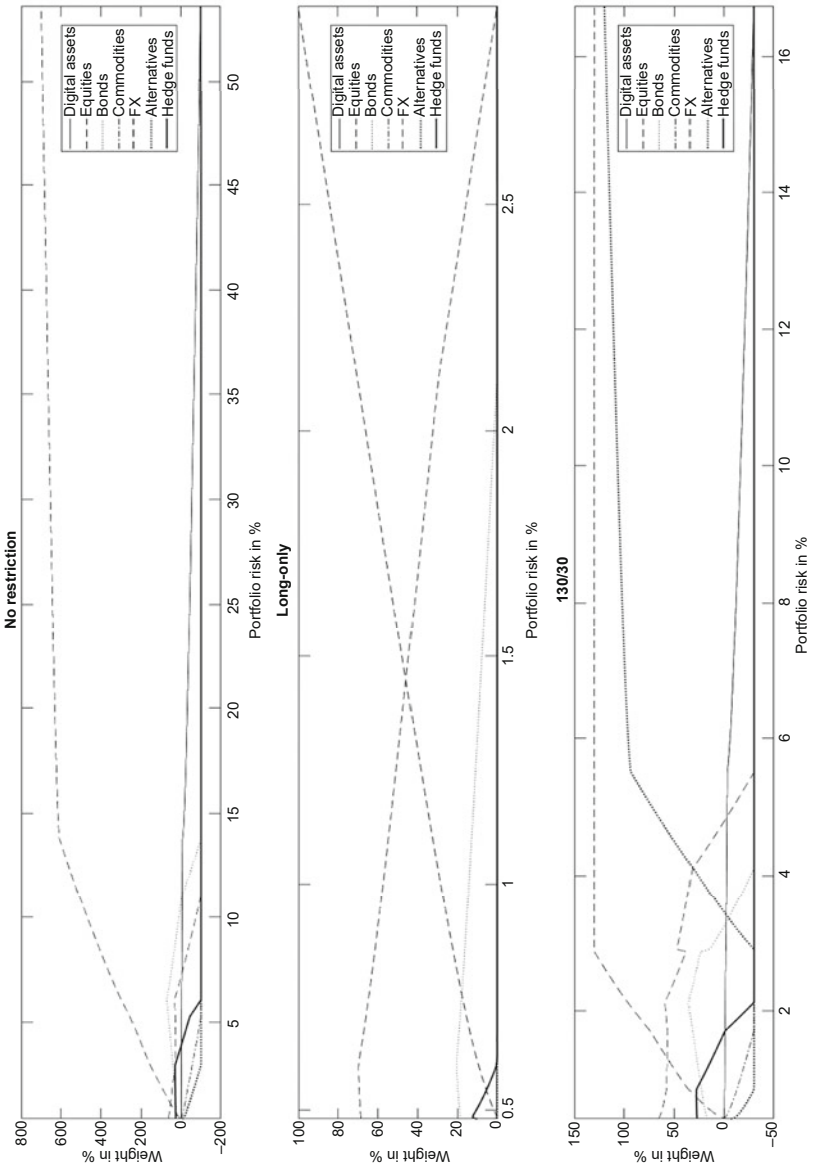


Fig. 6.6 Transition map with digital assets

I applied the same spanning tests as described by Zhou and Kan (2012) and Cao et al. (2017). The basis of their works was a very early version of spanning tests (Huberman and Kandel 1987; HK). They also used tests in line with Ferson et al. (1993) (FFK) and Bekaert and Urias (1996) (BU). The first step of the HK test is to run a simple OLS regression (Huberman and Kandel 1987):

$$\underbrace{\mathbf{r}_{N,t}}_{N\times 1} = \underbrace{\boldsymbol{\alpha}}_{N\times 1} + \underbrace{\boldsymbol{\beta}}_{N\times K}\underbrace{\mathbf{r}_{K,t}}_{K\times 1} + \underbrace{\boldsymbol{\epsilon}_t}_{N\times 1}, \tag{6.1}$$

where $\mathbf{r}_{N,t}$ is an N-vector of returns for each t of the test asset(s), $\boldsymbol{\alpha}$ is a vector, $\mathbf{r}_{K,t}$ is the K-vector of returns of the reference assets, and $\boldsymbol{\beta}$ the corresponding regression coefficient(s); $\boldsymbol{\epsilon}_t$ denotes the error term. Zhou and Kan (2012) stated that $E(\boldsymbol{\epsilon}_t)$ is an N-vector of zeros, and $E(\boldsymbol{\epsilon}_t \times \mathbf{r}'_{K,t})$ an $N \times K$ matrix of zeros. They also showed that

$$\boldsymbol{\alpha}_t = E(\mathbf{r}_{N,t}) - \boldsymbol{\beta} E(\mathbf{r}_{K,t}) \tag{6.2}$$

and

$$\boldsymbol{\beta} = \mathbf{V}_{NK}\mathbf{V}_{KK}^{-1}, \tag{6.3}$$

where $\mathbf{V}_{NK}$ and $\mathbf{V}_{KK}$ stem from the variance-covariance matrix VAR of the combined $N + K$ assets:

$$VAR(\mathbf{r}_{Kt}, \mathbf{r}_{Nt}) = \begin{bmatrix} \mathbf{V}_{KK} & \mathbf{V}_{KN} \\ \mathbf{V}_{NK} & \mathbf{V}_{NN} \end{bmatrix}. \tag{6.4}$$

Subsequently, Huberman and Kandel (1987) formulated the null hypothesis H_0 as

$$\boldsymbol{\alpha} = \mathbf{0}, \tag{6.5}$$

$$\boldsymbol{\delta} = \mathbf{0}, \quad \text{where } \boldsymbol{\delta} = \mathbf{1} - \boldsymbol{\beta}\mathbf{1}_k \tag{6.6}$$

to determine if $\mathbf{r}_{N,t}$ is in the span of $\mathbf{r}_{K,t}$. Here, $\boldsymbol{\alpha}$ is again an $N \times 1$ vector, $\boldsymbol{\beta}$ an $N \times K$ matrix, and $\mathbf{1}_k$ a K-vector of ones. If H_0 is not rejected, $\mathbf{r}_{N,t}$ lies in the span of $\mathbf{r}_{K,t}$ and can be replicated by the returns of the reference assets K. Thus, the inclusion of $\mathbf{r}_{N,t}$ would be inefficient and would not enhance the efficient frontier. The same idea can be recognized when solving the weights of N analytically for

the global minimum-variance Q_{w1} and the tangency portfolio Q_{w2} (Zhou and Kan 2012):

$$\begin{aligned} Q_{w1} &= \frac{\mathbf{Q}\mathbf{V}^{-1}\mathbf{1}_{N+K}}{\mathbf{1}'_{N+K}\mathbf{V}^{-1}\mathbf{1}_{N+K}} = \frac{[-\boldsymbol{\Sigma}^{-1}\boldsymbol{\beta}\boldsymbol{\Sigma}^{-1}]\mathbf{1}_{N+K}}{\mathbf{1}'_{N+K}\mathbf{V}^{-1}\mathbf{1}_{N+K}} \\ &= \frac{\boldsymbol{\Sigma}^{-1}(\mathbf{1}_N - \boldsymbol{\beta}\mathbf{1}_K)}{\mathbf{1}'_{N+K}\mathbf{V}^{-1}\mathbf{1}_{N+K}} = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\delta}}{\mathbf{1}'_{N+K}\mathbf{V}^{-1}\mathbf{1}_{N+K}}, \end{aligned} \tag{6.7}$$

$$Q_{w2} = \frac{\mathbf{Q}\mathbf{V}^{-1}\boldsymbol{\mu}}{\mathbf{1}'_{N+K}\mathbf{V}^{-1}\boldsymbol{\mu}} = \frac{[-\boldsymbol{\Sigma}^{-1}\boldsymbol{\beta}\boldsymbol{\Sigma}^{-1}]\boldsymbol{\mu}}{\mathbf{1}'_{N+K}\mathbf{V}^{-1}\boldsymbol{\mu}} = \frac{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_K - \boldsymbol{\beta}\boldsymbol{\mu}_N)}{\mathbf{1}'_{N+K}\mathbf{V}^{-1}\boldsymbol{\mu}} = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\alpha}}{\mathbf{1}'_{N+K}\mathbf{V}^{-1}\boldsymbol{\mu}}, \tag{6.8}$$

with $\mathbf{Q} = [\mathbf{0}_{N\times K}, \mathbf{I}_N]$ and $\mathbf{I}_N$ the $N \times N$ identity matrix. $\boldsymbol{\Sigma} = \mathbf{V}_{NN} - \mathbf{V}_{NK}\mathbf{V}_{KK}^{-1}\mathbf{V}_{KN}$, and $\boldsymbol{\mu}$ is the vector of expected returns of the combined $N + K$ or single N and K assets. From Eqs. 6.7 and 6.8 it becomes clear that the weight of test asset N equals 0, given $\boldsymbol{\alpha} = \mathbf{0}$ or $\boldsymbol{\delta} = \mathbf{0}$. This means that N would not be included in a combined portfolio of $N + K$ assets, and thus the efficient frontier remains unchanged. Through linear combination, the complete efficient frontier can be composed from only two points (i.e., portfolios) located on the efficient frontier. If the weight of N is 0 for those two portfolios, it is 0 for any other combination.

H_0 is then tested using a likelihood ratio (LR) test by applying a more general notion of Eq. 6.1:

$$\underbrace{\mathbf{Y}}_{T\times N} = \underbrace{\mathbf{X}}_{T\times K+1}\underbrace{\mathbf{B}'}_{N\times K+1} + \underbrace{\mathbf{E}}_{T\times N}, \tag{6.9}$$

where $\mathbf{Y}$ corresponds to $\mathbf{r}_{N,t}$, $\mathbf{X}$ is equal to $[\mathbf{1}, \mathbf{r}'_{K,t}]$, $\mathbf{B}$ is given by $[\boldsymbol{\alpha}, \boldsymbol{\beta}]'$, and $\mathbf{E}$ is the error term. Zhou and Kan (2012) defined the likelihood functions as follows:[4]

$$\hat{\mathbf{B}} = [\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\beta}}]' = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{Y}), \tag{6.10}$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{T}(\mathbf{Y} - \mathbf{X}\hat{\mathbf{B}})'(\mathbf{Y} - \mathbf{X}\hat{\mathbf{B}}). \tag{6.11}$$

H_0 can then be rearranged to

$$\boldsymbol{\Theta} = \mathbf{0}_{2\times N}, \tag{6.12}$$

with $\boldsymbol{\Theta} = [\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\delta}}]'$. Due to $\boldsymbol{\Theta} = \mathbf{A}\mathbf{B} + \mathbf{C}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & \mathbf{0}'_K \\ \mathbf{0} & -\mathbf{1}'_K \end{bmatrix} \tag{6.13}$$

[4] The following equations are reproduced in a slightly adjusted form from Zhou and Kan (2012).

and

$$\mathbf{C} = \begin{bmatrix} \mathbf{0}'_N \\ \mathbf{1}'_N \end{bmatrix}, \tag{6.14}$$

the maximum likelihood estimator $\hat{\mathbf{\Theta}}$ is given by

$$\hat{\mathbf{\Theta}} = \mathbf{A}\hat{\mathbf{B}} + \mathbf{C}. \tag{6.15}$$

Adding the auxiliary variables $\hat{\mathbf{G}}$ and $\hat{\mathbf{H}}$, the LR test can be performed for λ (see below), where

$$\hat{\mathbf{G}} = T\mathbf{A}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{A}' \tag{6.16}$$

and

$$\hat{\mathbf{H}} = \mathbf{\Theta}\hat{\mathbf{\Sigma}}^{-1}\mathbf{\Theta}' \tag{6.17}$$

such that

$$LR = T\sum_{i=1}^{2}\ln(1+\lambda_i) \sim \chi^2_{2N}, \tag{6.18}$$

$$W = T(\lambda_1 + \lambda_2) \sim \chi^2_{2N}, \tag{6.19}$$

$$LM = T\sum_{i=1}^{2}\left(\frac{\lambda_i}{1+\lambda_i}\right) \sim \chi^2_{2N}, \tag{6.20}$$

with LR, W, and LM following a χ^2 distribution. W represents a Wald test, LM is a Lagrange multiplier, and λ_i is the eigenvalue of $\hat{\mathbf{H}}\hat{\mathbf{G}}^{-1}$. To circumvent the assumption of a χ^2-distributed LR, FFK and BU apply a generalized method of moments, inspired by Hansen (1982) and Hansen and Jagannathan (1991).[5]

In our case, $\mathbf{r}_N$ is the return vector of an equally weighted digital asset index, and $\mathbf{r}_K$ is the matrix of the specific reference asset returns of the asset class under consideration. Additionally, I followed the approach of Glas and Poddig (2018) and included the single digital assets as described in Table 5.5 as single test assets. Table 6.1 reports the p-values for the three spanning tests (HK, FFK, and BU) for all the test assets and seven portfolios of reference assets.

[5] For more information refer to the original works of Ferson et al. (1993) and Bekaert and Urias (1996).

Table 6.1 Mean–variance spanning tests

Test	Index	BTC	USDT	ETH	XRP	LTC	EOS	MIOTA	DASH	XEM	DOGE	WAVES	TRX	ADA	XMR
Equities															
HK	4.03	**9.37**	**42.34**	**8.30**	2.18	3.18	3.26	0.26	**9.52**	**7.07**	1.43	2.54	**6.07**	2.57	4.01
	0.13	0.01	0.00	0.02	0.34	0.20	0.20	0.88	0.01	0.03	0.49	0.28	0.05	0.28	0.13
FFK	4.16	**10.12**	**76.51**	**9.40**	2.22	3.27	3.79	0.26	**10.45**	**7.77**	1.44	2.68	**8.67**	3.03	4.18
	0.12	0.01	0.00	0.01	0.33	0.20	0.15	0.88	0.01	0.02	0.49	0.26	0.01	0.22	0.12
BU	3.90	**8.70**	**25.83**	**7.36**	2.14	3.10	2.82	0.26	**8.70**	**6.45**	1.41	2.41	4.42	2.20	3.85
	0.14	0.01	0.00	0.03	0.34	0.21	0.24	0.88	0.01	0.04	0.49	0.30	0.11	0.33	0.15
Bonds															
HK	0.07	2.64	**6.45**	**6.75**	1.56	1.36	0.21	0.50	**7.67**	**10.98**	0.99	2.52	1.57	1.53	2.67
	0.97	0.27	0.04	0.03	0.46	0.51	0.90	0.78	0.02	0.00	0.61	0.28	0.46	0.47	0.26
FFK	0.07	2.70	**7.02**	**7.47**	1.59	1.37	0.21	0.51	**8.26**	**12.73**	1.00	2.66	1.72	1.68	2.74
	0.97	0.26	0.03	0.02	0.45	0.50	0.90	0.78	0.02	0.00	0.61	0.26	0.42	0.43	0.25
BU	0.07	2.59	**5.95**	**6.12**	1.54	1.34	0.21	0.49	**7.13**	**9.54**	0.98	2.39	1.44	1.39	2.60
	0.97	0.27	0.05	0.05	0.46	0.51	0.90	0.78	0.03	0.01	0.61	0.30	0.49	0.50	0.27
Commodities															
HK	0.41	4.05	**26.35**	**9.03**	0.95	0.96	—	**13.26**	3.68	**8.86**	1.67	1.21	—	—	2.75
	0.81	0.13	0.00	0.01	0.62	0.62	—	0.00	0.16	0.01	0.43	0.55	—	—	0.25
FFK	0.41	4.19	**37.64**	**10.34**	0.96	0.97	—	**24.24**	3.82	**9.98**	1.70	1.24	—	—	2.83
	0.81	0.12	0.00	0.01	0.62	0.62	—	0.00	0.15	0.01	0.43	0.54	—	—	0.24
BU	0.41	3.92	**19.15**	**7.93**	0.94	0.95	**22.00**	**8.03**	3.56	**7.90**	1.65	1.18	**8.96**	**8.89**	2.67
	0.81	0.14	0.00	0.02	0.62	0.62	0.00	0.02	0.17	0.02	0.44	0.55	0.01	0.01	0.26

FX															
HK	0.49	3.05	**6.01**	**5.90**	2.81	1.70	1.03	3.47	3.67	2.34	0.51	**7.28**	**12.76**	1.95	3.28
	0.78	0.22	0.05	0.05	0.25	0.43	0.60	0.18	0.16	0.31	0.78	0.03	0.00	0.38	0.19
FFK	0.49	3.13	**6.50**	**6.44**	2.88	1.72	1.08	4.03	3.81	2.41	0.51	**8.51**	**28.14**	2.21	3.40
	0.78	0.21	0.04	0.04	0.24	0.42	0.58	0.13	0.15	0.30	0.77	0.01	0.00	0.33	0.18
BU	0.49	2.98	**5.57**	**5.41**	2.74	1.67	0.98	3.01	3.55	2.27	0.51	**6.28**	**6.82**	1.73	3.18
	0.78	0.23	0.06	0.07	0.25	0.43	0.61	0.22	0.17	0.32	0.78	0.04	0.03	0.42	0.20
Alternative investments															
HK	0.91	1.74	**11.20**	**4.99**	2.72	0.86	1.80	2.17	3.57	3.74	1.77	0.86	**6.76**	1.19	**5.30**
	0.63	0.42	0.00	0.08	0.26	0.65	0.41	0.34	0.17	0.15	0.41	0.65	0.03	0.55	0.07
FFK	0.92	1.76	**12.98**	**5.37**	2.78	0.87	1.96	2.38	3.70	3.93	1.80	0.88	**10.08**	1.28	**5.59**
	0.63	0.41	0.00	0.07	0.25	0.65	0.38	0.30	0.16	0.14	0.41	0.64	0.01	0.53	0.06
BU	0.90	1.71	**9.74**	**4.64**	2.65	0.86	1.66	1.99	3.45	3.56	1.74	0.85	**4.75**	1.11	**5.02**
	0.64	0.42	0.01	0.10	0.27	0.65	0.44	0.37	0.18	0.17	0.42	0.65	0.09	0.58	0.08
Hedge funds															
HK	0.90	3.86	1.51	2.44	1.68	0.50	—	—	4.23	**4.52**	1.40	**8.51**	—	—	4.02
	0.64	0.15	0.47	0.30	0.43	0.78	—	—	0.12	0.10	0.50	0.01	—	—	0.13
FFK	0.90	3.98	1.54	2.53	1.70	0.50	—	—	4.41	**4.80**	1.42	**10.22**	—	—	4.19
	0.64	0.14	0.46	0.28	0.43	0.78	—	—	0.11	0.09	0.49	0.01	—	—	0.12
BU	0.89	3.74	1.48	2.35	1.65	0.49	—	—	4.07	4.26	1.38	**7.17**	—	—	3.86
	0.64	0.15	0.48	0.31	0.44	0.78	—	—	0.13	0.12	0.50	0.03	—	—	0.14

(continued)

Table 6.1 (continued)

Test	Index	BTC	USDT	ETH	XRP	LTC	EOS	MIOTA	DASH	XEM	DOGE	WAVES	TRX	ADA	XMR
Multi-asset															
HK	0.99	1.12	0.43	3.91	0.60	0.14	0.60	0.26	1.43	3.71	0.72	0.62	0.33	0.56	1.67
	0.61	0.57	0.81	0.14	0.74	0.93	0.74	0.88	0.49	0.16	0.70	0.73	0.85	0.75	0.43
FFK	1.00	1.14	0.44	4.14	0.60	0.14	0.62	0.27	1.45	3.90	0.73	0.63	0.34	0.58	1.70
	0.61	0.57	0.80	0.13	0.74	0.93	0.73	0.88	0.48	0.14	0.70	0.73	0.84	0.75	0.43
BU	0.98	1.11	0.43	3.69	0.60	0.14	0.59	0.26	1.41	3.54	0.72	0.62	0.33	0.54	1.65
	0.61	0.57	0.81	0.16	0.74	0.93	0.75	0.88	0.49	0.17	0.70	0.73	0.85	0.76	0.44

This table includes the test statistics with their respective p-values. Tests that reject H_0 (at the 10% level) are highlighted in bold text. Due to data issues, some test statistics appear to be unrealistically high and are therefore excluded from the table (denoted by "—")

Tether (USDT) and ether (ETH) are able to increase the efficient frontier when added to five different asset classes (equities, bonds, commodities, FX, and alternative investments). Iota (MIOTA) and eos (EOS) are capable of shifting the efficient frontier of a reference portfolio of commodities significantly to the upside. Dash (DASH) could be of interest for equity or bond investors, whereas NEM (XEM) would additionally enhance a hedge fund portfolio. Waves (WAVES) would benefit portfolios of currencies and hedge funds. Tron (TRX) would also add to equities, FX, commodities, and alternative investments. Cardano (ADA) also increases the efficient frontier for commodities, and monero (XMR) for alternative investments. These results, however, need to be put in context. Tether, for example, is an asset-backed virtual asset and holds an intrinsic value of 1 USD at any point. Thus, its return volatility is very low (since it fluctuates around 1), which might explain the frequent rejection of H_0. For the other digital coins, their data history is relatively short in general, and even shorter for some specific assets. The outcomes of the multi-asset evaluation is in line with Chuen et al. (2018) and Glas and Poddig (2018). The inclusion of an (equally weighted) market index of digital assets results in no statistically significant upward shift of the efficient frontier, in this specific analysis not even for a single reference asset class.

It remains to be seen if these results apply to mean–variance optimization only, or if another portfolio construction approach yields similar results.

6.5 Equal Risk Contribution

A relatively young alternative to mean–variance optimization was introduced by Maillard et al. (2010). The aim of risk parity, or equal risk contribution (ERC), is not to reduce overall portfolio variance. ERC aims to equate the relative risk carried by each asset held in a portfolio. In the end, all assets included in a portfolio should contribute the same amount of risk to the overall portfolio variance. The starting point of this framework is to construct, e.g., an equally weighted portfolio to determine the absolute portfolio risk $\sigma(\mathbf{w}_A)$:[6]

$$\sigma(\mathbf{w}_A) = \sqrt{\mathbf{w}'_A \boldsymbol{\Sigma} \mathbf{w}_A} = \Sigma_i w_i^2 \sigma_i^2 + \Sigma_i \Sigma_{j \neq i} w_i w_j \sigma_{ij}, \tag{6.21}$$

with Σ being the variance-covariance matrix, σ_i^2 and w_i the variance and weight of asset i, and σ_{ij} the covariance of assets i and j; $\mathbf{w}_A$ (see Eq. 6.22) is an $N \times 1$ vector of asset weights. The marginal risk contribution w_i to $\sigma(\mathbf{w}_A)$ for each asset i is the

[6] For a full derivation of the ERC approach, please refer to Maillard et al. (2010). The following equations are reproduced in a slightly different form from Maillard et al. (2010).

first derivative of the risk metric under consideration with respect to the portfolio weight w_i:

$$MRC_i = \frac{\partial \sigma(\mathbf{w}_A)}{\partial w_i} = \frac{w_i \sigma_i^2 + \Sigma_{j \neq i} w_j \sigma_{ij}}{\sigma(\mathbf{w}_A)}. \tag{6.22}$$

Thus, the overall portfolio risk is simply the sum of absolute risk contributions,

$$ARC_i = \sum_{i=1}^{N} \frac{\partial \sigma(\mathbf{w}_A)}{\partial w_i} w_i. \tag{6.23}$$

To calculate the absolute risk contributions, the following problem needs to be solved:

$$w^* = \arg\min \text{Var}(ARC), \tag{6.24}$$

subject to the constraints

$$\sum_{i=1}^{N} w_i = 1 \qquad \text{for all } i = 1, \ldots, N \tag{6.25}$$

and the constraint for long-only portfolios,

$$0 \leq w_i \leq 1. \tag{6.26}$$

Following this framework, I compared the weights of the reference portfolios (K) to the weights of the reference portfolios including the equally weighted market index of digital assets ($N+K$). In line with Glas and Poddig (2018) and the previous sections, this analysis is an in-sample analysis only. Table 6.2 summarizes the results for the seven reference portfolios.

Figure 6.7 plots the weights for the multi-asset portfolio of reference assets (K) and compares it with the inclusion of digital assets ($N + K$). In this specific case, digital assets enter the asset allocation with a portfolio weight of 0.45%. This is the second lowest of all reference portfolios. The weight is even lower for a portfolio comprised of foreign exchange rates and hedge funds (see Table 6.2).[7]

All in all, digital assets enter the asset allocation in different asset classes with relatively low weights when applying an ERC approach. Consequently, the returns and standard deviations of the portfolios including digital assets do not differ significantly from portfolios ignoring digital assets, although the return/risk ratios decrease slightly. Thus, the results from the previous sections can be deemed robust

[7] The weight plots for all the other K assets can be found in Appendix E.

Table 6.2 Summary ERC portfolios

	Multi-asset	Equities	Bonds	Commodities	FX	Alternative	Hedge funds
Weight (%)	0.45	2.23	1.33	1.58	0.38	1.97	0.38
Return K (%)	0.10	0.70	0.09	−0.30	0.11	0.25	0.10
Return $K + N$ (%)	0.10	0.67	0.08	−0.30	0.11	0.24	0.10
Risk K in %	0.69	2.90	1.43	2.87	0.54	2.24	0.86
Risk $K + N$ (%)	0.76	3.30	1.56	2.96	0.59	2.47	0.92
Return/risk K	0.15	0.24	0.06	−0.10	0.20	0.11	0.12
Return/risk $K + N$	0.13	0.20	0.05	−0.10	0.19	0.10	0.11

This table summarizes the results of the ERC portfolios. Weight (%) is the weight of the digital asset index within the respective portfolio of reference assets. Returns K and $K+N$ (%) describe the in-sample returns with the respective standard deviations ("risk") in the corresponding lines below. Return/risk is the return–risk ratio

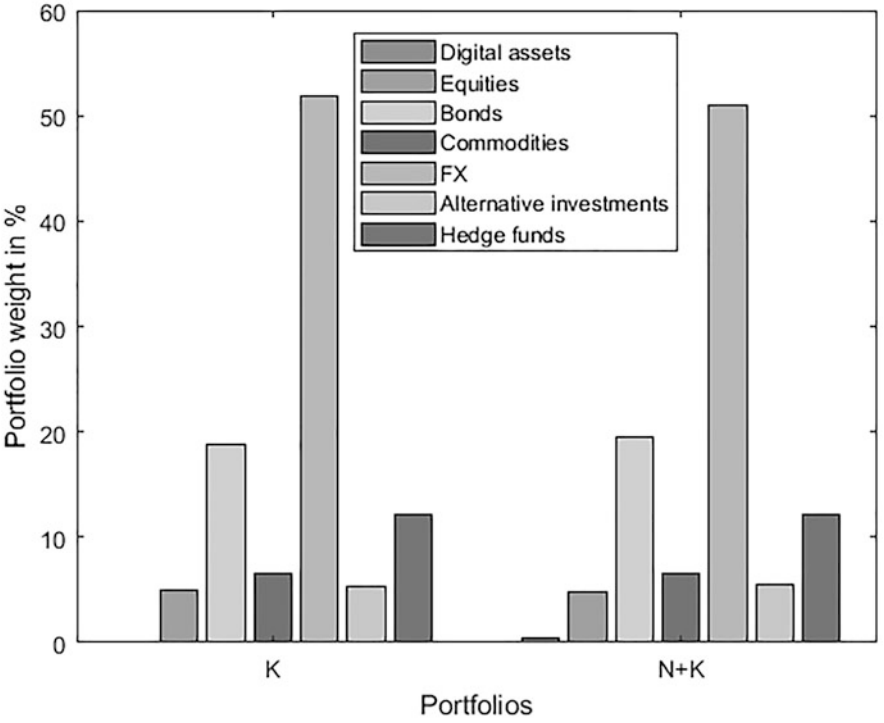


Fig. 6.7 A comparison of ERC weights, multi-asset

since digital assets are not capable of enhancing several portfolios of reference assets for another portfolio construction approach.

6.6 Summary

Therefore, inclusion of digital assets in a multi-asset portfolio or a portfolio comprised of assets from various single asset classes is not recommended. Or at least, it does not significantly enhance the initial portfolio selection. That finding is confirmed by applying a Markowitz (1952) and Maillard et al. (2010) portfolio construction approach as well as by several mean–variance spanning tests. This, however, might be due to a very short data history and the still comparatively high variance of digital assets.

Acknowledging the findings of this chapter almost prohibits any other analysis concerned with digital assets in combination with one or several other (traditional) asset classes. Consequently, I need to focus on the asset class digital assets even more, to answer the two central research questions of this book. The first question is about the legitimacy of digital assets as a standalone asset class. The spanning tests would negate such a claim, since digital assets are in the span of basically all investigated reference portfolios. As such, digital assets could not be considered a standalone asset class. The second question, regarding investments in digital assets, is not answered with this analysis. In combination with other asset classes, an investment is ambiguous. Focusing on an investment universe of digital assets only might help to resolve this issue.

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Chapter 7
On the Effectiveness of Technical Indicators in Digital Assets



7.1 The Use of Technical Indicators

A controversial approach in the finance literature is the application of technical indicators or technical analysis in general. Practitioners rely heavily on such tools, but research has not yet agreed on a final conclusion regarding their effectiveness. Technical analysis makes predictions of future prices of an asset based on the past performance information. Therefore, it can be regarded a time-series instrument.

On the other hand, with hundreds of investable digital assets, it is still difficult to obtain two things. It is almost impossible to determine (i) the *"fair" value* of a digital asset, and consequently (ii) an asset selection based on *fundamental data* (compared to equities) is not applicable. As such, it is even more important to generate buy and sell decisions through other methods.

Generally, technical analysis is used for exactly such decisions. Technical analysis can be divided into two different areas. Chart patterns like head-and-shoulders, which are hardly quantifiable, are part of the broader conception of chart analysis. Other parts of technical analysis are called technical indicators, including indicators like MACD or RSI, and can be expressed via mathematical formulas. Depending on the indicator's specific value, buy and sell decisions can be derived. Technical indicators, however, do not make any proposition concerning the fair value of an asset.

Regardless of that, they can be used for timing purposes. Timing solves the issue, if it is economically reasonable to buy or sell a financial instrument at a given point in time. Such timing decisions, however, are controversial in the investment literature. Up to now, no clear evidence has been found for significantly higher or lower returns when applying technical indicators. For instance, Fang et al. (2014) investigated 93 technical indicators based on the S&P500. They arrived at the conclusion that stock returns cannot be predicted by technical indicators. Similarly, Rosillo et al. (2013) found no clear evidence for the success of technical indicators applied to the Spanish stock market. Shynkevich (2016) focused on

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technical indicators applied to the US bond market and concluded that bond prices were possible to predict; the predictability, however, decreased over time.

Several other studies have focused on emerging markets, which are prone to higher market inefficiencies. Pavlov and Hurn (2012) tested moving averages in the Australian stock market and constructed a promising contrarian strategy. Moreover, positive results emerged for the Indian stock market (Mitra 2011) and Greek stocks (Vasiliou et al. 2006), as well as in Taiwan, Thailand, and Mexico (Ratner and Leal 1999). Similar results applied to FX markets (Menkhoff and Taylor 2007). Regarding digital assets, Detzel et al. (2018) and Corbet et al. (2019) used simple moving averages to construct a successful strategy for bitcoin. Caporale and Plastun (2019) arrived at the conclusion that overreactions cannot successfully be exploited for the largest digital assets. Huang et al. (2019) also found strong predictive power based on machine learning techniques using 124 technical indicators to predict narrow ranges of bitcoin prices. Other works regarding technical indicators in digital assets, which tested several indicators and/or digital assets at once, were very limited (to the best of my knowledge).

It is therefore interesting to extend the discussion on the effectiveness of technical indicators to the young and emerging asset class of digital assets. That is the object of investigation for this chapter. I investigate the effectiveness of seven widely accepted technical indicators (RSI, MACD, Bollinger bands (BB), stochastic (STO), directional movement index (DMI), Ichimoku (GOC), and volume at time (VAT)) based on a large sample of more than 1600 virtual currencies. Thereby, I arrive at the conclusion that only one technical indicator appears to be successful when applied to digital assets.

7.2 Data

The data sample used in this chapter comprises all digital currencies actively traded at the end of June 2018 that were listed on the leading data website for cryptocurrencies, www.coinmarketcap.com.[1] This exactly corresponds to the dataset used in the previous chapters. For each of the $N = 1602$ virtual currencies available with data, I used end-of-day data for the period from April 28, 2013, to June 30, 2018 ($T = 1890$ days). The data includes information on the open, close, high, and low prices, as well as trading volume and market capitalization, all in USD. www.coinmarketcap.com aggregates the price information from various exchanges for digital currencies and weighs this data according to the daily trading volume on the respective exchanges to determine a daily price. For BTC alone, the largest virtual coin, price information from approximately 400 different local exchanges is used. To make it easier to interpret the results, I also downloaded the end-of-day data of all stocks listed in the S&P500 over the same period from

[1] Please refer to Sects. 5.2 and 5.3 for more information.

Thomson Reuters Datastream. I used the Datastream constituent list "LS&PCOMP" for this purpose. I also used the standard data cleaning procedure according to Ince and Porter (2006).

7.3 Methodology

7.3.1 Calculating Technical Indicators

In order to make a statement on the effectiveness (i.e., the achievement of excess returns) of technical indicators in digital assets, I examined seven of the world's most frequently used technical indicators. These were selected on the basis of a representative survey by Bloomberg (Ciana 2011, p. 16). Between 2005 and 2010, Bloomberg conducted a survey among the users of the Bloomberg terminal and asked for their preference of seven widely used technical indicators.[2] The results were that 44.4% of the respondents indicated that they rely on RSI, 22.0% on MACD, 12.0% used Bollinger Bands, 9.3% mentioned STO, 4.9% DMI, 4.5% GOC, and 2.9% VAT. Since these seven indicators describe the majority of the world's most widely used technical indicators, the following analysis examines their effectiveness in the new and emerging asset class of digital assets.

Since there are innumerable possible settings for the calculation of the various indicators, I limited myself to the standard settings mentioned by Murphy (2006), which are also used by Bloomberg, for instance. Table 7.1 contains the formulas for calculating the seven indicators, as well as information on the (standard) time windows used.

7.3.2 Interpretation

The interpretation of the individual indicators is discussed in more detail below. In order not to create unnecessary confusion at this point, I will mainly adhere to the interpretation of the corresponding indicator values as presented in the current literature. The individual rules and setting characteristics were more or less heuristically defined in the literature. It is not part of this chapter to vary or discuss the different setting possibilities. I concentrate more on the question of whether excess returns can be achieved through the use of common technical indicators when applied to the digital asset market. Likewise, technical indicators can still be divided into two groups. Some indicators such as RSI or stochastic aim to detect a trend reversal. Other indicators such as MACD, DMI, and GOC try to follow

[2] Other technical indicators were not mentioned or taken into account by this survey, or by Ciana (2011).

Table 7.1 Technical indicator formulas

Indicator	Calculation formula	Setting
RSI[a]	$RSI = 100 - \frac{100}{1+RS}$, where $RS = \frac{\lvert\text{Average positive daily returns in } t\rvert}{\lvert\text{Average negative daily returns in } t\rvert}$	$t = 14$ days
MACD[b]	$MACD_{n_1,n_2} = EMA(P_t, n_1) - EMA(P_t, n_2)$, $Signal = EMA(MACD_t, n_3)$, where $EMA = \alpha P_t + (1-\alpha)EMA_{t-1}$, $\alpha = \frac{2}{n_3+1}$	$n_1 = 12$ days, $n_2 = 26$ days, $n_3 = 9$ days
Bollinger bands	$Bollinger_t^{\text{Mid}} = SMA_t = \frac{1}{n}\Sigma_{i=0}^{n-1} W_{t-i}$ $Bollinger_t^{\text{Up(Down)}} = SMA_t + (-)\, k\sigma_t$	$n = 20$ days, $k = 2$
Stochastic	$\%K = 100 \times \frac{P-L}{H-L}$, $\%D = \frac{1}{n}(\%K_i + \%K_{i-1} + \%K_{i-2})$	$t = 14$ days, $n = 3$
DMI[c]	$DI^+ = \max[H_t - H_{t-1}, 0]$, $DI^- = \max[L_t - L_{t-1}, 0]$ $DMI = 100 \times \frac{DI^+ - (DI^-)}{DI^+ + DI^-}$	$t = 14$ days
GOC	$Tenkan = \frac{1}{2}\max[H_t, H_{t-1}, \ldots, H_{t-n}] + \min[L_t, L_{t-1}, \ldots, L_{t-n}]$ $Kijun = \frac{1}{2}\max[H_t, H_{t-1}, \ldots, H_{t-k}] + \min[L_t, L_{t-1}, \ldots, L_{t-k}]$ $Senkou_1 = \frac{1}{2}(Tenkan + Kijun)$ $Senkou_2 = \frac{1}{2}\max[H_t, H_{t-1}, \ldots, H_{t-l}] + \min[L_t, L_{t-1}, \ldots, L_{t-l}]$ $Chikou = P_{t-m}$	$n = 9$ days, $k = 26$ days, $l = 52$ days, $m = 26$ days
VAT	$VAT = \frac{1}{n}\sum_{i=0}^{n-1} VOL_{t-i}$	$n = 20$ days

[a] If the average of the negative daily returns is 0, the RSI would be 100

[b] The calculation formula for α also applies to the moving averages n_2 and n_1

[c] To avoid a misunderstanding: in this case DI^- is subtracted from DI^+ in the counter. The denominator values of DI^- and DI^+ are summed

Here, P is the current price of a cryptocurrency and H and L the daily high or low prices over the last t days. *EMA* is the exponentially weighted moving average of the digital asset P at time t in the previous time window n_1 or n_2, respectively. α is a smoothing factor, which can take values between 0 and 1. *SMA* is the simple moving average at time t based on the price of a digital currency P. The upper (lower) Bollinger band is calculated based on the middle band. The (realized) standard deviation σ of the price of the asset under consideration over the past time window n is added or subtracted by a factor k from the value of the middle band. *VOL* is the trading volume (units of accounts traded $\times$ price of a digital asset) on day t, expressed in USD

a trend as long as possible and are therefore also referred to as trend-following indicators. Bollinger bands can represent both, depending on the interpretation. The same applies to VAT, which was conceived as a hybrid and must always be interpreted as a function of price. The values of the indicators can thus be interpreted as follows.

Since the RSI can only assume values of 0 to 100, two extreme zones are defined. In the range between 0 and 30, an asset is considered oversold. The range above 70 is referred to as overbought. This allows us to derive very simple trading recommendations. If the RSI breaks through the 30-point level from below, a buy signal is generated, as the oversold phase is considered to have ended from this point on. The same can also be applied to the overbought area: a sell signal is generated when the RSI breaches the 70-point level from above.

The MACD can be interpreted in a similar manner. It typically oscillates around the zero line. A highly positive deviation is usually interpreted as an overbought phase and a negative deviation as an oversold phase. However, the signal is only generated when the signal line crosses the MACD line. As soon as the MACD crosses the signal line from below, a buy signal is generated. A cut from above to below the line yields a sell signal.

Bollinger bands can be interpreted in various ways (Murphy 2006). In order to illustrate the applicability of Bollinger bands as trend-following and trend-reversal indicators, I will refer to both of them as such in the following. I call the trend-reversal indicator a "reversal strategy." Here, it is assumed that prices reverse (at least in the short term) as soon as the closing price cuts through the upper (lower) Bollinger band from top to bottom (bottom to top). The price target subsequently is the middle or lower (upper) band. In contrast, Bollinger bands can also be used as trend-following indicator (further referred to as a "trend-following strategy"). A buy (sell) signal is generated when the closing price is higher (lower) than the upper (lower) Bollinger band. Only when the price returns to the range between the upper and lower bands is the position closed. Another interpretation is not to open (close) a new trading position until a new signal has been generated, which is also utilized in this chapter.

Like the RSI, stochastic also moves in an interval between 0 and 100. By adding a signal line ($\%D$), signals can be generated in different ways. Similar to the RSI, two extreme zones can be defined. In the case of stochastic, however, these are between 80 and 100 for the overbought area and between 0 and 20 for the oversold area. In order to distinguish the stochastic signals from the RSI, I generated slightly different trading signals for the stochastic indicator, which are also described in Murphy (2006). These are a combination of the RSI and MACD methodologies. As soon as the $\%D$ line is crossed by the faster $\%K$ line in an extreme range from below (above) to above (below), a buy signal (in the range between 0 and 20) or a sell signal (in the range between 80 and 100) is generated.

There are also comparable interpretation possibilities for the DMI. One can use the DMI itself as a signal line and generate a buy or sell signal as soon as the DMI shows a positive or negative slope. Since the slope is not always clear (e.g., a short uptrend during a long-term downtrend), I used an alternative trading rule (Ciana 2011). This one is mainly dependent on DI^+. As soon as DI^+ crosses the DI^- line from below, a buy signal is generated. Similarly, a sell signal is generated when DI^+ crosses the DI^- line from above. This is also very similar to the MACD.

Ichimoku consists of five lines and accordingly its interpretation possibilities are manifold. However, I followed a simple rule to generate signals that is also described by Ciana (2011). As soon as the Tenkan line crosses the slower Kijun line from below (above), a buy (sell) signal results. However, these signals can also be set in relation to the other three lines and an aggregated signal from several partial signals could be used. Since the interpretation possibilities are almost unlimited, I constrained myself to an easily understandable interpretation, which strongly follows that of the DMI.

VAT is probably one of the simplest indicators to calculate and interpret. It assumes that price movements must always be supported by the trading volume. This means that if the price has risen compared to the previous day, the trading volume must also have risen compared to the previous day's volume or the average trading volume in order to generate a buy signal. If this relationship is contrary (increasing price but decreasing volume), a sell signal is generated. Additionally, a sell signal occurs if the price development is negative compared to the previous day and the trading volume increases at the same time. If the volume falls with price, this is more likely to be interpreted as a buy signal, as the consolidation seems to be complete. The average trading volume can be used over the last 5, 20, or even 60 days (Ciana 2011). I applied the average of the last 20 days in this analysis, since all other indicators are calculated over a similar period, ensuring better comparability.

The following analysis is guided by the above-described standard interpretations of the seven technical indicators and their various trading signals, which are presented in more detail in Table 7.2. These signals can be used to perform three different analyses. One is to examine only buy signals (long only), another is to examine only sell signals, and a third is to include both together. Since digital assets cannot yet be sold short (except for a few), an analysis of the sell signals alone is less relevant. Therefore, I limit the analyses to buy signals, which are practically realizable, and for the sake of completeness to a combination of buy and sell signals (long–short), in order to draw additional conclusions.

Table 7.2 Trading signals from technical indicators

Indicator	Trading rule	Hold until
RSI	Buy = $RSI_{t-1} < 30$ & $RSI_t > 30$	Sell
	Sell = $RSI_{t-1} > 70$ & $RSI_t < 70$	Buy
MACD	Buy = $MACD_{t-1} <$ signal line & $MACD_t >$ signal line	Sell
	Sell = $MACD_{t-1} >$ signal line & $MACD_t <$ signal line	Buy
Bollinger bands (reversal)	Buy = $P_{t-1} <$ lower band & $P_t >$ lower band	Sell
	Sell = $P_{t-1} >$ upper band & $P_t <$ upper band	Buy
Bollinger bands (trend following)	Buy = $P_{t-1} <$ upper band & $P_t >$ upper band	Sell
	Sell = $P_{t-1} >$ lower band & $P_t <$ lower band	Buy
Stochastic	Buy = $\%K_{t-1} \leq 20$ & $\%D_{t-1} > \%K_{t-1}$ & $\%K_t \leq 20$ & $\%D_t \leq \%K_t$	Sell
	Sell = $\%K_{t-1} \geq 80$ & $\%D_{t-1} < \%K_{t-1}$ & $\%K_t \geq 80$ & $\%D_t \geq \%K_t$	Buy
DMI	Buy = $DI^+_{t-1} < DI^-_{t-1}$ & $DI^+_t > DI^-_{t-1}$	Sell
	Sell = $DI^+_{t-1} > DI^-_{t-1}$ & $DI^+_t < DI^-_{t-1}$	Buy
GOC	Buy = $Tenkan_{t-1} < Kijun_{t-1}$ & $Tenkan_t > Kijun_t$	Sell
	Sell = $Tenkan_{t-1} > Kijun_{t-1}$ & $Tenkan_t < Kijun_t$	Buy
VAT	Buy = $P_{t-1} < P_t$ & $VAT_{t-1} < VOL_t$	Sell
	Sell = $P_{t-1} > P_t$ & $VAT_{t-1} < VOL_t$	Buy

Refer to Table 7.1 for the calculation of the single indicators

7.4 Evaluation of Technical Indicator Performance

In order to measure the quality of the trading strategies I used various key figures, which are described in more detail below. In total, I applied seven different measures of quality. The best known of these might be the calculation of the cumulative return or profit. Here, the daily continuous (log) returns are accumulated in order to be able to give a general statement about the success or otherwise of the various indicators. As soon as a technical indicator generates a buy signal, the log yield of the next day is added. For a sell signal, the yield of the next day is subtracted (related to a long–short strategy). When a sell signal occurs in a long-only strategy, the position is closed and the strategy awaits the next signal to enter a long position. In the last available data point, the total cumulative yield per indicator can be measured. This sum is calculated for each digital currency. In a final step, the mean is calculated from the cumulative returns of all digital assets in order to obtain a final average cumulative return across all virtual currencies. This number does not account for the different sample length of the assets under consideration.

In a second step, I looked at the average daily return across all digital coins per technical indicator. I calculated an equally weighted daily return across all assets every day. The returns for all days are averaged in order to make a statement about the average (daily) return. The average daily return thus takes into account various sample sizes. This number is also used to calculate a risk-adjusted ratio. The average return is put in relation to the risk (the standard deviation of the average daily returns). The larger this ratio, the more return can be generated per unit of risk taken (Sharpe 1975).

The profit factor makes a similar statement. Here, the sum of the profits is set in relation to the (absolute) sum of the losses. The focus is normally on the respective time series of an asset. However, since I examined about 1600 different virtual currencies, I calculated the sum of all positive cumulative returns in relation to the sum of all negative cumulative returns.

The maximum drawdown, i.e., the maximum loss within the investment horizon, is again based on the equally weighted daily returns across all crypto assets. Thereby, the maximum loss within the time series from the highest price to the next lowest price is calculated (Acar and James 1997). Figure 7.1 depicts the price chart of terracoin, one of the first cryptocurrencies, to illustrate the concept of maximum drawdown. This is the largest loss, from the highest price of an asset to the lowest price that follows. In the case of terracoin, the maximum price is about 1.50 USD per terracoin and the subsequent low is a fraction of a cent (0.0009 USD per terracoin), representing a loss of almost 100%. However, I did not calculate the maximum drawdown on the basis of individual digital assets, but on the basis of the average daily returns for each of the technical indicators considering all assets examined. Such price swings are not unusual for virtual currencies, which is why the values listed in the following tables appear quite plausible. Finally, I calculated a success rate. This indicates how many of the examined digital coins have a positive cumulative return in relation to the number of all examined assets. The higher the

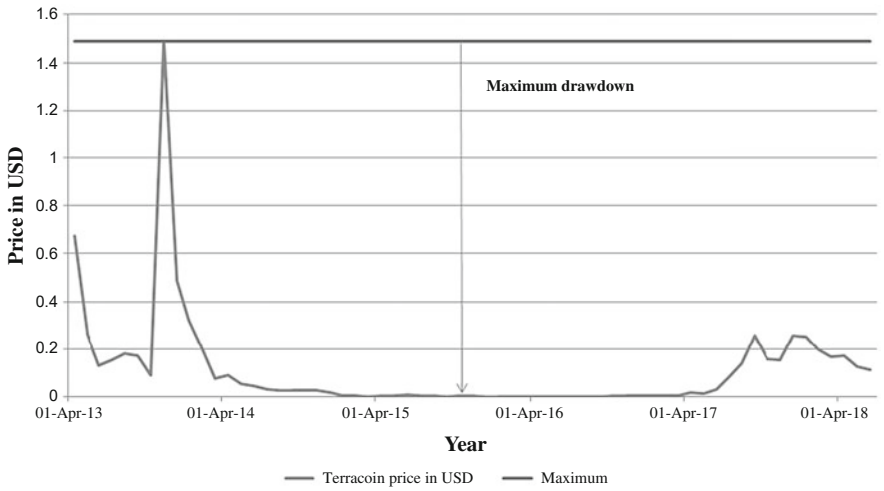


Fig. 7.1 Terracoin prices in USD

number, the more promising the technical indicator in question, since it delivers positive results for a large number of examined assets. On the other hand, a low number indicates low effectiveness of the respective indicator.

All the values described are aggregated values over the entire data sample and take into account transaction costs of 0.25% per trade. Transaction costs for cryptocurrencies on one of the largest European exchanges for digital assets range from 0.10% to 0.25% per purchase or sale.[3] Thus, I used a conservative transaction cost estimate by subtracting a return of 0.25% per trade (in the long-only case). More precisely, since any trade signal results in the closing of an existing position and the opening of a new one, transactions costs need to be considered twice (i.e., 0.50% per trade signal) for long–short strategies. However, bid–ask spreads were not taken into account as the corresponding data is not available. Taking into account transaction costs, however, yields the same outcomes (with lower average returns). When running the same analysis without transaction costs,[4] the findings remain the same.

7.5 Empirical Results

The results of the long-only and long–short analyses are presented in more detail in the following subsections. I compare the corresponding trading strategies with the returns of a simple buy-and-hold strategy. I also indicate how many of the 1602

[3] See "Unified fee schedule," https://www.bitstamp.net/fee_schedule/, accessed on May 30, 2018.

[4] The results are reported below.

digital assets examined show a positive trading performance for each indicator used (success rate). It is difficult to interpret the average number of trading signals at this point because not all digital currencies have uniform data availability (in the sense of data points).

7.5.1 Long-Only

The analysis of the buy signals (see Table 7.3) alone indicates very clear results. If only the success rate is considered, no clear superiority can be observed in comparison to a simple buy-and-hold strategy (excluding Bollinger bands). This buy-and-hold strategy invests in all available digital assets as early as available and holds them until the last available data point.[5]

All other technical indicators aim to generate a higher return through timing. The worst instrument for this is the use of MACD. It obtains a success rate of 28.84% only and achieves a cumulative return of −366.65%. VAT and STO are in a similarly bad range. With a success rate of 42.82%, GOC is the second most successful of the technical indicators examined (trailing the two Bollinger band variants) and is at eye level with a simple buy strategy. DMI and RSI yield similar results, and the success rate of the DMI is comparable to that of the benchmark strategy, though with DMI generating a significantly lower cumulative return. The success rate for Bollinger bands (reversal) is significantly higher than for a simple buy-and-hold strategy, while that of the Bollinger bands (trend following) is only slightly higher.

This is also reflected in the other key figures. While an equally weighted portfolio consisting of all virtual currencies generates a total return of −40.36%, the returns of almost all technical indicators (except Bollinger bands and GOC) are lower. These figures result from the accumulation of the daily returns and the subsequent calculation of the mean. This, and the use of continuous returns, can also result in values significantly smaller than −100%. Bollinger bands (reversal) generates a clearly positive cumulative return and Bollinger bands (trend following) still achieves a positive cumulative return compared to a simple buy-and-hold strategy. Considering mean daily returns only results in a stronger outcome for just one technical indicator, Bollinger bands (reversal).

Glas (2019) shows that the universe of cryptocurrencies is driven by a few large digital currencies only and that there is no clear connection between individual virtual coins. Thus, positive returns compensate for other negative returns, resulting in an average return close to 0% (Glas 2019). The negative returns of the individual indicators also lead to negative, risk-adjusted returns, as well as a profit factor of less than 1. The latter measures the ratio between the generated profits and losses. Only (both) Bollinger bands result in a profit factor greater than 1, which means a significantly higher profit per unit of loss incurred. With regard to the maximum

[5] The values described in the text below are taken from analysis *not* including transaction costs.

Table 7.3 Results for long-only technical indicators, digital assets

	RSI	MACD	Bollinger (reversal)	Bollinger (trend following)	STO	DMI	GOC	VAT	Buy and hold
No transaction costs									
Cumulative return (%)	−73.01	−366.65	45.97	10.27	−348.14	−84.18	−34.06	−305.53	−40.36
Daily return (%)	−0.53	−1.36	0.47	−0.41	−1.70	−0.60	−0.58	−2.25	−0.08
Standard deviation (%)	17.08	19.69	15.99	19.42	18.55	19.72	21.27	20.01	18.88
No. of trades	10.00	45.72	7.89	8.48	81.38	31.09	26.23	161.68	1.00
Return/risk	−0.03	−0.07	0.03	−0.02	−0.09	−0.03	−0.03	−0.11	−0.00
Profit factor	0.32	0.10	2.08	1.12	0.08	0.45	0.69	0.09	0.66
Max. drawdown (%)	99.78	100.00	94.82	99.40	100.00	99.99	100.00	100.00	99.33
Success rate (%)	29.90	28.84	54.06	43.20	24.66	36.89	42.82	29.59	36.64
Transaction costs									
Cumulative return (%)	−75.23	−377.00	44.34	8.23	−368.25	−90.82	−40.38	−345.50	−40.61
Daily return (%)	−0.54	−1.40	0.46	−0.42	−1.80	−0.62	−0.61	−2.58	−0.08
Standard deviation (%)	17.09	19.71	15.97	19.42	18.62	19.74	21.27	20.14	18.88
No. of trades	10.00	45.72	7.89	8.48	81.38	31.09	26.23	161.68	1.00
Return/risk	−0.03	−0.07	0.03	−0.02	−0.10	−0.03	−0.03	−0.13	−0.00
Profit factor	0.31	0.09	2.03	1.09	0.07	0.42	0.64	0.07	0.66
Max. drawdown (%)	99.80	100.00	94.99	99.42	100.00	99.99	100.00	100.00	99.33
Success rate (%)	29.65	28.34	53.93	42.95	22.22	36.14	41.95	25.66	36.64

Cumulative return is the average log-cumulative return across all digital assets for the corresponding technical indicator. Continuous returns are defined in the range from $-\infty$ to $+\infty$, which means that even values significantly smaller than −100% can occur. Average daily return refers to the average returns of all digital assets for the corresponding technical indicator in the time series of returns. Standard deviation is the standard deviation of the average daily returns. The risk/return ratio is also calculated from these two values. Average number of trades denotes the mean value of the number of all trades per virtual currency. Profit factor is the sum of all positive cumulative returns in relation to the sum of all negative cumulative returns. The maximum drawdown refers to the average daily returns. The success rate measures the number of all digital assets with a positive cumulative return compared to the number of all examined assets

drawdown, Bollinger bands (reversal) achieves the best results among the technical indicators. However, a value of 94.82% still means almost a complete loss during the corresponding investment horizon.

If one compares these results with the stocks from the S&P500 (see Table 7.4) over the same period of time, the results clearly differ. None of the seven technical indicators achieves a cumulative return higher than the benchmark strategy. However, most of them still yield positive (cumulative and mean) returns. MACD, STO, DMI, and VAT result in negative returns after taking transaction costs into account. Both the profit factors and the success rates are significantly higher in equities. The maximum drawdown, additionally, is significantly lower. As such, technical indicators appear to work better for equities (which might be due to their positive mean monthly returns—see Table 5.6) than for digital assets,[6] but they are not capable of outperforming a buy-and-hold strategy. The only indicator that significantly outperforms its benchmark in digital assets are both Bollinger band variants.

7.5.2 Long–Short

Since the analysis of long signals alone does not support the use of technical indicators in digital assets (except for Bollinger bands), I performed a second analysis as a robustness test. This included both buy and sell signals for the timing of virtual currencies and thus assumed short positions. Table 7.5 shows the results for the seven technical indicators in a long–short framework, analogous to Tables 7.3 and 7.4. Due to the higher losses of the corresponding indicators, the other key figures also deteriorate. Bollinger bands in both variations are the only technical indicator with a positive cumulative return. GOC continues to achieve the second-best results (behind both Bollinger bands), but its cumulative return (profit or loss) is clearly negative at −27.76%. The MACD, STO, and VAT indicators deteriorate to −693.55%, −661.49%, and −601.43%, respectively. DMI is again very close to the RSI results. The success rates increase across the board, which indicates a slightly positive contribution of the short signals. The use of the RSI leads to positive cumulative returns in 32.02% of all the digital assets examined. The other 67.98%, on the other hand, generate negative cumulative returns. The success rate of the Bollinger bands (reversal) strategy is stable at 57.12%. Only Bollinger bands improve to a cumulative return of 106.14% and 54.00%. The average daily returns, on the other hand, do not change significantly, which also applies to the return/risk ratios. The addition of short positions continues to halve (or double) the profit factor for most technical indicators.

[6] The success rate of the buy-and-hold strategy in digital assets already shows that only 36.64% of the whole universe achieve positive cumulative returns. That translates into almost two thirds of digital assets losing value.

Table 7.4 Results for long-only technical indicators, equities

	RSI	MACD	Bollinger (reversal)	Bollinger (trend following)	STO	DMI	GOC	VAT	Buy and hold
No transaction costs									
Cumulative return (%)	28.42	18.61	36.07	18.54	22.18	27.03	23.59	13.79	54.62
Daily return (%)	0.06	0.03	0.07	0.02	0.03	0.02	0.03	0.04	0.04
Standard deviation (%)	1.61	1.47	1.63	1.42	1.48	1.47	1.44	1.62	1.51
No. of trades	30.64	104.39	28.84	28.48	251.22	144.60	59.82	465.27	1.00
Return/risk	0.04	0.02	0.04	0.02	0.02	0.02	0.02	0.02	0.03
Profit factor	9.12	4.02	11.27	4.09	4.44	5.33	4.81	3.62	16.34
Max. drawdown (%)	21.18	16.38	20.88	12.36	17.80	18.63	13.62	16.60	17.57
Success rate (%)	83.96	72.48	86.93	68.32	72.28	73.86	71.88	71.49	86.14
Transaction costs									
Cumulative return (%)	20.85	−7.42	28.98	11.55	−40.65	−8.97	8.54	−102.48	54.37
Daily return (%)	0.04	−0.01	0.06	0.01	−0.06	−0.01	0.01	−0.29	0.04
Standard deviation (%)	1.62	1.52	1.61	1.41	1.55	1.50	1.45	1.76	1.51
No. of trades	30.64	104.39	28.84	28.48	251.22	144.60	59.82	465.27	1.00
Return/risk	0.03	−0.01	0.03	0.01	−0.04	−0.01	0.00	−0.17	0.03
Profit factor	5.31	0.59	7.50	2.33	0.08	0.60	1.71	0.00	16.34
Max. drawdown (%)	22.83	27.27	22.21	14.70	53.33	27.62	18.90	97.36	17.57
Success rate (%)	77.43	38.42	82.38	60.00	14.85	39.21	56.44	0.00	86.14

Cumulative return is the average, log-cumulative return across all stocks for the corresponding technical indicator. Continuous returns are defined in the range from $-\infty$ to $+\infty$, which means that even values significantly smaller than -100% can occur. Average daily return refers to the average returns of all stocks for the corresponding technical indicator in the time series of returns. Standard deviation is the standard deviation of the average daily returns. The risk/return ratio is also calculated from these two values. Average number of trades denotes the mean value of the number of all trades per stock. Profit factor is the sum of all positive cumulative returns in relation to the sum of all negative cumulative returns. The maximum drawdown refers to the average daily returns. The success rate measures the number of all stocks with a positive cumulative return compared to the number of all examined assets

Table 7.5 Results for long–short technical indicators, digital assets

	RSI	MACD	Bollinger (reversal)	Bollinger (trend following)	STO	DMI	GOC	VAT	Buy and hold
No transaction costs									
Cumulative return (%)	−118.87	−693.55	106.14	54.00	−661.49	−128.29	−27.76	−601.43	−40.36
Daily return (%)	−0.29	−0.86	0.25	−0.10	−0.92	−0.31	−0.16	−1.66	−0.08
Standard deviation (%)	18.43	18.83	18.36	18.56	18.61	18.88	18.89	18.84	18.88
No. of trades	20.00	91.45	15.79	16.96	162.76	62.18	26.23	161.68	1.00
Return/risk	−0.02	−0.05	0.01	−0.01	−0.05	−0.02	0.01	−0.09	−0.00
Profit factor	0.33	0.09	2.44	1.63	0.07	0.43	0.83	0.07	0.66
Max. drawdown (%)	99.42	100.00	95.07	86.01	100.00	99.98	99.44	100.00	99.33
Success rate (%)	32.02	42.01	57.12	53.56	34.96	44.63	57.87	35.21	36.64
Transaction costs									
Cumulative return (%)	−128.33	−735.66	99.02	46.55	−742.05	−154.10	−52.12	−702.75	−40.61
Daily return (%)	−0.31	−0.94	0.23	−0.11	−1.11	−0.36	0.12	−2.01	−0.08
Standard deviation (%)	18.45	18.89	18.33	18.55	18.72	18.93	18.91	19.03	18.88
No. of trades	20.00	91.45	15.79	16.96	162.76	62.18	26.23	161.68	1.00
Return/risk	−0.02	−0.05	0.01	−0.01	−0.06	−0.02	0.01	−0.11	−0.00
Profit factor	0.30	0.08	2.30	1.53	0.05	0.36	0.71	0.05	0.66
Max. drawdown (%)	99.60	100.00	95.21	87.58	100.00	99.99	99.73	100.00	99.33
Success rate (%)	31.09	39.45	56.74	52.43	28.78	41.14	55.68	28.78	36.64

Cumulative return is the average, log-cumulative return across all digital assets for the corresponding technical indicator. Continuous returns are defined in the range from $-\infty$ to $+\infty$, which means that even values significantly smaller than −100% can occur. Average daily return refers to the average returns of all digital assets for the corresponding technical indicator in the time series of returns. Standard deviation is the standard deviation of the average daily returns. The risk/return ratio is also calculated from these two values. Average number of trades denotes the mean value of the number of all trades per virtual currency. Profit factor is the sum of all positive cumulative returns in relation to the sum of all negative cumulative returns. The maximum drawdown refers to the average daily returns. The success rate measures the number of all digital assets with a positive cumulative return compared to the number of all examined assets

The maximum losses (drawdowns) improve slightly in contrast to the long-only approach. Due to the arbitrary scalability of digital assets, these virtual coins can also have prices of several decimal places. The price of some digital assets is so small that prices are described in "Satoshi," the smallest possible unit for digital assets—a Satoshi corresponds to the 100,000,000th part of a bitcoin (0.00000001 BTC) and is named after the alleged bitcoin inventor Satoshi Nakamoto. As soon as a virtual currency with a value of, e.g., 0.01 BTC crashes to 1 Satoshi, the maximum drawdown in the tables would amount to 100%, which is actually a rounded value. The remaining values do not change noticeably compared to the previous analysis.

Overall, it can be said that the Bollinger bands (especially the reversal strategy) achieve the best results of the seven technical indicators examined. For the other indicators, however, there is no significant improvement compared to a simple buy-and-hold strategy, which generally does not support the use of technical indicators in digital assets. Taking into account actual tradeable bid–ask spreads, the described returns should also deteriorate somewhat.

The analysis of technical indicators in equities also deteriorates significantly after the addition of short signals (see Table 7.6). Bollinger bands (reversal), RSI, and DMI are the only indicators with a positive cumulative return (not taking into account transaction costs). Also, the profit factor decreases drastically across all indicators. The same applies to the maximum drawdown and the success rate. This is in contrast to digital assets, which increase success rates but suffer from some severe losses. The general finding of worse results after the addition of short signals, on the other hand, applies to both asset classes.

These results are comparable with the existing literature on technical indicators in other asset classes. Here, for example, the work of Park and Irwin (2007) should be highlighted, who concluded that the use of technical indicators in developed markets is not promising. Nazário et al. (2017), however, came to the opposite conclusion. In general, it can be stated that some technical indicators (depending on the setting) can function well in specific asset classes, and the majority of the remaining indicators cannot. However, technical indicators in digital assets seem to be further evidence against the use of such indicators.

In the case of digital assets, another phenomenon needs to be pointed out. Glas (2019) showed that only a few investment strategies, such as short-term reversal or illiquidity, lead to positive results in digital currencies. The Bollinger bands reversal strategy is comparable to the short-term reversal strategy and could therefore be explained by this effect. Bollinger bands trend following, on the other hand, achieves only slightly better results than a simple buy-and-hold strategy. According to Jegadeesh and Titman (1993), there is also a short-term reversal effect in the stock market, which could explain the results of the Bollinger bands reversal strategy for the S&P500. Overall, I can particularly support the conclusion of Fang et al. (2014), who did not proclaim the predictability of the S&P500 based on technical indicators.

Table 7.6 Results for long–short technical indicators, equities

	RSI	MACD	Bollinger (reversal)	Bollinger (trend following)	STO	DMI	GOC	VAT	Buy and hold
No transaction costs									
Cumulative return (%)	6.02	−16.14	20.04	−15.03	−10.19	3.96	−7.44	−4.38	54.62
Daily return (%)	0.00	−0.01	0.02	−0.01	−0.01	0.00	−0.01	−0.01	0.04
Standard deviation (%)	1.51	1.52	1.51	1.52	1.51	1.51	1.51	1.51	1.51
No. of trades	61.27	208.77	57.67	56.95	502.44	289.21	59.82	465.27	1.00
Return/risk	0.00	−0.01	0.01	−0.01	−0.01	0.00	−0.00	−0.00	0.03
Profit factor	1.37	0.44	2.75	0.46	0.64	1.24	0.67	0.73	16.34
Max. drawdown (%)	8.10	19.92	7.49	15.54	14.83	19.77	12.56	23.17	17.57
Success rate (%)	55.64	35.05	68.12	35.05	41.58	52.28	44.36	44.16	86.14
Transaction costs									
Cumulative return (%)	−24.34	−120.30	−8.12	−43.17	−261.20	−139.79	−67.20	−292.88	54.37
Daily return (%)	−0.02	−0.09	−0.01	−0.03	−0.20	−0.11	−0.05	−0.45	0.04
Standard deviation (%)	1.54	1.61	1.48	1.51	1.68	1.61	1.55	1.77	1.51
No. of trades	61.27	208.77	57.67	56.95	502.44	289.21	59.82	465.27	1.00
Return/risk	−0.01	−0.06	−0.00	−0.02	−0.12	−0.07	−0.03	−0.25	0.03
Profit factor	0.26	0.00	0.64	0.13	0.00	0.00	0.04	0.00	16.34
Max. drawdown (%)	25.71	70.77	14.71	36.46	93.04	76.32	52.07	99.74	17.57
Success rate (%)	29.50	2.18	45.15	18.22	0.00	1.98	9.50	0.00	86.14

Cumulative return is the average, log-cumulative return across all stocks for the corresponding technical indicator. Continuous returns are defined in the range from $-\infty$ to $+\infty$, which means that even values significantly smaller than -100% can occur. Average daily return refers to the average returns of all stocks for the corresponding technical indicator in the time series of returns. Standard deviation is the standard deviation of the average daily returns. The risk/return ratio is also calculated from these two values. Average number of trades denotes the mean value of the number of all trades per stock. Profit factor is the sum of all positive cumulative returns in relation to the sum of all negative cumulative returns. The maximum drawdown refers to the average daily returns. The success rate measures the number of all stocks with a positive cumulative return compared to the number of all examined assets

7.6 Summary

In summary, the literature is divided on the use of technical indicators. Since digital assets are a relatively young and aspiring asset class, there are not many studies available that investigate the use of technical indicators in virtual currencies.

I considered the complete universe of digital assets available at the end of June 2018. Thus, I investigated the profitability of the seven technical indicators RSI, MACD, stochastics, DMI, Ichimoku, volume at time, and Bollinger bands (reversal and trend following) and compared these indicators to a simple buy-and-hold strategy using a long-only and a long–short approach. In both cases, Bollinger bands (in particular the reversal strategy) generated the best results and achieved a (clearly) higher return than the benchmark even after transaction costs. For stocks included in the S&P500, similar results were obtained for a long–short analysis. For equities, however, the analyses did not yield sufficiently better results.

Taking into account a possible survivorship bias in the digital asset data and actually tradeable bid–ask spreads, the results obtained should deteriorate. The results for the Bollinger bands reversal strategy can also be relativized by a short-term reversal effect in digital assets (the same applies to equities).

Since this chapter and the previous one have yielded rather disappointing results, I again need to switch to another approach. As multi-asset allocation and a time-series method of timing digital assets results in negative results, it remains to be seen whether certain characteristics or effects exist in the cross-section of digital asset returns.

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Chapter 8
Investment Styles in Digital Assets



8.1 Characteristics and the Cross-Section of Returns

In the early days of the asset pricing literature, researchers applied portfolio sorts to explore the cross-section of (equity) returns. In the 1970s, this approach was used to investigate whether the predictions of the CAPM hold in empirical tests. Since the ex ante beta of an asset is not observable, and data issues make tests difficult for single assets, portfolio sorts combine multiple assets into a portfolio of assets to mitigate such issues. Thereby, the number of test assets is reduced significantly, but data quality increases tremendously. One outcome of these tests is the so-called low-beta anomaly (Black et al. 1972), which proclaims the same or even higher average returns for low-beta stocks when compared to high-beta stocks.

Since then, research has uncovered hundreds of effects and anomalies in the cross-section of stock, bond, commodity, FX, and also mutual and hedge fund returns. Predictors used for this type of analysis are often referred to as "characteristics." Asness et al. (2013, 2015) broke those multi-asset findings down to four investment styles based on characteristics, which apparently work in several different asset classes. These are momentum, value, defensive, carry, and, to some extent, illiquidity.[1]

Motivated by their work, such effects have been investigated in the returns of digital assets as well. Hubrich (2017) tested the investment styles momentum, carry, and value in 11 of the largest digital coins. Based on this rather small data sample, he arrived at the existence of all three in digital assets. Rohrbach et al. (2017) explored momentum in different datasets of currencies (including digital assets) and arrived at a strong time-series and cross-sectional momentum effect in an even smaller sample of only seven digital assets. Glas (2019) tested momentum, reversal, illiquidity,

[1] Illiquidity is hardly implementable for institutional investors, which is why Asness et al. (2015) did not take this style into further consideration.

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defensive, and size and arrived at significant illiquidity, size, and reversal effects in the cross-section of about 1500 digital assets. He also showed that there is no link between traditional asset classes like stocks or FX and the asset class of digital assets.

8.2 Investment Styles Methodology and Data

Inspired by this work, I extended the approach of Glas (2019) in several ways. I added more characteristics to my tests, with a longer (in the sense of time) and broader (in the sense of the number of digital assets) dataset of digital assets. The data is the same as in the previous chapters; more details are given in Chap. 5. Table 8.1 describes the calculation methodology of the investment styles included in this (and subsequent) analyses. The table also references the original study connected to the respective characteristic.

To make the results more accessible, I again compare them to other traditional asset classes with datasets of comparable size (i.e., about the same number of assets and the same number of months). These asset classes are equities and bonds. All three datasets are denoted in USD. Both traditional asset classes serve as rough orientation only, in comparison to digital assets. They are not intended to show unequivocal findings that are valid for the asset class as a whole. Since the time period under investigation and the limited data access (see below) provide just a snapshot of the asset classes equities and bonds, the subsequent findings should be treated with caution. Moreover, it would not be possible for FX and commodities (two asset classes seemingly closer to digital assets) to obtain a comparable number of assets.

For equities, I downloaded the monthly and daily close, total return, turnover by value, and market capitalization data for all stocks contained in the list "LSPGLOBL" from Thomson Reuters Datastream. This list includes all stocks of the S&P Global 1200 index, which represents about 70% of the worldwide equity market capitalization.[2] I could not obtain the same data for the MSCI world index, which would match the size of the digital asset dataset even better, due to restricted data access. For bonds, there is no reasonable list available on Datastream. Therefore, I limited myself to all USD-denominated US corporate bonds of all ratings issued between 2013 and 2018. That approach yielded a similar unbalanced dataset as for digital assets and included around $N = 1100$ bonds on average. For this asset class, however, I could not obtain turnover data.

[2] See "S&P Global 1200," https://us.spindices.com/indices/equity/sp-global-1200, accessed on December 23, 2018.

Table 8.1 Investment style calculation methodology

Factor	Initially introduced by	Calculation in this work	Sort direction
Reversal	De Bondt and Thaler (1985)	Returns from $t-12$ to $t-3$	Down
Short-term reversal	Jegadeesh and Titman (1993)	Returns from $d-30$ to $d-1$	Down
Amihud illiquidity	Amihud (2002)	Absolute daily return divided by daily trading volume, averaged over the last 252 days	Up
Volume	Datar et al. (1998)	Log of monthly turnover by value	Down
Low volatility	Blitz and van Vliet (2007)	Realized volatility from $t-12$ to $t-1$	Down
Short-term low vol	Glas (2019)	Realized volatility from $d-30$ to $d-1$	Down
Low beta	Black et al. (1972)	Realized beta from $t-12$ to $t-1$	Down
		Benchmark is the equally weighted return of the respective dataset	
Short-term low beta	Glas (2019)	Realized beta from $d-30$ to $d-1$	Down
		Benchmark is the equally weighted return of the respective dataset	
Momentum	Jegadeesh and Titman (1993)	Returns from $t-12$ to $t-2$	Up
Size	Banz (1981)	Log of market capitalization	Down
Mispricing factor	Stambaugh and Yuan (2016)	Rolling alphas of a regression against the equally weighted returns of the respective dataset	Up
		Rolling window from $t-12$ to $t-1$	
Idiosyncratic volatility	Ang et al. (2006)	Volatility of the residuals of a regression against the equally weighted returns of the respective dataset	Down
		Rolling window from $t-12$ to $t-1$	
Explained variation	Stambaugh and Yuan (2016)	Rolling R^2 of a regression against the equally weighted returns of the respective dataset	Up
		Rolling window from $t-12$ to $t-1$	
Nominal price	Blume and Husic (1973)	Log of the nominal USD price of each digital asset	Up

Since the time periods applied in the original works are too long to derive meaningful portfolio sorts in digital assets, I limited all lookback periods in line with Glas (2019) to 12 months or 30 days (for short-term strategies). t indicates the usage of monthly data, whereas d means daily data. Direction of sorts states the order of sorting: "Up" means an ascending order whereas "Down" means a descending order. I sorted the characteristics into quintiles; P05 subsequently invests in assets at the end of the sorted list and P01 sells the assets on top of the sorted list. All investment styles are rebalanced monthly

8.3 Investment Style Results

For the calculation of the investment styles I did not take into account transaction costs. I used the different characteristics at time t to sort the respective assets into quintiles at t. The returns were then calculated at the end of $t + 1$ based on the portfolios constructed in t. The direction of sorts is indicated in Table 8.1. The average return of a portfolio is the equally weighted return of all assets included in the respective portfolio. All returns are monthly returns. The return of the hedge portfolio is the difference between P05 (long) and P01 (short). Table 8.2 presents the results for the investment styles in the asset classes digital assets, equities, and bonds. Here, I investigated different settings, applying a dataset including all available assets and three others where I followed the idea of Fama and French (1992, 1993) to control for different size groups.

The results presented in Table 8.2 are very much in line with the findings of Glas (2019). The illiquidity proxies Amihud's illiquidity ratio and simple turnover data yield significant hedge portfolio returns in almost all datasets. Illiquidity cannot be found in the largest digital assets. As documented by Glas (2019), sorting for high Amihud ratios results in negative average returns, as opposed to low turnover, which yields positive average returns. This result indicates some differences in the usage of daily versus monthly digital asset data. Similar differences exist when looking at the short-term variants of the reversal, low-volatility (low vol), and low-beta characteristics. Although investment styles based on these characteristics do not yield significant hedge portfolio returns, differences in the sign of the returns can be acknowledged. Short-term reversal is weaker than in the work of Glas (2019), with only small digital assets yielding significant hedge portfolio returns.

Additional characteristics, which are not investigated by Glas (2019), are based on the findings of Blume and Husic (1973) and Stambaugh and Yuan (2016). Stambaugh and Yuan (2016) used rankings from 11 stock anomalies to form two mispricing factors. Since these 11 anomalies are not applicable to digital assets, and literature on characteristics and effects in digital assets is scarce, I followed a more basic approach. I simply took the value of the different alphas from a rolling univariate time-series regression as the sorting criterion. The independent variable in that case is inspired by the CAPM and is the equally weighted[3] return of all available digital assets. An alpha of 0 thus indicates a good fit, whereas high positive or negative alphas lead to the assumption that the market cannot explain the returns of individual digital assets. From the same regression, I estimated idiosyncratic volatility (idio risk), which is the standard deviation of residuals, and "Explained" is the R^2 of the regression analysis. "Mispricing" (the alpha from the regression) as a sorting criterion yields significant hedge portfolio returns when assuming all and only the smallest digital coins. Highly positive alphas, however, lead to negative returns, and highly negative alphas to positive returns, which also indicates

[3] Stambaugh and Yuan (2016) originally used a value-weighted benchmark. However, because of the dominance of bitcoin, I prefer an equally weighted benchmark.

Table 8.2 Multi-asset investment style outcomes

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Digital assets														
All														
Mean (%)	2.90	1.30	−9.64	11.11	−0.47	−1.53	−2.73	−2.89	−2.70	7.12	−10.20	2.55	4.19	−17.57
t-stat	(1.17)	(0.36)	(−2.84)[c]	(3.79)[c]	(−0.12)	(−0.60)	(−0.30)	(−0.93)	(−1.22)	(1.29)	(−4.08)[c]	(1.32)	(2.26)[b]	(−3.68)[c]
SD (%)	17.56	28.06	24.44	21.14	27.84	24.40	67.96	25.79	15.63	43.55	16.84	13.55	12.87	37.54
Big														
Mean (%)	1.33	−4.00	−5.00	−1.17	−8.70	−2.45	−3.06	−1.12	−1.20	−4.53	0.44	1.91	−0.07	−1.46
t-stat	(0.36)	(−1.50)	(−1.67)[a]	(−0.43)	(−1.80)[a]	(−0.87)	(−0.49)	(−0.28)	(−0.35)	(−1.33)	(−0.20)	(0.91)	(−0.03)	(−0.59)
SD (%)	26.96	21.33	21.69	20.05	34.40	22.11	45.88	30.99	24.36	23.40	15.67	14.73	14.60	27.11
Medium														
Mean (%)	0.61	5.33	−9.66	8.12	−1.17	0.35	−0.55	5.46	2.27	7.06	−1.31	0.72	−2.32	−14.97
t-stat	(0.11)	(0.92)	(−2.95)[c]	(2.46)[b]	(−0.35)	(0.07)	(0.11)	(1.12)	(0.77)	(1.63)	(−0.62)	(0.34)	(1.71)[a]	(−2.72)[c]
SD (%)	21.88	44.74	23.56	24.11	23.72	41.89	38.69	38.15	23.59	39.62	14.79	14.97	15.69	47.04
Small														
Mean (%)	7.04	21.05	−9.85	18.57	1.42	−7.64	−0.32	−7.56	−8.25	6.48	−15.65	5.46	−0.20	−31.67
t-stat	(1.08)	(3.22)[c]	(−2.28)[b]	(4.20)[c]	(0.28)	(−1.40)	(0.19)	(−1.41)	(−1.35)	(1.15)	(−3.39)[c]	(1.31)	(1.16)	(−5.60)[c]
SD (%)	39.93	33.86	31.24	31.79	35.81	42.36	53.07	38.51	38.11	41.87	31.57	29.15	17.15	46.46

(continued)

Table 8.2 (continued)

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Equities														
All														
Mean (%)	−0.19	−0.03	0.38	0.25	0.77	0.55	0.67	0.19	0.25	0.56	0.18	0.21	0.02	−0.04
t-stat	(−0.38)	(−0.12)	(1.71)[a]	(0.95)	(1.40)	(1.33)	(0.83)	(0.42)	(0.52)	(1.75)[a]	(0.38)	(1.31)	(0.10)	(−0.13)
SD (%)	3.49	2.25	1.57	2.04	3.94	3.26	6.06	3.54	3.47	2.54	3.35	1.12	1.50	2.33
Big														
Mean (%)	−0.08	0.14	0.41	−0.05	0.18	0.24	0.27	0.06	0.06	0.01	0.13	−0.04	0.35	0.32
t-stat	(−0.17)	(0.49)	(1.46)	(−0.16)	(0.31)	(0.51)	(0.34)	(0.11)	(0.13)	(0.03)	(0.29)	(−0.21)	(1.24)	(1.02)
SD (%)	3.42	2.23	2.02	2.50	4.26	3.62	6.11	4.38	3.59	2.61	3.19	1.30	1.97	2.49
Medium														
Mean (%)	−0.12	−0.14	−0.00	0.03	0.91	0.68	0.48	0.23	0.17	0.08	0.46	−0.07	0.11	0.10
t-stat	(−0.24)	(−0.39)	(−0.02)	(0.12)	(1.73)	(1.61)	(0.79)	(0.51)	(0.34)	(0.51)	(0.88)	(−0.44)	(0.58)	(0.36)
SD (%)	3.58	2.86	1.98	1.80	3.77	3.32	4.51	3.62	3.60	1.24	3.67	1.17	1.36	2.09

Small														
Mean (%)	−0.19	−0.03	0.45	0.39	0.88	0.70	0.59	0.12	0.33	1.18	0.17	0.24	−0.02	−0.24
t-stat	(−0.40)	(−0.09)	(1.96)[a]	(1.66)	(1.51)	(1.46)	(0.78)	(0.23)	(0.70)	(5.99)[c]	(0.33)	(1.09)	(−0.06)	(−0.70)
SD (%)	3.37	2.43	1.63	1.86	4.19	3.75	5.65	3.87	3.36	1.56	3.63	1.56	1.97	2.66
Bonds														
All														
Mean (%)	−0.38	−0.50	—	—	0.66	0.47	0.01	0.24	0.38	0.00	0.40	0.08	−0.16	0.19
t-stat	(−0.88)	(−1.84)[a]	—	—	(1.66)	(1.36)	(0.03)	(0.84)	(0.83)	(0.00)	(0.97)	(0.57)	(−1.16)	(0.59)
SD (%)	3.10	2.10	—	—	2.85	2.70	3.40	2.24	3.27	0.82	2.93	1.00	0.95	2.58
Big														
Mean (%)	−0.03	−0.05	—	—	0.42	0.27	0.21	0.29	−0.00	−0.12	0.21	0.14	−0.05	−0.02
t-stat	(−0.08)	(−0.19)	—	—	(1.23)	(1.00)	(0.57)	(1.15)	(−0.00)	(−1.12)	(0.54)	(1.08)	(−0.34)	(−0.07)
SD (%)	2.90	2.14	—	—	2.46	2.09	2.79	2.02	2.92	0.87	2.77	0.93	1.04	2.28
Medium														
Mean (%)	−0.09	−0.66	—	—	1.19	0.86	0.45	0.18	0.23	−0.38	0.88	0.21	−0.30	0.42
t-stat	(−0.15)	(−1.79)[a]	—	—	(2.29)[b]	(1.92)[a]	(0.89)	(0.51)	(0.37)	(−1.81)[a]	(1.58)	(0.84)	(−1.36)	(0.90)
SD (%)	4.26	2.88	—	—	3.70	3.47	3.84	2.70	4.43	1.69	3.93	1.73	1.55	3.73
Small														
Mean (%)	−0.29	−0.36	—	—	0.76	0.59	0.09	0.20	0.38	−0.27	0.37	0.06	−0.20	0.17
t-stat	(−0.59)	(−1.13)	—	—	(1.89)[a]	(1.72)[a]	(0.21)	(0.69)	(0.74)	(−1.62)	(0.80)	(0.41)	(−1.22)	(0.48)
SD (%)	3.58	2.45	—	—	2.88	2.67	3.35	2.29	3.70	1.32	3.30	1.08	1.13	2.82

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level

The columns represent the investment styles under consideration, with "st" denoting the short-term variants. The remainder of the table is separated by asset classes; within each asset class, results are reported for different size groups. "Big" is the largest size tercile, "Medium" the second largest, and "Small" denotes the smallest size tercile in every month *t*. "Mean" is the equally weighted monthly average hedge portfolio return in %. *t*-stat is the *t*-statistics of the mean in parentheses. "SD" represents the standard deviation of the volatility of the monthly mean returns in %

a reversal effect. An effect based on idiosyncratic volatility cannot be found, and the R^2 values yield weaker results compared to alphas. The results, on the other hand, are not robust when controlling for small digital assets. Nominal price, which assumes higher returns for assets with relatively high prices, results in significantly negative hedge portfolio returns for almost all but the largest digital assets. This indicates a low price effect in digital assets.

These findings apply neither to equities nor bonds. In almost all datasets, the same investment styles yield insignificant outcomes except for some cases. This could be due to the relatively short data history (when compared to the standard asset pricing and investment literature), or the specific period of time under investigation.[4] The (positive) sign for most investment styles in equities and bonds, however, is in line with its expectation derived from the investment styles literature. Only both reversal strategies yield negative mean monthly returns, which is contrary to the academic literature. This finding, on the other hand, could also be due to the shorter calculation period of reversal, which is usually computed over the last 36 to 60 months. Interestingly, the signs of the same investment styles in digital assets are contrary to their counterparts in equities and bonds. Nevertheless, Glas (2019) reported comparable results for the traditional asset classes under investigation.

Next to the significance of hedge portfolio returns, it was necessary to investigate whether the characteristics proxy the same effect(s) or if the characteristics are independent from each other. Therefore, I applied a simple correlation analysis based on the returns of the single investment styles comprised of all available assets (see Table 8.3).[5] For digital assets, only size exhibits relatively high positive correlation coefficients to short-term reversal, volume, and a negative coefficient to nominal price. Also, reversal and explained (variation) exhibit a positive correlation coefficient larger than 50%. Volume and nominal price appear to be negatively correlated as well. Low beta and low volatility are obviously positively correlated to some extent. All remaining characteristics do not show especially high or low correlation coefficients.

In equities, the returns of reversal are naturally negatively correlated to the returns of momentum and mispricing alphas, whereas momentum and mispricing alphas are positively correlated. Volume and nominal price exhibit a strongly negative correlation coefficient, and low volatility and mispricing alphas yield a positive correlation coefficient. This is somewhat unexpected, since mispricing alphas appear to proxy momentum, and thus low volatility is also positively correlated with momentum, although to a lesser extent. Low beta and low volatility are again relatively similar. Nominal price and size are also negatively correlated. This is not very surprising, since smaller stocks usually carry lower nominal prices.

In corporate bonds, the outcomes are even more extreme. Reversal is negatively correlated with almost any other investment style. The correlation coefficients of low

[4] Tests with the same assets but a longer history (starting at the end of 1989) yield outcomes considered as typical from the equity and bond literature. Those results are not reported here.

[5] The results for the different size groups can be found in Appendix F.

Table 8.3 Multi-asset investment style correlations

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Digital assets														
Reversal	100.00	−9.56	20.94	27.99[b]	−9.24	−15.46	−18.08	19.99	−82.35	40.35[c]	−36.14[b]	−10.64	52.23[c]	−22.58
Reversal st	−9.56	100.00	−4.04	19.00	2.86	−31.62[b]	12.63	−25.92[b]	−2.81	74.43[c]	−21.80	−13.43	2.13	−51.83[c]
Amihud	20.94	−4.04	100.00	9.67	0.68	−17.60	4.98	15.07	−21.87	25.05[a]	−32.47[b]	−24.34[a]	22.15	27.53[b]
Volume	27.99[b]	19.00	9.67	100.00	−2.80	−35.59[c]	19.22	−3.54	−25.44[a]	79.86[c]	−42.48[c]	0.23	19.23	−64.10[c]
Low vol	−9.24	2.86	0.68	−2.80	100.00	53.53[c]	71.44[c]	41.80[c]	0.14	−23.04	25.32[a]	0.37	24.27[a]	44.89[c]
Low vol st	−15.46	−31.62[b]	−17.60	−35.59[c]	53.53[c]	100.00	14.44	38.33[c]	14.62	−41.59[c]	24.56[a]	25.59[a]	−12.51	38.42[c]
Low beta	−18.08	12.63	4.98	19.22	71.44[c]	14.44	100.00	42.93[c]	16.66	4.09	19.52	−11.56	6.65	9.15
Low beta st	19.99	−25.92[b]	15.07	−3.54	41.80[c]	38.33[c]	42.93[c]	100.00	−13.53	−35.44[c]	6.47	20.54	16.12	25.25[a]
Momentum	−82.35[c]	−2.81	−21.87	−25.44[a]	0.14	14.62	16.66	−13.53	100.00	−36.04[b]	35.57[b]	7.86	−44.82[c]	11.88
Size	40.35[c]	74.43[c]	25.05[a]	79.86[c]	−23.04	−41.59[c]	4.09	−35.44[c]	−36.04[b]	100.00	−45.74[c]	−27.14[a]	23.49	−77.12[c]
Mispricing	−36.14[b]	−21.80	−32.47[b]	−42.48[c]	25.32[a]	24.56[a]	19.52	6.47	35.57[b]	−45.74[c]	100.00	−7.89	−19.02	20.16
Idio risk	−10.64	−13.43	−24.34[a]	0.23	0.37	25.59[a]	−11.56	20.54	7.86	−27.14[a]	−7.89	100.00	−1.14	−3.29
Explained	52.23[c]	2.13	22.15	19.23	24.27[a]	−12.51	6.65	16.12	−44.82[c]	23.49	−19.02	−1.14	100.00	−10.02
Nominal	−22.58	−51.83[c]	27.53[b]	−64.10[c]	44.89[c]	38.42[c]	9.15	25.25[a]	11.88	−77.12[c]	20.16	−3.29	−10.02	100.00
Equities														
Reversal	100.00	5.26	−47.78[c]	−3.80	−53.79[c]	−48.74[c]	−31.09[b]	−27.25[b]	−98.75[c]	5.79	−93.03[c]	6.23	−7.51	−38.45[c]
Reversal st	5.26	100.00	−25.81[a]	−14.15	−47.24[c]	−43.55[c]	−42.64[c]	−30.80[b]	−6.37	−15.71	−27.05[a]	−15.59	12.06	−9.70
Amihud	−47.78[c]	−25.81[a]	100.00	28.44[b]	42.89[c]	46.19	16.70[a]	25.75[c]	49.16[a]	22.21[c]	52.37	−10.95	5.67	41.25[c]
Volume	−3.80	−14.15	28.44[b]	100.00	26.78[a]	31.72[c]	13.55	25.22[a]	3.74	90.03[c]	9.36	6.19	18.04	−75.52[c]
Low vol	−53.79[c]	−47.24[c]	42.89[c]	26.78[a]	100.00	93.98[c]	78.71[c]	75.85[c]	54.41[c]	21.31	62.52[c]	−14.99	7.62	23.49
Low vol st	−48.74[c]	−43.55[c]	46.19[c]	31.72[b]	93.98[c]	100.00	69.40[c]	79.67[c]	48.75[c]	26.52[b]	58.43[c]	−15.08	9.60	17.26
Low beta	−31.09[b]	−42.64[c]	16.70	13.55	78.71[c]	69.40[c]	100.00	65.48[c]	34.35[b]	7.79	43.03[c]	−8.15	16.77	6.83
Low beta st	−27.25[b]	−30.80[b]	25.75[a]	25.22[a]	75.85[c]	79.67[c]	65.48[c]	100.00	28.62[b]	15.69	39.28[c]	−23.51	14.25	7.05

(continued)

Table 8.3 (continued)

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Momentum	−98.75[c]	−6.37	49.16[c]	3.74	54.41[c]	48.75[c]	34.35[b]	28.62[b]	100.00	−5.79	94.93[c]	−5.77	11.63	38.79[c]
Size	5.79	−15.71	22.21[a]	90.03[c]	21.31	26.52[b]	7.79	15.69	−5.79	100.00	1.33	8.54	12.84	−71.59[c]
Mispricing	−93.03[c]	−27.05[a]	52.37[c]	9.36	62.52[c]	58.43[c]	43.03[c]	39.28[c]	94.93[c]	1.33	100.00	3.48	12.80	38.95[c]
Idio risk	6.23	−15.59	−10.95	6.19	−14.99	−15.08	−8.15	−23.51	−5.77	8.54	3.48	100.00	16.13	−11.74
Explained	−7.51	12.06	5.67	18.04	7.62	9.60	16.77	14.25	11.63	12.84	12.80	16.13	100.00	−16.26
Nominal	−38.45[c]	−9.70	41.25[c]	−75.52[c]	23.49	17.26	6.83	7.05	38.79[c]	−71.59[c]	38.95[c]	−11.74	−16.26	100.00
Bonds														
Reversal	100.00	46.02[c]	—	—	−70.13[c]	−69.21[c]	−46.86[c]	1.28	−98.68[c]	−8.72	−90.75[c]	3.79	45.95[c]	−88.55[c]
Reversal st	46.02[c]	100.00	—	—	−46.48[c]	−42.07[c]	−46.23[c]	6.78	−47.94[c]	−17.68	−68.94[c]	6.92	10.01	−62.83[c]
Amihud	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Volume	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Low vol	−70.13[c]	−46.48[c]	—	—	100.00	97.38[c]	84.89[c]	34.62[c]	71.67[c]	39.24[c]	75.55[c]	−21.34	−22.04	78.88[c]
Low vol st	−69.21[c]	−42.07[c]	—	—	97.38[c]	100.00	81.23[c]	48.04[c]	70.09[c]	42.38[c]	73.54[c]	−23.72[c]	−22.38	74.90[c]
Low beta	−46.86[c]	−46.23[c]	—	—	84.89[c]	81.23[c]	100.00	48.16[c]	49.20[c]	47.11[c]	59.99[c]	−22.83	2.47	56.34[c]
Low beta st	1.28	6.78	—	—	34.62[c]	48.04[c]	48.16[c]	100.00	−3.08	13.89	−3.46	−41.16[c]	12.63	−5.12
Momentum	−98.68[c]	−47.94[c]	—	—	71.67[c]	70.09[c]	49.20[c]	−3.08	100.00	12.20	93.05[c]	−0.04	−43.52[c]	90.46[c]
Size	−8.72	−17.68	—	—	39.24[c]	42.38[c]	47.11[c]	13.89	12.20	100.00	16.60	5.75	18.89	15.04
Mispricing	−90.75[c]	−68.94[c]	—	—	75.55[c]	73.54[c]	59.99[c]	−3.46	93.05[c]	16.60	100.00	3.17	−31.98[b]	95.86[c]
Idio risk	3.79	6.92	—	—	−21.34	−23.72[a]	−22.83	−41.16[c]	−0.04	5.75	3.17	100.00	43.22[c]	−3.50
Explained	45.95[c]	10.01	—	—	−22.04	−22.38	2.47	12.63	−43.52[c]	18.89	−31.98[b]	43.22[c]	100.00	−38.24[c]
Nominal	−88.55[c]	−62.83[c]	—	—	78.88[c]	74.90[c]	56.34[c]	−5.12	90.46[c]	15.04	95.86[c]	−3.50	−38.24[c]	100.00

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
The columns and lines represent the investment styles under consideration, with "st" denoting the short−term variants. The numbers are the correlation coefficients in %. The table reports the outcomes for all available assets

volatility and low beta are the highest of all three asset classes. Momentum is again highly correlated to low volatility and mispricing alphas, as well as to nominal price. Size in bonds, on the other hand, shows no high positive or negative correlation coefficients to other investment styles, which might be due to the (relatively) static nature of issuance sizes. Market values of stocks or digital assets, though, are able to vary dynamically, which is not true for bonds. Nominal price and mispricing alphas co-vary highly with each other. The same applies to mispricing alphas and low volatility, which remains very surprising.

Based on his results, Glas (2019) formulated a four-factor asset pricing model for digital assets.[6] This model includes market, size, illiquidity (as measured by turnover), and short-term reversal as explanatory variables for the returns of digital assets. Short-term reversal and size are less significant in this analysis than in the work of Glas (2019). Illiquidity, however, still produces comparable outcomes. Since the correlation analysis shows that size, in particular, is correlated with several other characteristics, I investigate the model developed by Glas (2019) a bit further.

8.4 Independence of Investment Styles

With the introduction of the Fama–French five-factor model, Fama and French (2015) not only constructed hedge portfolios (comparable to Table 8.2) and investigated correlations (comparable to Table 8.3), but they also tested for the independence of the factors used in their asset pricing model. Therefore, they ran several OLS regressions based on the five factors from their asset pricing model to determine the independence of each single factor. Thus, they regressed each single factor on the remaining factors. By doing so, they concluded that value (*HML*) is a redundant factor in the five-factor model. I ran the same analysis on the independent variables identified by Glas (2019) (see Table 8.4). The results for each size group can be found in Appendix G.

As indicated by the correlation matrix in Table 8.3, the investment style size is related to several other styles. Table 8.4 shows a significant positive relationship of size to volume and short-term reversal. Looking at volume as the dependent variable yields a significant intercept, which implies some independence from the remaining factors. The same does not apply to size or short-term reversal. Market, on the other hand, seems not to be dependent on investment styles or vice versa. Therefore, size might not be a necessity in this model, since it is subsumed by volume and short-term reversal. The relationship between size and volume exists in equities as well. At this point, it needs to be noted that I am not aiming to develop a "new" asset pricing model for equities or bonds. I am referencing the digital asset results to both traditional asset classes using the same setting and starting position. Thereby, a similar relationship between volume and size emerges for both digital assets and

[6] More information regarding this model can be found in Chap. 9.

Table 8.4 Multi-asset investment style independence

Style	Intercept	Market	Size	Volume	Short-term reversal	Adjusted R^2 (%)
Digital assets						
Market	−0.06	—	0.52	−0.24	0.19	3.84
	(−1.09)	—	(1.48)	(−0.68)	(0.63)	
Size	−0.02	0.08	—	0.81	0.31	68.70
	(−1.00)	(1.48)	—	(9.39)[c]	(2.71)[c]	
Volume	0.06	−0.04	0.79	—	−0.17	63.53
	(2.95)[c]	(−0.68)	(9.39)[c]	—	(−1.39)	
Short-term reversal	0.02	0.04	0.41	−0.22	—	13.73
	(0.94)	(0.63)	(2.71)[c]	(−1.39)	—	
Equities						
Market	0.01	—	0.20	−0.26	0.36	5.05
	(2.04)[b]	—	(0.61)	(−0.67)	(2.37)[b]	
Size	0.00	0.03	—	1.11	−0.04	80.55
	(1.45)	(0.61)	—	(15.62)[c]	(−0.68)	
Volume	−0.00	−0.03	0.73	—	0.01	80.48
	(−0.76)	(−0.67)	(15.62)[c]	—	(0.20)	
Short-term reversal	−0.00	0.25	−0.18	0.07	—	6.56
	(−0.43)	(2.37)[b]	(−0.68)	(0.20)	—	
Bonds						
Market	−0.00	—	−0.93	—	0.11	45.32
	(−1.30)	—	(−6.47)[c]	—	(1.94)[a]	
Size	−0.00	−0.45	—	—	0.01	41.81
	(−1.19)	(−6.47)[c]	—	—	(0.23)	
Short-term reversal	−0.00	0.55	0.10	—	—	5.92
	(−1.44)	(1.94)[a]	(0.23)	—	—	

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
This table shows the regression coefficients of a multivariate regression against a specific investment style. Below the coefficient, the *t*-statistics is reported in parentheses. The investment styles are based on a dataset comprised of all available assets of the respective asset class

equities. Short-term reversal in equities, however, contributes to explaining market returns, although only to a very small extent. The same effect can be observed in bonds, with a stronger relationship between market and size.

When controlling for different size groups, the results do not differ dramatically. For digital assets, the same effects are observable in a group of the largest digital assets only (see Table G.1 in Appendix G). In medium-sized digital assets these outcomes weaken. In the smallest digital assets, an effect similar to bonds can be observed. Next to the already described relationships, a significant positive relationship between market and size returns emerges.

The adjusted R^2 values, on the other hand, still do not exceed values of 40%, which indicates only little explanatory power. Taking into account all available digital assets (see Table 8.4) results in R^2 values of c.69% for size as the dependent variable and about 64% for volume as the dependent variable. Those numbers increase to c.80% in equities. In bonds, the highest adjusted R^2 is 45.32%, including one independent variable less. Except for the largest stocks in this sample, the relatedness between size and volume disappears for medium-sized and small stocks. The results also weaken for bonds when applied to differently sized bond issuances.

8.5 Summary

Using a longer and broader data sample compared to Glas (2019), this section reviewed the success or failure of different investment styles. Glas (2019) arrived at promising illiquidity, size, and reversal investment styles in digital assets. I can confirm the result for illiquidity, but not for size and reversal. Size indeed produces mainly positive mean monthly average returns; however, they are not (any longer) statistically significant. The same applies to (short-term) reversal. By including four more investment styles, I arrived at significant mispricing alpha and nominal price characteristic premia.

While the outcomes of Sect. 8.4 are not of great concern for equities and bonds, the results point to some issues for the model described by Glas (2019). Multicollinearity in regression models can make results less reliable. Also, the less- (or even in-) significant hedge portfolio returns based on an updated dataset leave the robustness of the model in doubt. Therefore, I investigate these issues further in the next chapter. Chapter 9 aims to explore risk factors in digital assets, and whether such factors are priced significantly. Thus, I not only apply time series but also apply cross-sectional regressions (see Sect. 2.2) to identify possible risk factors and/or characteristics in the returns of digital assets. In that case, I focus solely on digital assets since similar or even identical analyses have been extensively described for other asset classes in the existing literature.

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Chapter 9
Risk Factors in Digital Assets



9.1 Risk Factors and Digital Assets

Following on from the previous chapter, I extend the search for risk factors in digital assets by several means. First, I use statistical tools like principal component analysis to condense the information contained in the digital asset dataset. After investigating the principal components more deeply, it emerges that this first approach yields no statistically significant results. Therefore, as a second step I consider fundamental as well as other data (see Sect. 5.4.3) to intensify the search for risk factors. As that approach still results in rather disappointing outcomes, I turn to the characteristics versus covariances story as a last step. By doing so, I find three characteristics which tend to explain digital asset returns while being priced significantly in the underlying data.

Digital assets in general are an emerging asset class which mainly started out based on the success of BTC. Compared to other asset classes like stocks or bonds digital assets are rather small, with a combined value of c.321 billion USD and about 1600 tradeable assets;[1] 810 other digital assets, however, have already passed away.[2] The overall numbers had changed by mid-2020, with 5705 tradeable assets,[3] valued at a total market capitalization of c.271 billion USD, and 1928 dead coins.[4] Due to only light to no regulation, digital asset markets are comparable to the Wild West. Exchanges and personal wallets are hacked, markets are manipulated (Griffin and Shams 2020), and funds are raised for projects or tokens which are (obviously)

[1] See "Total market capitalization," https://www.coinmarketcap.com/charts, accessed on June 10, 2018.

[2] See "Full list," https://deadcoins.com/, accessed on June 10, 2018.

[3] See "Total market capitalization," https://www.coinmarketcap.com/charts, accessed on July 8, 2020.

[4] See "Full list," https://deadcoins.com/, accessed on July 8, 2020.

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scam. The large holdings of early cryptocurrency enthusiasts (also referred to as "whales") are not beneficial for the worldwide trading volume, which results in a relatively illiquid digital asset market (in comparison to the global FX market, for instance). The large number of local digital asset exchanges further worsens this issue. With (relatively) low trading volume comes high volatility, discouraging institutional investors from entering this young asset class. A much more important problem for digital assets is their speculative character. Since no fundamental data (like balance sheet information) exists for blockchains or digital assets in general, investors cannot reasonably come up with a "fair" or "fundamental" value for one specific digital token or the whole market. For equity or commodity tokens such a problem might not exist since there should be an underlying asset which could be valued easily. However, the legal situation is unclear so far, which is why an "equity" token does not necessarily need to carry the same rights as a stock. Thus, I aim to overcome the issue of valuation by searching for (priced) risk factors, which explains the cross-section of digital assets.

Inspired by Fama and French (1993) and Fung and Hsieh (1997), I find evidence for two to three risk factors in the returns of digital assets. These factors, however, are not priced significantly, and thus the development of a formal asset pricing model is not (yet) feasible for digital assets. One explanation for this finding might be a very high aggregate idiosyncratic variance. Nevertheless, I find indications of some priced characteristics (especially volume and nominal price) but not priced covariances, which is similar to the asset pricing literature of the 1970s and adds to recent discussion—see Daniel and Titman (1997), among others.

This chapter therefore contributes to the literature in several ways. To my knowledge, I am one of the first to empirically investigate possible risk factors in the returns of digital assets. These risk factors might eventually lay the groundwork for future asset pricing models specific to digital assets. Moreover, this chapter shows that previous research connected to other asset classes can be transferred to this young and emerging asset class, too. It does not, however, yield clear results. Third, I find first indications in favor of the characteristics story in a young and emerging asset class (see Sect. 2.3 for a more detailed review).

Since there is still much uncertainty surrounding the (price/return) behavior of digital assets, I aim to investigate the same questions finance researchers encountered decades ago. Is there a common source or risk factor(s) that can explain the cross-section of digital asset returns? Is the dependency (β) priced robustly and significantly? If not, are there other sources of explanatory variables, like digital asset specific characteristics themselves, which solve this issue? By investigating these questions I could find links to similar results from other asset classes, which would be (first) evidence for market efficiency in digital assets. It would also document links to other more established asset classes. If I do not arrive at related results, digital assets could be fairly in line with the random-walk theory and therefore justifiably called a "Wild West" asset class (if the term "asset class" is still applicable to digital assets).

To conduct such a wide-ranging analysis I needed to utilize a lot of different data and different statistical methods.[5] To follow the approach of Fung and Hsieh (1997) and Lustig et al. (2011) I used a large sample of digital assets, which is analogous to the previous chapters (Panel A of Table 5.9). Thus, I applied principal component analysis (PCA) and partial least squares (PLS) to the dataset of digital asset returns to extract a certain number of possible risk factors. After investigating these factors (in terms of asset pricing tests), I extended the analysis in multiple directions to find other (possibly priced) risk factors. This is why I needed to include additional data from several sources. The data ranges from fundamental data (Panels B and C of Table 5.9) to hedge fund data (Panel E of Table 9.3) to investment styles in digital assets (Panel B of Table 9.2), which stem from Chap. 8. Based on this data I investigated whether the betas with regard to the obtained data or the characteristics themselves are priced (only Panel B of Table 9.2). Thereby, it emerges that some characteristics appear to be priced significantly, but not their covariances. The analyses described in this chapter are broken down into different steps:

1. Run PCA and PLS to extract possible risk factors.[6]
2. Investigate the betas with regard to those risk factors.
3. Consider further the covariances versus characteristics issue.

9.2 Identification of Risk Factors: A Short Comparison

In Chap. 1 I mentioned different classifications of digital assets. One would need to account for them when developing an asset pricing model or doing asset pricing research in general. Assuming there were different clusters that differentiate themselves in their return behavior, several asset pricing models for each cluster would need to be developed. In Sect. 5.5, however, I could not arrive at different clusters in the returns of digital assets; I therefore aim to develop a very general model for all available digital assets. Beforehand, I tested existing asset pricing models specific to digital assets and arrived at comparably poor results. Consequently, it is even more important to (try to) identify risk factors in the returns of digital assets.

Analogous to the previous chapter, the literature has already identified several (assumed) risk factors and set up models to describe the behavior of digital assets. Hayes (2017), for example, constructed a fundamental model to explain the prices (not returns) of 66 digital assets. His cross-sectional model includes the factors computational power (giga hashes per second, *GHS*), coins found per minute

[5] The data used is described in Chap. 5, and in Sect. 9.4.3.

[6] A new approach (Kelly et al. 2019), allowing for latent factors and time-varying loadings, might solve several issues connected to the principal component analysis described in this chapter. Thus, the reported results could serve as a "benchmark" for the instrumented principal component analysis, which I leave for future research.

Table 9.1 Regression results for existing models

	All			Mineable			Non-mineable		
	Eq. 9.1	Eq. 9.2	Eq. 9.3	Eq. 9.1	Eq. 9.2	Eq. 9.3	Eq. 9.1	Eq. 9.2	Eq. 9.3
alpha	−30.55	−10.26	5.81	−22.64	−9.87	−5.55	−35.85	−10.27	14.75
SA (%)	15.99	7.60	9.90	13.77	6.79	10.98	17.78	7.66	8.89
$\bar{\beta}_1$	1.3761	−0.1647	1.5406	0.8907	−0.2562	1.0072	1.7130	−0.1136	1.9557
$\bar{\beta}_2$	−0.1482		2.3564	−0.1042		1.9005	−0.1783		2.7545
$\bar{\beta}_3$			−1.6018			−1.2193			−1.9299
$\bar{\beta}_4$			−1.5374			−0.6829			−2.1909
R^2	26.29	12.06	47.43	15.85	7.89	31.68	33.54	14.94	58.49

This table shows the regression results for the three asset pricing models above. I ran these models against each available digital asset in the three datasets all, mineable, and non-mineable. Thus, the values in the tables are averaged over all assets. Alpha, significant alphas (SA), and R^2 are percentage values. The value of significant alphas is the number of all alphas significant at the 10% level divided by the number of all digital assets. β_1 to β_4 are the averaged regression coefficients of all digital assets

(*CPM*), and a dummy variable for the algorithm employed (*ALGO*), where *ALGO* is 0 for digital assets which employ SHA-256 and 1 for scrypt. A more formal description can be given as

$$\ln(Price_{it}) = \alpha_i + \beta_1 \ln(GHS_t) + \beta_2 \ln(CPM_t) + \beta_3 ALGO_t + \epsilon_{it}. \tag{9.1}$$

In the subsequent analysis (see Table 9.1) I do not include the dummy variable *ALGO* since I could not obtain the corresponding data. Wheatley et al. (2018) aimed to predict BTC price bubbles based on the value of a network (Metcalfe's law). Here, the value of a network is approximated by the number of active addresses (theoretically the number of active users). The time-series regression model can thus be written as

$$\ln(Price_i) = \alpha_i + \beta_1 \ln(NAA_t) + \epsilon_{it}, \tag{9.2}$$

with *NAA* the number of active addresses, which is proxied in the following analysis by the number of unique bitcoin wallet addresses.[7] A third, empirically motivated, model was described by Glas (2019), who arrived at results similar to Chap. 8. This model includes a market factor (*CMKT*) and three apparently significant factor-mimicking portfolios—size (*CSMB*), illiquidity (*CILQ*), and short-term reversal (*CSTR*)—to explain the returns (r) of digital assets:

$$r_{it} = \alpha_i + \beta_1 CMKT_t + \beta_2 CSMB_t + \beta_3 CILQ_t + \beta_4 CSTR_t + \epsilon_{it}. \tag{9.3}$$

I compared these three models with regard to the data obtained. To avoid the problem of spurious regressions due to non-stationary price data, I did not take the

[7] That number might also be available for other coins associated with their own blockchain, but I stuck to the same data as used by Wheatley et al. (2018) to ensure comparable results.

prices of digital assets, see Eqs. 9.1 and 9.2, as the dependent variable, but their returns.[8] Thus, I ran the models described above as time-series regressions against each of the digital tokens in my dataset. I also used the log monthly change of the independent variables (the right-hand side of the equations) in Eqs. 9.1 and 9.2. Otherwise, the factors in both models would not be stationary and the models themselves would no longer be consistent.

Table 9.1 reports the averaged results (alpha, beta, and R^2) for each model and states the percentage value of digital assets with a significant alpha (significant at the 10% level). This analysis is meant for illustrative purposes only, to show the extent to which the models are able to explain the cross-section of digital asset returns. The alphas of the models in Eqs. 9.1 and 9.2, for example, cannot be interpreted as excess returns since both models include non-tradeable fundamental factors. These alphas are only reported for the sake of completeness. The percentage values of significant alphas are relatively high in all three models, ranging from 7% to 18%. A good fit of a model would be indicated by values around 10% (which would be in line with its expected value). Values above or below 10% indicate models which are not consistent with this expectation. On the other hand, all three models are able to explain the cross-section of digital asset returns only to a comparatively small extent. The Hayes (2017) and Wheatley et al. (2018) models achieve an average R^2 of 16%–34% and 8%–15%, respectively. The Glas (2019) model explains about 32%–58%, which is higher than the other models, since it also includes twice as many independent variables.[9]

Since Chap. 8 of this book and Glas (2019) arrived at similar results, this first test of asset pricing models for digital assets yields relatively weak results. Although Chap. 8 lays the groundwork for possible factors in digital assets, it has been shown that a regression model based on such factors is relatively weak in explaining digital asset returns (see Table 9.1). Therefore, it is necessary to search for and identify (other) common risk factors in the returns of digital assets, which might be used for asset pricing models in subsequent research. To do so, I apply different statistical methods, which are described in more detail below.

[8] Time-series regressions on prices are especially prone to an effect called "spurious regressions." This is mostly due to non-stationary (in)dependent variables. This means the regression result will show a linear relationship where normally no relationship exists. That is also the case for variables moving in trends. Therefore, regressions on returns usually circumvent the problem of non-stationarity. The residuals of a regression of returns on, e.g., monthly changes of the independent variable should additionally be checked for autocorrelation by using the Durbin–Watson test. For more information regarding spurious regressions, see Granger and Newbold (1974).

[9] Asset pricing models for other (traditional) asset classes achieve clearly higher values. Fama and French (1993), for example, reported R^2 of about 90% on average. Carhart (1997) arrived at R^2 between 78% and 97%. Similar R^2 values are reported by Lustig et al. (2011) for FX markets.

9.3 Spectral Decomposition in More Detail

Due to the unbalanced nature of digital asset datasets, I applied spectral decomposition to circumvent this data issue, particularly for correlation and covariance matrices. Spectral decomposition, which is used several times in this chapter, is described in more detail in Rebonato and Jaeckel (2000). This method is applied frequently to correct a covariance or correlation matrix calculated from incomplete or bad data (similar to digital asset data) to make it positive-semidefinite, which is a requirement for the subsequent factor analysis (PCA) and describes an $N \times N$ matrix $\mathbf{A}$ with $\mathbf{x}^{\mathrm{T}}\mathbf{A}\mathbf{x} \geq 0$ for all non-zero vectors $\mathbf{x}$. Thus, all the eigenvalues need to be greater than or equal to 0. In particular for correlation matrices based on digital asset data with very heterogeneous starting and ending points, the initially computed matrices (i.e., pairwise-computed correlation or covariance matrices) are not positive-semidefinite. Therefore, Rebonato and Jaeckel (2000) proposed the following steps to correct such a matrix.[10]

Assume the right-hand eigensystem $\mathbf{E}$ of a real $N \times N$ matrix $\mathbf{M}$. The associated eigenvalues are denoted as λ_i:

$$\mathbf{M}\mathbf{E} = \mathbf{\Lambda}\mathbf{E}, \qquad \text{with } \mathbf{\Lambda} = \mathrm{diag}(\lambda_i). \tag{9.4}$$

The first step is to calculate all the eigenvalues and eigenvectors $\mathbf{e}_i$ of $\mathbf{M}$. Then, define all the non-zero elements of the diagonal matrix $\mathbf{\Lambda}'$ as

$$\mathbf{\Lambda}' : \lambda'_i = \begin{cases} \lambda_i & \lambda_i \geq 0, \\ 0 & \lambda_i < 0, \end{cases} \tag{9.5}$$

and set any negative λ_i to 0. Afterwards, multiply the eigenvectors $\mathbf{e}_i$ with their corrected λ'_i. To scale the diagonal entries of $\mathbf{M}$ back to unit length, all non-zero elements of the diagonal scaling matrix $\mathbf{T}$ are defined with respect to the eigensystem $\mathbf{E}$:

$$t_i = \left[\sum_{j=1}^{n} e_{i,j}^2 \lambda'_j \right]^{-1}, \tag{9.6}$$

where $e_{i,j}$ is the (i, j)th value of $\mathbf{E}$ and λ'_j the (j, j)th value of $\mathbf{\Lambda}'$. The scaling matrix $\mathbf{T}$ thus becomes a diagonal matrix with t_i on its diagonal:

$$\mathbf{T} = \mathbf{I}_n(\mathbf{1}_n^{\mathrm{T}} \otimes \mathbf{t}). \tag{9.7}$$

[10] Most of the equations in this section are reproduced in slightly adjusted form from Rebonato and Jaeckel (2000).

Arranging the results as $\mathbf{B}'$,

$$\mathbf{B}' := \mathbf{E}\sqrt{\mathbf{\Lambda}'}, \tag{9.8}$$

$\mathbf{B}$ can then be constructed by normalizing $\mathbf{B}'$ to unit length:

$$\mathbf{B} := \sqrt{\mathbf{T}}\mathbf{B}' = \sqrt{\mathbf{T}}\mathbf{E}\sqrt{\mathbf{\Lambda}'}. \tag{9.9}$$

The new (correlation) matrix $\hat{M}$, by definition, becomes positive-semidefinite with unit diagonal elements:[11]

$$\hat{\mathbf{M}} := \mathbf{B}\mathbf{B}^{\mathrm{T}}. \tag{9.10}$$

9.4 PCA and PLS in More Detail

With adjusted covariance matrices I was able to apply principal component analysis. PCA, or factor analysis, is a common approach described in the literature to extract and identify a possible number of risk factors—see, for instance, Fung and Hsieh (1997) and Lustig et al. (2011). The basic idea of PCA is to reduce the dimensionality of a dataset without losing too much variation within the data. Thus, it is helpful to look for factors (principal components) which are similar (i.e., high correlation) for a large number of assets. If there is a common variation in the data, PCA is able to extract the relevant principal components, which are also orthogonal to each other. These principal components, however, are only a statistical description rather than actual risk factors.

The principal components would need to be mapped to tradeable risk factors in a second step, if they are significant. It is also of great importance that the identified risk factors are tradeable assets or strategies. If that is not the case, the alphas from a time-series regression can no longer be interpreted easily. In that case, the risk factors would have to be replicated by some sort of strategy or tradeable assets. Following the approach of Fung and Hsieh (1997) and Lustig et al. (2011), though, comes with one serious problem. To run a proper PCA the number of assets of a dataset, N, should be less than the number of observations T—see Connor (1984) for an in-depth description of this issue. One solution at this point is to form portfolios to reduce N. That method emerged in early asset pricing studies which found beta estimation to be prone to high uncertainty.[12]

[11] Obtaining the covariance matrix would require the multiplication of $\hat{\mathbf{M}}$ by the respective standard errors.

[12] See Chap. 2 for more details.

Another approach, PLS regression, which is similar to PCA, not only uses the dependent variables but also considers external, independent variables. PLS is also known as a bivariate factor model, which tries to maximize the variance of the dependent variables by finding covariance commonalities in both datasets (independent and dependent data). It is also especially suitable to use if the (independent) data suffers from a $T < N$ issue.

9.4.1 PCA

The principal components (PCs) from such an analysis are simply linear combinations of the original data. I applied PCA to the return data of all digital assets in my dataset to identify risk factors "within" digital assets. The first PC usually explains most of the variation in the data. The second PC is orthogonal to the first, and explains the second highest variation. The same applies to any subsequent PC. The start of a PCA is mostly the calculation of a positive-semidefinite covariance matrix $\mathbf{C}$ (see Sect. 9.3) based on a de-meaned matrix of returns $\mathbf{Z}$. Thus, $\mathbf{C}$ can be calculated as[13]

$$\mathbf{C} = \mathbf{Z}^{\mathrm{T}}\mathbf{Z}. \tag{9.11}$$

Through eigendecomposition, $\mathbf{C}$ can be decomposed into eigenvalues and eigenvectors,

$$\mathbf{C} = \mathbf{V}\mathbf{D}\mathbf{V}^{\mathrm{T}}, \tag{9.12}$$

where $\mathbf{V}$ is an $N \times N$ matrix including the eigenvectors from $\mathbf{C}$ in each column, and $\mathbf{D}$ is an $N \times N$ diagonal matrix which includes the eigenvalues of $\mathbf{C}$ on its diagonal and 0 otherwise. If there are i eigenvalues which are significantly larger than the $N - i$ other eigenvalues, $\mathbf{C}$ can be approximated with only i principal components,

$$\mathbf{C} \approx \mathbf{V}_i\mathbf{D}_i\mathbf{V}_i^{\mathrm{T}}, \tag{9.13}$$

with $\mathbf{V}_i$ being an $N \times i$ matrix of eigenvectors and D_i an $i \times i$ diagonal matrix of the i largest eigenvalues. Subsequently, $\mathbf{V}_i^{\mathrm{T}}$ can be interpreted as a matrix of factor loadings similar to a simple OLS regression with i independent variables. The principal component or factor values $\mathbf{F}$ can then be calculated as

$$\mathbf{F} = \mathbf{Z}\mathbf{V} \tag{9.14}$$

[13] The steps described below are reproduced in a slightly adjusted form from Poddig et al. (2015).

or, with respect to Eq. 9.13 as[14]

$$\mathbf{F}_i = \mathbf{Z}\mathbf{V}_i. \tag{9.15}$$

All principal components are orthogonal to each other (and thus uncorrelated). The elements (eigenvalues) on the diagonal of $\mathbf{D}$ can be interpreted as variances of the principal components. Therefore, the PC with the highest eigenvalue contains the highest explanatory power. The factor values $\mathbf{F}$ can also be interpreted as risk factors (in the sense of an asset pricing (factor) model). All the $\mathbf{F}$ are uncorrelated with each other, and thus the variance of each $\mathbf{F}$ equals the eigenvalues of $\mathbf{D}$:

$$\mathbf{F}^{\mathrm{T}}\mathbf{F} = (\mathbf{Z}\mathbf{V})^{\mathrm{T}}\mathbf{Z}\mathbf{V} = \mathbf{V}^{\mathrm{T}}(\mathbf{Z}^{\mathrm{T}}\mathbf{Z})\mathbf{V} = \mathbf{V}^{\mathrm{T}}\mathbf{C}\mathbf{V} = \mathbf{V}^{\mathrm{T}}(\mathbf{V}\mathbf{D}\mathbf{V}^{\mathrm{T}})\mathbf{V} = \mathbf{I}\mathbf{D}\mathbf{I} = \mathbf{D}. \tag{9.16}$$

Since $\mathbf{C}$ is a symmetric matrix, the matrix of eigenvectors $\mathbf{V}$ is unitary. That is why the identity matrix $\mathbf{I}$ can be written as $\mathbf{I} = \mathbf{V}^{\mathrm{T}}\mathbf{V}$. With $\mathbf{D}$ being a diagonal matrix, all eigenvalues of $\mathbf{D}$ are pairwise covariances of the principal components, which all equal 0. The values on the diagonal of $\mathbf{D}$, however, can be interpreted as variances of the principal components. To calculate the relative explanatory power $\mathbf{p}$, simply take the fraction of each eigenvalue λ_i and the sum of all eigenvalues λ_k:

$$p_i = \frac{\lambda_i}{\sum_{k=1}^{N} \lambda_k}. \tag{9.17}$$

The initial matrix $\mathbf{Z}$ can then be replicated by

$$\mathbf{F}\mathbf{V}^{\mathrm{T}} = \mathbf{Z}\mathbf{V}\mathbf{V}^{\mathrm{T}} = \mathbf{Z}\mathbf{I} = \mathbf{Z} \quad \text{or} \quad \mathbf{Z} \approx \mathbf{F}_i\mathbf{V}_i^{\mathrm{T}}. \tag{9.18}$$

Additional information regarding PCA can be found in Jolliffe (2002) and Abdi and Williams (2010).

9.4.2 PLS

Partial least squares analysis is similar to PCA (Abdi 2010; Haenlein and Kaplan 2004). The advantage of PLS over PCA is that it directly estimates the principal components from a set of independent variables (like risk factors) which (might) explain the covariance structure of the dependent variables best. Thereby, the independent variables can suffer from a $T < N$ issue, since N will be reduced to only some principal components. This is why I used all the different external sources from Tables 9.2 and 9.3 as independent variables. Consequently, by running

[14] The calculation of $\mathbf{F}$ would not work if $\mathbf{Z}$ contains missing values. Therefore, I replaced all missing values of $\mathbf{Z}$ with 0, since $\mathbf{Z}$ was already de-meaned.

Table 9.2 Data composition, complete

#	Variable	Source	Start	Mean	*t*-stat	SD (%)	# obs.
Panel A: Digital assets							
	All	[1]	April 2013	−6.65	−1.49	35.06	63
	Mineable	[1]	April 2013	−6.39	−1.39	36.15	63
	Not mineable	[1]	April 2013	−5.23	−1.42	28.74	63
Panel B: Investment styles in digital assets (based on CoinMarketCap data)							
1	Value-weighted index	[1]	April 2013	3.53	0.98	28.23	63
2	Reversal	[2]	April 2014	1.10	0.34	23.18	51
3	Short-term reversal	[2]	April 2013	1.33	0.43	24.22	62
4	Amihud illiquidity	[2]	April 2013	−9.74	−2.95*	23.77	51
5	Volume	[2]	April 2013	11.14	3.80*	21.13	62
6	Low volatility	[2]	April 2014	0.66	0.16	29.13	51
7	Short-term low vol	[2]	April 2013	−1.15	−0.48	24.05	62
8	Low beta	[2]	April 2014	−9.94	−1.03	71.38	51
9	Short-term low beta	[2]	April 2013	−0.55	−0.15	28.48	62
10	Momentum	[2]	April 2014	−0.71	−0.23	21.44	51
11	Size	[2]	April 2013	6.81	1.34	40.06	63
12	BTC straddle (PMTS)	[3]	June 2013	11.45	2.00	44.40	60
13	BTC lookback straddle (PTFS)	[3]	June 2013	8.73	1.65	41.01	60
14	Mispricing factor	[4]	April 2014	−10.63	−4.58*	15.46	51
15	Idiosyncratic volatility	[5]	April 2014	2.77	1.32	14.70	51
16	Explained variation	[4]	April 2014	4.74	2.49*	13.19	51
17	Nominal price	[6]	April 2014	−17.27	−3.92*	34.61	63
Panel C: BTC fundamental data							
18	Average block size	[7]	April 2013	0.57	14.32*	22.97	63
19	Transactions per day	[7]	April 2013	164,749.30	13.71*	21.51	63
20	Mempool size	[7]	April 2016	24,280,399.51	4.13*	156.49	27
21	Bitcoins in circulation	[7]	April 2013	14,620,313.49	63.75*	0.31	63
22	Blockchain size	[7]	April 2013	66,782.71	10.25*	1.63	63
23	Transactions per block	[7]	April 2013	1062.69	13.66*	22.69	63
24	Orphaned blocks	[7]	April 2013	0.41	3.92*	—	63
25	Median confirmation time	[7]	April 2013	9.40	23.05*	29.85	63
26	Hash rate	[7]	April 2013	4,498,508.58	4.05*	24.16	63
27	Difficulty	[7]	April 2013	581,464,638,188.91	4.07*	21.82	63
28	Mining revenue	[7]	April 2013	4,510,743.45	5.24*	36.16	63

(continued)

Table 9.2 (continued)

#	Variable	Source	Start	Mean	*t*-stat	SD (%)	# obs.
29	Total transaction fees (BTC)	[7]	April 2013	84.23	5.42*	49.12	63
30	Cost % of transaction volume	[7]	April 2013	2	10.91*	36.35	63
31	Cost per transaction	[7]	April 2013	26.34	7.52*	30.52	63
32	Unique addresses	[7]	April 2013	323,414.14	13.4*	18.97	63
33	Transactions rate	[7]	April 2016	3.24	15.3*	37.67	27
34	Mempool transaction count	[7]	April 2016	19,488.93	3.96*	140.21	27
35	Mempool size growth	[7]	April 2016	1383.08	13.39*	45.86	27
36	Transaction volume ex popular addresses	[7]	April 2013	155,335.03	12.93*	21.43	63
37	Transaction volume ex long chains	[7]	April 2013	103,941.89	12.03*	18.65	63
38	Total output value	[7]	April 2013	1,646,596.89	9.05*	53.60	63
39	Estimated transaction value	[7]	April 2013	214,931.26	17.93*	34.28	63
40	Bitcoin wallet users	[7]	April 2013	7,996,149.52	8.38*	5.45	63
41	Number of unspent transactions output	[7]	April 2013	31,372,758.37	12.96*	5.36	63

This table describes the data used in the course of this chapter. I report the exact data source and the start of the time periods with available data as well as the mean, the *t*-statistics of the mean, the standard deviation (SD), and number of observations (months with available data). * denotes significance at the 5% level. For Panel A, all values are based on the returns of an equally weighted "market" index constructed over all available assets. For the investment styles and hedge fund data (Panels B and E), the values are also reported in terms of returns. All other statistics (Panels C and D) are based on the variable itself

Sources: [1] www.coinmarketcap.com; [2] Glas (2019); [3] Fung and Hsieh (2001); [4] Stambaugh and Yuan (2016); [5] Ang et al. (2006); [6] Blume and Husic (1973); [7] www.blockchain.info

PCA and PLS I was able to look for possible "internal" and "external" risk factors. PLS predicts dependent variables stored in an $N \times K$ matrix $\mathbf{Y}$ by the $N \times J$ matrix of independent variables $\mathbf{X}$.[15] Therefore, principal components (see above) from $\mathbf{X}$ are estimated with the goal of explaining as much of the covariance as possible between $\mathbf{X}$ and $\mathbf{Y}$. Following Eq. 9.18, $\mathbf{X}$ can also be written as

$$\mathbf{X} = \mathbf{F}\mathbf{V}^{\mathrm{T}}, \qquad \text{with } \mathbf{V}^{\mathrm{T}}\mathbf{V} = \mathbf{I}. \tag{9.19}$$

Additionally, two vectors $\mathbf{t}$ and $\mathbf{u}$ must be found which satisfy the criteria of $\mathbf{u}^{\mathrm{T}}\mathbf{u} = 1$, $\mathbf{t}^{\mathrm{T}}\mathbf{t} = 1$, and $\max \mathbf{b} = \mathbf{t}^{\mathrm{T}}\mathbf{u}$. Here, $\mathbf{t}$ and $\mathbf{u}$ can be defined as $\mathbf{t} = \mathbf{X}\mathbf{w}$ and $\mathbf{u} = \mathbf{F}\mathbf{g}$, where $\mathbf{w}$ and $\mathbf{g}$ are two vectors of weights, to create a linear combination of

[15] The equations related to PLS are reproduced in slightly adjusted form from Abdi (2010).

Table 9.3 Data composition, complete (2)

#	Variable	Source	Start	Mean	*t*-stat	SD (%)	# obs.
Panel D: ETH fundamental data							
42	Total ETH supply	[1]	July 2015	87,169,020.14	58.64*	0.24	36
43	Total number of transactions	[1]	July 2015	242,028.5	27.32*	—	36
44	Block gas usage	[1]	July 2015	2,260,259.28	4.82*	39.54	36
45	Total gas usage	[1]	July 2015	12,194,590,624.36	4.66*	40.09	36
46	Average block usage	[1]	July 2015	32.11	5.40*	49.74	36
47	Hashrate	[1]	July 2015	30,942.89	1.19	224.72	36
48	Mining reward	[1]	July 2015	26,781.69	30.98*	13.28	36
49	Block mining reward	[1]	July 2015	24,755.42	26.10*	12.72	36
50	Uncle mining reward	[1]	July 2015	1737.33	15.72*	33.44	36
51	Fee mining reward	[1]	July 2015	288.97	4.02*	61.19	36
52	Distinct miners	[1]	July 2015	77.25	6.65*	20.45	36
53	Mining revenue	[1]	July 2015	77.92	10.47*	39.35	36
54	Blocks per day	[1]	July 2015	5523.67	40.55*	18.06	36
55	Uncles per day	[1]	July 2015	590.83	9.81*	37.93	36
56	Block time	[1]	July 2015	15.64	27.96*	18.61	36
57	Difficulty	[1]	July 2015	919,689,797,743,802.9	4.48*	45.53	36
58	Block size	[1]	July 2015	8215.22	5.21*	26.05	36
59	Block gas limit	[1]	July 2015	4,948,129.36	13.94*	111.18	36
60	New accounts	[1]	July 2015	38,670.06	4.24*	116.25	36
61	Total accounts	[1]	July 2015	7,772,960.86	3.73*	52.03	36
Panel E: Hedge fund factors							
62	Bond lookback straddle	[2]	April 2013	−3.06	−1.70	14.14	63
63	Currency lookback straddle	[2]	April 2013	−1.53	−0.61	19.69	63
64	Commodity lookback straddle	[2]	April 2013	−0.74	−0.39	14.85	63
65	Short-term interest rate lookback straddle	[2]	April 2013	−12.00	−6.25*	15.11	63
66	Stock index lookback straddle	[2]	April 2013	−4.74	−2.52*	14.81	63
67	Equity market factor	[3]	April 2013	1.05	2.51*	3.29	63

(continued)

Table 9.3 (continued)

#	Variable	Source	Start	Mean	*t*-stat	SD (%)	# obs.
68	Size spread factor	[3]	April 2013	−0.02	−0.07	2.45	63
69	Bond market factor	[3]	April 2013	0.82	0.72	9.04	63
70	Credit spread factor	[3]	April 2013	0.11	0.23	3.59	63
71	Emerging market factor	[4]	April 2013	0.27	0.42	5.04	63

This table describes the data used in the course of this chapter. I report the exact data source and the start of the time periods with available data, as well as the mean, the *t*-statistics of the mean, the standard deviation (SD) and number of observations (months with available data). * denotes significance at the 5% level. For Panel A, all values are based on the returns of an equally weighted "market" index constructed over all available assets. For the investment styles and hedge fund data (Panels B and E), the values are also reported in terms of returns. All other statistics (Panels C and D) are based on the variable itself
Sources: [1] www.etherchain.org; [2] Fung and Hsieh (2001); [3] Fung and Hsieh (2004); [4] Fung and Hsieh (2006)

X and **F** which maximizes the covariance between both. After the first latent vector is determined, it is subtracted from **X** and **F** and reiterated until **X** becomes a null matrix (see Step 5 of the algorithm below).[16] When the final optimal weights are found, **Y** (or more specifically $\hat{\mathbf{Y}}$) can simply be replicated by

$$\hat{\mathbf{Y}} = \mathbf{FBG}^{\mathrm{T}}, \tag{9.20}$$

where **B** is a diagonal matrix containing **b** on the diagonal, and **G** is a matrix of all weight vectors **g**. The iterative PLS procedure can be summarized in several steps. The iteration stops once **X** becomes a null matrix.

1. Obtain **F** from a PCA on **X**.
2. Estimate **w**, calculate **t** and estimate **g**, and calculate **u**:
   a. $\mathbf{w} \propto \mathbf{X}^{\mathrm{T}}\mathbf{u}$ (i.e., estimation of **X** weights).
   b. $\mathbf{t} \propto \mathbf{Xw}$ (i.e., estimation of **X** scores).
   c. $\mathbf{g} \propto \mathbf{F}^{\mathrm{T}}\mathbf{t}$ (i.e., estimation of **Y** weights).
   d. $\mathbf{u} = \mathbf{Fg}$ (i.e., estimation of **Y** scores).
3. If **t** has converged, get **b** as $\mathbf{b} = \mathbf{t}^{\mathrm{T}}\mathbf{u}$. If **t** has not converged, run steps 2a–d again.
4. The factor loadings for **X** are then given by $\mathbf{p} = \mathbf{X}^{\mathrm{T}}\mathbf{t}$.
5. Deflate the matrices **X** and **F** by $\mathbf{X} = \mathbf{X} - \mathbf{tp}^{\mathrm{T}}$ and $\mathbf{F} = \mathbf{F} - \mathbf{btg}^{\mathrm{T}}$.
6. Store **b** in the diagonal matrix **B**.
7. Store **g** in the weight matrix **G**.
8. Estimate $\hat{\mathbf{Y}}$ by $\mathbf{FBG}^{\mathrm{T}}$.

[16] For more details please refer to the algorithm section in Abdi (2010), or see the description below.

In this way, in line with PCA, **F** can be interpreted as a score matrix which is weighted by **B**, and **G** can be seen as the matrix of factor loadings.

9.4.3 Update: Fundamental Data

To identify these different (risk) factors, I extended my dataset in several directions. I not only used the already described digital asset data, but I additionally added fundamental and other data. Thus, next to the data shown in Sect. 5.4, I added straddles and lookback straddles on BTC, which are inspired by the optionality as described in the hedge fund literature, and is due to the highly asymmetric return profile of digital assets. Thus, I aimed to capture this return behavior via option strategies.[17] Since I am interested in digital asset specific risk factors only, the prices of all digital assets were converted to BTC prices to limit the impact of the BTC/USD exchange rate fluctuation (see Sect. 5.3.2). This is necessary since most digital assets are quoted against BTC, but the prices on www.coinmarketcap.com are converted to USD.[18] This conversion thus includes two highly volatile digital coins which I aimed to avoid by converting the prices back to BTC prices. By doing so, I reduced the dependence of an equally weighted market index on the BTC/USD exchange rate fluctuation. BTC is the only digital coin which remains quoted against the USD; BTC prices would otherwise be equal to a vector of 1s only.[19] I additionally excluded every asset with fewer than six monthly observations to make the covariance, and thus the PCA estimation, less volatile (this reduced the number of digital assets to $N = 1263$). As a robustness check, I report all results based on the original USD data in Appendix H. In short, the results are robust for BTC as well as USD data.

The calculation of investment styles in Panel B, which could be used as factors in an asset pricing model, is necessary for the step after the calculation of the principal components.[20] Since the first step of the analysis yields no clearly priced (assumed) risk factors (in terms of principal components), I extended the scope by investigating

[17] For more information, see Fung and Hsieh (2001).

[18] See "Methodology – (1) Price (Market Pair)," https://coinmarketcap.com/methodology/, accessed on November 15, 2018.

[19] That approach does not change the results at all, since the analyses in this chapter are based on the returns of digital assets and not their prices. I also ran all analyses with USD-denominated prices and the results did not change materially (see Appendix H).

[20] See Chap. 8 for more details. Due to the slightly different dataset in this chapter, the values in Table 9.2 diverge from Table 8.2. Although this chapter utilizes BTC-denominated prices, the differences between Chaps. 8 and this chapter are relatively small. This is due to the formation of hedge portfolios. In particular, in Chap. 8 the long part of the hedge portfolio is intrinsically long the BTC/USD exchange rate. The short portfolio, on the other hand, would be short BTC/USD and thus cancel each other out. Therefore, the results of both chapters are an indication of comparably robust results, since the dataset used in this chapter is somewhat smaller in size (i.e., the number of assets under consideration).

the covariances and characteristics of other possible risk factors (Panels B to E of Tables 9.2 and 9.3).[21] Therefore, I included several types and sources of data. As already mentioned, I included different (possible) risk factors based on digital asset characteristics (Panel B of Table 9.2).

Since traditional asset classes seem to have no impact on digital assets (Glas 2019), I added the seven factors described by Fung and Hsieh (2001, 2004), plus two additional trend-following factors (short-term interest rate and equity) and an additional emerging markets factor in line with Fung and Hsieh (2006), to the "search" variables (Panel E). Trend-following risk factors were obtained from David Hsieh's website. The data to construct the non-trend-following hedge fund risk factors came from Thomson Reuters Datastream. For the construction, I used the same Datastream code as described on David Hsieh's website.[22] I also downloaded the bond risk factors from the St. Louis FED website.[23]

9.4.4 Extracting Risk Factors

For digital coins with a relatively short (data) history, the number of assets is larger than the number of monthly return observations. That implies issues for some statistical methods. A possible solution for this problem is to run PCA, PLS, or to construct random portfolios to reduce the number of assets and thus the idiosyncratic variation (Connor and Korajczyk 1986).

That is why I conducted two similar PCA analyses. One, where I ran a Monte Carlo simulation to construct 30 portfolios containing 50 randomly picked digital assets each[24] 10,000 times and used these portfolios for the PCA.[25] Based on that PCA, I calculated the betas with regard to the first three PCs (see Table 9.4). Additionally, I ran rolling FM regressions and a Gibbons–Ross–Shanken (GRS) test (Gibbons et al. 1989) for each of the 10,000 simulations. The GRS test investigates whether the alphas ($\boldsymbol{\alpha}$) of a time-series regression are jointly distributed close to zero (i.e., $H_0 \colon \alpha_{ip} = 0$). If that were the case, the asset pricing model under consideration has done a good job in explaining the dependent asset returns. Should the alphas not

[21] Panels C and D mostly feature non-stationary data.

[22] See "Hedge Fund Data Library," http://faculty.fuqua.duke.edu/~dah7/DataLibrary/TF-FAC.xls, accessed on June 18, 2018.

[23] See "FRED Economic Data," https://fred.stlouisfed.org/series/DGS10 and https://fred.stlouisfed.org/series/DBAA, accessed on July 2, 2018.

[24] These numbers were chosen intentionally. The number of portfolios (30) times the number of assets within each portfolio (50) is fairly close to the overall number of assets in the dataset used. Thus, each digital asset should be included in a randomly picked portfolio at least once per simulation. The assets within the portfolios are equally weighted. For more information regarding Monte Carlo simulation see Cochrane (2001).

[25] For more information regarding principal component analysis see Jolliffe (2002) and Abdi and Williams (2010), or refer to the previous sections.

Table 9.4 Explained variance by principal component

	All			Mineable			Non-mineable		
PC	PLS	Random	Missing	PLS	Random	Missing	PLS	Random	Missing
PC 1	23.73	62.14	20.91	29.07	56.18	61.40	24.19	38.24	46.30
PC 2	22.07	7.65	7.88	18.99	8.30	5.28	22.01	11.20	8.44
PC 3	9.61	5.21	6.98	6.24	5.85	2.57	7.68	8.18	5.42
PC 4	6.61	3.99	5.69	6.19	4.57	2.27	6.79	6.60	5.15
PC 5	6.47	3.22	4.56	5.15	3.75	1.81	4.68	5.50	4.22
PC 6	5.82	2.68	4.06	5.11	3.15	1.73	4.41	4.66	3.20
PC 7	3.36	2.26	3.51	4.93	2.68	1.70	3.75	3.98	2.94
PC 8	2.89	1.94	3.02	2.79	2.31	1.39	3.53	3.41	2.45
PC 9	2.80	1.66	2.77	2.52	2.00	1.24	3.51	2.93	2.10
PC 10	1.83	1.44	2.63	2.36	1.73	1.21	2.44	2.53	1.77
Sum	85.20	92.20	62.01	83.35	90.51	80.61	82.97	87.22	82.00

All values are percentage values. PLS denotes the principal components based on a PLS regression, whereas the random and missing PCs both stem from a PCA

be distributed close to zero, the asset pricing model under consideration might not be a suitable model for the respective dependent variables. The GRS test statistic is given by (Cochrane 2001; Gibbons et al. 1989)

$$\frac{T-N-K}{N}\left(1+\mathbf{E}_T(\mathbf{f})'\hat{\boldsymbol{\Omega}}^{-1}\mathbf{E}_T(\mathbf{f})\right)^{-1}\hat{\boldsymbol{\alpha}}'\hat{\boldsymbol{\Sigma}}_{\epsilon}^{-1}\hat{\boldsymbol{\alpha}} \sim F_{N,T-N-K}, \tag{9.21}$$

where N is the number of assets, K represents the number of factors, T is the sample length, $\hat{\boldsymbol{\Sigma}}_{\epsilon}$ denotes the covariance matrix of the residuals of a time-series regression, and $\hat{\boldsymbol{\Omega}} = \frac{1}{T}\sum_{t=1}^{T}[\mathbf{f}_t - \mathbf{E}_T(\mathbf{f})][\mathbf{f}_t - \mathbf{E}_T(\mathbf{f})]'$. Thereby, $\mathbf{E}_T(\mathbf{f})$ is the sample mean of the returns of the (independent) factors. $F_{N,T-N-K}$ represents the F-distribution with N and $T-N-K$ degrees of freedom. Afterwards, I averaged the results of all 10,000 simulations to arrive at one final result.

The second approach ignores the $T < N$ issue and runs a PCA on the available data. This approach, however, suffers from a high number of missing values. Therefore, I avoided missing values by calculating the covariance matrix pairwise. Since this covariance matrix is no longer positive semi-definite, I needed to apply spectral decomposition to circumvent this issue.[26] Finally, I also applied PLS to extract principal components. The PLS takes into account all digital assets as well as all the variables stated in Tables 9.2 and 9.3 (Panels B to E). This enabled me to make two steps at once. First, I extracted possible (risk) factors by applying a PCA on (assumed) important drivers of digital assets, and second, I explained the digital asset returns by these principal components. The first step is not possible for a simple PCA, since it does not take into account external or independent variables.

[26] See Sect. 9.3 for further details.

However, one major issue remains. A PLS depends on good data quality (i.e., at best no missing values). In my case, both sides (dependent and independent variables) suffered from a high number of missing values. Therefore, to run a proper PLS the missing values needed to be taken care of. To do this, I applied the same methods as described above, namely spectral analysis and PCA. PCA can be used not only to identify risk factors but also to allow replication of data based on the different principal components. Based on a spectral analyzed covariance matrix I ran a normal PCA to identify all possible principal components. From the PCA, missing values can be replicated by multiplying the factor (principal component) values by the factor loadings and then adding the mean return (or in this case, also the average monthly change) of each single asset, variable, or characteristic. Here, only the missing values are "recreated" and added to the dataset. All the other already known values remain unaffected. This approach was conducted once and independently for both the dependent (digital asset returns) and independent (all variables from Tables 9.2 and 9.3) data matrices. Although that procedure creates a small look-ahead bias (since future values are not known in the past), it is a necessary step to conduct the PLS; otherwise, it could not be run with missing values. That bias, on the other hand, should not be too great, since this is a static analysis and aims to identify any possible risk factor in the data.

Thus, the results of the random approach (i.e., the Monte Carlo simulation), in particular, indicate a good reduction of the idiosyncratic volatility, with relatively high explanatory power for PC1 (see Table 9.4). PLS and "missing" (which is the PCA) yield similar results, though with a higher explanatory power for the PLS.

Table 9.4 lists the explained variances for the first 10 principal components (PCs) in all three datasets. As can be seen from the PLS and both PCA approaches, it appears that the universe of digital assets is being shaped by two to three risk factors (PC 1 to PC 3). This analysis, however, tells us no more than that there are about three risk factors of importance. It does not tell us anything about their type or nature (except for the PLS, which yields no clear indication of more or less important exogenous variables in this specific case).

9.5 Are There Priced Risk Factors in the Returns of Digital Assets?

To be considered a true risk factor, the risk factor or the betas with regard to a risk factor need to be priced sufficiently in the underlying data. The asset pricing literature proposes different tests at this point. The first and most intuitive might be to run a simple time-series regression and to look at R^2. This test can also be seen as some form of APT or multi-factor asset pricing model. That is why I used the following time-series regression based on the first three PCs as described in Table 9.4:

$$r_i = \alpha + \beta_1 \text{PC1} + \beta_2 \text{PC2} + \beta_3 \text{PC3} + \epsilon, \tag{9.22}$$

with r_i being the returns of digital asset i, and PC1 to PC3 the different principal components from the analysis in Sect. 9.4.4. By doing so, I assumed that the PCs are already tradeable assets (even though that is not possible for the PLS principal components). Further, it is important that all the alphas of the test assets are jointly indistinguishable from zero. This is commonly investigated by using the GRS test. This test, however, can only be conducted when the condition $T > N$ holds. In this case the test assets N are clearly larger than T, which is why I needed to follow another approach. Thus, I ran the same analysis as described in Sect. 9.2, where I simply counted the number of significant alphas. This means that, when assuming a confidence level of 90%, one will find 10% of the alphas of all digital assets to be significant under the null hypothesis. The same also applies to a randomly generated return series, for instance. Therefore, to conclude that the alphas are jointly distributed close to zero (in the sense of the GRS test), the number of digital assets with a significant alpha from a time-series regression should be close to 10%. Otherwise, the hypothesis of jointly zero alphas can be rejected. Another test is to run cross-sectional regressions to see if the betas with regard to a risk factor exhibit a statistically significant coefficient. This test is also known as an FM test (Fama and MacBeth 1973). Consequently, I ran the following regression on the 12-month rolling betas[27] of every digital asset with regard to the possible risk factors as measured by PCs 1 to 3:

$$r_{it+1} = \lambda_{i0} + \lambda_1 d_{1it} + \cdots + \lambda_j d_{jit} + \epsilon_{it}, \tag{9.23}$$

with r_{it+1} the return of digital asset i at time $t+1$, d_{jit} the rolling 12-month beta with regard to a principal component from an in-sample analysis,[28] and ϵ the error term. The t-value t_{λ_j} of the lambdas is calculated as

$$t_{\lambda_j} = \sqrt{N}\frac{\bar{\lambda_j}}{\sigma_{\lambda_j}}, \tag{9.24}$$

with $\bar{\lambda}$ the average lambda, and $\sigma_{\lambda_j}/\sqrt{N}$ the standard error of the average of the lambdas. I estimated the rolling beta (β) of digital asset i with regard to a principal component (PC) as

$$\beta_{it} = \frac{1}{12}\sum_{k=0}^{11}\frac{COV(r_{it-k}, r_{\mathrm{PC}t-k})}{VAR(r_{\mathrm{PC}t-k})}, \tag{9.25}$$

[27] The asset pricing literature usually considers a timeframe of 36–60 months. Due to the short digital asset data I needed to deviate from this number to be able to conduct a reasonable analysis.

[28] Of course, an out-of-sample analysis would be more appropriate at this point. The in-sample analysis, however, already includes all available data and should yield the "correct" principal components.

with r_{it} the return of digital asset i at time t, and r_{PCt} the value (i.e., return)[29] of the principal component in t. Due to gaps in the data and very volatile betas, I averaged the betas over the same 12 months to arrive at more reliable results.

As shown in Table 9.5, the possible risk factors in the form of various principal components are not sufficiently priced (see the FM section). R^2 ranges from about 25% to 73%. This is clearly lower than the comparable R^2 values reported by Fama and French (1993), Fung and Hsieh (1997), Carhart (1997), or Lustig et al. (2011) for other asset classes. The betas, on the other hand, are comparable (in absolute magnitude) for all three approaches.

For the missing approach, the PCs are obtained from an unbalanced dataset and are regressed against all available digital assets. Since many of them have a relatively short data history, the betas become relatively small. Thus, the averaged betas in Table 9.5 reflect this issue. A similar outcome applies to the random approach. On the other hand, the significant alphas are fairly close to 10% for PCA and PLS, which could be an indication that those models would be a good fit for digital assets. The random approach, though, yields clearly higher to almost no significant alphas (for all and non-mineable assets, respectively). Also, the betas of the second and third PCs are very small. Only the first PC seems to have explanatory power (which can also be observed in Table 9.4). The GRS test values of the Monte Carlo PCA also indicate a bad fit of that model, which is in line with the regression results. When looking at the rolling FM regressions, I also find no evidence for sufficiently priced betas in either univariate or multivariate settings. This result leads to the conclusion that there might exist two to three risk factors; the covariances of the assumed risk factors from the PLS and PCA analyses, however, are not priced.[30] Thus, I need to focus on other possible sources of risk factors.

9.6 Covariances or Characteristics

The discussion of covariances versus characteristics is almost as old as the CAPM. Researchers found that the market beta of individual assets was not priced significantly. The relationship between betas and returns also appeared to be too flat, which led to confusion. As such, two opinions emerged, one supported by empiricists hinting at methodological weaknesses in testing the beta (the true market beta cannot be observed, and as such the beta estimation is prone to many influences, also known as the "errors in variables" problem) and the other proposed by behaviorists. The behaviorists mainly explained factors like HML (high minus low) by investor over- and underreaction. They also stated that the beta with regard to some characteristic should not be priced, but the characteristic itself (Daniel and Titman 1997). A recent

[29] Only the PLS factor values cannot be interpreted as returns.

[30] Which might also be due to the relatively short dataset.

Table 9.5 Regression results and Fama–MacBeth coefficients

	All			Mineable			Non-mineable		
PC	PLS	Random	Missing	PLS	Random	Missing	PLS	Random	Missing
Regression results									
alpha	−6.42	−6.67	−16.15	−9.88	−6.47	−10.44	−9.45	−5.03	−13.22
SA	8.23	29.79	11.16	8.78	6.54	7.78	7.39	0.60	13.27
$\bar{\beta}_1$	0.2141	0.1873	−0.0216	−0.0231	0.1876	0.3500	−0.5152	0.1801	−0.1531
$\bar{\beta}_2$	−1.2935	0.0002	0.0167	−0.8234	−0.0004	0.0456	0.4493	0.0001	−0.0445
$\bar{\beta}_3$	−1.9633	−0.0001	−0.0078	−0.0326	0.0001	0.0181	−1.1387	0.0000	−0.0129
R^2	47.50	72.26	47.75	24.59	67.34	38.58	48.46	53.40	59.72
Fama–MacBeth									
Univariate									
PC 1	−0.0041	0.2179	−0.4184	−0.0085	0.2215	0.1276	0.0065	0.1119	−0.0965
	(−0.25)	(0.85)	(−0.92)	(−0.47)	(0.86)	(1.11)	(0.38)	(0.59)	(−0.58)
PC 2	−0.0010	0.0003	−0.0350	−0.0021	−0.0016	−0.3240	0.0125	0.0008	0.0138
	(−0.06)	(0.01)	(−0.29)	(−0.12)	(−0.02)	(−1.00)	(0.68)	(0.01)	(0.07)
PC 3	−0.0072	−0.0011	0.1019	0.0080	0.0003	−0.0485	−0.0065	−0.0018	−0.0269
	(−0.44)	(−0.02)	(0.40)	(0.26)	(0.00)	(−0.36)	(−0.31)	(−0.02)	(−0.23)
Multivariate									
PC 1	0.0096	0.2049	−0.3964	−0.0083	0.2209	0.2672	−0.0002	0.1549	−0.6404
	(0.16)	(0.37)	(−0.59)	(−0.29)	(0.48)	(0.60)	(−0.01)	(0.65)	(−1.00)
PC 2	0.0058	0.0021	−0.0654	−0.0064	0.0010	0.5360	−0.0262	0.0001	−0.9235
	(0.19)	(0.02)	(−0.29)	(−0.24)	(0.01)	(0.43)	(−0.81)	(0.00)	(−0.85)
PC 3	−0.0109	−0.0015	0.1394	0.0533	0.0003	0.1020	0.0039	−0.0010	−0.3411
	(−0.13)	(−0.02)	(0.33)	(0.94)	(0.00)	(0.40)	(0.14)	(−0.01)	(−0.93)
GRS									
F-stat		650.14			434.85			75.02	
p-Value		(0.00)			(0.00)			(0.00)	

This table shows the regression results of the identified principal components. I ran these regressions against each available digital asset in the dataset. Thus, the values in the tables are averaged over all assets. Alpha, significant alphas (SA), and R^2 are percentage values. The SA value is the number of all alphas significant at the 10% level divided by the number of all digital assets. β_1 to β_3 are the averaged regression coefficients of all digital assets. Further, I show the Fama–MacBeth regression coefficients for the first three PCs as described in Table 9.4. Below the coefficients I report the respective t-statistics in parentheses. Moreover, I report the GRS test statistics for the Monte Carlo simulation approach: F-stat denotes the averaged GRS test statistics, and their corresponding p-values are given in parentheses below. Data is given in BTC terms, since that analysis focuses on an "internal" perspective. Nonetheless, I report the outcomes of the same analysis based on USD prices in Appendix H

work which also summarizes and contributes to this discussion is Fieberg et al. (2016b).

9.6.1 Testing the Covariance and Characteristics Model

Inspired by that controversy, I ran an additional analysis based on all variables in Panels B–E of Tables 9.2 and 9.3. Fama and MacBeth (1973) proposed a two-step procedure. First, the (rolling) betas with regard to an assumed risk factor need to be estimated. This beta comes from a simple univariate regression model. Then,

Fama and MacBeth (1973) ran a cross-sectional regression in every month t. In that case, asset returns are the dependent variable and the previously estimated betas the independent variable. The results were then averaged over all months to arrive at one final result. By following this method, it can be determined whether the betas connected to some risk factor are priced significantly and, as such, represent a priced risk premium.

I also calculated the rolling 12-month univariate betas of every digital asset with respect to the 71 variables described in Tables 9.2 and 9.3. Equipped with 71 matrices of rolling betas, I applied two approaches: a pooled OLS panel regression, and uni- and multivariate rolling FM regressions. For both approaches I ran an analysis based on the betas with regard to the 71 variables, to check if the covariance theory holds in digital assets. An additional analysis aims to investigate the characteristics theory. Therefore, I did not use the betas as the independent variable, but the characteristics themselves. However, I could not obtain enough fundamental data (in the sense of Panels C and D) for every digital asset in the dataset. Thus, I was forced to limit the characteristics model investigation to the characteristics from Panel B. These were variables 2 to 17 (see Table 9.2) and a dummy variable for digital assets, equal to 1 when connected to its own blockchain and 0 otherwise. Variables 12 and 13 were no longer based on BTC only, but calculated for every single digital asset in the dataset.

To identify possible risk drivers in digital asset returns, I applied a full enumeration of the betas or the characteristics. This means I tested every possible combination of variables, starting from just one, over combinations of two to six[31] variables. For every combination, I then ran a multivariate (or univariate in the case of only one variable) panel regression and a rolling FM regression. This "brute force" algorithm tested any of the possible combinations and ran a panel and FM regression for each enumeration. The panel regression model is specified as follows:

$$r_{it+1} = \alpha + \mathbf{Variables}'_{it}\mathbf{v} + Dummy_{it}d + \epsilon_{it+1}, \tag{9.26}$$

with r_{it+1} the return of digital asset i at time $t+1$, a variables vector $\mathbf{Variables}_{it}$ from Panels B–E (see Tables 9.2 and 9.3) containing different betas with respect to a possible risk factor (for the covariance model) or the characteristics themselves (for the characteristics model). $Dummy_{it}$ is the dummy variable for mineable ($Dummy = 1$) or non-mineable ($Dummy = 0$) digital assets, and the error term is given by ϵ_{it+1}. I calculated the standard errors (SE) as cluster-robust standard errors clustered along the time dimension, which is the most similar approach to FM regressions.[32] For the

[31] Choosing a higher number at this point would not necessarily yield better results, but would dramatically increase the computation time.

[32] Using two-way cluster-robust standard errors at this point could lead to less reliable results. Two-way standard errors could lead to non-positive-semidefinite covariance matrices which would need to be corrected by using spectral decomposition. Doing so very often might distort the results and make them less reliable. That would also significantly increase the computation time.

FM regressions I used the same model as described in Eq. 9.23. The detailed results of the panel regressions can be found in Appendix I.

9.6.2 Empirical Results

Both approaches (pooled OLS and rolling FM regressions) yielded only slightly different results. For the analysis focused on covariances there were only one or two significant "fundamental" variables, which can be deemed not reliable since both exhibit tremendous data issues (for example, orphaned blocks).[33] When focusing on characteristics I arrived at 14 (reversal, short-term reversal, Amihud's illiquidity, volume, low volatility, short-term low volatility, low beta, short-term low beta, momentum, size, straddle, lookback straddle, nominal price, and blockchain dummy) different significant variables in total that exhibit a significantly priced lambda at least once.[34] The full enumeration of both panel regression and rolling FM regression could not come up with more than five significant variables at once in a multivariate framework (see Appendix I). Since I do not know which of the models in Appendix I is the "right" one, I follow the approach of Green et al. (2017) to further reduce the number of preselected variables. As such, I needed to control for each of the characteristics in the subsequent tests to determine the characteristics carrying independent information.

In general, a number of 14 out of 17 variables (see Panel B of Table 9.2) already points at evidence for priced characteristics. In contrast, the same analysis yielded literally no significantly priced covariance (beta), even though I considered 54 more variables. From the 14 characteristics, I aimed to select the variables with independent information by following the approach of Lewellen (2015) and Green et al. (2017).

Green et al. (2017) proposed several steps in order to identify independent risk factors.

First, they calculated variance inflation factors (VIFs) and excluded variables with a VIF > 7.[35]

In step 2, Green et al. (2017) added each single characteristic to a standard asset pricing model and compared the results from a univariate analysis (characteristic under consideration) and a multivariate analysis (factors or characteristics from a well-known asset pricing model like the Fama–French three-factor-model + the characteristic under consideration). Only characteristics that remain significant when added to the standard asset pricing model were considered robust.

In the third step, all characteristics were included as independent variables in a large multivariate FM regression analysis.

[33] Due to basically no significant outcomes covariance results are not included in Appendix I.

[34] In different multivariate models. Results are reported in Appendix I.

[35] I calculated the VIFs as the diagonal of the inverse of the correlation matrix.

The significant variables from steps 2 and 3 were then taken into consideration for step 4, which was inspired by Lewellen (2015). Here, Green et al. (2017) aimed to investigate the economic importance of the characteristics. This is achieved by forming hedge portfolios based on predicted returns, as shown by Lewellen (2015). Therefore, past FM regression coefficients based on the identified characteristics were used to predict the returns of each individual asset in $t + 1$. Thus, the predicted returns of a (multivariate) FM regression were used as a sorting criterion from which decile (in this case quintile) portfolios were formed. In the case of relatively similar or even higher hedge portfolio returns of the filtered characteristics (when compared to all available characteristics), the filtered characteristics can be deemed economically significant. The interpretation of the last step's results, however, has many degrees of freedom since no clear "rules" exist here.

In the fifth and final step, Green et al. (2017) split their data in 2003 to test for robustness before and after different regulatory as well as market infrastructural changes. This, however, is not possible for my dataset since it starts after 2003. I also did not run the second step because no commonly accepted asset pricing model has yet been found for digital assets. Table 9.6 depicts the results of the first (VIF column) and third steps (all other columns).

Based on VIFs larger than 7, I excluded reversal (2) and momentum (10) from further analysis. Also, I did not run the rolling multivariate FM regression for the non-mineable dataset, since it is mainly comprised of ERC-20 tokens, which mostly emerged in 2016 and 2017. Thus, the history of that dataset is too short for a meaningful cross-sectional analysis. Further data issues became apparent for variable 12 (straddles), which leads to unrealistically high FM coefficients in both datasets comprised of tiny digital assets only. Only three characteristics remain significant (in bold) when controlling for all other characteristics in a multivariate FM regression. This, however, applies only to the dataset containing all available digital assets. (Low) volume (5) is priced significantly and with the correct sign (i.e., in line with the "anomaly" documented in equity markets); it also produces significant hedge portfolio returns (see Tables 8.2 and 9.2), which is a requirement in the asset pricing literature to be considered some kind of effect or anomaly. Short-term low beta (9), on the other hand, exhibits a positive sign, indicating a higher compensation for high short-term beta assets. Tables 8.2 and 9.2 are sorted for low betas, which results in negative, but not significant, hedge portfolio returns. Thus, this finding might lead to the conception of a classic CAPM-like relationship between risk and beta in digital assets. The same applies to the longer-term beta variant, however in a weaker form. These findings are also in line with Glas (2019), who found negative (but not significant) hedge portfolio returns for the low-beta characteristic in digital assets. The prices of digital assets themselves (17) are also priced significantly. The negative sign of the FM coefficient indicates a low (nominal) price effect in digital assets. When sorting for nominal prices, I also document significant hedge portfolio returns (see Tables 8.2 and 9.2). Other characteristics, namely low volatility (6), low beta (8), and size (11), exhibit significant FM coefficients in other datasets, too. Low volatility and size additionally exhibit the correct expected sign (as known from the investment styles literature).

Table 9.6 VIFs and multivariate FM regression results

		All			Mineable		
Var	VIF	All	All but tiny	Tiny	All	All but tiny	Tiny
2	8.57	—	—	—	—	—	—
		—	—	—	—	—	—
3	2.71	−0.0139	−0.0206	−0.1487	−0.0069	−0.0150	−0.1694
		(−0.93)	(−1.31)	(−1.51)	(−0.30)	(−0.52)	(−1.53)
4	1.47	−0.0099	−0.0056	−0.0048	−0.0073	−0.0135	−0.0109
		(−0.69)	(−0.31)	(−0.23)	(−0.32)	(−0.47)	(−0.80)
5	1.62	**−0.0105**	−0.0056	**−0.0547**	−0.0011	−0.0012	−0.0656
		(−2.64)	(−1.32)	**(−5.33)**	(−0.19)	(−0.17)	(−1.31)
6	2.64	−0.0022	−0.0099	**−0.4479**	0.0957	−0.0406	−0.2161
		(−0.04)	(−0.27)	**(−3.04)**	(1.09)	(−0.72)	(−1.04)
7	2.43	0.0388	0.1651	−0.2628	0.3316	0.1857	−0.2618
		(0.27)	(1.17)	(−0.76)	(1.84)	(0.96)	(−0.33)
8	1.96	0.0690	0.0718	0.0101	0.0316	0.0786	**0.2246**
		(1.44)	(1.55)	(0.11)	(0.65)	(1.42)	**(2.08)**
9	1.40	**0.0242**	0.0146	0.0449	0.0107	0.0083	0.0065
		(2.69)	(1.62)	(1.90)	(0.70)	(0.51)	(0.23)
10	8.69	—	—	—	—	—	—
		—	—	—	—	—	—
11	5.41	0.0118	0.0033	**−0.0872**	−0.0019	−0.0074	**−0.1093**
		(1.75)	(0.49)	**(−2.22)**	(−0.22)	(−0.85)	**(−2.31)**
12	3.19	0.8429	0.9275	−141,697.0501	0.9641	−0.3727	794,968.0340
		(0.68)	(0.76)	(−1.08)	(1.01)	(−0.48)	(1.64)
13	3.72	0.0137	0.0079	0.2113	0.0119	0.0243	0.0823
		(0.43)	(0.25)	(1.01)	(0.16)	(0.29)	(0.20)
17	3.36	**−0.0068**	−0.0057	**−0.0270**	−0.0057	−0.0009	−0.0414
		(−2.16)	(−1.88)	**(−2.23)**	(−1.95)	(−0.31)	(−1.04)
Dummy	2.73	−0.0127	−0.0186	0.0527	—	—	—
		(−0.86)	(−1.25)	(0.97)	—	—	—

This table presents the VIFs and the FM regression results for the identified characteristics. The multivariate FM coefficients are reported for every characteristic with the respective t-value in parentheses below. The variables 2–17 are in line with the numbering in Table 9.2. "Dummy" is the blockchain dummy which takes values of 1 for digital assets connected to their own blockchain and 0 otherwise. VIF is the variance inflation factor for every variable over the full data history. To calculate the VIFs, dummy is the return spread between an equally weighted index of mineable digital assets and non-mineable assets. I exclude characteristics with VIF > 7 from the rolling multivariate FM regression analysis. The datasets "All" and "Mineable" include all digital assets and only digital assets with their own blockchain, respectively. I do not consider non-mineable digital assets here since these emerged mainly since 2016, which leaves too short a data history. However, I split the dataset in line with Green et al. (2017). "All" again includes all digital assets within the former mentioned categories. "All but tiny" excludes all digital assets with market capitalizations smaller than the 20th percentile of the respective dataset. "Tiny," on the other hand, contains only digital assets with market capitalizations equal to or less than the 20th percentile of the respective dataset. T-values equal or larger than |2| are highlighted in bold

The sign of low beta is in line with its short-term variant. However, these effects only appear in datasets comprised of the smallest digital assets. When controlling for tiny digital assets, the effects disappear.

To determine if the characteristics volume, short-term low beta, and nominal price are also economically significant, I ran a last test proposed by Green et al. (2017), which is based on the work of Lewellen (2015) (i.e., step 4). If the hedge portfolio returns based on predicted returns from rolling univariate FM regressions, using each of the three aforementioned characteristics (volume, short-term low beta, and nominal price), do not perform significantly worse than a prediction based on all 12 available characteristics, they can be considered economically significant (assuming positive and significant hedge portfolio returns). Therefore, I used the uni- and multivariate FM coefficients over the last 12 months in t to predict the returns in $t+1$. Then, the predictions served as sorting criteria from which I formed quintile portfolios. Acknowledging the results of Table 9.6, I applied one dataset comprised of all available digital assets and another one where I controlled for tiny digital assets.

From Table 9.7 several things can be learned. Volume (5) and size (11) appear to be reasonably priced (but less robust) risk factors, which can be utilized to predict the returns of digital assets. I arrived at comparable results when excluding tiny digital assets. Also, nominal price (17) produces significantly higher hedge portfolio returns. Sorting for low (log nominal) prices only (not the FM coefficient return estimates) yields positive and significant hedge portfolio returns as well (see Tables 8.2 and 9.2). Also, a combination of size, illiquidity, and nominal price results in statistically significant returns. Thus, the results from Tables 8.2, 9.2, 9.6, and 9.7 indicate the existence of low price, low volume (illiquidity), and to some extent low size effects in digital assets. That basically indicates that young and small digital assets carry higher returns than older and also larger digital assets. All in all, I can (partly) support the findings of Glas (2019). Volume and size indeed appear to be weak proxies for common risk factors in the cross-section of digital assets. The same applies to the prices of digital assets. Covariances, in general, seem not to be priced (yet).

9.7 Summary

Asset pricing in digital assets is still a difficult task. Data history is very short compared to traditional asset classes, and digital assets themselves suffer from various issues. The data quality also needs to be reviewed critically since many coins, tokens, or projects mainly appeared in the years 2016 and 2017. This leads to distorted datasets with clearly fewer assets in the years 2013 to 2016 than afterwards. Taking these problems into account, I do not aim to provide a work totally compliant with the asset pricing literature. Nevertheless, I aim to explore possible risk factors or characteristics in digital assets in a first step to lay the groundwork for further research.

Table 9.7 Portfolio sorts based on predicted returns

Long/short quintile portfolios	Mean (%)	*t*-stat	SD (%)	Return/risk	Skewness	Kurtosis
All						
12 characteristics	0.66	(0.14)	32.34	0.02	0.49	2.89
5, 6, 8, 9, 11, 17	3.89	(0.67)	40.12	0.10	0.91	4.58
5, 9, 17	11.12	(3.83)[c]	19.44	0.57	0.61	3.27
5, 6, 11, 17	5.62	(2.06)[b]	18.72	0.30	−0.58	4.31
8, 11	3.84	(0.51)	51.91	0.07	0.88	4.10
5 only	6.08	(1.97)[a]	21.20	0.29	−0.16	3.09
6 only	−1.37	(−0.32)	29.92	−0.05	−0.75	3.39
8 only	1.60	(0.16)	71.36	0.02	0.75	2.92
9 only	−1.16	(−0.49)	16.62	−0.07	0.32	2.82
11 only	5.23	(1.72)[a]	20.97	0.25	−0.04	3.87
17 only	14.53	(4.75)[c]	20.06	0.72	0.77	3.47
5, 17	10.55	(3.41)[c]	20.89	0.51	0.47	3.46
11, 17	11.75	(4.19)[c]	18.61	0.63	0.76	3.70
5, 11	6.40	(2.56)[b]	17.10	0.37	−0.13	4.39
5, 11, 17	6.10	(2.39)[b]	17.50	0.35	−0.17	4.14
All but tiny						
12 characteristics	4.03	(0.94)	29.72	0.14	0.86	3.27
5, 6, 8, 9, 11, 17	3.91	(0.75)	35.94	0.11	0.98	4.07
5, 9, 17	7.14	(2.55)[b]	19.08	0.37	0.57	3.93
5, 6, 11, 17	9.71	(3.39)[c]	19.33	0.50	0.29	3.44
8, 11	2.09	(0.30)	47.88	0.04	1.05	3.53
5 only	11.12	(3.77)[c]	19.77	0.56	−0.10	2.54
6 only	2.72	(0.61)	30.91	0.09	0.55	4.00
8 only	−4.63	(−0.51)	63.50	−0.07	0.30	3.63
9 only	1.96	(0.89)	15.23	0.13	−0.20	2.88
11 only	18.29	(3.63)[c]	33.83	0.54	−0.06	2.78
17 only	11.68	(3.81)[c]	20.51	0.57	1.13	4.79
5, 17	7.58	(2.44)[b]	21.29	0.36	0.41	3.78
11, 17	9.35	(3.02)[c]	21.03	0.44	0.54	3.64
5, 11	10.33	(3.47)[c]	20.07	0.51	−0.54	2.55
5, 11, 17	11.01	(3.69)[c]	20.03	0.55	−0.44	2.51

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
This table presents the return moments of a dollar-neutral long/short portfolio. The hedge portfolio is formed by going long for the top quintile and short for the bottom quintile from a portfolio sort based on predicted returns. The returns are predicted in accordance with Lewellen (2015). I sort for high predicted returns in all cases. Quintile portfolios are equally weighted and based on the dataset containing all available digital assets or all but tiny digital assets. For the FM coefficients I use the average FM coefficient over a rolling 12-month lookback period to predict the returns. "Mean" is the monthly percentage hedge portfolio return. "*t*-stat" is the *t*-value of the mean monthly returns in parentheses. "SD" is the standard deviation of mean monthly hedge portfolio returns in %. I calculated the risk-adjusted return "return/risk" by dividing the return by the standard deviation. "Skewness" and "kurtosis" are the third and fourth return moments

By doing so, I arrive at two major conclusions. First, PLS and PCA point at two or three risk factors in digital assets. Betas with respect to these principal components, on the other hand, are not priced significantly. The same applies to covariances with respect to other possible risk factors in Panels B–E. Second, following the debate of characteristics versus covariances, I provide first indications for priced characteristics but not covariances. This finding is comparable to the early asset pricing literature of the 1960s and 1970s. In particular, I find the characteristics volume, size, and nominal price to be priced at the earliest in the cross-section of digital assets. The results, though, are not robust to other specifications. This also supports the work of Glas (2019) only partially. I could find no evidence for a priced short-term reversal characteristic, nor priced market betas (except for short-term betas). Putting all of these outcomes into context, it seems not yet feasible to develop an asset pricing model specific to digital assets.

Asset pricing in digital assets in general is still in the early stages. It is not possible to draw general conclusions yet, due to a very short data history. Regulatory interventions could also fundamentally (re)shape the behavior of the asset class within a short period of time. Another matter of importance is the explanatory power of the market factor. I have the impression of a vast influence of the market on single digital assets, although it is not priced clearly in the cross-section. Additionally, almost any approach from the investment or asset pricing literature is suitable to be applied to digital assets. Research in this asset class has only begun, and more work is necessary to arrive at a clearer picture.

Now that several tests have been performed for digital assets, which have resulted in relatively unconvincing outcomes (especially Chaps. 6 and 7), we need to see whether the remaining findings up to this point can be transformed into profitable trading strategies. Chapter 8 explored this issue already; however, it might not be implementable in real-world trading. Therefore, in the next chapter I test several approaches which incorporate the findings of the previous chapters into a portfolio construction process that is not subject to short-selling constraints and also controls for tiny digital assets.

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Chapter 10
How to Incorporate Characteristics into the Portfolio Construction Process



10.1 Portfolio Construction

Recent studies (see Glas (2019) among others, and the previous chapters of this book) have uncovered the first indications of characteristics which are able to describe and explain the cross-section of digital asset returns. These characteristics include, in particular, trading volume and nominal price, and to a lesser extent (short-term) market beta, (short-term) reversal, and market capitalization.

The standard procedure in the asset pricing and investment styles literature to harvest such characteristic premia is to construct hedge portfolios based on a specific characteristic. A hedge portfolio commonly involves the ranking of assets by the desired asset characteristic. Subsequently, it is formed from the top (long portfolio) and bottom (short portfolio) percentiles of the assets under consideration. In asset classes like stocks, commodities, FX, or even bonds, the same approach can be applied by investors. In digital assets, however, short selling is almost impossible.[1]

Therefore, alternative (long-only) approaches that incorporate characteristics into the portfolio construction process need to be found for digital asset investors. There is an ongoing discussion in the literature on how best to combine several characteristics at once into a portfolio. Proponents of the integrated approach (Bender and Wang 2016; Clarke et al. 2016; Fitzgibbons et al. 2017) argue that it is more efficient to determine portfolio weights based on multiple characteristics and thus avoid the issue of holding assets with offsetting characteristic exposures. The mix approach simply weights equally the characteristic portfolios of importance and allows the repeated inclusion of the same asset in the portfolio. While the integrated

[1] CME, CBOE, and Nasdaq are the only regulated exchanges (of importance for institutional investors) which introduced bitcoin futures in 2017.

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approach seems to be more of interest for the research community, there is (to my knowledge) only one (Leippold and Rueegg 2017) peer-reviewed study in favor of the mix approach.[2] A more sophisticated framework is the parametric portfolio policy developed by Brandt et al. (2009), who determined the portfolio weight of an asset as a function of different characteristics; this is described in more detail below.

The aim of this chapter is twofold. First, I show whether the incorporation of characteristics is useful for portfolios consisting of digital assets only. Second, I compare different approaches of (long-only) characteristics investing. By doing so, I find that investors can indeed profit from taking characteristics into the portfolio formation process. The best tool for such an approach appears to be the integrated approach as described by Bender and Wang (2016).[3]

This chapter is structured as follows. Section 10.2 describes the methodology and data employed in this chapter. Section 10.3 presents the empirical results, and Sect. 10.4 summarizes the outcomes.

10.2 Methodology and Data

10.2.1 Benchmark Model

Recently, well-established portfolio optimization frameworks, for example Markowitz (1952), have been questioned. Duchin and Levy (2009) and Jacobs et al. (2014) found that heuristic weighting schemes like the $\frac{1}{N}$ rule yield comparable or even better out-of-sample results than the classical Markowitz (1952) optimization. This is especially of interest for this chapter. Thus, I used a naive portfolio as reference portfolio to which I compared all other portfolio construction methodologies. An equally weighted benchmark is an even better fit for the approaches described by Bender and Wang (2016). Their methods assume equally weighted starting weights, which is why the $\frac{1}{N}$ rule is the natural benchmark for these approaches (which are described in more detail below).

The weights of the equally weighted benchmark portfolio w_{EBM} are thus calculated as

$$w_{\text{EBM},t} = \frac{1}{N_t}, \tag{10.1}$$

where N is the total number of assets at time t.

[2] Fernandez-Perez et al. (2019) also question the superiority of the integrated over a simple mix approach.

[3] That, however, also depends on the chosen risk aversion γ.

Following Brandt et al. (2009) and Fieberg et al. (2016a), the approximative out-of-sample return of a portfolio r_P, net of trading costs, is given by[4]

$$r_{P,t} = \sum_{i=1}^{N_t} w_{i,t-1} r_{i,t} - |w_{i,t-1} - w_{i,t-2}| c_{i,t-1}, \tag{10.2}$$

with the log return $r_{i,t}$ of an asset i at time t and the asset weight of an asset held in the portfolio $w_{i,t}$. The proportional trading costs are denoted by $c_{i,t-1}$.[5] I subtracted transaction costs from both the benchmark and the other portfolio construction processes.

10.2.2 Mix of Characteristics and the Integrated Approach

As stated above, the $\frac{1}{N}$ rule is the benchmark of choice for the mix and integrated models. Therefore, I followed Bender and Wang (2016) in constructing both models. The weights in the mix model w_{MM} are given by

$$\mathbf{w}_{\text{MM}} = \frac{1}{N} \frac{\mathbf{C}_1 \circ \mathbf{w}_{\text{S}} + \mathbf{C}_2 \circ \mathbf{w}_{\text{S}} + \cdots + \mathbf{C}_n \circ \mathbf{w}_{\text{S}}}{\mathbf{C}_1' * \mathbf{w}_{\text{S}} + \mathbf{C}_2' * \mathbf{w}_{\text{S}} + \cdots + \mathbf{C}_n' * \mathbf{w}_{\text{S}}}, \tag{10.3}$$

where $\mathbf{C}_1$ to $\mathbf{C}_n$ are the multipliers (rank) of the N characteristics under consideration, and $\mathbf{w}_{\text{S}}$ is a vector of the starting weights. In Eq. 10.3, $\circ$ denotes elementwise multiplication and $*$ represents matrix multiplication. For the mix portfolio, $\mathbf{w}_{\text{S}}$ is described by Eq. 10.1. However, for robustness reasons I also report the results for value-weighted starting weights. When sorting assets into quintiles, the multiplier can take values from 1 to 5 (with 1 being the lowest and 5 the highest rank), dependent on the location of the respective asset. In line with Bender and Wang (2016), I calculated a *cross-sectional* z-score to determine the ranks (i.e. to determine each asset's rank, the assets were sorted into quintiles based on standardized characteristics and not the raw characteristic). According to Bender and Wang (2016) using standardization not only preserves cross-sectional distribution characteristics but also makes both approaches more comparable to the parametric portfolio policy (see Sect. 10.2.3), which makes use of standardization as well. The weights $\mathbf{w}_{\text{MM}}$ are then rescaled to sum to 100%.

[4] The following formula only applies approximately for continuous returns.

[5] Trading costs for digital assets amount to 0.25% for smaller investors at the cryptocurrency exchange Bitstamp (see "Unified fee schedule," available at https://www.bitstamp.net/fee_schedule/, accessed 23 November, 2018) and 0.26% at Kraken (see "Fee schedule," available at https://www.kraken.com/en-us/help/fees, accessed 23 November, 2018). Since institutional investors with higher trading volume would normally not be placed in such a fee tier, I assume very conservative transaction costs of 0.25% per trade.

The calculation of the integrated portfolio weights $\mathbf{w}_{\text{IM}}$ is slightly different to the mix model:

$$\mathbf{w}_{\text{IM}} = \frac{[\frac{1}{N}(\mathbf{C}_1 + \mathbf{C}_2 + \cdots + \mathbf{C}_n)] \circ \mathbf{w}_{\text{S}}}{[\frac{1}{N}(\mathbf{C}_1 + \mathbf{C}_2 + \cdots + \mathbf{C}_n)]' * \mathbf{w}_{\text{S}}}. \tag{10.4}$$

Here again, the portfolio weights are rescaled to sum to 100%. The main difference between the approaches, however, is that the mix model directly combines all the characteristics (and their weights) under consideration in one portfolio. The integrated portfolio, on the other hand, is constructed "bottom up," which means the ranks are taken into account at the same time, since individual security weights are determined based on the total rank (Bender and Wang 2016). Equations 10.3 and 10.4 are thus comparable to a rank-based approach. Each asset is assigned a specific rank, dependent on the relative characteristic.

To calculate the out-of-sample performance of the approaches described in Eqs. 10.3 and 10.4, I multiplied the weights in $t-1$ by the respective returns, net of transaction costs, in t (see Eq. 10.2).

10.2.3 Parametric Portfolio Policy

For the last model, which is inspired by Brandt et al. (2009), I again apply mostly the same specifications as described by Fieberg et al. (2016a). This approach does not minimize portfolio risk, like Markowitz (1952), but it maximizes the expected utility u of a portfolio's returns r_P at time $t+1$ based on the number of assets N in the data sample under consideration. Thereby, I assumed constant relative risk aversion preferences and a relative risk aversion (γ) value of 5, in line with prior studies:[6]

$$u(r_{P,t+1}) = \frac{(1 + r_{P,t+1})^{1-\gamma}}{1-\gamma}, \qquad \gamma \neq 1. \tag{10.5}$$

The basic problem (maximization of the expected utility) then becomes

$$\max_{\{w_{i,t}\}_{i=1}^{N_t}} E_t[u(r_{P,t+1})] = E_t\left[u\left(\sum_{i=1}^{N_t} w_{i,t} r_{i,t+1}\right)\right]. \tag{10.6}$$

[6] The following equations are reproduced in a slightly adjusted form from Fieberg et al. (2016a) and Brandt et al. (2009). These studies also applied a relative risk aversion (γ) value of 5.

The parametrized optimal portfolio weights $w_{i,t}$ of an asset i at time t are given as a function of an asset's cross-sectionally standardized characteristics x:

$$w_{i,t} = f(x_{i,t}; \boldsymbol{\theta}), \tag{10.7}$$

with the linear portfolio weight function

$$w_{i,t} = \bar{w}_{i,t} + \frac{1}{N_t}\boldsymbol{\theta}^{\mathrm{T}} x_{i,t}, \tag{10.8}$$

where $\boldsymbol{\theta}$ is a vector of coefficients, which needs to be estimated. Brandt et al. (2009) stated that the parametrization results in stable $\boldsymbol{\theta}$s over time. Assuming the characteristics x contain full information of the joint return distribution, the $\boldsymbol{\theta}$s maximize an investor's conditional expected utility at any time t. Since the $\boldsymbol{\theta}$s are constant over time, the investor's unconditional expected utility is also maximized:

$$\max_{\boldsymbol{\theta}} E[u(r_{P,t+1})] = E\left[u\left(\sum_{i=1}^{N_t} f(x_{i,t}; \boldsymbol{\theta}) r_{i,t+1}\right)\right]. \tag{10.9}$$

As an illustration, the coefficients $\boldsymbol{\theta}$ can be maximized by

$$\max_{\boldsymbol{\theta}} \frac{1}{T}\sum_{t=0}^{T-1} u(r_{P,t+1}) = \frac{1}{T}\sum_{t=0}^{T-1} u\left(\sum_{i=1}^{N_t} f(x_{i,t}; \boldsymbol{\theta}) r_{i,t+1}\right). \tag{10.10}$$

In the linear case (see Eq. 10.8) the coefficients can be maximized according to

$$\max_{\boldsymbol{\theta}} \frac{1}{T}\sum_{t=0}^{T-1} u\left(\sum_{i=1}^{N_t} (\bar{w}_{i,t} + \frac{1}{N_t}\boldsymbol{\theta}^{\mathrm{T}} x_{i,t}) r_{i,t+1}\right). \tag{10.11}$$

The standard errors for $\boldsymbol{\theta}$ were bootstrapped as described by Fieberg et al. (2016a). In this chapter I applied an expanding-window portfolio optimization and estimated new θ_j for every date t to obtain new weights w_i. Due to the unbalanced data, I limited the $\boldsymbol{\theta}$s to between -10 and 10. The calculation of the out-of-sample performance was in line with the other models presented above. The optimization is also highly dependent on high data availability. Therefore, the data was cleaned before the optimization, which might result in a slightly different dataset than used for the mix and integrated approaches.

10.2.4 Data

For the following analysis I used the same data as in the previous chapters. I took into consideration monthly close, trading volume, and market capitalization data

Table 10.1 Descriptive statistics

	Panel A			Panel B		
Descriptive statistics	log(price)	log(turnover)	log(size)	log(price)	log(turnover)	log(size)
Mean	−3.15	9.01	14.14	−2.68	9.65	14.85
SD	3.47	3.76	3.02	3.35	3.58	2.56
Skewness	−0.04	0.29	−0.09	−0.26	0.24	0.15
Kurtosis	4.24	2.16	2.65	4.49	2.37	2.52
25th percentile	−5.37	5.89	11.78	−4.44	6.90	12.86
Median	−2.89	8.67	14.27	−2.36	9.44	14.79
75th percentile	−1.08	11.87	16.50	−0.57	12.25	16.73

This table reports the mean, standard deviation (SD), skewness, kurtosis, 25th percentile, 75th percentile, and median for the characteristics in this chapter. All characteristics are measured in USD. Log(price) is the nominal price characteristic, log(turnover) represents trading volume, and log(size) describes the size characteristic. Panel A is comprised of 1263 digital coins as stated above, and Panel B excludes digital assets smaller than the 20% percentile every month

between April 2013 and June 2018 ($t = 63$ months).[7] All values are in USD since an investor usually wants to convert their digital asset holdings to fiat currency at some point. In line with Chap. 9, digital assets with fewer than six observations were excluded, which reduced the number of assets to $N = 1263$.[8] The initial estimation of the parametric portfolio policy was based on the first 12 months of the dataset. Therefore, out-of-sample returns were calculated starting 12 months later, from April 2014 to June 2018, for all approaches (i.e. a 12 month shorter data history than in the prior chapters).

The characteristics considered for this analysis were selected based on Glas (2019) and Chaps. 8 and 9. Glas (2019) arrived at four important characteristics in digital assets: market beta, trading volume, size, and short-term reversal. Market beta and short-term reversal are not robust when controlling for other characteristics (see Chap. 9). Chapter 9 also adds the characteristic nominal price, next to volume and size, which are able to describe and predict the cross-section of digital asset returns. Nominal price is the log of the price of a virtual coin at time t. Volume, which proxies illiquidity, is the log of the trading volume in USD at time t. The same applies to size, which is the log of the market capitalization. Table 10.1 reports descriptive statistics for the three characteristics considered in this analysis.

Since many digital coins are too small (in size) to be relevant for institutional investors, I constructed a second dataset which excludes the smallest digital assets (Panel B). This dataset is motivated by similar research in the asset class of equities. Fama and French (2015) defined micro-cap stocks as stocks which are located in the smallest quintile of market values. Green et al. (2017) used the same definition and constructed a dataset from which they excluded all micro-cap stocks to control

[7] For more information, see Chaps. 5, 8, and 9.

[8] I also ran the same analysis with other datasets, as described in the next chapter.

for effects which might be caused by only the smallest stocks. I followed both interpretations and constructed Panel B by excluding digital assets smaller than the 20% percentile every month. This procedure reduced the number of assets to a monthly average of $N = 286$. Table 10.1 contains descriptive statistics for both data panels. Panel A describes the dataset including all available digital assets and Panel B contains only the largest digital assets, excluding the smallest market capitalization quintile. As a risk-free rate I used the one-month treasury bill rate, downloaded from the St. Louis Fed (Code: DGS1MO) over the same period of time.[9]

10.3 Empirical Results

To answer the questions of whether it is reasonable to incorporate characteristics into the portfolio construction process and how to do so, I compared three portfolio construction methodologies involving characteristics to a naive benchmark portfolio; I investigated two different datasets and applied different starting weights as well. The datasets were comprised of all available digital assets (Panel A), and digital assets excluding the smallest digital assets (Panel B). I applied an equally weighted and a market capitalization weighted benchmark, which both served as the respective starting weights. Table 10.2 reports the results for Panel A. For the benchmark, mix, and integrated approaches I skipped the first 12 months, since the parametric portfolio policy estimates its parameters in that period of time, which is not taken into account for performance measurement.

10.3.1 Panel A

An equally weighted benchmark based on Panel A yields a monthly average return of -0.55%. This number needs to be outperformed to achieve a positive and, at best, a statistically significant Jensen's alpha.[10] The characteristics mix and integration approaches yield relatively similar statistics (see Table 10.2). As described in the literature, the integrated approach is somewhat superior to the mix approach. Characteristics integration yields an average monthly return of 1.23%, whereas a characteristics mix performs 27 basis points worse. Both are additionally able to achieve significant alphas, with 1.51% (mix) and 1.79% (integrated). The adjusted R^2 for both approaches is very high, but they are still able to outperform an equally weighted benchmark.

[9] See "1-Month Treasury Constant Maturity Rate," available at https://fred.stlouisfed.org/series/DGS1MO#0, accessed 27 November, 2018.

[10] For more information regarding the Jensen's alpha regression, see Sect. 2.1.2.

Table 10.2 Empirical results, Panel A

	Equally weighted				Capitalization weighted			
Statistics	BM	Mix	Integrated	Parametric	BM	Mix	Integrated	Parametric
θ_{Nominal}				2.07				−0.08
θ_{Volume}				−0.74				−0.20
θ_{Size}				10.00				0.06
Std(θ_{Nominal}) (%)				64.27				4.28
Std(θ_{Volume}) (%)				418.22				13.16
Std(θ_{Size}) (%)				0.01				6.31
$Varcoeff_{\text{Nominal}}$				0.31				−0.54
$Varcoeff_{\text{Volume}}$				−5.64				−0.67
$Varcoeff_{\text{Size}}$				0.01				1.06
$\bar{w}_i$ (%)	0.08	0.31	0.31	0.94	0.33	0.33	0.42	0.87
min w_i (%)	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00
max w_i (%)	0.89	3.54	3.64	21.69	93.85	16.48	18.80	86.53
$\mu(r)$ (%)	−0.55	0.96	1.23	−2.57	7.46	8.04	8.11	4.06
$\sigma(r)$ (%)	28.81	29.57	29.75	30.35	26.70	31.13	28.89	24.12
$\mu(r)/\sigma(r)$	−0.02	0.03	0.04	−0.08	0.28	0.26	0.28	0.17
$\overline{Nominal}$	−0.01	−0.01	−0.01	−0.01	0.01	−0.01	−0.01	0.00
$\overline{Volume}$	0.01	0.01	0.01	0.02	0.02	0.02	0.03	0.02
$\overline{Size}$	0.02	0.02	0.02	0.04	0.02	0.04	0.04	0.03
α (%)		1.51	1.79	−2.04		1.58	1.84	−2.57
T(α)		2.53	2.83	−1.60		0.52	0.69	−3.78
P(α)		0.01	0.01	0.12		0.61	0.49	0.00
β_{BM}		1.02	1.02	0.98		0.87	0.84	0.88
Adjusted R^2 (%)		97.92	91.12	84.29		54.33	59.48	96.26
CER		−0.14	−0.17	0.09		−0.21	−0.15	−0.06

This table reports the mean θs of the parametric portfolio policy for the three characteristics nominal price, volume, and size. The next three lines represent the standard deviations of the θs, and the following three lines are the variation coefficients calculated as $\frac{\sigma_\theta}{\mu_\theta}$, where σ is the standard deviation of θ and μ is the mean of θ. These values are taken from the out-of-sample portfolio optimization using an expanding window. The next three lines include information regarding the portfolio weights, describing the average, minimum, and maximum weights. Then, performance information is described: $\mu(r)$ is the average return, $\sigma(r)$ is the standard deviation of returns, and additionally the return/risk relationship is reported. Average characteristics of the portfolios are included as $\overline{Nominal}$, $\overline{Volume}$, and $\overline{Size}$. The last five lines contain information regarding the Jensen's alpha regression, comparing the different strategies to the respective equally or market capitalization weighted benchmark. The regression can be read as $r_i - r_f = \alpha_i + \beta_i(r_{BM} - r_f) + \epsilon_i$, where r_i is the return of one of the respective strategies, r_f is the risk-free rate, and r_{BM} is the return of the benchmark model. Moreover, I report the t- and p-values of α, as well as the β and the adjusted R^2. CER describes the difference between the certainty-equivalent returns of the respective portfolio construction approach and the benchmark portfolio using a power utility function (see Eq. 10.5) and a relative risk aversion (γ) value of 5

The parametric portfolio policy performs clearly worse, with a mean return of −2.57%. It also exhibits a slightly higher return volatility compared to the other approaches. Thus, the reward–risk ratio is also the lowest of all three portfolio construction methodologies involving characteristics. The parametric portfolio policy also assigns more importance to the characteristics size and volume than the mix or integrated approaches. Its alpha is also negative, at −2.04%. The certainty equivalent return (CER), however, is the highest for the optimization approach. The CER is a measure to compare the returns of different investment approaches. It is calculated as the difference of the certainty-equivalent returns of the respective portfolio construction approach (mix, integration, or parametric portfolio policy) and the benchmark portfolio using a power utility function (see Eq. 10.5) and a relative risk aversion (γ) value of 5.

When using a market capitalization weighted benchmark (and starting weights) the results remain unchanged. The mix and integrated approaches still result in positive average monthly returns. Also, their Jensen's alphas are higher than for the parametric portfolio policy, but statistically insignificant. The parametric portfolio policy, on the other hand, yields an average return of 4.06%. Therefore, it underperforms its benchmark by 2.57 percentage points (i.e. Jensen's alpha). For an equally weighted benchmark, the minimum and maximum asset weights within the portfolio are not as widely spread (due to the unbalanced dataset); the weights diverge more in the market capitalization weighted case. While it is highest for the benchmark with 93.85% (which is the weight of bitcoin at the beginning of the sample), the parametric portfolio policy assigns a maximum weight of 86.53%, whereas the remaining portfolio construction processes arrive at maximum weights of 16.48% (mix) and 18.80% (integrated).

Thus, it can be said that the parametric portfolio policy is much less able to cope with diverse starting weights, which results in positive average returns for only one initial starting weight, but that does not necessarily translate into outperforming the benchmark. Due to a relatively conservative γ, the parametric portfolio policy might not use its full capabilities, and thus yields rather disappointing results. Also, the weight of bitcoin dominates the parametric policy portfolio, which ultimately yields less favorable outcomes when using value weights. The mix and integrated approaches, on the other hand, are able to handle different starting weights, but produce statistically significant alphas when applied to equally weighted starting weights only.

10.3.2 Panel B

Due to the heterogeneity of digital assets and the low market capitalization of the smallest assets in this investment universe, I applied a second dataset, which excludes virtual tokens smaller than the 20th market capitalization percentile. Institutional investors (and perhaps even private investors) would not be able to invest in these digital assets due to liquidity and also legal/exchange constraints.

The outcomes based on a dataset free of the smallest digital coins, however, do not change significantly.

The characteristics mix and the integrated framework again perform better or similar to the parametric portfolio policy in both cases (in terms of mean return). The parametric portfolio policy achieves lower (and statistically significant) alphas when applying equally weighted starting weights (see Table 10.3) and value-weighted starting weights. By eliminating the smallest digital coins, the average returns of the mix and integrated approaches increase (decrease) with constant or slightly decreasing standard deviations for the equal (market capitalization) starting weights. The alphas for the characteristics mix and integration are higher when measured against an equally weighted benchmark. However, they become statistically insignificant. As for Panel A, the parametric portfolio policy underperforms an equally weighted benchmark (in the sense of alpha). When excluding the smallest digital coins, the alpha for market capitalization weights decreases to -2.56%, which is statistically significant. Thus, while the mix and integrated approaches do slightly worse when using another panel, the parametric portfolio policy again does not yield any benefit when using naive starting weights.

10.4 Summary

This chapter ultimately aims at incorporating the prior findings of this book into three portfolio construction processes for digital assets. The previous chapters have shown that the characteristics (low) nominal price, illiquidity, and size are the clearest priced characteristics in the cross-section of digital asset returns. Covariances or other characteristics seem not to be priced in this asset class.

When compared to two simple benchmarks (equally weighted and market capitalization weighted), taking into account characteristics for investments in digital assets indeed yields positive Jensen's alphas (even after controlling for transaction fees). The success of the respective portfolio construction process is also robust for the applied starting weights. While the characteristics mix and characteristics integration are superior to both benchmarks, when using Panel A the parametric portfolio policy yields no positive alpha for any of the initial weights. In line with the literature, the integrated approach performs slightly better than a mix of characteristics. Here again, on the other hand, the results are clearly driven by the smallest digital assets. When excluding young assets with low market capitalizations, the results tend to become insignificant.

Since most of my analyses are based on a relatively unbalanced dataset, the results might be somehow distorted by that issue. To overcome claims of unreliable data and thus weak results, I ran the most important analyses from the previous chapters again on a more balanced dataset; the details are presented in the next chapter.

Table 10.3 Empirical results, Panel B

	Equally weighted				Capitalization weighted			
Statistics	BM	Mix	Integrated	Parametric	BM	Mix	Integrated	Parametric
θ_{Nominal}				3.38				−0.08
θ_{Volume}				−6.06				−0.12
θ_{Size}				9.99				−0.05
Std(θ_{Nominal}) (%)				140.29				3.32
Std(θ_{Volume}) (%)				120.18				4.42
Std(θ_{Size}) (%)				0.05				5.88
$Varcoeff_{\text{Nominal}}$				0.42				−0.42
$Varcoeff_{\text{Volume}}$				−0.20				−0.38
$Varcoeff_{\text{Size}}$				0.04				−1.11
$\bar{w}_i$ (%)	0.10	0.41	0.41	1.25	0.41	0.41	0.41	1.02
min w_i (%)	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00
max w_i (%)	1.11	9.44	9.42	27.65	93.85	14.95	14.84	86.39
$\mu(r)$ (%)	0.41	1.10	1.16	−1.79	7.45	3.99	4.00	4.01
$\sigma(r)$ (%)	29.02	27.04	27.02	28.99	26.70	25.48	25.46	24.03
$\mu(r)/\sigma(r)$	0.01	0.04	0.04	−0.06	0.28	0.16	0.16	0.17
$\overline{Nominal}$	−0.01	−0.02	−0.02	−0.01	0.01	−0.02	−0.02	0.00
$\overline{Volume}$	0.02	0.03	0.03	0.03	0.02	0.04	0.04	0.03
$\overline{Size}$	0.03	0.06	0.06	0.06	0.03	0.06	0.06	0.04
α (%)		0.73	0.79	−2.19		−1.70	−1.68	−2.56
T(α)		0.87	0.95	−1.92		−0.75	−0.75	−3.80
P(α)		0.39	0.34	0.06		0.46	0.46	0.00
β_{BM}		0.91	0.91	0.91		0.76	0.76	0.88
Adjusted R^2 (%)		95.21	95.19	92.12		63.03	63.07	96.26
CER		0.12	0.12	0.07		−0.04	−0.04	−0.06

This table reports the mean θs of the parametric portfolio policy for the three characteristics nominal price, volume, and size. The next three lines represent the standard deviations of the θs, and the following three lines are the variation coefficients calculated as $\frac{\sigma_\theta}{\mu_\theta}$, where σ is the standard deviation of θ and μ is the mean of θ. These values are taken from the out-of-sample portfolio optimization using an expanding window. The next three lines include information regarding the portfolio weights, describing the average, minimum, and maximum weights. Then, performance information is described: $\mu(r)$ is the average return, $\sigma(r)$ is the standard deviation of returns, and additionally the return/risk relationship is reported. Average characteristics of the portfolios are included as $\overline{Nominal}$, $\overline{Volume}$, and $\overline{Size}$. The last five lines contain information regarding the Jensen's alpha regression, comparing the different strategies to the respective equally or market capitalization weighted benchmark. The regression can be read as $r_i - r_f = \alpha_i + \beta_i(r_{BM} - r_f) + \epsilon_i$, where r_i is the return of one of the respective strategies, r_f is the risk-free rate, and r_{BM} is the return of the benchmark model. Moreover, I report the t- and p-values of α, as well as the β and the adjusted R^2. CER describes the difference between the certainty-equivalent returns of the respective portfolio construction approach and the benchmark portfolio using a power utility function (see Eq. 10.5) and a relative risk aversion (γ) value of 5

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Chapter 11
Are the Results Robust and Still Valid?



11.1 Findings from Another Dataset

Between the original time of writing of this book and its publication, the results appear to be a bit "out of date." Therefore, I obtained another dataset, totally independent from www.coinmarketcap.com. The intention with this second dataset was to prove that the described results are correct, since it includes a much larger set of digital assets than the previous dataset. It is also longer (in terms of data history) and does not suffer from a survivorship bias (see details below). Since the second source of data is not connected to www.coinmarketcap.com at all, it serves as an additional (data) robustness check.

The new source of data is a website similar to www.coinmarketcap.com, but it offers its services totally free of charge. The site, www.coingecko.com, additionally collects data on digital assets which have already passed away and are listed on www.deadcoins.com. A list of all digital coins included in the dataset can be found in Appendix J. Thus, I downloaded almost the same data as before (close, market capitalization, and turnover) from www.coingecko.com between the end of April 2013 and the end of July 2020.[1] Unfortunately, I am not aware of any (daily) high or low price information, which means that the outcomes of some technical indicators reported in Chap. 7 can neither be replicated nor verified with this second dataset.

In order to obtain a deeper understanding of this new dataset, I visualize the data by graphs and tables below. In total, the CoinGecko dataset includes $N = 7680$ digital assets covering a data history of $t = 88$ months. To put this number into context, the number of digital assets listed on www.coinmarketcap.com on July 31, 2020 was 5862,[2] while www.deadcoins.com listed 1928 deceased virtual tokens on

[1] The exact date of the download of the historical data is August 2, 2020.

[2] See "Cryptocurrencies," https://coinmarketcap.com/all/views/all/, accessed on July 31, 2020.

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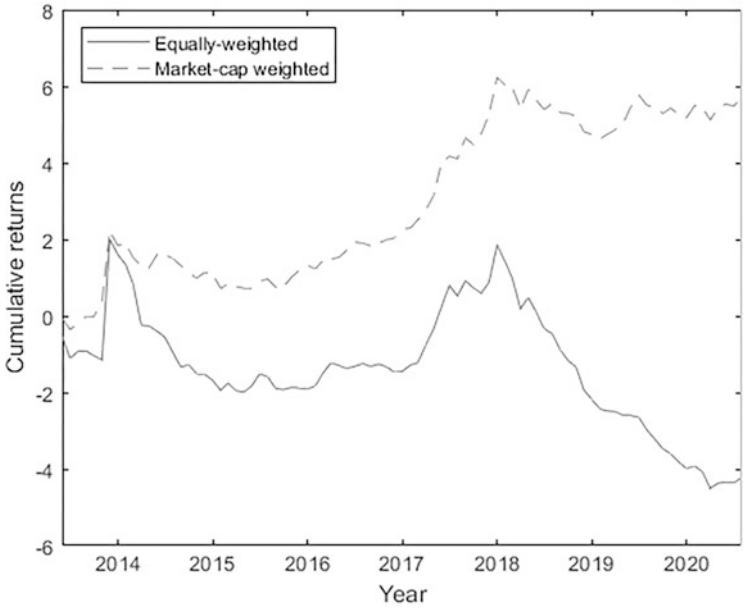


Fig. 11.1 Equal versus market capitalization weights

its website.[3] That means the total universe of digital assets was made up of 7790 virtual tokens (5862 alive coins + 1928 dead coins). Therefore, the new dataset (i.e., Panel A) obtained from www.coingecko.com should be almost free of any survivorship bias, since it includes 7680 digital assets in total.[4] That is almost five times as broad as and about 1.4 times as long as the initial CoinMarketCap dataset.

Following the previous analyses, I excluded digital assets with less than 6 months of close price observations. That reduced the number of assets to $N = 5120$, which is still more than three times as broad as the original dataset. Figure 11.1 shows the equally weighted and market capitalization weighted returns of the www.coingecko.com dataset. Figure 11.2 depicts the number of digital assets over time. Both plots are fairly in line (until mid-2018) with the graphs in Chap. 5. After the market crash in 2018/2019, the market capitalization weighted cumulative returns held relatively steady, whereas an equally weighted benchmark clearly lost value. The number of digital assets, on the other hand, rose further to slightly more than 5000 virtual coins at the end of July 2020 (when excluding digital assets with less than 6 months of observations). Figure 11.3 plots the number of monthly observations for every of the 5120 digital assets in that new dataset. As can be seen there, the average number of observations over all digital assets is $t = 20.03$ months.

[3] See "Full List," https://deadcoins.com/, accessed on August 2, 2020.

[4] That is just 110 virtual coins less than all assets listed on www.coinmarketcap.com and www.deadcoins.com.

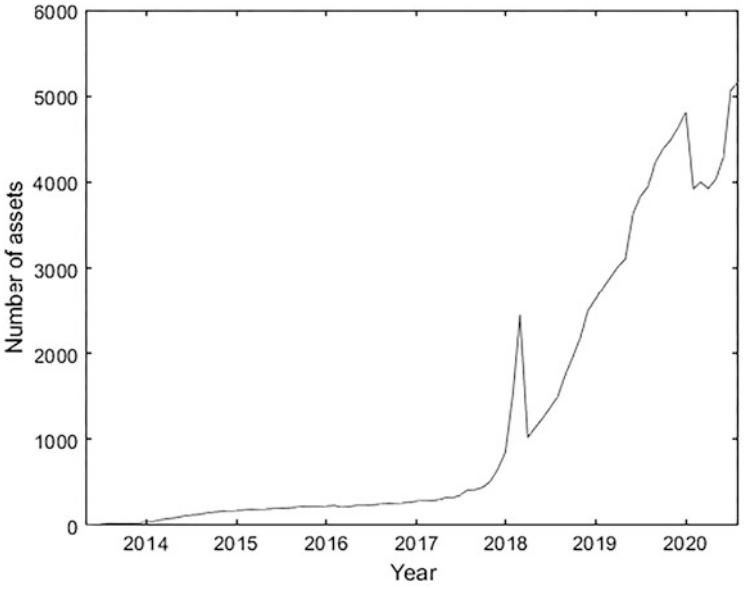


Fig. 11.2 Number of assets over time

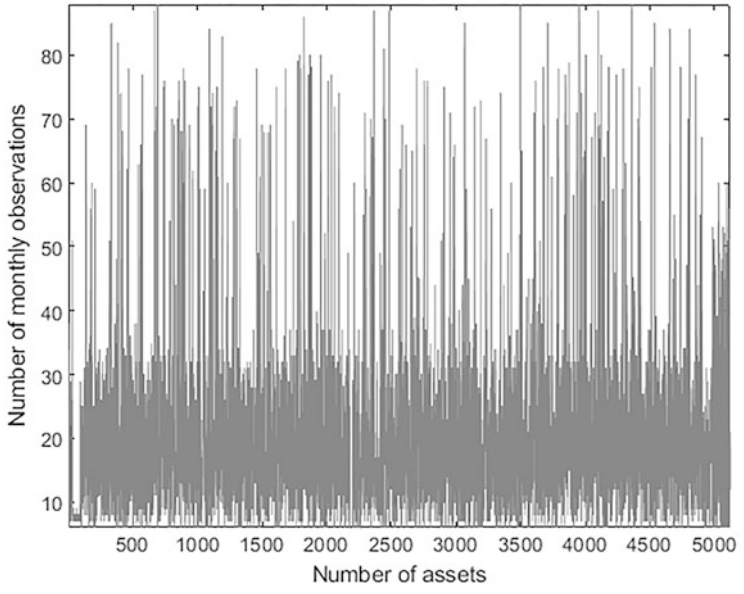


Fig. 11.3 Number of monthly observations per asset

Table 11.1 Descriptive statistics, CoinGecko dataset

Descriptive statistics	Return (%)	log(price)	log(turnover)	log(size)
Panel A				
Mean	−4.86	−5.60	6.18	12.99
SD	46.17	4.14	4.78	2.93
Skewness	3.85	−0.29	−0.09	−0.54
Kurtosis	28.14	8.09	2.95	6.88
25th percentile	−25.66	−7.91	3.14	11.09
Median	−7.50	−5.65	6.13	13.07
75th percentile	6.73	−3.41	9.63	14.94
Panel B				
Mean	−0.69	−4.60	6.98	13.45
SD	46.23	3.64	3.99	2.35
Skewness	3.78	−0.23	0.11	1.30
Kurtosis	27.70	4.15	5.19	6.11
25th percentile	−21.46	−6.66	4.99	11.98
Median	−3.30	−4.54	6.85	13.16
75th percentile	12.99	−2.46	8.85	14.41

This table reports the mean, standard deviation (SD), skewness, kurtosis, 25th percentile, 75th percentile, and median for the characteristics of the newly obtained www.coingecko.com data panels. All characteristics in this table are measured in USD, and all the analyses in this chapter are also conducted in USD. Return is the equally weighted monthly return of all assets contained in the dataset. Log(price) is the nominal price characteristic, log(turnover) represents the trading volume, and log(size) describes the size characteristic. The panel is comprised of 5120 digital coins

When looking at this second dataset in more detail (see Table 11.1), there do not seem to be structural differences with respect to the www.coinmarketcap.com dataset. Comparing the statistics of the www.coingecko.com dataset to the statistics of the full www.coinmarketcap.com dataset (see Table 5.4) does not yield any obvious data issues (except for the issues discussed in Chap. 5). The main difference is the slightly more negative mean return of the www.coingecko.com dataset, which is due to the weak equally weighted market index performance of digital assets in 2018 and 2019.

Another issue stems from the "explosion" of ERC-20 and other tokens that were mainly created from 2016/2017 onward. The dataset used in the previous chapters of this book might just be too unbalanced to arrive at significant and reliable results. Therefore, to weaken claims of data issues, I constructed a more balanced dataset (i.e., Panel B) based on the data obtained from www.coingecko.com, which I used to test the findings of the previous chapters again. Should the results be robust using this additional higher-quality dataset, the outcomes of this book are well supported. On the contrary, should the results differ completely from the previous chapters, the findings of this work can be deemed unreliable. The appendices also provide various robustness tests: Appendices B and H repeat the analyses from Chaps. 5

and 9 with data denominated in USD only. The results there are in line with the previous findings, but the dataset still suffers from a lot of missing values.

Unfortunately, it is not yet possible to construct a totally perfect data sample of digital assets covering all relevant information for a sufficient number of virtual coins without any missing values. That is mainly due to the relatively small number of virtual coins after the introduction of bitcoin in 2009. Thus, I needed to find the right balance between a sufficient number of virtual coins and a long data history. As shown in Fig. 5.3, the number of assets grew exponentially after 2017. At the start of the sample, only a few hundred virtual coins existed. Since I used portfolio sorts in many of my analyses, which mainly involved the formation of quintile portfolios, I aimed to design a new dataset containing at least 100 to 150 virtual tokens. This means that every quintile is then comprised of roughly 20 to 30 digital assets, which ensures sufficient diversification such that one quintile is not dominated by a specific single asset. The trade-off, however, is to decrease the total number of observations, since not enough digital assets existed at the beginning of that sample.

Consequently, Panel B used for this robustness section is comprised of $N = 125$ digital assets with a minimum of $T = 65$ or more monthly observations. Increasing the minimum T would result in a lower N, and vice versa. This combination (125 assets with 65 or more observations) thus seems like a good balance between a sufficient number of assets and ensuring enough data observations. The drawback of this approach is a relatively pronounced survivorship bias in the data. Nonetheless, to make this newly created data sample even more accessible, I report key descriptive statistics in Table 11.1. All of these 125 assets stem from the original www.coingecko.com dataset as described above. They also feature the same period of time (April 2013 to July 2020) but are screened according to the above-mentioned criteria.

Since the original and new datasets share a common behavior, the outcomes of the main analyses of this book based on the second data panels should either strengthen the reported results or call them into question. Therefore, I follow the same structure as in the previous chapters in repeating the main analyses of this study based on the www.coingecko.com data panels.

11.2 Technical Indicators Based on a Second Dataset

Due to the lack of high and low price information, the comparison with the results of Chap. 7 is a bit flawed. Not all of the eight technical indicators can be calculated with the www.coingecko.com dataset. The technical indicators *not* involving high or low price information are RSI, MACD, and VAT—not even half of the technical indicators investigated before. Nonetheless, I report the results of the three possible technical indicators (next to a simple buy-and-hold strategy) below.

Basically, the outcomes of the three technical indicators based on both www.coingecko.com datasets do not change dramatically. The cumulative, and thus daily, returns of the long-only technical indicators (see Table 11.2) are even more negative

Table 11.2 CoinFecko long-only technical indicators

	Panel A				Panel B			
	RSI	MACD	VAT	Buy and hold	RSI	MACD	VAT	Buy and hold
No transaction costs								
Cumulative return (%)	−128.82	−658.63	−421.44	−224.73	−231.58	−1526.45	−1089.14	−7.03
Daily return (%)	−0.77	−3.06	−5.93	−0.23	−0.20	−1.49	−2.24	−0.01
Standard deviation (%)	32.59	33.89	31.90	34.11	21.62	23.60	23.37	23.33
# of trades	9.75	69.49	148.24	1.00	47.64	188.74	741.84	1.00
Return/risk	−0.02	−0.09	−0.19	−0.01	−0.01	−0.06	−0.10	−0.00
Profit factor	0.27	0.03	0.04	0.20	0.23	0.03	0.03	0.96
Max. drawdown (%)	100.00	100.00	100.00	99.98	99.85	100.00	100.00	98.53
Success rate (%)	33.61	16.29	17.09	25.21	31.20	15.20	15.20	56.80
Transaction costs								
Cumulative return (%)	−130.68	−666.48	−455.17	−224.98	−242.30	−1570.04	−1254.08	−7.28
Daily return (%)	−0.78	−3.08	−6.27	−0.23	−0.21	−1.53	−2.57	−0.01
Standard deviation (%)	32.60	33.91	32.00	34.11	21.64	23.63	23.48	23.33
# of trades	9.75	69.49	148.24	1.00	47.64	188.74	741.84	1.00
Return/risk	−0.02	−0.09	−0.20	−0.01	−0.01	−0.06	−0.11	−0.00
Profit factor	0.27	0.03	0.03	0.20	0.21	0.03	0.01	0.96
Max. drawdown (%)	100.00	100.00	100.00	99.98	99.88	100.00	100.00	98.53
Success rate (%)	31.95	15.27	13.07	25.21	29.60	14.40	8.80	56.80

Cumulative return is the average, log-cumulative return across all digital assets for the corresponding technical indicator. Continuous returns are defined in the range from $-\infty$ to $+\infty$, which means that even values significantly smaller than −100% can occur. Average daily return refers to the average returns of all digital assets for the corresponding technical indicator in the time series of returns. Standard deviation is the standard deviation of the average daily returns. The risk/return ratio is also calculated from these two values. The average number of trades denotes the mean value of the number of all trades per virtual currency. Profit factor is the sum of all positive cumulative returns in relation to the sum of all negative cumulative returns. The maximum drawdown refers to the average daily returns. The success rate measures the number of all digital assets with a positive cumulative return compared to the total number of assets examined

than reported in Chap. 7. This observation is in line with the buy-and-hold return, which is also more negative (due to the extended sample period; see Fig. 11.1). Also, the number of trades, maximum drawdowns, and profit factors are comparable to (or even worse than) the findings reported in Chap. 7. Only the success rates clearly differ: the success rate of long-only RSI remains stable at about 30% for both panels and also after accounting for transaction fees. That compares to a success rate of 29.90% as reported in Chap. 7 (see Table 7.3). The success rates for MACD and VAT are almost down by half, at 16.29% and 15.20% (for Panels A and B not involving transaction costs) and 17.09% and 15.20% (also for Panels A and B not involving transaction costs), respectively. The same success rates described in Chap. 7 are 28.84% for MACD and 29.59% for VAT.

The most interesting outcome of the long–short analysis (see Table 11.3) is the improved result of RSI (for Panel A). The results of the same indicator worsen for Panel B, which indicates some dependence of the results obtained on the dataset used. Whereas the long–short results worsened in the previous analyses, RSI (in Panel A) improves compared to the long-only analysis. The cumulative return is still negative but amounts to a loss of only -51% as opposed to -128.82% using a long-only framework. In line with Chap. 7, the results for MACD and VAT are twice as bad as the long-only analysis (that also applies to RSI in Panel B). For RSI, the number of trades is comparable to the findings in Chap. 7. That also applies to VAT. Only MACD exhibits more signals than reported in Chap. 7. The profit factors decrease for MACD and VAT, which is not true for RSI. The long–short profit factor of RSI (in Panel A) is more than twice as high as the respective profit factor reported in Chap. 7. With a higher profit factor, the success rate also improves for RSI. The success rates for MACD and VAT are clearly below the results of Chap. 7. In general, the results reported here do not differ dramatically from the prior findings. That already hints at the robustness of the results of the analyses described before.

Unfortunately, it is not possible to investigate Bollinger bands with this newly obtained dataset. Bollinger bands are the best performing technical indicators in Chap. 7 (in terms of (positive) returns). The outcomes of the three reported technical indicators of this chapter, on the other hand, appear to be robust and in line with the findings of Chap. 7. Although the long–short performance of RSI improves compared to the long-only analysis (for Panel A only), technical indicators are not a profitable investment strategy when applied to alternative datasets of digital assets independent of the original data source.

11.3 Larger Cross-Section, Better Findings?

Since most of this book focuses on cross-sectional analyses, I also used the newly obtained dataset to form portfolios based on well-known characteristics. I report the hedge portfolio returns of 14 characteristics (see Table 11.4) that have already been described in more detail in Chap. 8 (see Table 8.1). The direction of sorts remains the same as before.

Table 11.3 CoinGecko long–short technical indicators

	Panel A				Panel B			
	RSI	MACD	VAT	Buy and hold	RSI	MACD	VAT	Buy and hold
No transaction costs								
Cumulative return (%)	−51.00	−1092.70	−810.98	−224.73	−460.91	−3051.86	−2257.85	−7.03
Daily return (%)	−0.01	−2.74	−4.79	−0.23	−0.23	−1.61	−2.34	−0.01
Standard deviation (%)	33.95	33.93	33.86	34.11	23.26	23.26	23.33	23.33
# of trades	19.51	138.98	148.24	1.00	95.28	377.49	741.84	1.00
Return/risk	0.00	−0.08	−0.14	−0.01	−0.01	−0.07	−0.10	−0.00
Profit factor	0.74	0.05	0.03	0.20	0.21	0.02	0.02	0.96
Max. drawdown (%)	99.96	100.00	100.00	99.98	99.86	100.00	100.00	98.53
Success rate (%)	43.95	22.77	17.05	25.21	21.60	12.80	11.20	56.80
Transaction costs								
Cumulative return (%)	−58.14	−1134.09	−897.29	−224.98	−503.85	−3226.49	−2679.25	−7.28
Daily return (%)	−0.01	−2.83	−5.18	−0.23	−0.25	−1.69	−2.76	−0.01
Standard deviation (%)	33.97	33.99	34.02	34.11	23.28	23.33	23.49	23.33
# of trades	19.51	138.98	148.24	1.00	95.28	377.49	741.84	1.00
Return/risk	−0.00	−0.08	−0.15	−0.01	−0.01	−0.07	−0.12	−0.00
Profit factor	0.71	0.04	0.02	0.20	0.19	0.02	0.01	0.96
Max. drawdown (%)	99.98	100.00	100.00	99.98	99.92	100.00	100.00	98.53
Success rate (%)	43.14	19.86	11.58	25.21	20.80	10.40	4.80	56.80

Cumulative return is the average, log-cumulative return across all digital assets for the corresponding technical indicator. Continuous returns are defined in the range from $-\infty$ to $+\infty$, which means that even values significantly smaller than -100% can occur. Average daily return refers to the average returns of all digital assets for the corresponding technical indicator in the time series of returns. Standard deviation is the standard deviation of the average daily returns. The risk/return ratio is also calculated from these two values. The average number of trades denotes the mean value of the number of all trades per virtual currency. Profit factor is the sum of all positive cumulative returns in relation to the sum of all negative cumulative returns. The maximum drawdown refers to the average daily returns. The success rate measures the number of all digital assets with a positive cumulative return compared to the total number of assets examined

Table 11.4 CoinGecko digital asset investment styles

	Mean (%)	*t*-stat	SD (%)	Ret/risk	Skewness	Kurtosis
Panel A						
Reversal	2.51	(1.17)	18.55	0.14	0.54	5.78
Reversal st	10.88	(4.10)[c]	24.55	0.44	−0.47	3.98
Amihud	7.85	(2.88)[c]	23.96	0.33	0.71	6.00
Volume	8.87	(3.37)[c]	23.27	0.38	0.78	6.00
Low vol	7.21	(1.79)[a]	35.12	0.21	0.26	6.84
Low vol st	2.40	(0.61)	29.98	0.08	0.09	3.92
Low beta	−3.73	(−0.40)	83.25	−0.04	0.17	2.90
Low beta st	−0.60	(−0.23)	23.92	−0.03	−0.23	4.90
Momentum	−7.06	(−2.74)[c]	22.19	−0.32	−1.28	7.15
Size	15.29	(3.75)[c]	38.05	0.40	2.64	20.82
Mispricing	−15.28	(−6.43)[c]	19.52	−0.78	−0.47	4.06
Idio risk	1.24	(0.69)	15.54	0.08	0.52	4.17
Explained	1.77	(1.19)	17.16	0.10	−0.95	3.68
Nominal	−22.28	(−6.96)[c]	29.79	−0.75	0.01	6.91
Panel B						
Reversal	2.98	(1.16)	22.17	0.13	0.29	4.43
Reversal st	10.38	(2.65)[c]	36.34	0.29	0.57	5.56
Amihud	4.49	(1.37)	28.90	0.16	0.37	4.40
Volume	5.15	(1.79)[a]	25.58	0.20	0.79	4.81
Low vol	−2.60	(−0.84)	26.99	−0.10	−0.52	4.72
Low vol st	−5.31	(−1.57)	33.78	−0.16	0.02	4.00
Low beta	−1.40	(−0.19)	66.02	−0.02	0.06	2.94
Low beta st	0.24	(0.02)	31.18	0.01	0.09	2.65
Momentum	−9.17	(−2.69)[c]	29.37	−0.31	−0.66	4.07
Size	12.22	(2.79)[c]	40.77	0.30	2.19	16.81
Mispricing	−16.84	(−4.37)[c]	32.45	−0.52	−0.69	3.27
Idio risk	0.99	(0.50)	17.01	0.06	−1.04	5.01
Explained	0.91	(0.41)	19.20	0.05	−0.61	3.62
Nominal	−13.00	(−3.55)[c]	34.16	−0.38	−0.46	5.42

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
The rows represent the investment styles under consideration, with "st" denoting the short-term variants. Mean is the equally weighted monthly average hedge portfolio return in %. *t*-stat is the *t*-statistics of the mean in parentheses. Standard deviation (SD) represents the volatility of the monthly mean returns in %. Ret/risk is the mean divided by the standard deviation. Skewness and kurtosis are the third and fourth return moments

Interestingly, the results remain relatively constant for this analysis as well. Nonetheless, there are some surprises. Short-term reversal, for instance, yields a significantly positive mean monthly hedge portfolio return of 10.88% and 10.38% for Panels A and B, respectively. When using the full www.coinmarketcap.com dataset, it only yields an insignificant 1.30%. Glas (2019), on the other hand, found a

significant short-term reversal style as well, which again hints at differing results for different datasets and varying periods of time. The standard deviation (in Panel A) is in line with the result reported in Chap. 8 and exhibits a slightly negative skewness with a relatively normal kurtosis. These return moments are more pronounced for the short-term reversal returns based on the narrower Panel B. Mean monthly hedge portfolio returns of volume are positive again as well, with monthly average returns of 8.87% in Panel A, significant at the 1% level, and a statistically significant 5.15% for Panel B. The mean returns are comparable in magnitude to the outcomes reported in Chap. 8 and by Glas (2019).

What is really striking, however, is the positive and significant performance of the Amihud measure (in Panel A). It produces mean monthly hedge portfolio returns comparable to volume and is statistically significant at the 1% level as well. The same applies to the result based on Panel B, which is not statistically significant, though. That comes as a bit of surprise. With all else equal, the sign of the Amihud investment style return reverses from negative (see Chap. 8) to positive (see Table 11.4). At this point, it becomes apparent once more that some styles seem to be dependent on the applied dataset. With a much broader and longer dataset, the Amihud investment style produces positive *and* statistically significant returns, which is in line with the investment style literature.

A similar outcome applies to low volatility, which yields positive mean monthly hedge portfolio returns, but statistically significant only at the 10% level (in Panel A). I would deem that result not very reliable, since the sign of low volatility changes directions with different-sized classes (see Chap. 8), which is also reported in a slightly different form by Glas (2019). Also, that positive result disappears when looking at Panel B. Another not very surprising outcome is that of momentum. When using a larger dataset, the negative momentum returns, which have been reported before, are now actually statistically significant (at the 1% level for both panels). In that case, a larger or higher-quality dataset seems to assist in arriving at clearer results.

Also not very surprising are the outcomes of mispricing, size, and nominal price. All three exhibit strong statistically significant (at the 1% level) hedge portfolio returns (for both panels), which are even more pronounced in magnitude. That again indicates better performance of young and small capitalized digital assets as well as a reversal effect, which are some of the main findings of this book. In the next section, I again test whether these characteristics are able to predict cross-sectional returns and thereby control for the smallest digital assets. Up to now, the previously described outcomes were robust to the usage of two alternative data panels.

11.4 Portfolio Sorts Based on Predictions

It remains of interest to see whether the characteristics are actually priced. That is what I aimed for in Chap. 9. The outcome of that chapter was basically that only characteristics (if at all) but no covariances are priced. Of the 14 characteristics

investigated there, only two to five characteristics appear to be priced statistically significant.[5] Green et al. (2017) proposed a multivariate Fama and MacBeth (1973) test to see which characteristics are not subsumed by others. I applied the same analysis in a slightly adjusted form to the www.coingecko.com data below (see Table 11.5). I included all "traditional" characteristics as known from the equity (and other asset classes) literature. BTC straddles and lookback straddles were not significant in the prior analysis, which is why I included the mispricing factor, idiosyncratic volatility, and explained variation instead. Also, I could not obtain data on whether a digital asset is mineable or not.

Green et al. (2017) proposed calculating the VIFs of various characteristics (investment style returns) first and excluding all characteristics with values >7. All remaining characteristics can then be used in a multivariate FM regression. Since all the VIFs in Table 11.5 are smaller than 7, all characteristics are included in the regression framework. As can be seen in the table, the significance shown in Chap. 9 disappears across all size classes. This means that the already weak results cannot be supported from an asset pricing perspective.

Due to these weak results, I did not run the same analysis for Panel B. All the assets in Panel B are included in Panel A, which means more importance for the results obtained based on Panel A. Panel B is smaller in size and suffers from a survivorship bias; it is therefore not a representative dataset for that kind of analysis.

Next to the strict rules of the asset pricing literature, it is important to explore whether these characteristics can be applied as predictors for financial investment decisions. Therefore, the same FM regression (as shown in Table 11.5) can be used to predict the returns of each digital asset and form portfolios by going long on the quintiles with the highest predicted returns and shorting the lowest predicted returns (see Table 11.6). If one includes all 14 characteristics in such a regression framework, the results are disappointing. That strategy loses 3.19% per month on average and is not statistically significant. As reported in Chap. 9, portfolios formed on volume, size, and nominal price (and their combinations) yield positive and significant results. The same holds true for the CoinGecko Panel A. All three achieve statistically and economically significant average hedge portfolio returns in univariate and multivariate settings (variables 5, 11, and 17). The results are robust even after controlling for tiny digital assets and are in line with Chap. 9.

Repeating the same analysis for Panel B yields comparable outcomes (see Table 11.7). Including all 14 characteristics at once produces positive but statistically insignificant mean monthly returns. Combinations of volume, size, and nominal price, on the other hand, result in positive and statistically significant mean monthly returns (in multivariate cases). The results for volume (variable 5) are a bit weaker in a univariate framework, which means statistical significance is lost in Panel B. The outcomes for size and nominal price, though, remain the same as in Chap. 9. Also, variable 3 (short-term reversal) is able to significantly predict the returns of the assets in Panel B. Thus, some impressions regarding short-term

[5] Depending on the size of the digital assets investigated (see Table 9.6 for more details).

Table 11.5 CoinGecko variance inflation values and multivariate FM regression results

Variable	VIF	All	All but tiny	Tiny
2	3.13	0.0021	−0.0148	0.0317
		(0.04)	(−0.07)	(0.43)
3	2.08	−0.0407	−0.0430	0.1134
		(−0.88)	(−1.03)	(0.46)
4	1.40	−0.3042	4.7747	16.87
		(−0.15)	(0.15)	(0.27)
5	4.90	−0.0036	−0.0047	0.0044
		(−0.18)	(−0.08)	(0.72)
6	3.15	−0.0076	0.0486	−0.2031
		(−0.20)	(0.14)	(−0.55)
7	5.02	0.1530	0.1072	−0.1837
		(0.37)	(0.23)	(−0.10)
8	1.81	−0.0012	0.0022	−0.0389
		(−0.25)	(0.20)	(−0.20)
9	2.05	0.0160	−0.1064	0.2619
		(0.16)	(−0.13)	(0.72)
10	4.00	0.0109	0.0113	−0.1871
		(0.14)	(0.14)	(−0.58)
11	2.25	0.0089	0.0071	−0.0757
		(0.18)	(0.14)	(−0.10)
14	2.25	−0.626	−0.0119	0.8967
		(−0.43)	(−0.26)	(0.49)
15	1.40	−0.2838	−0.2812	0.6571
		(−0.27)	(−0.32)	(0.26)
16	1.23	0.1518	0.0908	0.6203
		(0.28)	(0.29)	(0.30)
17	5.01	−0.0043	−0.0029	0.0236
		(−0.20)	(−0.14)	(0.58)
Adjusted R^2		22.75	21.88	18.14

This table presents the VIFs and FM regression results for the characteristics identified. The multivariate FM coefficients are reported for every characteristic, with the respective t-value in parentheses below. Variables 2–17 are in line with the numbering in Table 9.2. VIF is the variance inflation factor for each variable over the full data history. I exclude characteristics with a VIF > 7 from the rolling multivariate FM regression analysis. Furthermore, I split the dataset (Panel A) in line with Green et al. (2017). "All" includes all digital assets. "All but tiny" excludes all digital assets with market capitalizations smaller than the 20th percentile of the dataset. "Tiny," on the other hand, only contains digital assets with market capitalizations less than or equal to the 20th percentile of the respective dataset. Adjusted R^2 is reported in %

Table 11.6 CoinGecko portfolio sorts based on predicted returns, Panel A

Long/short quintile portfolios	Mean (%)	*t*-stat	SD (%)	Return/risk	Skewness	Kurtosis
All						
14 characteristics	−3.19	(−1.20)	21.94	−0.15	−0.31	4.97
3, 4, 5, 6, 9, 10, 11, 17	0.28	(0.07)	32.78	0.01	−3.02	20.02
5, 9, 17	7.60	(3.89)[c]	16.45	0.46	1.98	10.68
5, 6, 11, 17	5.14	(2.68)[c]	16.26	0.32	0.07	3.58
8, 11	−1.53	(−0.39)	33.50	−0.05	0.79	4.81
3 only	1.19	(0.68)	11.56	0.10	−0.02	2.81
4 only	1.12	(0.42)	17.79	0.06	0.26	4.09
5 only	5.46	(2.42)[b]	16.95	0.32	0.63	4.92
6 only	4.18	(0.84)	39.21	0.11	−0.84	7.55
9 only	0.33	(0.22)	12.44	0.03	0.25	4.57
10 only	2.50	(1.04)	20.67	0.12	0.66	3.76
11 only	5.83	(2.61)[b]	16.67	0.35	0.40	3.15
17 only	10.41	(5.06)[c]	17.09	0.61	1.69	8.61
5, 17	8.81	(3.61)[c]	17.17	0.51	0.52	8.58
11, 17	7.65	(4.81)[c]	13.26	0.58	0.75	5.68
5, 11	3.86	(1.77)[a]	15.43	0.25	0.94	5.32
5, 11, 17	6.35	(3.77)[c]	14.17	0.45	0.90	6.03
All but tiny						
14 characteristics	−1.25	(−0.35)	29.66	−0.04	−3.33	22.40
3, 4, 5, 6, 9, 10, 11, 17	1.36	(0.33)	33.91	0.04	−2.32	17.27
5, 9, 17	3.64	(1.92)[a]	16.18	0.23	1.82	7.65
5, 6, 11, 17	4.75	(1.99)[b]	20.28	0.23	1.32	9.37
8, 11	0.35	(0.03)	34.43	0.01	0.51	3.00
3 only	0.57	(0.18)	11.69	0.05	0.01	3.76
4 only	1.66	(0.51)	21.77	0.08	1.34	7.82
5 only	5.16	(2.53)[b]	15.72	0.33	1.27	5.64
6 only	−4.78	(−1.02)	37.56	−0.13	−2.31	11.21
9 only	0.91	(0.24)	10.46	0.09	0.10	3.23
10 only	0.87	(0.40)	18.52	0.05	0.20	4.12
11 only	9.07	(2.31)[b]	30.33	0.30	0.58	5.07
17 only	5.35	(2.53)[b]	17.93	0.30	2.54	11.58
5, 17	4.20	(2.23)[b]	16.02	0.26	2.06	9.28
11, 17	7.42	(3.44)[c]	18.21	0.41	0.80	4.63

(continued)

Table 11.6 (continued)

Long/short quintile portfolios	Mean (%)	*t*-stat	SD (%)	Return/risk	Skewness	Kurtosis
5, 11	5.57	(1.88)[a]	22.02	0.25	1.01	5.58
5, 11, 17	5.79	(2.85)[c]	17.21	0.34	0.97	4.03

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
This table includes the return moments of a dollar-neutral long/short portfolio. The hedge portfolio is formed by going long for the top quintile and short for the bottom quintile from a portfolio sort based on predicted returns. The returns are predicted in accordance with Lewellen (2015). I sort for high predicted returns in all cases. The quintile portfolios are equally weighted. For the FM coefficients, I used the average FM coefficient over a rolling 12-month lookback period to predict the returns. "Mean" is the monthly hedge portfolio return in %. "*t*-stat" is the *t*-value of the mean monthly returns below in parentheses. "SD" is the standard deviation of the mean monthly hedge portfolio returns in %. I calculated the risk-adjusted return "return/risk" by dividing the return by the standard deviation. "Skewness" and "kurtosis" are the third and fourth return moments

reversal described in the previous chapters, and also in the findings of Glas (2019), are evident at this point once again.

11.5 Portfolio Construction Based on Characteristics

Even if the prior section weakens the asset pricing results of this book, it still supports the investment strategy findings. Thus, we need to check whether that also applies to long-only portfolio construction approaches. In Chap. 10, I applied four different long-only investing approaches: a simple buy-and-hold strategy, characteristics mix, characteristics integration (Bender and Wang 2016), and parametric portfolio policy (Brandt et al. 2009). All but the buy-and-hold strategy incorporate the characteristics identified above (volume, size, and nominal price) into the portfolio construction process. So, the aim is to outperform a naive benchmark (buy and hold) by the incorporation of characteristics.

As shown in Table 11.8, only the mix and integrated approaches are capable of outperforming the buy-and-hold benchmark when using the broader CoinGecko dataset (Panel A). For Panel B (see Table 11.9), the mix and integrated approaches result in statistically significant positive alphas as well. The parametric portfolio policy, on the other hand, achieves the highest CER for both starting weights based on Panel B.[6] The parametric portfolio policy alpha with regard to the cap-weighted starting weights is negative and statistically significant for both panels. The signs and values of the thetas are also similar to the results described in Chap. 10. Only

[6] I again used a relative risk aversion value (γ) of 5.

Table 11.7 CoinGecko portfolio sorts based on predicted returns, Panel B

Long/short quintile portfolios	Mean (%)	*t*-stat	SD (%)	Return/risk	Skewness	Kurtosis
All						
14 characteristics	2.97	(1.02)	24.08	0.12	0.35	4.50
3, 4, 5, 6, 9, 10, 11, 17	4.19	(1.29)	26.79	0.16	0.60	4.92
5, 9, 17	4.83	(1.73)[a]	23.74	0.20	0.81	5.51
5, 6, 11, 17	7.75	(2.70)[c]	24.36	0.32	0.31	3.00
8, 11	−2.99	(−0.56)	45.84	−0.07	0.16	3.07
3 only	6.87	(2.69)[c]	21.66	0.32	−0.15	2.58
4 only	−2.11	(−0.81)	22.44	−0.09	−0.16	4.94
5 only	3.70	(1.55)	20.38	0.18	0.49	4.04
6 only	4.35	(1.36)	27.40	0.16	0.33	3.69
9 only	1.38	(0.58)	20.57	0.07	0.31	3.62
10 only	6.10	(1.61)	32.28	0.19	1.25	5.25
11 only	7.74	(2.58)[b]	25.49	0.30	0.64	3.68
17 only	13.17	(4.02)[c]	27.50	0.48	1.54	7.99
5, 17	9.54	(3.41)[c]	23.59	0.40	1.56	7.51
11, 17	10.75	(4.03)[c]	22.39	0.48	0.67	5.11
5, 11	5.59	(2.08)[b]	22.85	0.24	0.24	2.71
5, 11 & 17	5.66	(1.95)[a]	24.68	0.23	0.28	3.78

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
This table presents the return moments of a dollar-neutral long/short portfolio. The hedge portfolio is formed by going long for the top quintile and short for the bottom quintile from a portfolio sort based on predicted returns. The returns are predicted in accordance with Lewellen (2015). I sorted for high predicted returns in all cases. The quintile portfolios are equally weighted. For the FM coefficients, I used the average FM coefficient over a rolling 12-month look-back period to predict the returns. "Mean" is the monthly hedge portfolio return in %. "*t*-stat" is the *t*-value of the mean monthly returns below in parentheses. "SD" is the standard deviation of mean monthly hedge portfolio returns in %. I calculated the risk-adjusted return "return/risk" by dividing the return by the standard deviation. "Skewness" and "kurtosis" are the third and fourth return moments

the magnitude of the average return, using the cap-weighted starting weights, is clearly less than that in Chap. 10.

Robust results can also be reported for characteristics mix and integration. As expected, characteristics integration outperforms the characteristics mix (based on Panel A), and both approaches achieve positive and statistically significant alphas over the naive benchmark as well. The outcomes of characteristics mix and integration are slightly better (in terms of statistical significance) for the capitalization weighted starting weights based on Panel B than reported in Chap. 10. The betas and adjusted R^2 are also comparable to the results of Chap. 10. The mix and integrated approaches perform better when applied to capitalization weights than that in Chap. 10, as well.

Table 11.8 CoinGecko characteristics investing in digital assets, Panel A

	Equally weighted				Capitalization weighted			
Statistics	BM	Mix	Integrated	Parametric	BM	Mix	Integrated	Parametric
θ_{Nominal}				0.35				−0.11
θ_{Volume}				−6.62				−0.33
θ_{Size}				9.99				0.02
Std(θ_{Nominal}) (%)				185.28				7.91
Std(θ_{Volume}) (%)				250.81				15.71
Std(θ_{Size}) (%)				0.50				13.34
$Varcoeff_{\text{Nominal}}$				5.36				−0.72
$Varcoeff_{\text{Volume}}$				−0.38				−0.47
$Varcoeff_{\text{Size}}$				0.00				7.26
$\bar{w}_i$ (%)	0.03	0.29	0.29	0.88	0.31	0.31	0.33	0.79
min w_i (%)	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
max w_i (%)	1.11	3.59	3.50	21.12	93.72	20.35	16.33	78.30
$\mu(r)$ (%)	−5.93	−3.72	−2.82	−6.97	4.98	10.83	11.38	1.57
$\sigma(r)$ (%)	27.99	28.69	28.54	29.49	26.11	27.94	27.60	23.44
$\mu(r)/\sigma(r)$	−0.21	−0.13	−0.10	−0.24	0.19	0.39	0.41	0.07
$\overline{Nominal}$	−0.00	−0.00	−0.00	−0.01	0.00	−0.01	−0.01	−0.00
$\overline{Volume}$	0.00	0.00	0.00	0.01	0.01	0.02	0.02	0.01
$\overline{Size}$	0.01	0.01	0.01	0.03	0.01	0.03	0.03	0.02
α (%)		2.25	3.12	−1.24		6.80	7.53	−2.74
T(α)		3.41	4.99	−1.24		3.16	3.40	−4.53
P(α)		0.00	0.00	0.22		0.00	0.00	0.00
β_{BM}		1.01	1.00	0.97		0.81	0.77	0.87
Adjusted R^2 (%)		96.15	96.50	91.84		56.46	52.82	95.10
CER		0.23	0.30	−0.02		0.07	0.11	−0.05

This table reports the mean θs of the parametric portfolio policy for the three characteristics, nominal price, volume, and size. The three lines below represent the standard deviation of the θs, and the next three lines are the variation coefficients calculated as $\frac{\sigma_\theta}{\mu_\theta}$, where σ is the standard deviation of θ and μ the mean of θ. The next three lines include information regarding the portfolio weights. In that case, average, minimum, and maximum weights are reported. Next, performance information is described. $\mu(r)$ is the average return, $\sigma(r)$ is the standard deviation of returns, and the return/risk relationship is also reported. Average characteristics of the portfolios are included as $\overline{Nominal}$, $\overline{Volume}$, and $\overline{Size}$. The last five lines contain information regarding Jensen's alpha regression where I compare the different strategies to the respective equally weighted or market capitalization weighted benchmark. The regression can be read as $r_i - r_f = \alpha_i + \beta_i(r_{BM} - r_f) + \epsilon_i$, where r_i is the return of one of the respective strategies, r_f is the risk-free rate, and r_{BM} is the return of the benchmark model. Moreover, I report the t- and p-values of α, as well as the β and the adjusted R^2. CER describes the difference between the certainty-equivalent returns of the respective portfolio construction approach and the benchmark portfolio using a power utility function (see Eq. 10.5) and a relative risk aversion (γ) value of 5

Table 11.9 CoinGecko characteristics investing in digital assets, Panel B

	Equally weighted				Capitalization weighted			
Statistics	BM	Mix	Integrated	Parametric	BM	Mix	Integrated	Parametric
θ_{Nominal}				2.10				−0.02
θ_{Volume}				−3.47				−0.18
θ_{Size}				10.00				0.02
Std(θ_{Nominal}) (%)				259.43				3.27
Std(θ_{Volume}) (%)				422.39				13.58
Std(θ_{Size}) (%)				0.00				9.47
$Varcoeff_{\text{Nominal}}$				1.24				−1.57
$Varcoeff_{\text{Volume}}$				−1.22				−0.75
$Varcoeff_{\text{Size}}$				0.00				4.46
$\bar{w}_i$ (%)	0.80	0.85	0.85	2.29	0.95	0.95	0.98	1.95
min w_i (%)	0.80	0.10	0.22	0.01	0.00	0.00	0.00	0.00
max w_i (%)	1.47	3.29	3.38	33.54	94.85	27.28	20.32	90.26
$\mu(r)$ (%)	−1.47	1.49	1.87	−3.40	4.34	10.16	9.58	2.33
$\sigma(r)$ (%)	28.40	28.52	28.67	27.70	24.33	32.81	32.60	22.34
$\mu(r)/\sigma(r)$	−0.05	0.05	0.07	−0.12	0.18	0.31	0.29	0.10
$\overline{Nominal}$	−0.04	−0.04	−0.04	−0.01	0.05	−0.04	−0.04	0.04
$\overline{Volume}$	0.06	0.05	0.05	0.16	0.16	0.08	0.08	0.16
$\overline{Size}$	0.11	0.11	0.11	0.22	0.19	0.14	0.14	0.19
α (%)		2.96	3.34	−2.05		5.87	5.26	−1.53
T(α)		6.68	7.76	−2.35		2.25	2.06	−2.97
P(α)		0.00	0.00	0.02		0.03	0.04	0.00
β_{BM}		0.99	1.00	0.93		0.99	1.00	0.90
Adjusted R^2 (%)		98.17	98.29	92.57		53.32	54.89	96.08
CER		0.19	0.24	0.56		−0.48	−0.15	−0.02

This table reports the mean θs of the parametric portfolio policy for the three characteristics, nominal price, volume, and size. The three lines below represent the standard deviation of the θs, and the next three lines are the variation coefficients calculated as $\frac{\sigma_\theta}{\mu_\theta}$, where σ is the standard deviation of θ and μ the mean of θ. The next three lines include information regarding the portfolio weights. In that case, average, minimum, and maximum weights are reported. Next, performance information is described. $\mu(r)$ is the average return, $\sigma(r)$ is the standard deviation of returns, and the return/risk relationship is also reported. Average characteristics of the portfolios are included as $\overline{Nominal}$, $\overline{Volume}$, and $\overline{Size}$. The last five lines contain information regarding Jensen's alpha regression, where I compare the different strategies to the respective equally weighted or market capitalization weighted benchmark. The regression can be read as $r_i - r_f = \alpha_i + \beta_i(r_{BM} - r_f) + \epsilon_i$, where r_i is the return of one of the respective strategies, r_f is the risk-free rate, and r_{BM} is the return of the benchmark model. Moreover, I report the t- and p-values of α, as well as the β and the adjusted R^2. CER describes the difference between the certainty-equivalent returns of the respective portfolio construction approach and the benchmark portfolio using a power utility function (see Eq. 10.5) and a relative risk aversion (γ) value of 5

It should also be mentioned that the mix approach outperforms the integrated approach slightly for capitalization weighted starting weights in Panel B. That was not the case in the previous analyses and seems to be another indication of the dependence of the obtained results on the dataset used. In this specific case, Panel B is a rather special dataset, since it includes only the oldest virtual coins.[7]

11.6 Summary

All in all, it can be said that the results described in Chaps. 1–10 remain valid today. By using another, broader and longer, dataset obtained from www.coingecko.com, I have shown that the main findings of this book continue to be robust two years after the main dataset ended. Comparison with the findings of Chap. 7 is difficult, since not all of the technical indicators can be calculated with the second dataset. The results of the remaining few indicators, however, point in the same direction. Also, the single investment style outcomes are in line with Chap. 8 and Glas (2019). Only the asset pricing findings (in terms of priced characteristics) are weakened by the new dataset. The three characteristics volume, size, and nominal price can still be used to construct profitable investment strategies and/or portfolios. A positive side effect of the broader dataset is the decrease in standard deviation of the strategies, since more digital assets are included in the respective portfolio sorts.

However, it can also be seen that average monthly returns decrease at some points due to the weak performance of the digital asset market after mid-2018. That, on the other hand, has no implications for the outcomes of this section. Next to the validity of the previously described results, it serves as a robustness test. With a second dataset, very different and independent from the first, it can be shown that the findings of this book are not dependent on just one special dataset or data source.[8] This is also a first positive sign for future researchers that results of diverse studies and analyses appear to be robust for different data sources providing data for this young asset class.

[7] At this point, a side note is important. As described by Taleb (2012), things that have existed for a long time tend to stay around for a long time in the future. This is also called the "Lindy effect." It basically states that the life expectancy of a thing, building, or technology is directly proportional to its age (as of today). If you take the pyramids of Egypt as an example, this would mean that they are prone to be still standing in thousands of years. If one applies the Lindy effect to digital assets, then coins like bitcoin are more likely to be around in the future than some ERC-20 tokens.

[8] Even though some results vary when using different datasets or panels.

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Chapter 12
Conclusion



This book empirically has explored several issues regarding asset pricing and investment styles in the young and emerging asset class of digital assets. For comparability reasons, I also set the results of the digital asset outcomes against more traditional asset classes such as equities, bonds, and others. Thereby, I have investigated several well-known and popular topics from the investment and asset pricing literature and applied them to digital assets. One main finding is that digital assets are still relatively heterogenous and shaped by the smallest digital coins. The return behavior of the asset class might also be closest to the random-walk model and cannot (or only partially) be explained by standard asset pricing models. This finding might, on the other hand, also be due to a limited data history and severe data issues.

12.1 Summary

After a theoretical and technical introduction, I described the largest and most prominent digital coins in more detail. Subsequently, I gave a first overview of the digital asset dataset applied in this book. The first empirical application (see Chap. 6) investigated a very basic question: Do digital assets add value when included in a multi-asset allocation framework? The very simple answer to this is no. None of the 15 test assets considered would constantly improve a multi-asset mean–variance portfolio. Only some single digital assets could improve the efficient frontier when added to a single asset class. The same finding applies to an equal risk portfolio construction approach.

Since the inclusion of digital assets into a multi-asset portfolio is not favorable, I focused on the asset class of digital assets itself to find promising investment strategies. A very popular approach of practitioners is technical analysis, which relies on past chart patterns or indicators based on historical data to arrive at

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investment decisions. Therefore, I considered seven different technical indicators and applied them on a large sample of digital assets and US stocks. Of all the technical indicators under consideration, Bollinger bands appear to be the most profitable. This finding, however, can be explained by a short-term reversal effect which has already been described by Glas (2019).

As the two previous time-series-based methods resulted in rather disappointing results, I switched the perspective to cross-sectional strategies. Therefore, I investigated 14 well-known characteristics in the cross-section of digital assets, equities, and bonds. The first results indicated a profitable illiquidity style (as measured by Amihud's (il)liquidity measure and turnover data), a promising mispricing characteristic, and a strong (low) nominal price effect. Those results, on the other hand, need to be read with care, due to a very short data sample. This issue can also be observed when looking at well-known characteristics in traditional asset classes. In equities and bonds, these characteristics produce only insignificant mean monthly returns over the same sample period.

In an attempt to validate the model described by Glas (2019), I again used the same 14 characteristics and several other data sources to search for (priced) risk factors in the returns of digital assets. By applying several statistical methods (like PCA or PLS), I could not arrive at any significant and robustly priced risk factor. Therefore, I dug dipper into the covariances versus characteristics discussion and found that three characteristics are priced, at least to some extent, in the returns of digital assets. These are turnover, nominal price, and the market capitalization of a digital coin. When controlling for the smallest assets within the respective dataset under consideration, on the other hand, this finding is not robust.

Having said that, the three characteristics would not satisfy the strong requirements of the asset pricing literature to be considered "true" risk factors. Thus, I used these findings for investment purposes to see if incorporating characteristics in the portfolio construction process yields any favorable outcome. As such, I applied three different investment approaches that take characteristics into account for portfolio construction and found that such approaches indeed lead to superior returns when measured against a naive benchmark. The outcomes of the respective approaches are also not dependent on the chosen starting weights. Furthermore, the findings described are mostly robust to transaction costs and seem to be practically implementable.

Due to the unbalanced data of digital assets, I needed to verify the results obtained. Therefore, I ran the main analyses of each chapter again with two higher-quality datasets from an alternative data source. One dataset is highly balanced and thus includes most of the oldest virtual coins. Nevertheless, that dataset contains assets of various market capitalizations, which makes that newly created dataset comparable to the previous ones. The other one is clearly broader and longer than the dataset used in most parts of this book. By using these two new samples, the previously described results appear relatively robust. The main findings are still valid today, although slightly less significant (in terms of statistical and economical significance).

Based on the two introductory questions that form the main motivation of this book, the answers seem pretty obvious now.

Can Digital Assets, with the Flagship Asset Bitcoin, Really be Considered a Serious Asset Class?
I would not say that digital assets can be considered a *serious* asset class, but it can be considered as its *own* asset class. Seriousness assumes certain rules and regulations which (still) do not yet exist for digital assets. Moreover, hacks, fraud, and other criminal activities are still a major issue for this asset class and need to be mitigated before it can be called a "serious" asset class. Also, basically all of my analyses have shown that digital assets are not connected to other asset classes and that digital assets cannot be explained by asset pricing models stemming from traditional asset classes or even digital assets themselves. Although well-known characteristics or risk factors have been shown not to exist in this new and emerging asset class, there are the first indications of priced characteristics like illiquidity, size, and nominal price, especially in the smallest virtual coins. Since it is difficult to explain the return behavior of digital assets otherwise, a random walk might be the closest explanation of digital asset returns yet.

Since it Is Possible to Trade Digital Assets, Does it Really Make Sense to Trade or Invest in These Assets?
For institutional multi-asset investors, investments into digital assets might not make very much sense, since digital assets are in the span of more traditional asset classes. For retail investors, an investment should be considered carefully, due to the continued lack of regulations and other concerns like exchange hacks, which could lead to a total loss of the initially invested amount. Sophisticated investors like hedge funds with the necessary technical and practical understanding could benefit from investments into that asset class. For decent investments, many different exchanges and wallets need to be included in the investment process, which requires a lot of maintenance and to some degree programming work. Shorting of digital assets is also not trivial currently and should be done by sophisticated accounts only. Due to the comparably small size of the asset class, (hedge) fund managers also need to consider the potential market impact of orders, which limits the size (and profitability) of eventual funds.

All in all, the findings described need to be put into further context. Due to the continually changing investment universe of digital assets, with the new creation and disappearance of virtual coins basically every day, the datasets for academic work vary constantly. This makes any results less comprehensible and reproducible. Furthermore, different data sources might report different values for the same virtual asset due to various timezones (digital assets can be traded 24 hours a day, every day of the year) or miscellaneous underlying data-gathering methods. Additionally, the length of the available data is still too short to make reliable and unequivocal statements. This issue can be particularly observed in the results of traditional asset classes over the same period of time. That is why this work is meant to be a first explorative investigation into the asset class of digital assets. By doing so, I find that basically none of the well-known effects and investment strategies can be applied to

digital assets, but at least three different important characteristics can be found in the smallest digital coins. This also holds true when utilizing higher-quality datasets.

12.2 Future Research

Based on all of the above-mentioned issues, future research with digital asset data can be devoted to many different areas. The first and probably most important issue to address is the construction of a common and "clean" database that can be accessed by all interested researchers. Since there is a huge turnover of available digital coins (dead and alive), this turnover needs to be captured and brought together in one place (see, for example, the www.coingecko.com dataset). This helps to avoid any possible survivorship bias in the data and makes results more reliable. There are also many opportunities to arbitrage digital assets on various exchanges. As such, it is even more difficult to determine a correct price for one asset at a given point in time. For statistically significant results, a longer data history seems to be necessary but is not currently available.

Assuming a suitable dataset, one of the most important fields of future research might be asset pricing in general and specifically the search for common risk factors in the returns of digital assets. This work has shown that three characteristics are priced at least to some extent in the smallest digital coins. Applying this finding to actual investment purposes yields better performance than a naive benchmark, even after taking into account transaction costs and implementation issues. With a longer data history and perhaps more sophisticated methods of data cleaning, results could become clearer for larger digital coins.

Also, the investment literature regarding digital assets is still in its early stages. For traditional asset classes, hundreds of characteristics or anomalies have been found and described over recent decades. The search has just started in cryptocurrencies, and there is still a long way to go to find promising strategies in this young and emerging asset class.

As Cochrane (2011) put it, it is not necessary to come up with more characteristics or anomalies, but it is important to reduce the known characteristics to arrive at a small set of factors that significantly explain the cross-section of asset returns. Since digital assets seem to be distinct from traditional asset classes, the first step is to identify characteristics. Only afterwards might it become necessary to condense the characteristics and information content to a few independent (risk) factors. Before that is going to happen, however, the construction of a reliable dataset is the first step for conclusive results.

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Appendix A
Digital Asset Data Details I

Below is a list of all the digital assets used in the analyses from www.coinmarketcap.com:

(1) 0chain (2) 0x (3) 0xbtc (4) 1337coin (5) 1world (6) 2give (7) 300-token (8) 42-coin (9) 808coin (10) 8bit (11) abjcoin (12) ac3 (13) accelerator-network (14) acchain (15) ace (16) aces (17) achain (18) acoin (19) acute-angle-cloud (20) adbank (21) adcoin (22) adhive (23) aditus (24) adshares (25) adtoken (26) advanced-internet-blocks (27) adx-net (28) adzcoin (29) aelf (30) aeon (31) aerium (32) aeron (33) aeternity (34) agrello-delta (35) aichain (36) aidcoin (37) aidoc (38) aidos-kuneen (39) aigang (40) aion (41) airswap (42) airtoken (43) akuya-coin (44) alax (45) alis (46) all-sports (47) allion (48) allsafe (49) alphabitcoinfund (50) alphacat (51) alqo (52) altcoin-alt (53) amber (54) amlt (55) ammo-reloaded (56) amon (57) amsterdamcoin (58) anarchistsprime (59) animation-vision-cash (60) animecoin (61) anoncoin (62) antimatter (63) apex (64) aphelion (65) apis (66) appcoins (67) apr-coin (68) apx (69) aquariuscoin (70) aragon (71) arbit (72) arbitracoin (73) arbitragect (74) arcblock (75) arcticcoin (76) ardor (77) argentum (78) argus (79) aricoin (80) ark (81) artbyte (82) artex-coin (83) asch (84) aseancoin (85) asiacoin (86) asiadigicoin (87) aston (88) astro (89) atbcoin (90) atlant (91) atmcoin (92) atmos (93) atn (94) atomic-coin (95) attention-token-of-media (96) auctus (97) audiocoin (98) augur (99) aurora (100) aurora-dao (101) auroracoin (102) aurumcoin (103) authorship (104) autonio (105) avatarcoin (106) aventus (107) avoncoin (108) axiom (109) axpire (110) b2bcoin (111) b2bx (112) b3coin (113) babb (114) banca (115) bancor (116) bank-coin (117) bankcoin (118) bankera (119) bankex (120) banyan-network (121) basic-attention-token (122) bastonet (123) bata (124) bean-cash (125) beatcoin (126) bee-token (127) belacoin

See Chap. 5.

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(128) belugapay (129) benjirolls (130) berncash (131) betacoin (132) betterbetting
(133) bezant (134) bezop (135) biblepay (136) bibox-token (137) bigbom
(138) bigone-token (139) bigup (140) billionaire-token (141) binance-coin
(142) biobar (143) biocoin (144) biotron (145) bipcoin (146) birds (147) bismuth
(148) bit-tube (149) bit20 (150) bitair (151) bitasean (152) bitbar (153) bitbase
(154) bitbay (155) bitbtc (156) bitclave (157) bitcloud (158) bitcny (159) bitcoal
(160) bitcoin (161) bitcoin-21 (162) bitcoin-atom (163) bitcoin-cash
(164) bitcoin-diamond (165) bitcoin-god (166) bitcoin-gold (167) bitcoin-green
(168) bitcoin-interest (169) bitcoin-planet (170) bitcoin-plus (171) bitcoin-private
(172) bitcoin-red (173) bitcoin-scrypt (174) bitcoindark (175) bitcoinfast
(176) bitcoinx (177) bitcoinz (178) bitconnect (179) bitcore (180) bitcrystals
(181) bitcurrency (182) bitdeal (183) bitdegree (184) bitdice (185) biteur
(186) bitgem (187) bitgold (188) bitmark (189) bitok (190) bitquark (191) bitqy
(192) bitradio (193) bitrent (194) bitsend (195) bitserial (196) bitshares
(197) bitshares-music (198) bitsilver (199) bitsoar (200) bitstar (201) bitstation
(202) bitsum (203) bitswift (204) bittwatt (205) bitusd (206) bitvolt (207) bitwhite
(208) bitzeny (209) blackcoin (210) blackmoon (211) blakecoin (212) blakestar
(213) blazecoin (214) blazercoin (215) blitzpredict (216) block-array
(217) blockcat (218) blockcdn (219) blocklancer (220) blockmason
(221) blockmesh (222) blocknet (223) blockpool (224) blockport (225) blocktix
(226) blockv (227) bloomtoken (228) blox (229) bluecoin (230) bluzelle
(231) bnktothefuture (232) bnrtxcoin (233) bodhi (234) bodhi-eth (235) bolenum
(236) bolivarcoin (237) bonpay (238) boolberry (239) boostcoin (240) boscoin
(241) bottos (242) bounty0x (243) boutspro (244) bowhead (245) bowscoin
(246) brahmaos (247) brat (248) bread (249) breakout (250) breakout-stake
(251) briacoin (252) bridgecoin (253) britcoin (254) bt2-cst (255) btcmoon
(256) btctalkcoin (257) bubble (258) budbo (259) bullion (260) bulwark
(261) bumbacoin (262) bunnycoin (263) burst (264) buzzcoin (265) byteball
(266) bytecent (267) bytecoin-bcn (268) bytom (269) c-bit (270) c20
(271) cabbage (272) cachecoin (273) californium (274) callisto-network
(275) campuscoin (276) canada-ecoin (277) candy (278) cannabiscoin
(279) cannacoin (280) cannation (281) canyacoin (282) cappasity (283) capricoin
(284) carboncoin (285) cardano (286) cargox (287) cartaxi-token (288) carvertical
(289) cashaa (290) cashbet-coin (291) cashcoin (292) casinocoin (293) catcoin
(294) cazcoin (295) ccore (296) ceek-vr (297) centrality (298) centurion (299) cfun
(300) chaincoin (301) chainium (302) chainlink (303) chancoin (304) change
(305) chatcoin (306) cheapcoin (307) chesscoin (308) chips (309) chronobank
(310) chronologic (311) cindicator (312) circuits-of-value (313) civic (314) ckusd
(315) clams (316) clearpoll (317) cloakcoin (318) cloud (319) clubcoin
(320) cobinhood (321) coffeecoin (322) cofound-it (323) coimatic-2
(324) coimatic-3 (325) coin (326) coin2-1 (327) coinfi (328) coinlancer
(329) coinmeet (330) coinonat (331) coinonatx (332) coinpoker (333) colossusxt
(334) colu-local-network (335) comet (336) commerceblock
(337) commodity-ad-network (338) compucoin (339) comsa-eth (340) comsa-xem
(341) concoin (342) condensate (343) connectjob (344) consensus

(345) constellation (346) content-and-ad-network (347) content-neutrality-network (348) contractnet (349) copytrack (350) corion (351) cortex (352) coss (353) couchain (354) counterparty (355) coupecoin (356) covesting (357) cpchain (358) crave (359) cream (360) creatio (361) creativecoin (362) credence-coin (363) creditbit (364) credits (365) credo (366) crevacoin (367) cropcoin (368) crowd-machine (369) crowdcoin (370) crown (371) crycash (372) cryptaur (373) cryptcoin (374) crypterium (375) crypto (376) cryptocarbon (377) cryptojacks (378) cryptonex (379) cryptonite (380) cryptopay (381) cryptopiafeeshares (382) cryptoping (383) crystal-clear (384) cube (385) cubits (386) curecoin (387) curriculum-vitae (388) cvcoin (389) cybereits (390) cybermiles (391) cybervein (392) cyder (393) dadi (394) daex (395) dai (396) dalecoin (397) daneel (398) dao-casino (399) daostack (400) darcrus (401) darsek (402) dascoin (403) dash (404) dashcoin (405) data (406) databits (407) datacoin (408) datarius-credit (409) datawallet (410) datum (411) datx (412) daxxcoin (413) debitcoin (414) debitum-network (415) decent (416) decent-bet (417) decentraland (418) decentralized-machine-learning (419) decision-token (420) decred (421) deepbrain-chain (422) deeponion (423) delphy (424) denarius-dnr (425) dent (426) dentacoin (427) dero (428) desire (429) dether (430) deuscoin (431) deutsche-emark (432) devery (433) deviantcoin (434) dew (435) diamond (436) digibyte (437) digicube (438) digifinextoken (439) digipulse (440) digital-developers-fund (441) digital-money-bits (442) digital-rupees (443) digitalcoin (444) digitalnote (445) digitalprice (446) digitex-futures (447) digix-gold-token (448) digixdao (449) dignity (450) dimcoin (451) dimecoin (452) dinastycoin (453) distributed-credit-chain (454) district0x (455) divi (456) dix-asset (457) dmarket (458) dnotes (459) docademic (460) dock (461) dogecoin (462) dollar-online (463) dollarcoin (464) domraider (465) donationcoin (466) dopecoin (467) dorado (468) dotcoin (469) doubloon (470) dovu (471) draftcoin (472) dragon-coins (473) dragonchain (474) dreamcoin (475) dropil (476) droxne (477) drp-utility (478) dubaicoin-dbix (479) dutch-coin (480) dynamic (481) dynamic-trading-rights (482) dynamiccoin (483) e-coin (484) e-dinar-coin (485) e-gulden (486) ea-coin (487) eaglecoin (488) earth-token (489) ebitcoin-cash (490) eboostcoin (491) ebtcnew (492) eccoin (493) echolink (494) ecobit (495) ecocoin (496) edgeless (497) edrcoin (498) edu-coin (499) educare (500) education-ecosystem (501) effect-ai (502) eggcoin (503) egretia (504) eidoo (505) einsteinium (506) ejoy (507) elastic (508) elastos (509) elcoin-el (510) electra (511) electrifyasia (512) electroneum (513) electronic-pk-chain (514) elementrem (515) elixir (516) ellaism (517) eltcoin (518) elysium (519) embercoin (520) embers (521) emerald (522) emercoin (523) emphy (524) encrypgen (525) encryptotel (526) encryptotel-eth (527) endor-protocol (528) energitoken (529) energo (530) energycoin (531) engine (532) enigma-project (533) enjin-coin (534) entcash (535) envion (536) eos (537) eosdac (538) equal (539) equitrader (540) erc20 (541) ergo (542) eroscoin (543) eryllium (544) escoro (545) espers (546) eternity (547) ethbet (548) ethbits (549) ether-zero (550) etherdelta-token (551) etherecash (552) ethereum (553) ethereum-blue (554) ethereum-classic (555) ethereum-dark

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(831) korecoin (832) kronecoin (833) kubera-coin (834) kucoin-shares
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(858) library-credit (859) life (860) lightchain (861) lightning-bitcoin (862) linda
(863) linker-coin (864) linkeye (865) linx (866) lisk (867) litebar (868) litebitcoin
(869) litecoin (870) litecoin-cash (871) litecoin-plus (872) litecoin-ultra
(873) litecred (874) litedoge (875) live-stars (876) local-world-forwarders
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(882) loopring (883) loopring-neo (884) loyalcoin (885) luna-coin (886) lunyr
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(902) mao-zedong (903) marijuanacoin (904) mark-space (905) marscoin
(906) martexcoin (907) marxcoin (908) masari (909) master-swiscoin
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(928) mercury-protocol (929) metal (930) metalcoin (931) metaverse
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(940) minexcoin (941) mintcoin (942) mirq (943) mithril (944) mixin
(945) mktcoin (946) moac (947) mobilego (948) mobius (949) modum
(950) moeda-loyalty-points (951) moin (952) mojocoin (953) molecular-future
(954) monaco (955) monacocoin (956) monacoin (957) monero
(958) monero-classic (959) monero-original (960) monetaryunit (961) monetha

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(1054) opus (1055) oraclechain (1056) orbitcoin (1057) origami
(1058) origin-sport (1059) origintrail (1060) ormeus-coin (1061) ost
(1062) oxycoin (1063) oyster (1064) oyster-shell (1065) pabyosi-coin-special
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Appendix B
Clustering Based on USD Prices

B.1 Euclidean Distances

See Figs. B.1, B.2, B.3 and B.4.

See Chap. 5.

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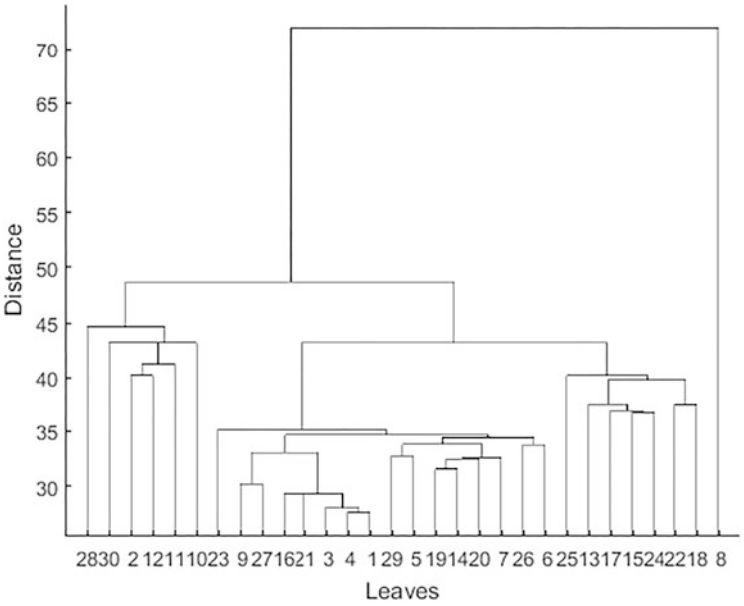


Fig. B.1 Dendrogram shortest distance USD

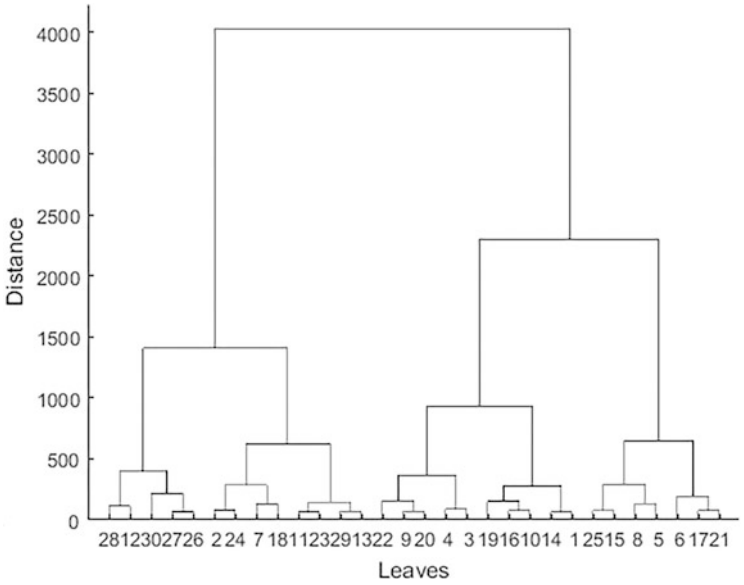


Fig. B.2 Dendrogram squared distance USD

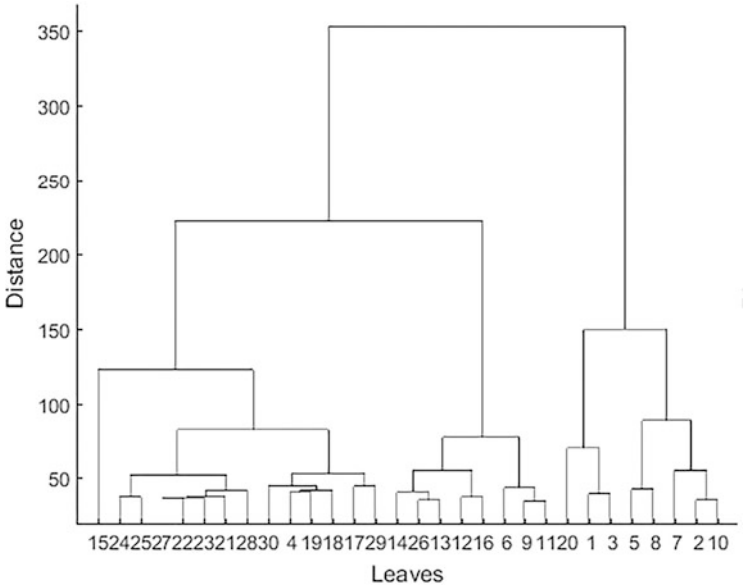


Fig. B.3 Dendrogram weighted distance USD

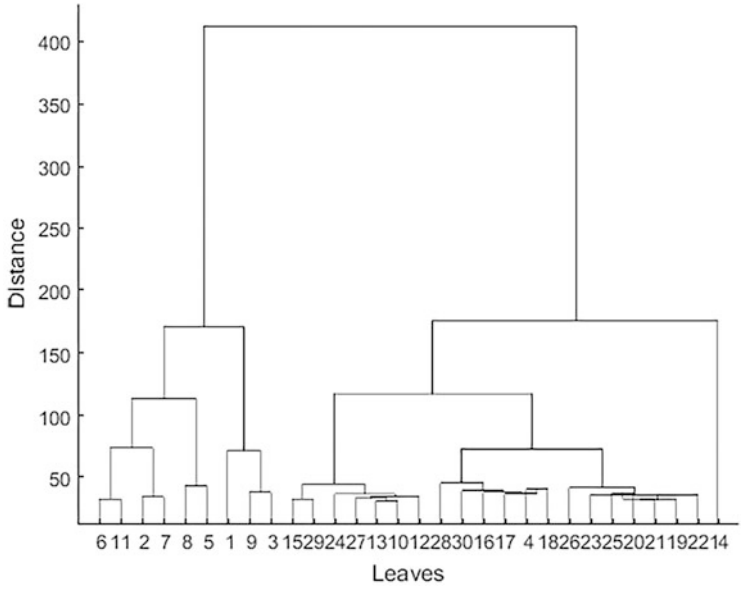


Fig. B.4 Dendrogram centroid distance USD

B.2 Correlations

See Figs. B.5, B.6, B.7 and B.8.

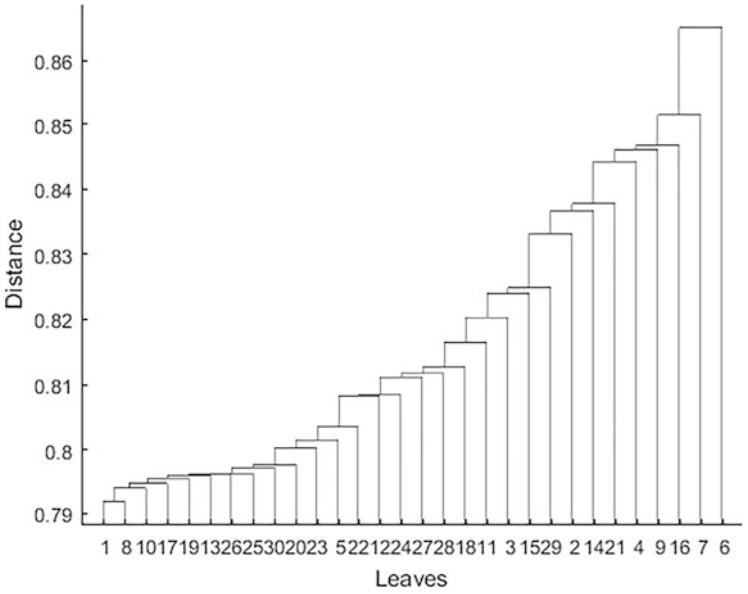


Fig. B.5 Dendrogram shortest distance (2) USD

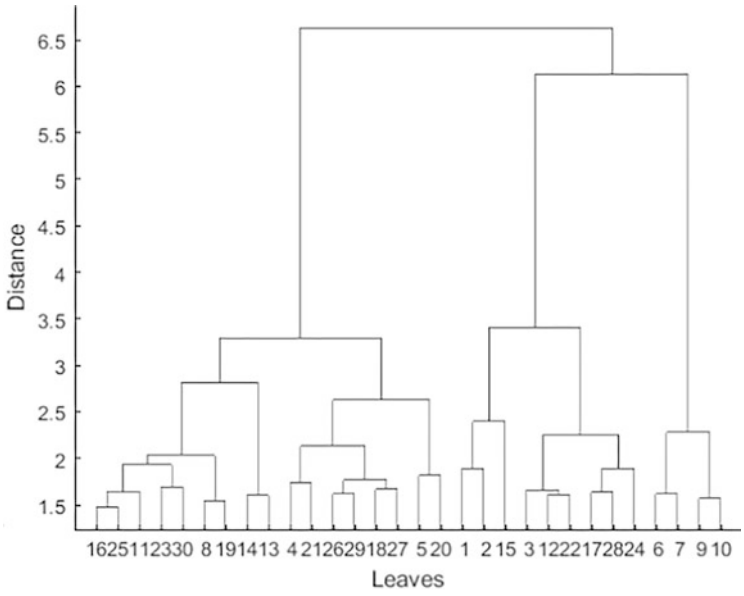


Fig. B.6 Dendrogram squared distance (2) USD

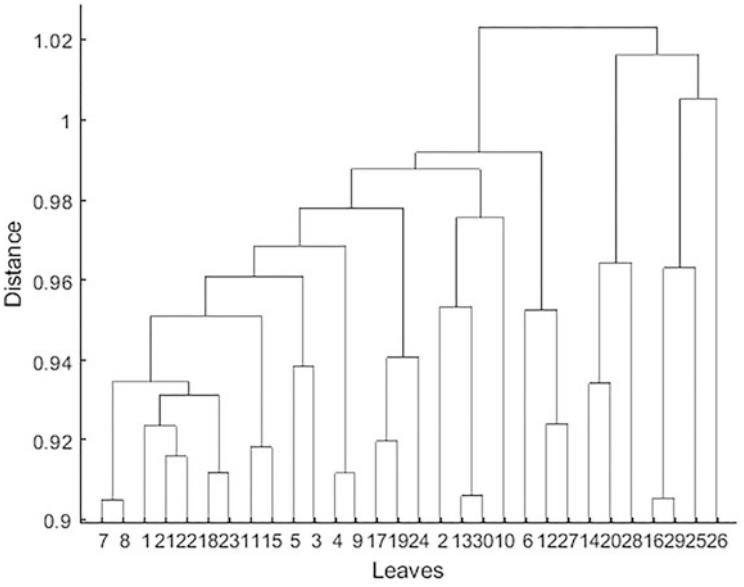


Fig. B.7 Dendrogram weighted distance (2) USD

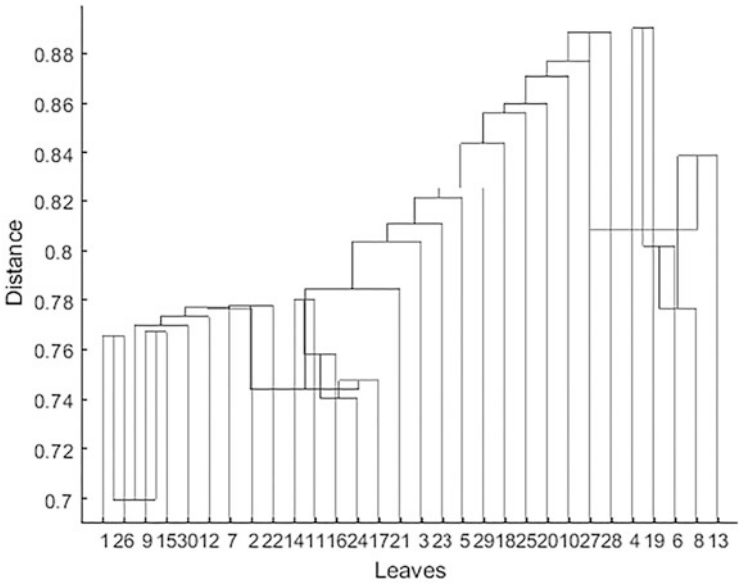


Fig. B.8 Dendrogram centroid distance (2) USD

B.3 Idiosyncratic Risks

See Figs. B.9 and B.10.

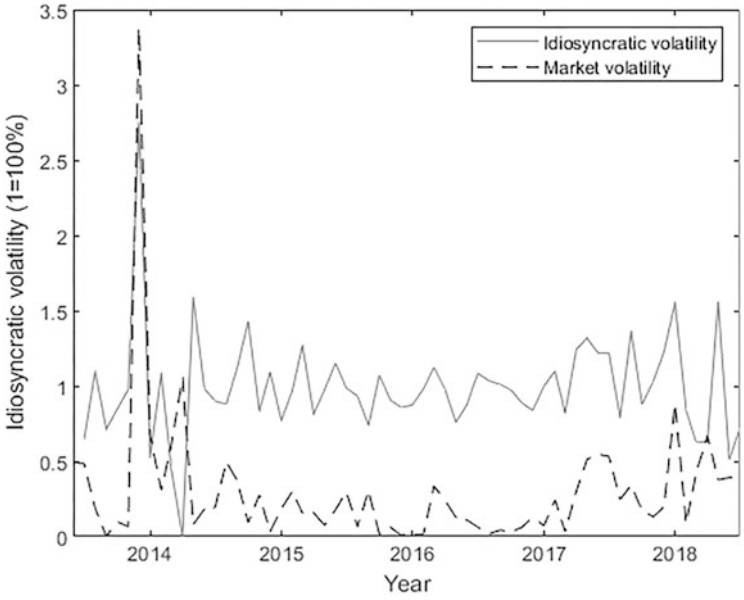


Fig. B.9 IR original data USD

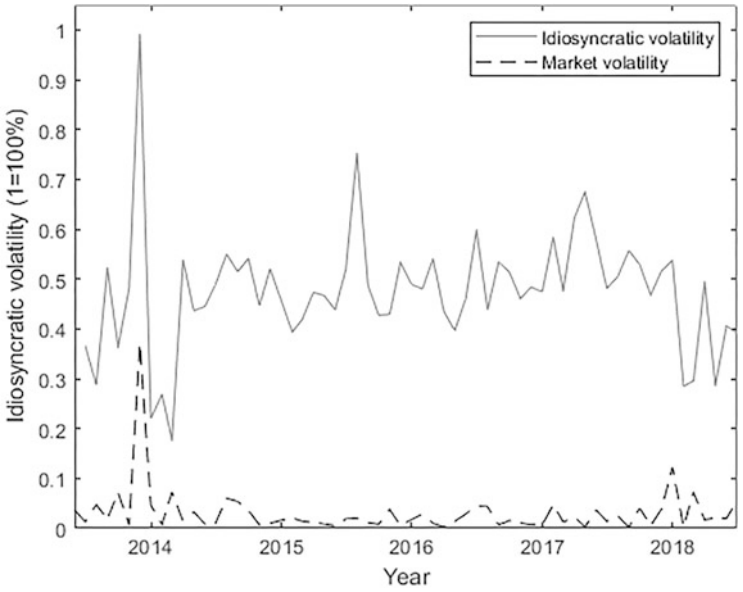


Fig. B.10 IR condensed data USD

Appendix C
Efficient Frontiers and Transition Maps

C.1 Equities

In some of the following figures, the 130/30 portfolio lies on the unconstrained frontier and thus is not observable. For the long-only portfolio, the optimization sometimes yields just two portfolios (Figs. C.1 and C.2).

See Chap. 6.

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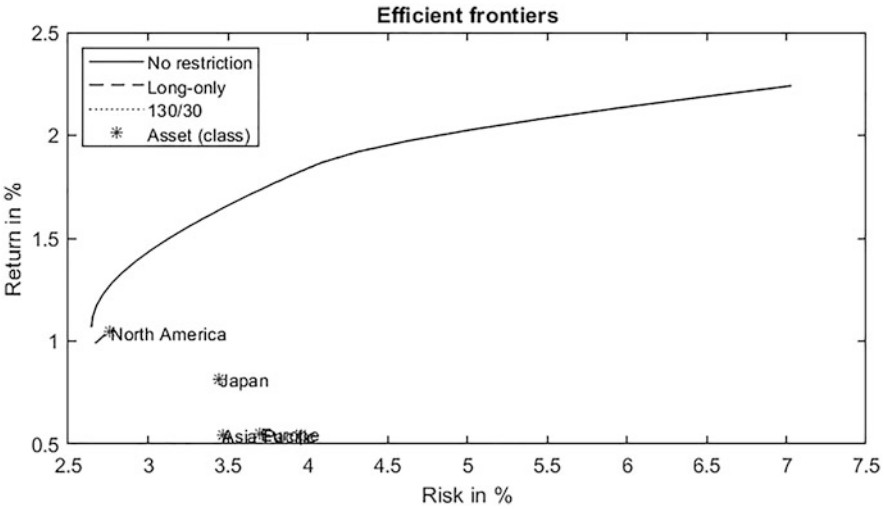


Fig. C.1 Efficient frontiers, equities

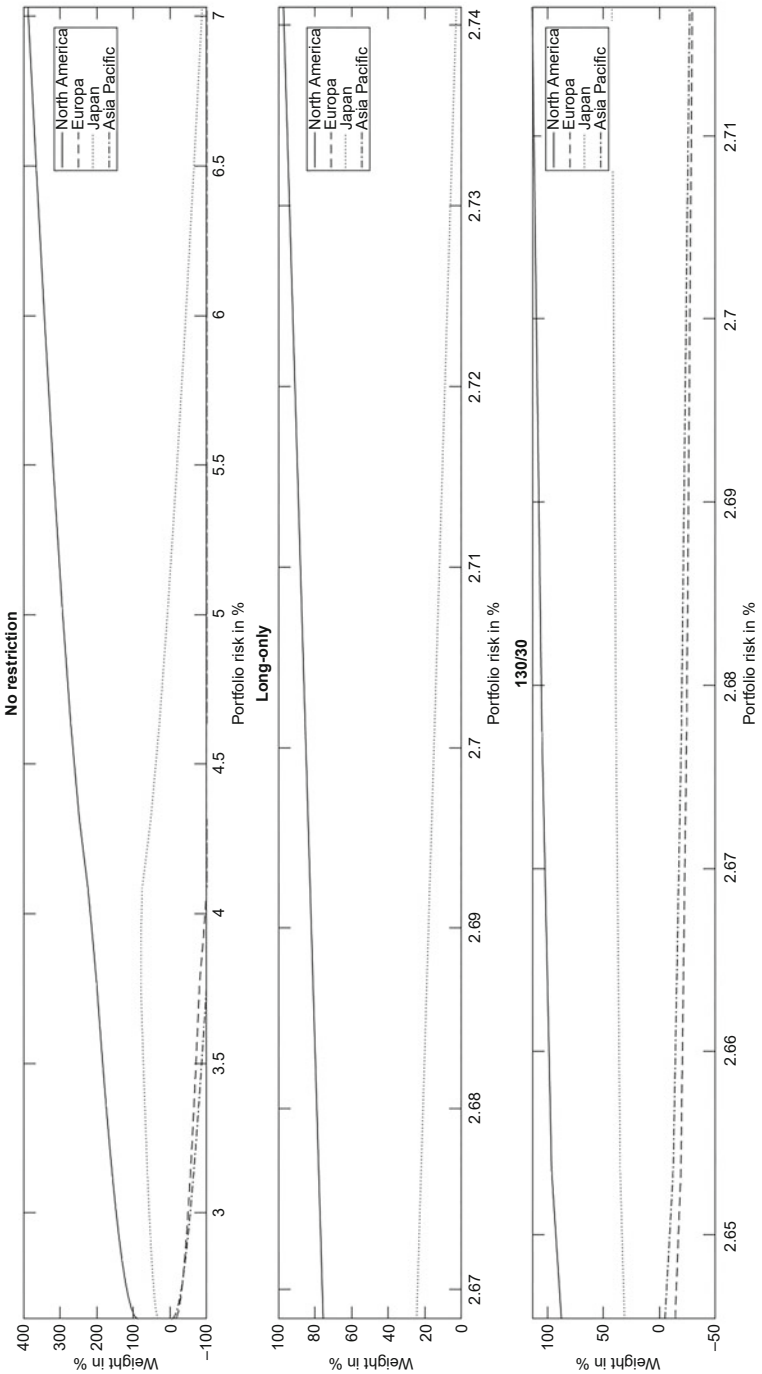


Fig. C.2 Transition map equities

C.2 Bonds

See Figs. C.3 and C.4.

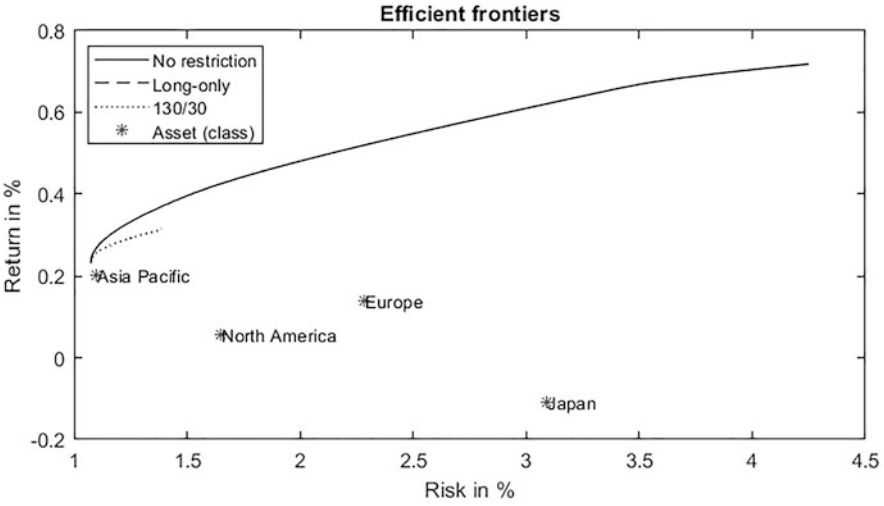


Fig. C.3 Efficient frontiers, bonds

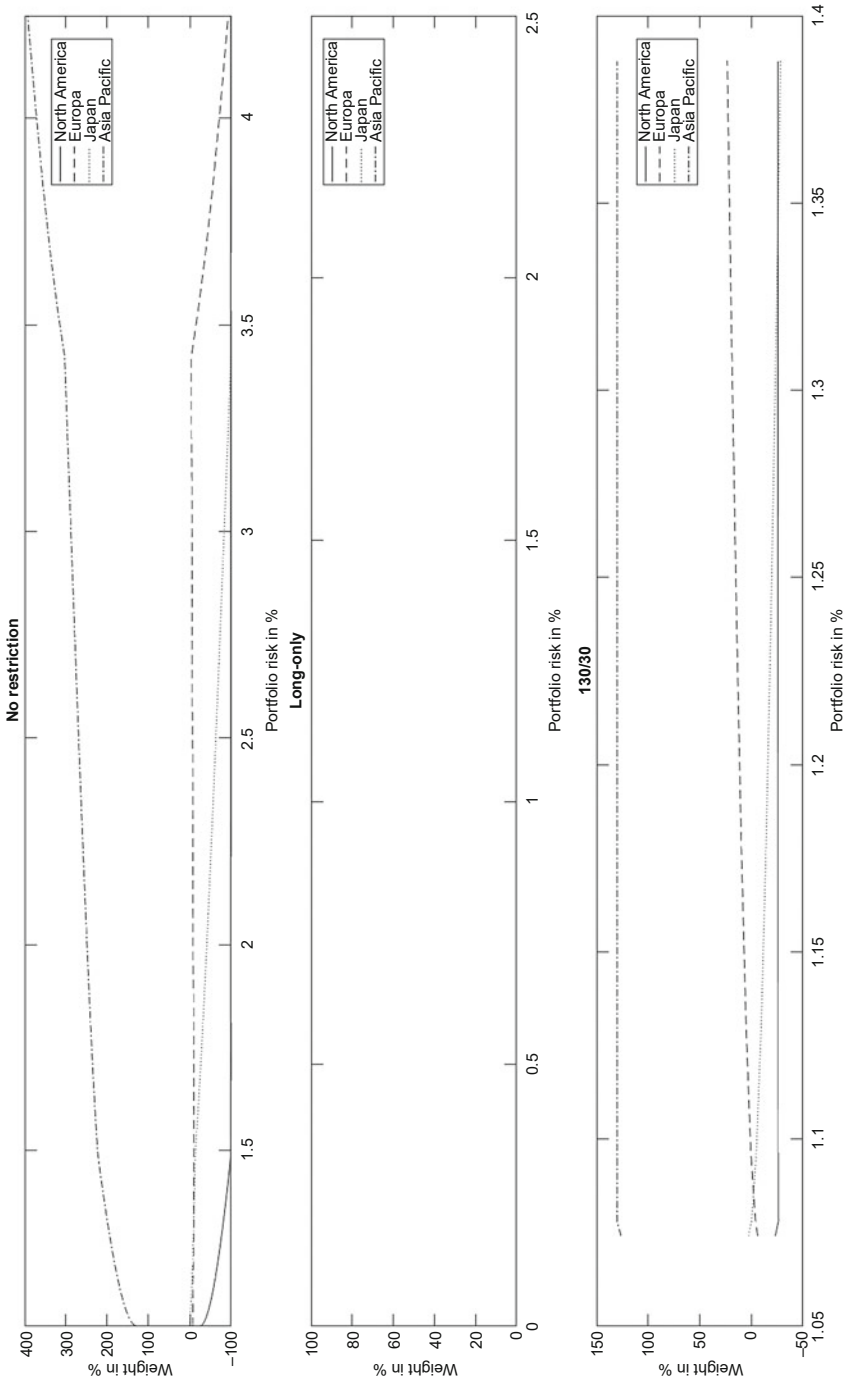


Fig. C.4 Transition map bonds

C.3 Commodities

See Figs. C.5 and C.6.

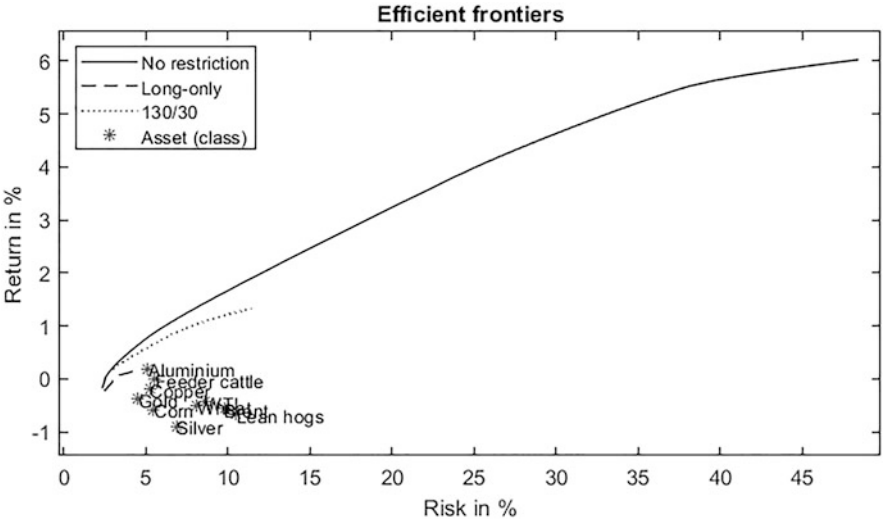


Fig. C.5 Efficient frontiers, commodities

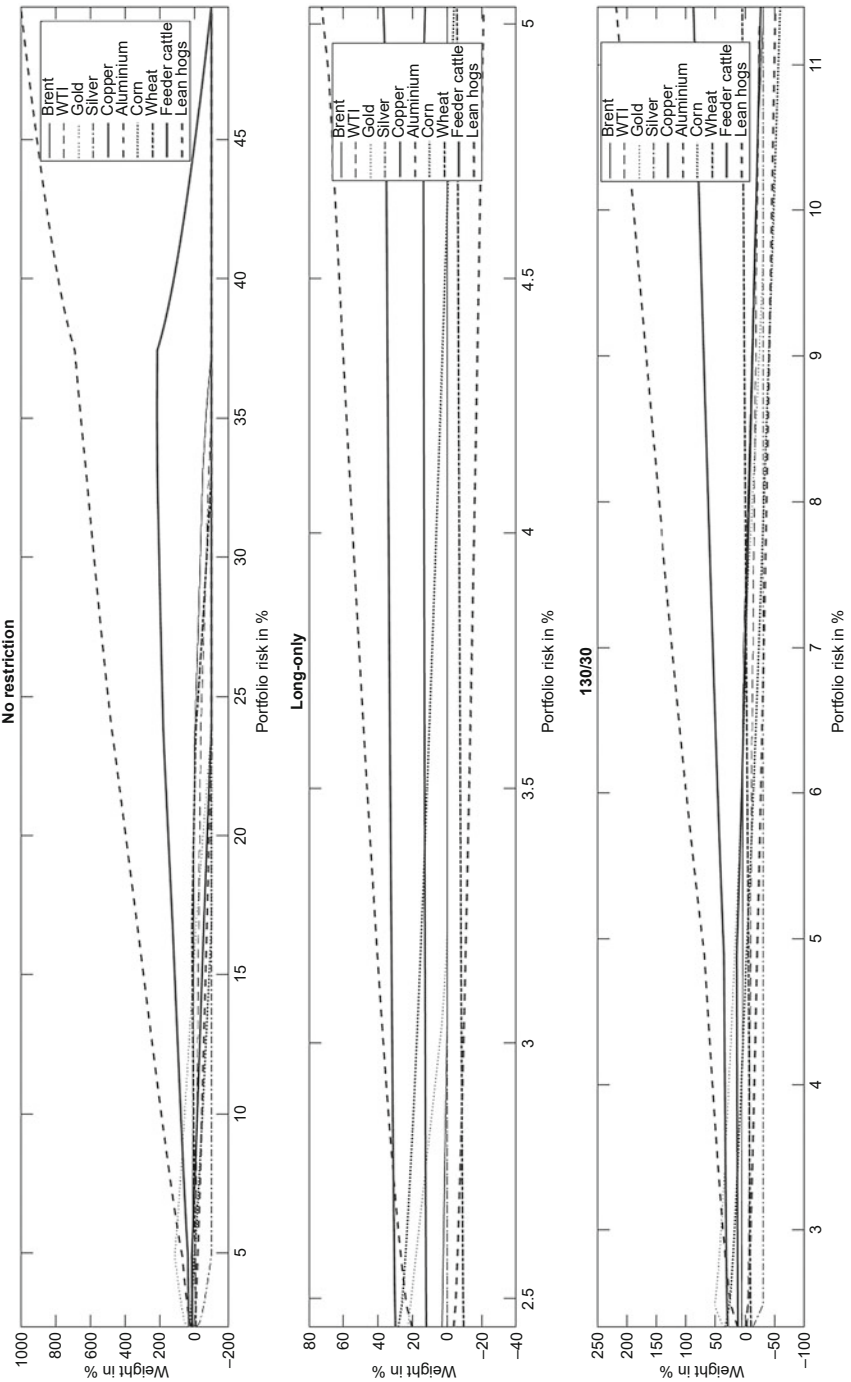


Fig. C.6 Transition map commodities

C.4 FX

See Figs. C.7 and C.8.

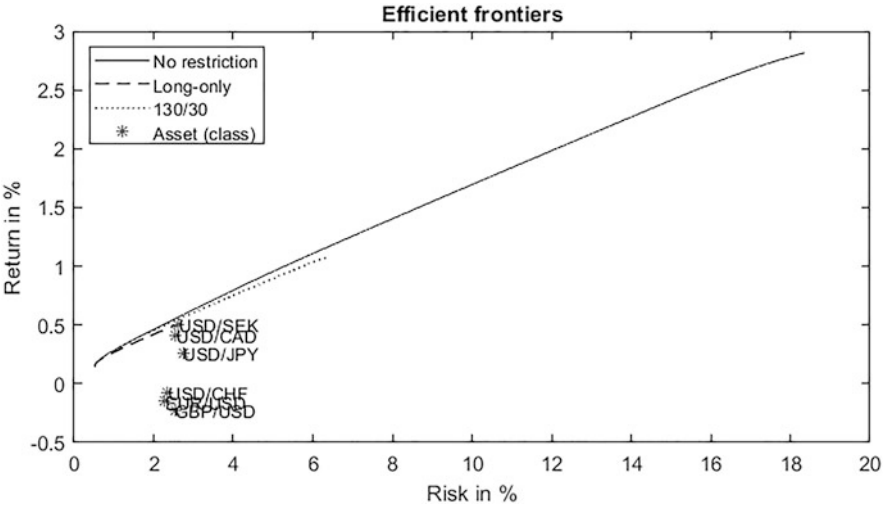


Fig. C.7 Efficient frontiers, FX

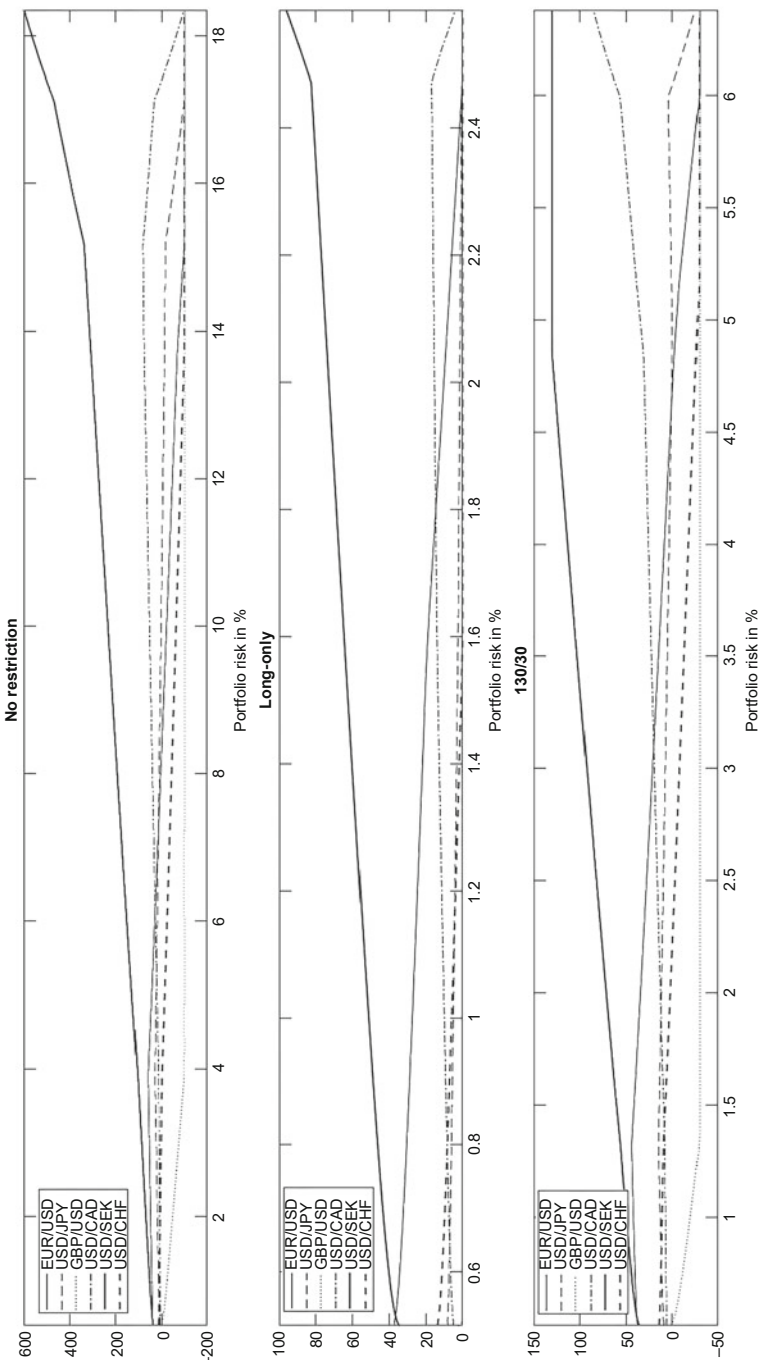


Fig. C.8 Transition map FX

C.5 Alternative Investments

See Figs. C.9 and C.10.

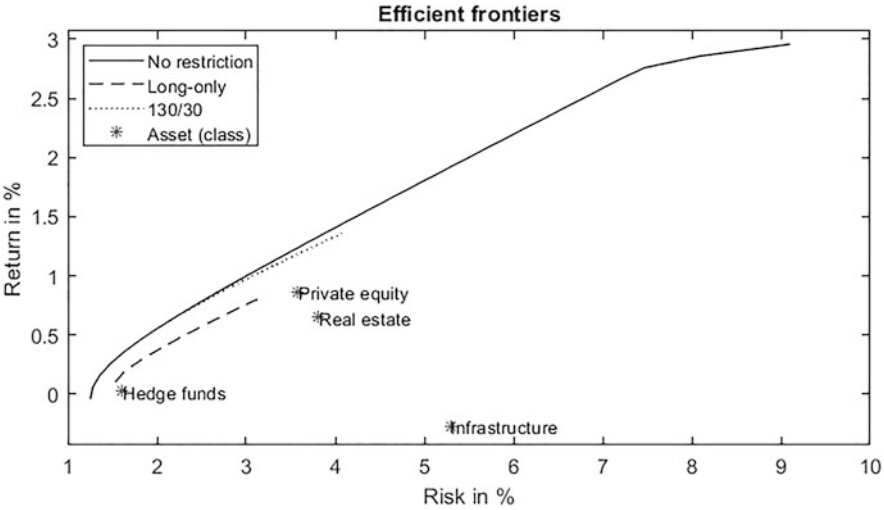


Fig. C.9 Efficient frontiers, alternatives

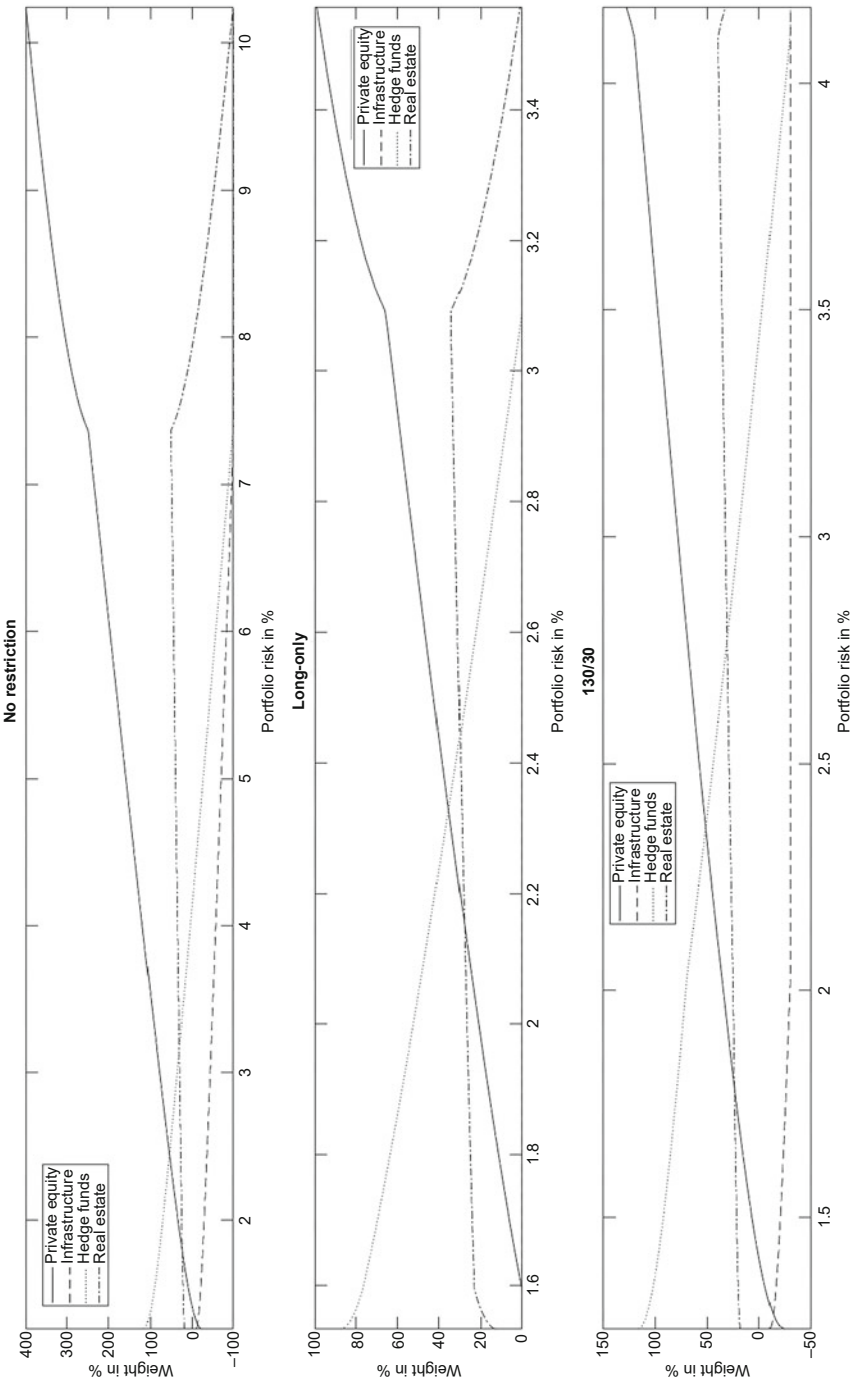


Fig. C.10 Transition map alternatives

C.6 Hedge Funds

See Figs. C.11 and C.12.

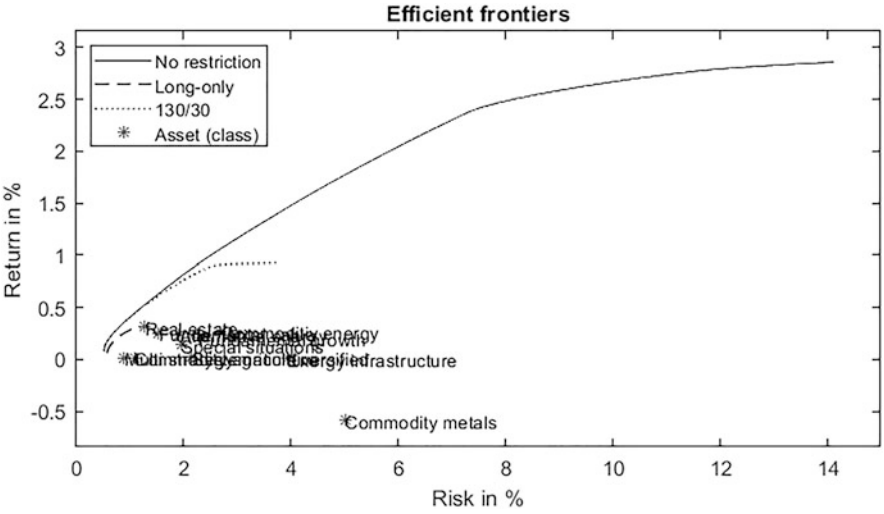


Fig. C.11 Efficient frontiers, hedge funds

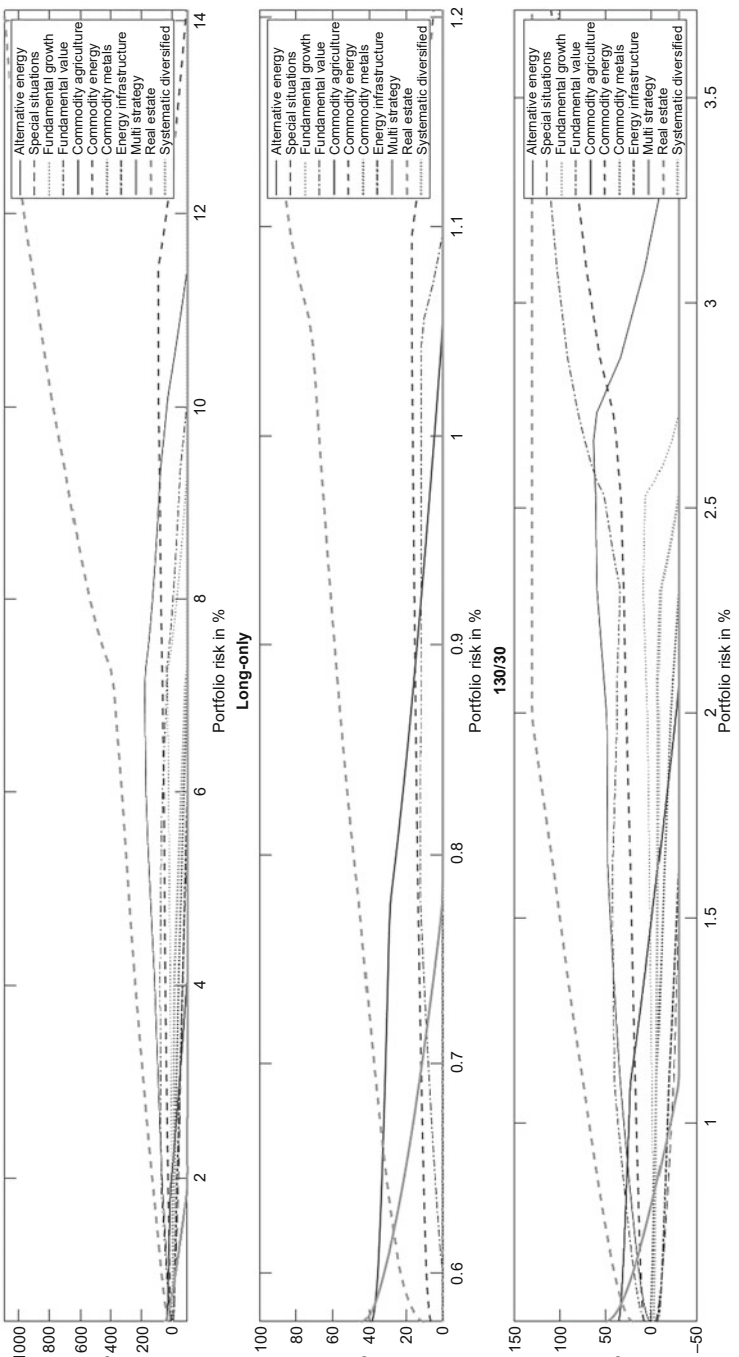


Fig. C.12 Transition map hedge funds

Appendix D
Portfolios Including Digital Assets

D.1 Equities

See Figs. D.1, D.2, D.3 and D.4.

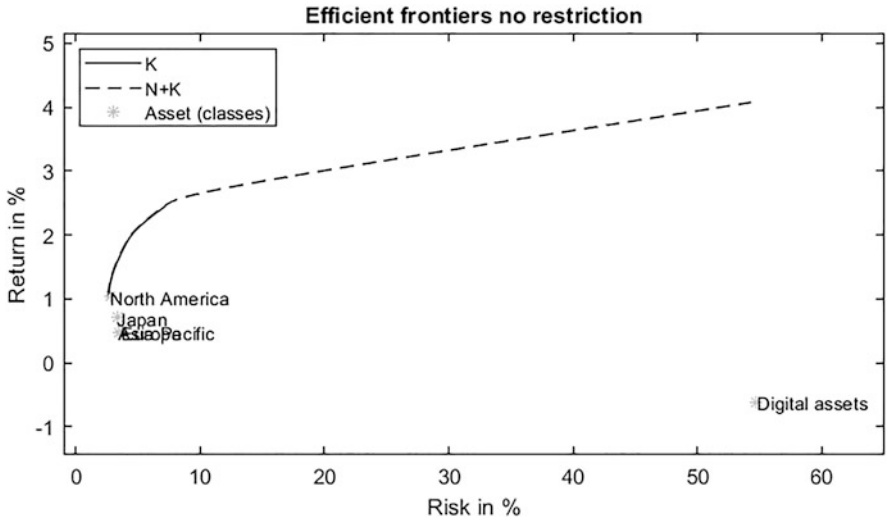


Fig. D.1 Efficient frontiers equities with digital assets, no restriction

See Chap. 6.

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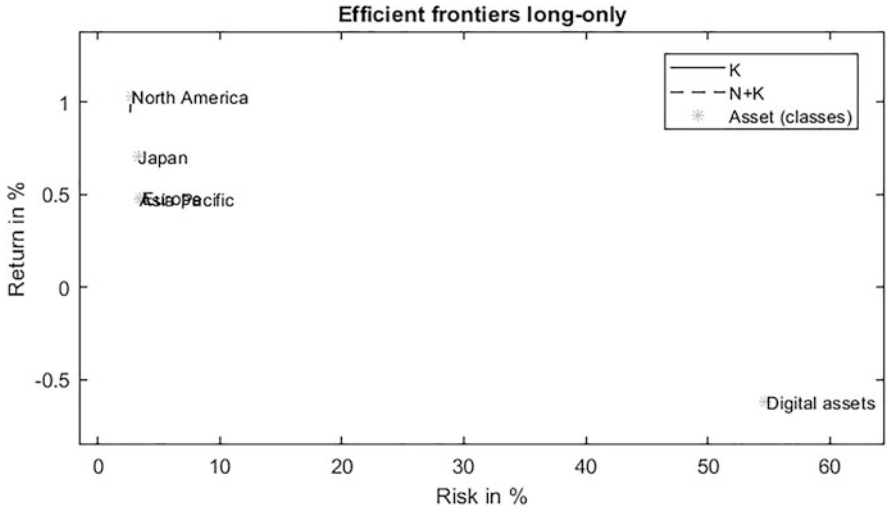


Fig. D.2 Efficient frontiers equities with digital assets, long-only

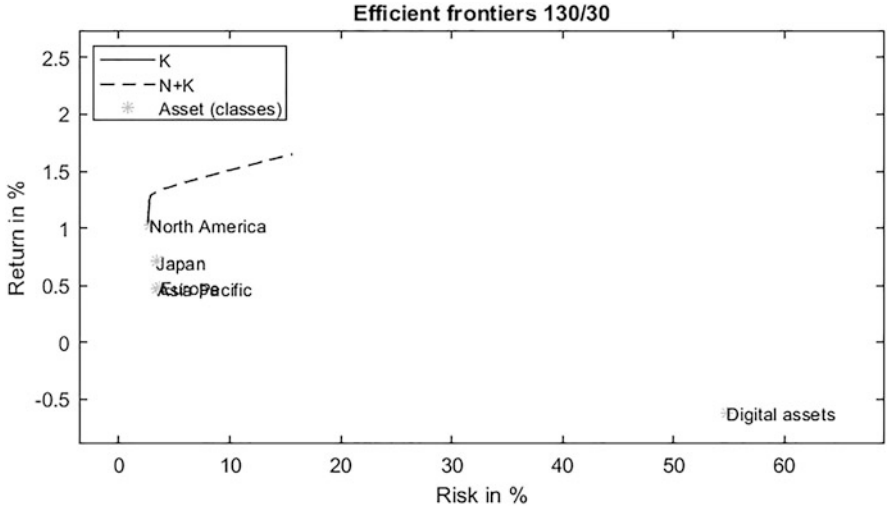


Fig. D.3 Efficient frontiers equities with digital assets, 130/30

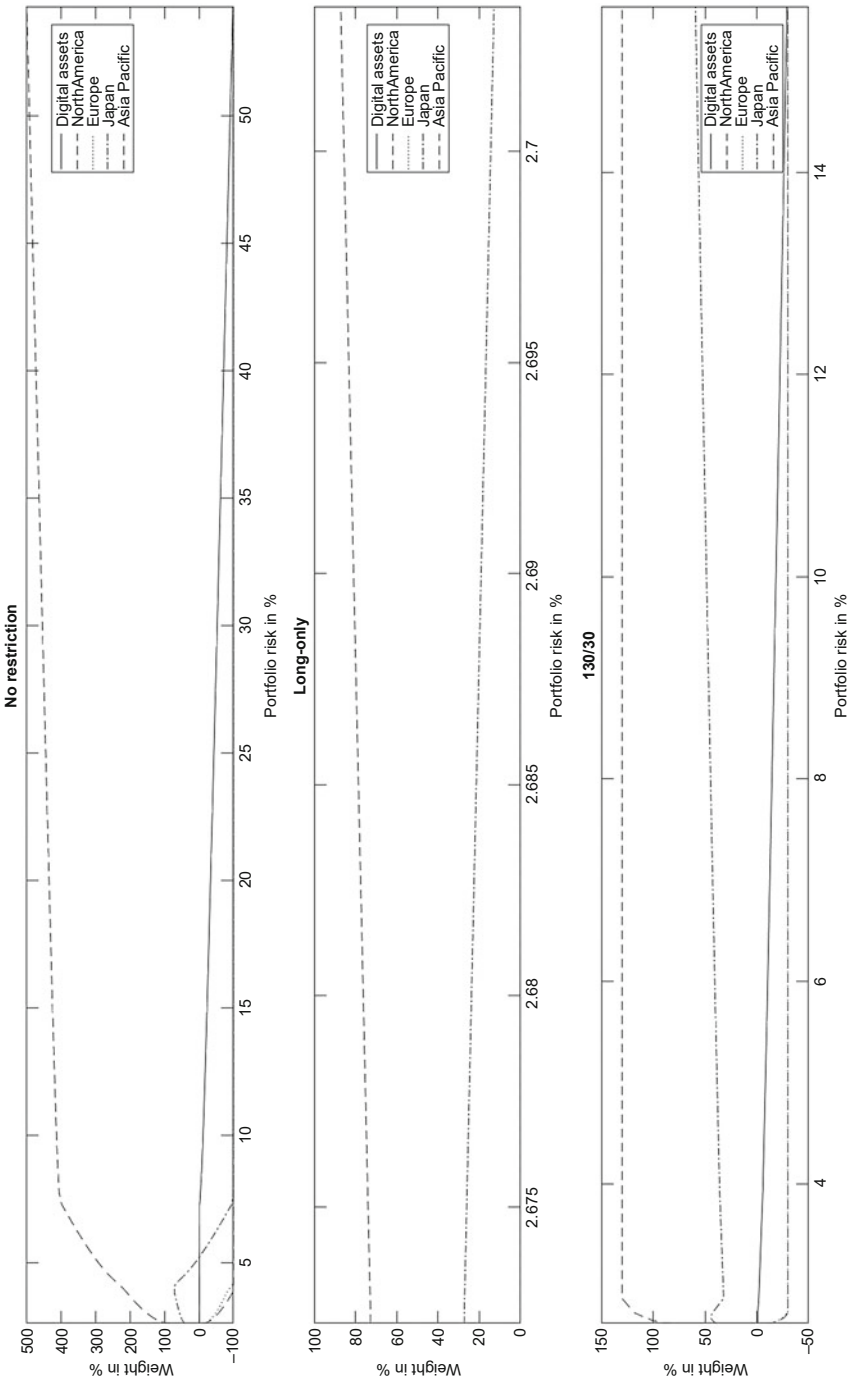


Fig. D.4 Transition map equities with digital assets

D.2 Bonds

See Figs. D.5, D.6, D.7 and D.8.

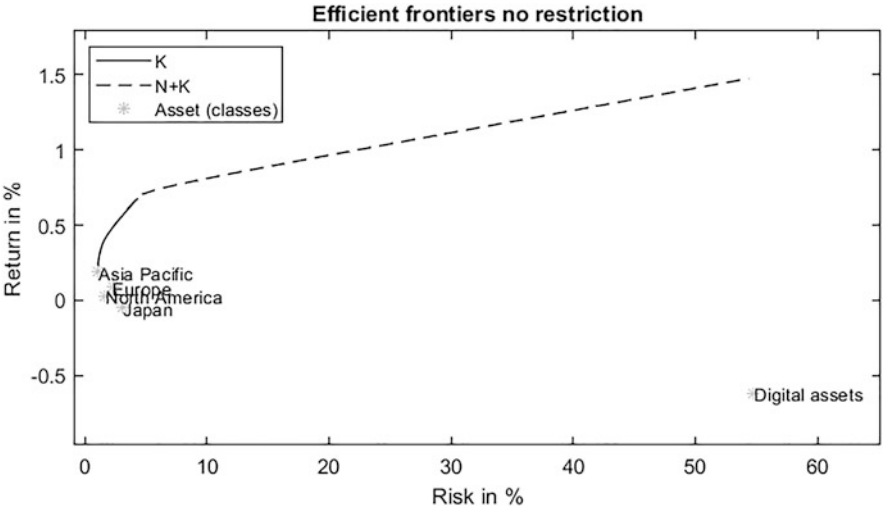


Fig. D.5 Efficient frontiers bonds with digital assets, no restriction

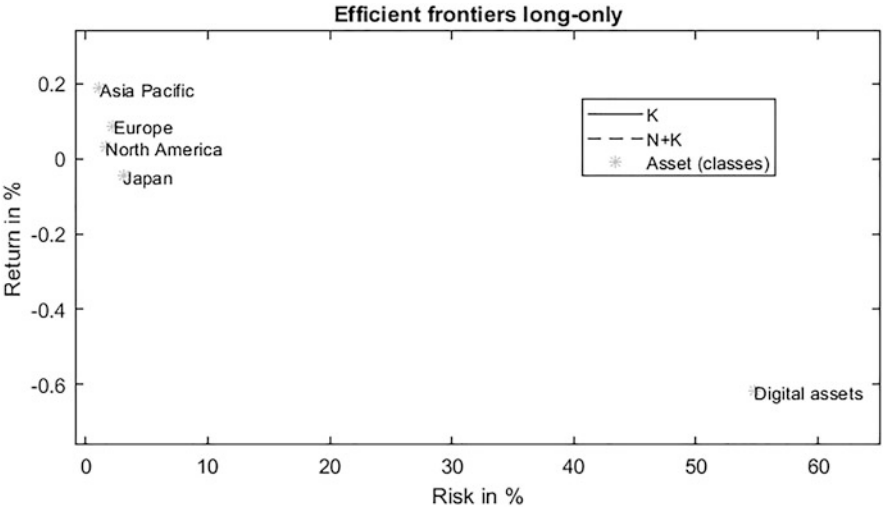


Fig. D.6 Efficient frontiers bonds with digital assets, long-only

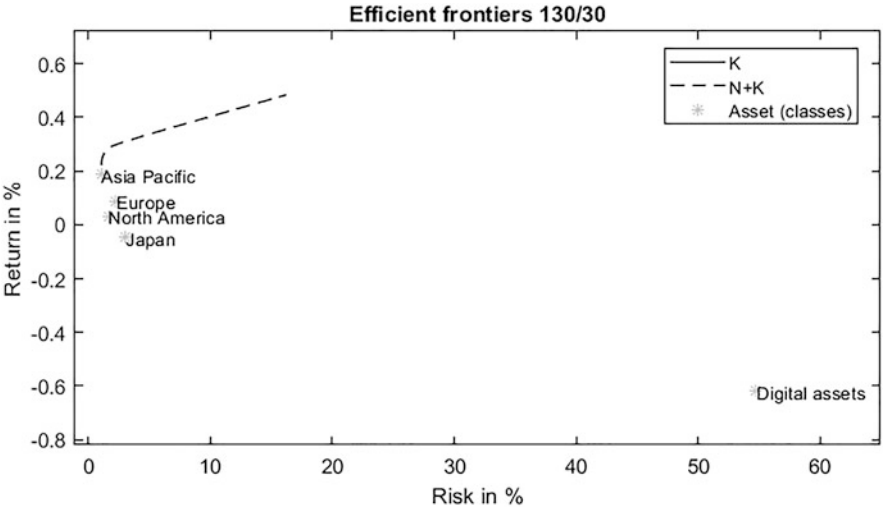


Fig. D.7 Efficient frontiers bonds with digital assets, 130/30

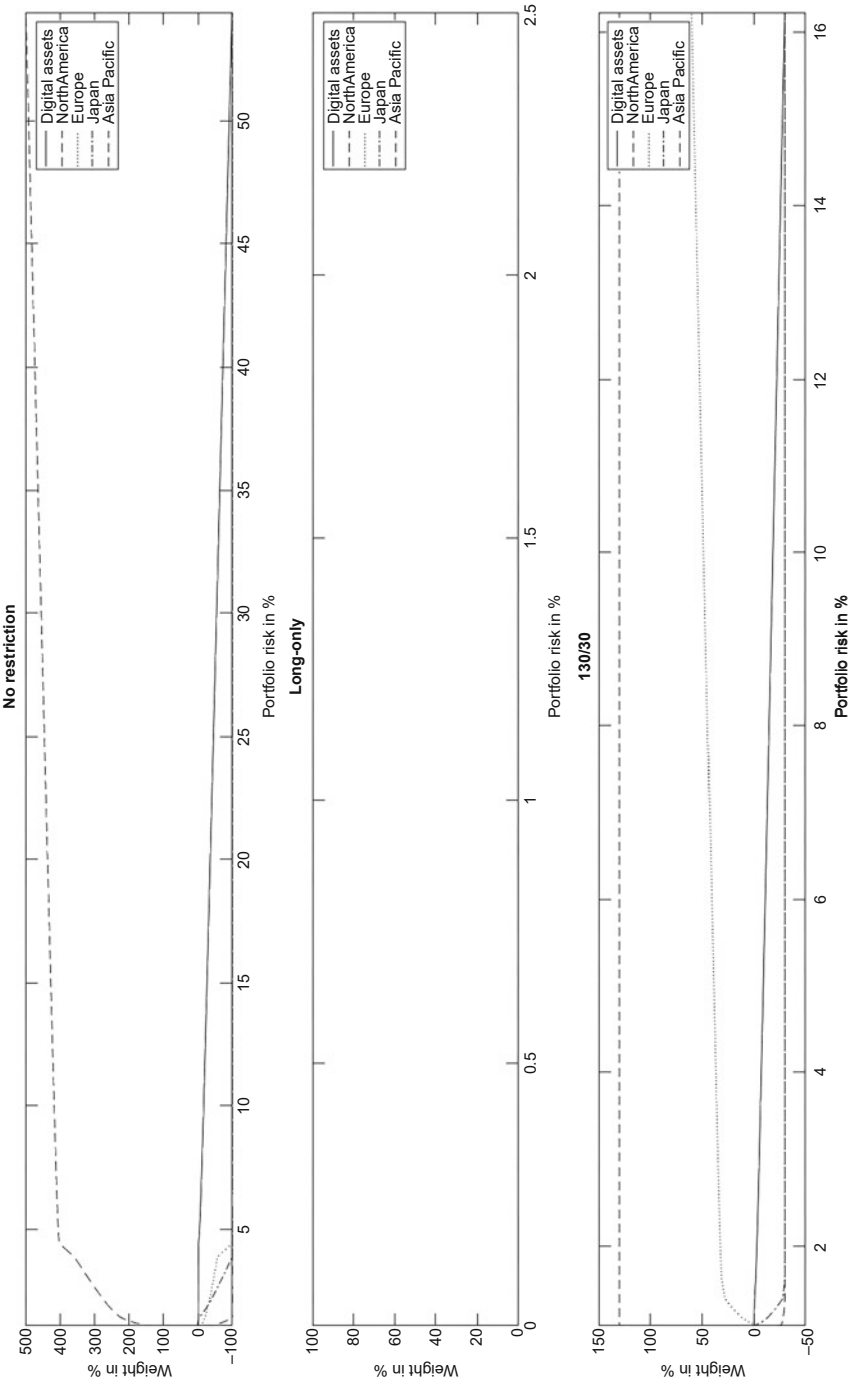


Fig. D.8 Transition map bonds with digital assets

D.3 Commodities

In Fig. D.10 digital assets are not included in the asset allocation, and thus the efficient frontiers for K and $N + K$ are the same (Figs. D.9, D.10, D.11 and D.12).

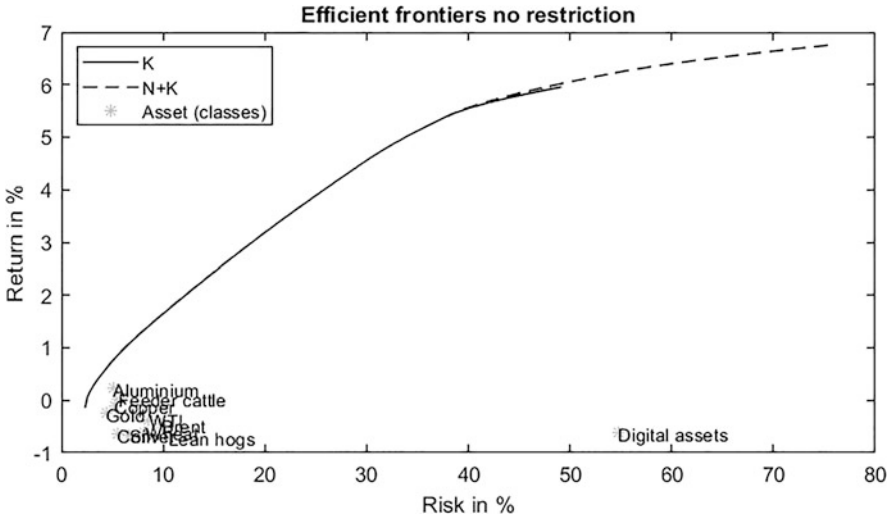


Fig. D.9 Efficient frontiers commodities with digital assets, no restriction

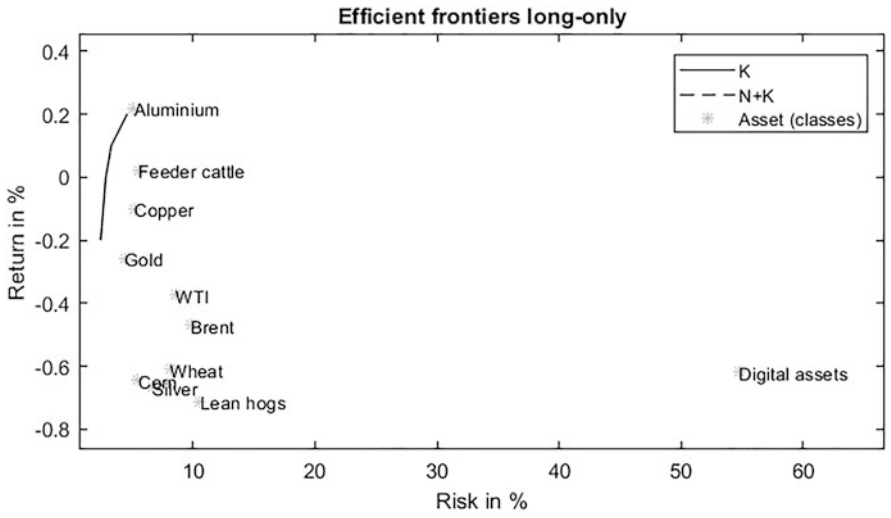


Fig. D.10 Efficient frontiers commodities with digital assets, long-only

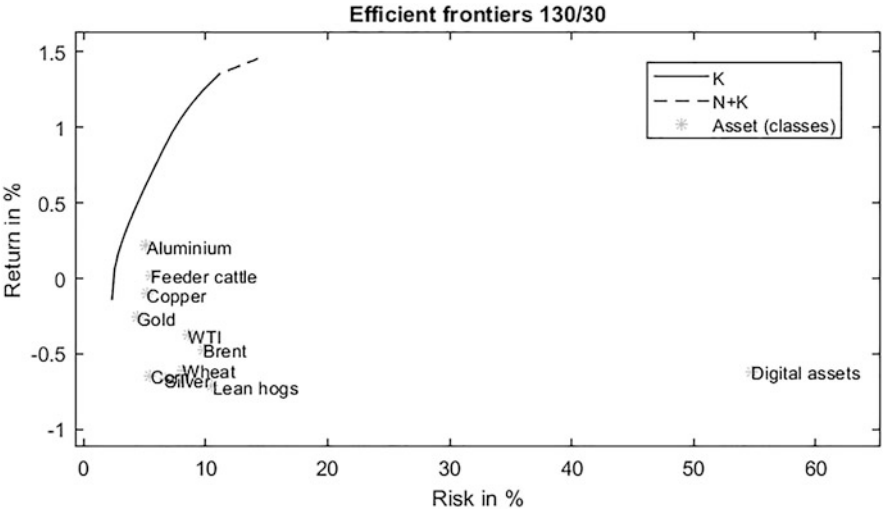


Fig. D.11 Efficient frontiers commodities with digital assets, 130/30

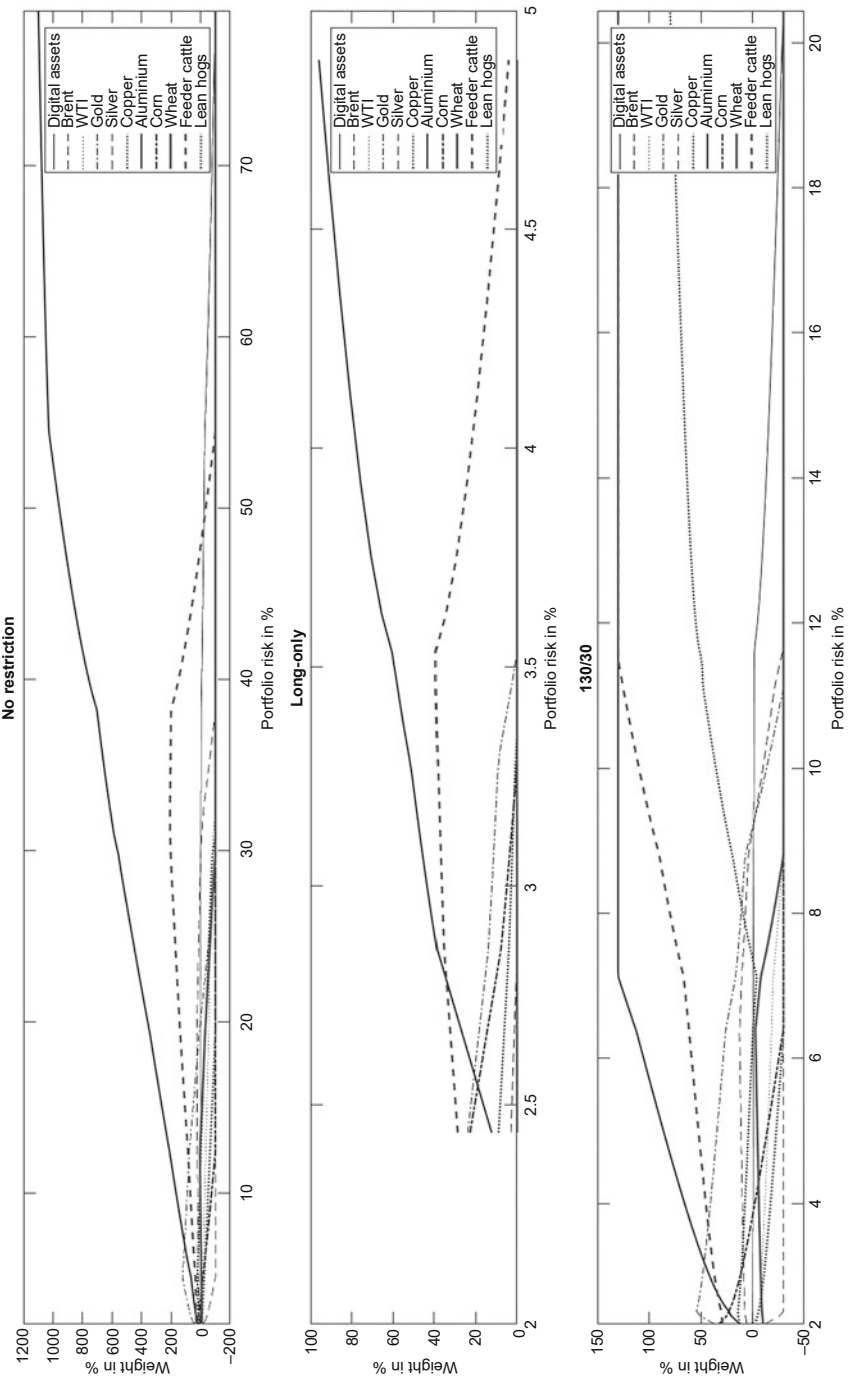


Fig. D.12 Transition map commodities with digital assets

D.4 FX

In Fig. D.14 digital assets are not included in the asset allocation, and thus the efficient frontiers for K and $N+K$ are the same (Figs. D.13, D.14, D.15, and D.16).

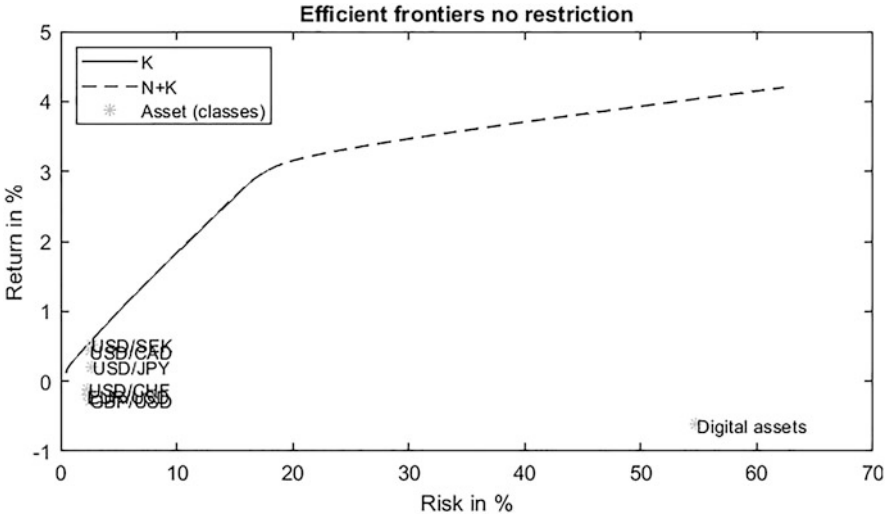


Fig. D.13 Efficient frontiers FX with digital assets, no restriction

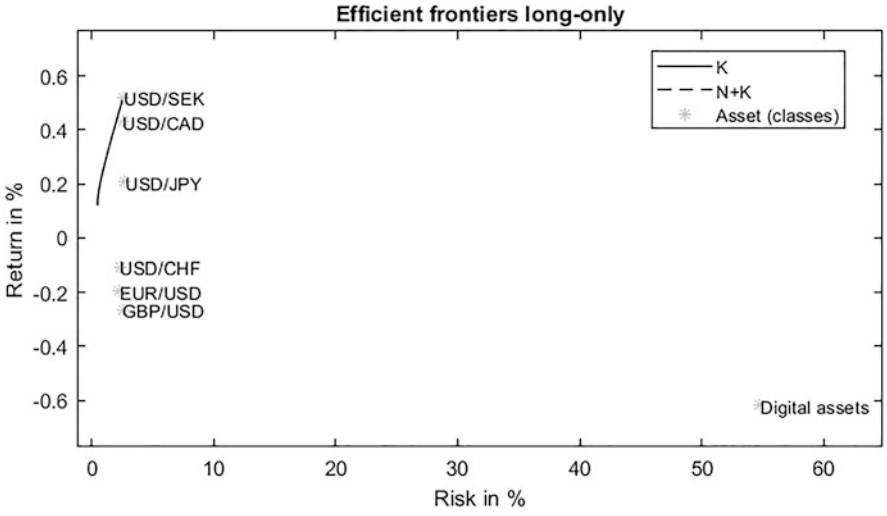


Fig. D.14 Efficient frontiers FX with digital assets, long-only

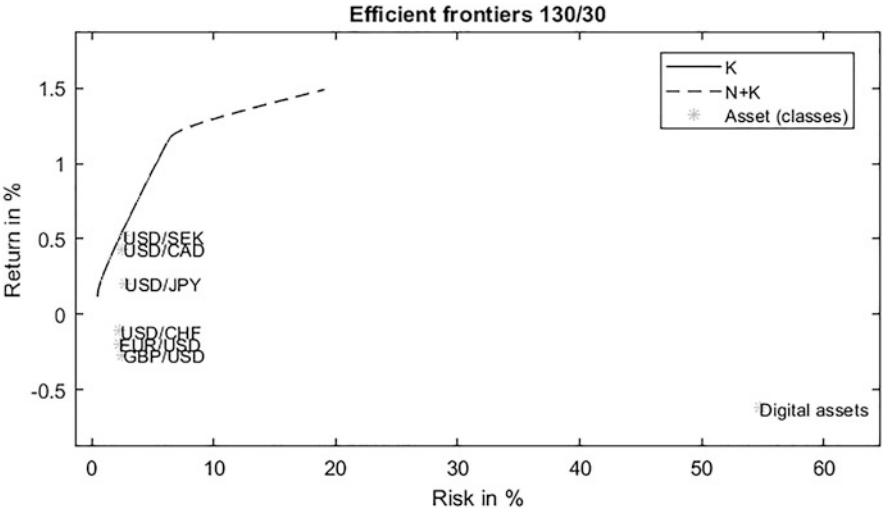


Fig. D.15 Efficient frontiers FX with digital assets, 130/30

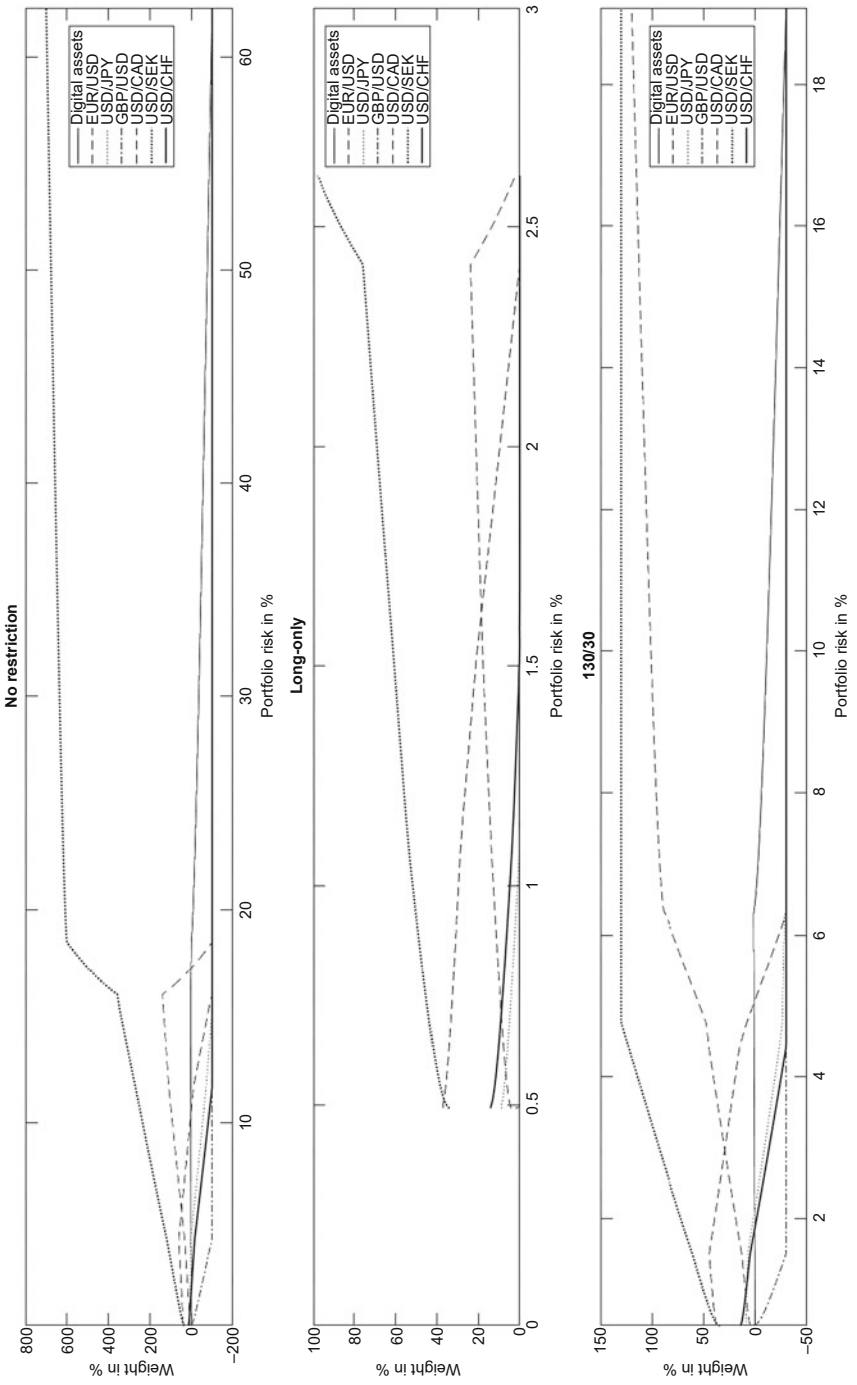


Fig. D.16 Transition map FX with digital assets

D.5 Alternatives

In Fig. D.18 digital assets are not included in the asset allocation, and thus the efficient frontiers for K and $N + K$ are the same (Figs. D.17, D.18, D.19 and D.20).

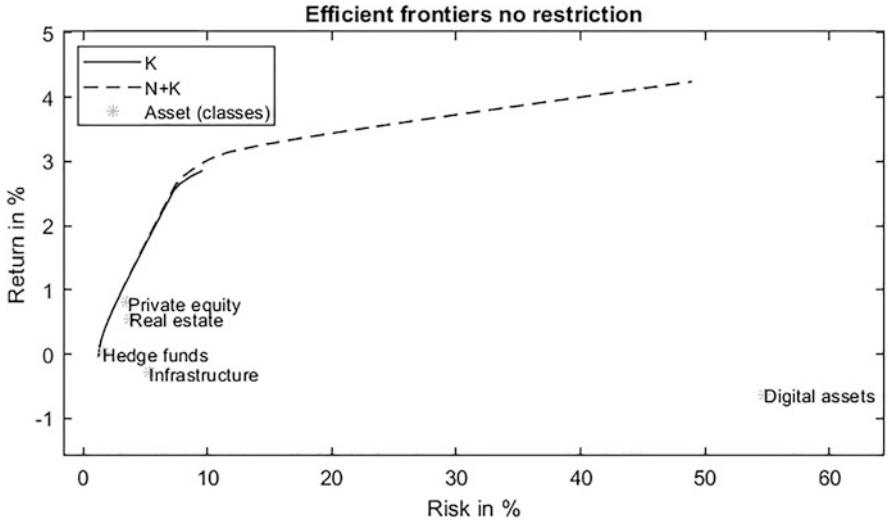


Fig. D.17 Efficient frontiers alternative investments with digital assets, no restriction

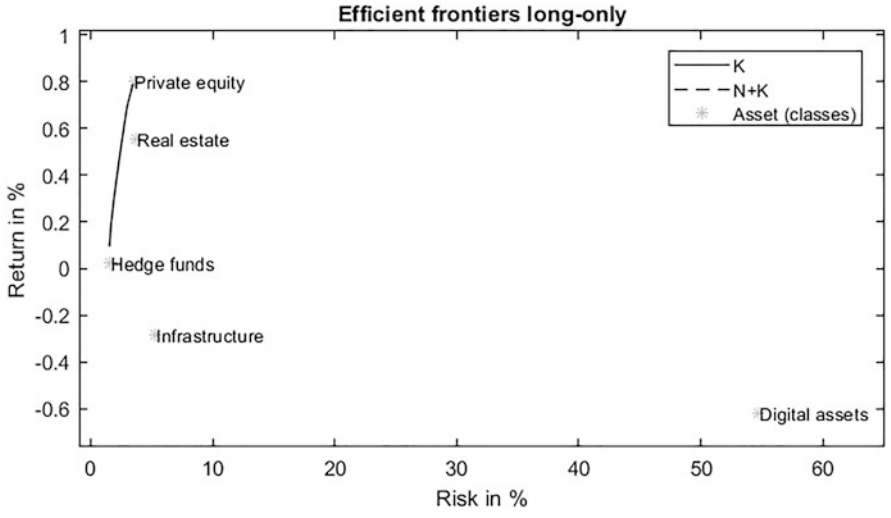


Fig. D.18 Efficient frontiers alternative investments with digital assets, long-only

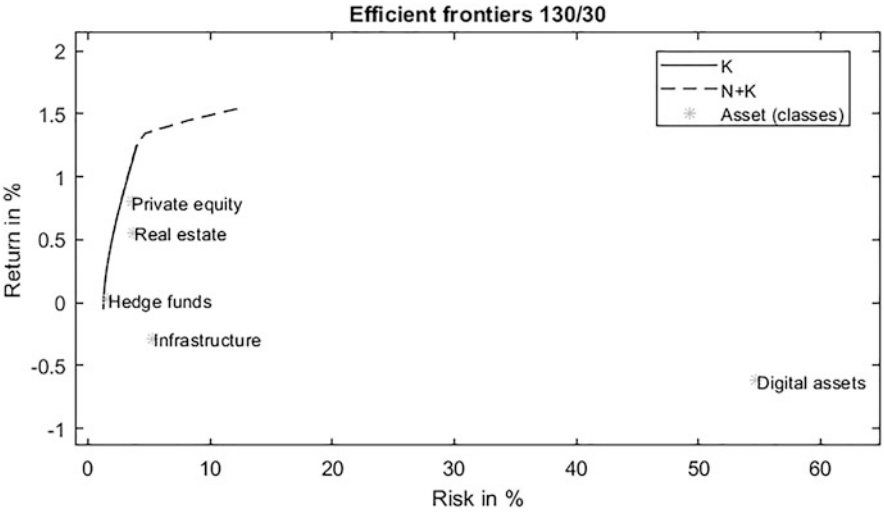


Fig. D.19 Efficient frontiers alternative investments with digital assets, 130/30

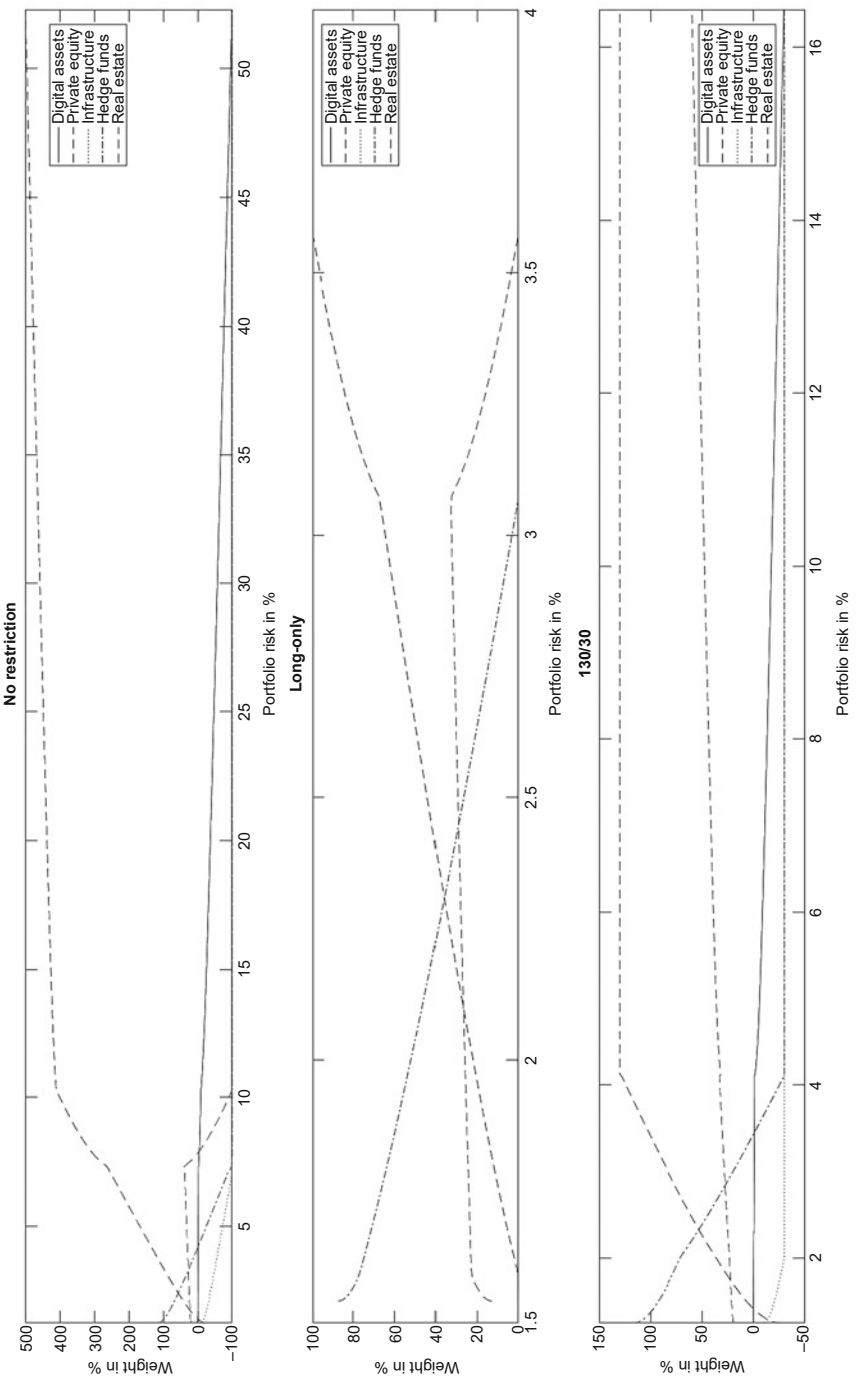


Fig. D.20 Transition map alternative investments with digital assets

D.6 Hedge Funds

In Fig. D.22 digital assets are not included in the asset allocation, and thus the efficient frontiers for K and $N + K$ are the same (Figs. D.21, D.22, D.23 and D.24).

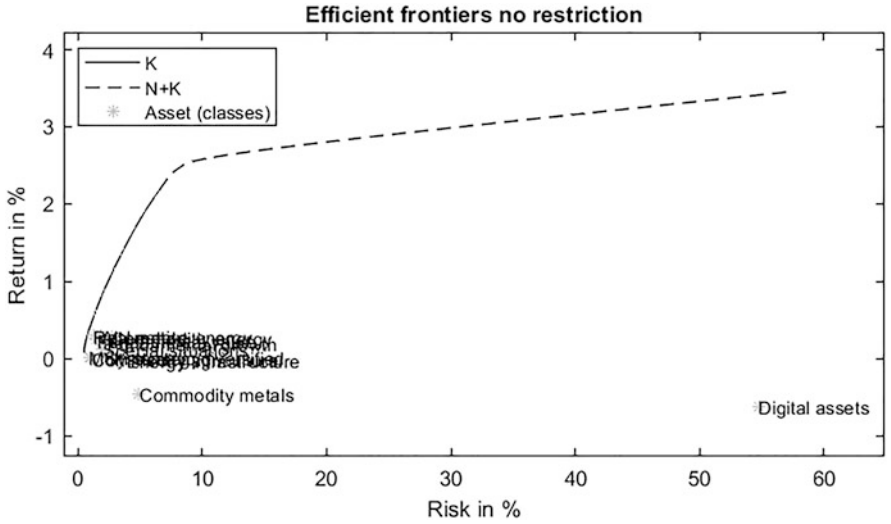


Fig. D.21 Efficient frontiers hedge funds with digital assets, no restriction

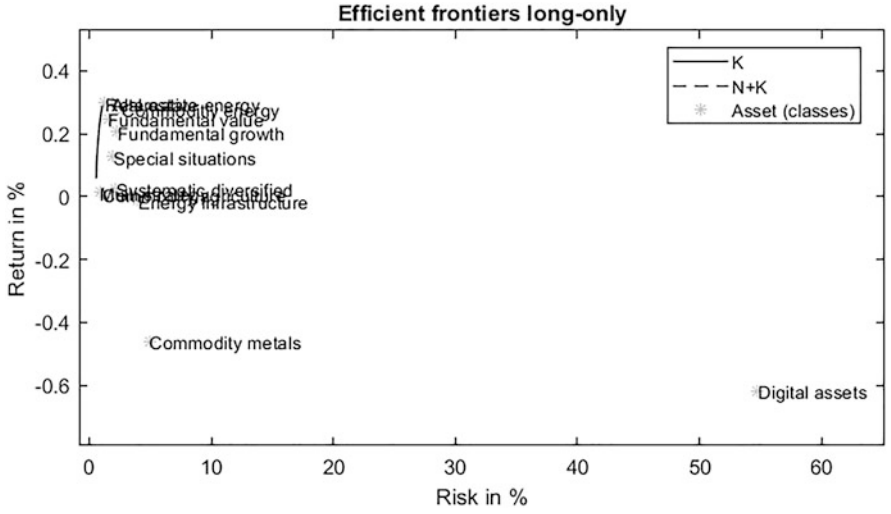


Fig. D.22 Efficient frontiers hedge funds with digital assets, long-only

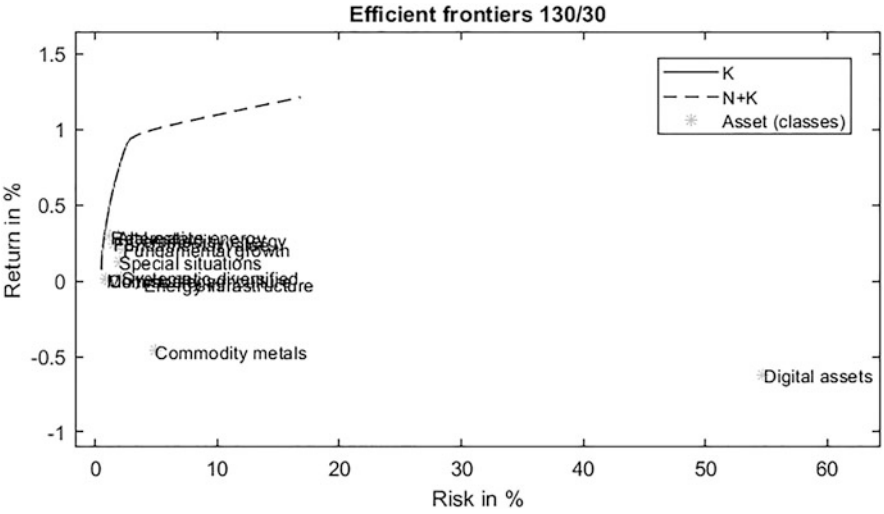


Fig. D.23 Efficient frontiers hedge funds with digital assets, 130/30

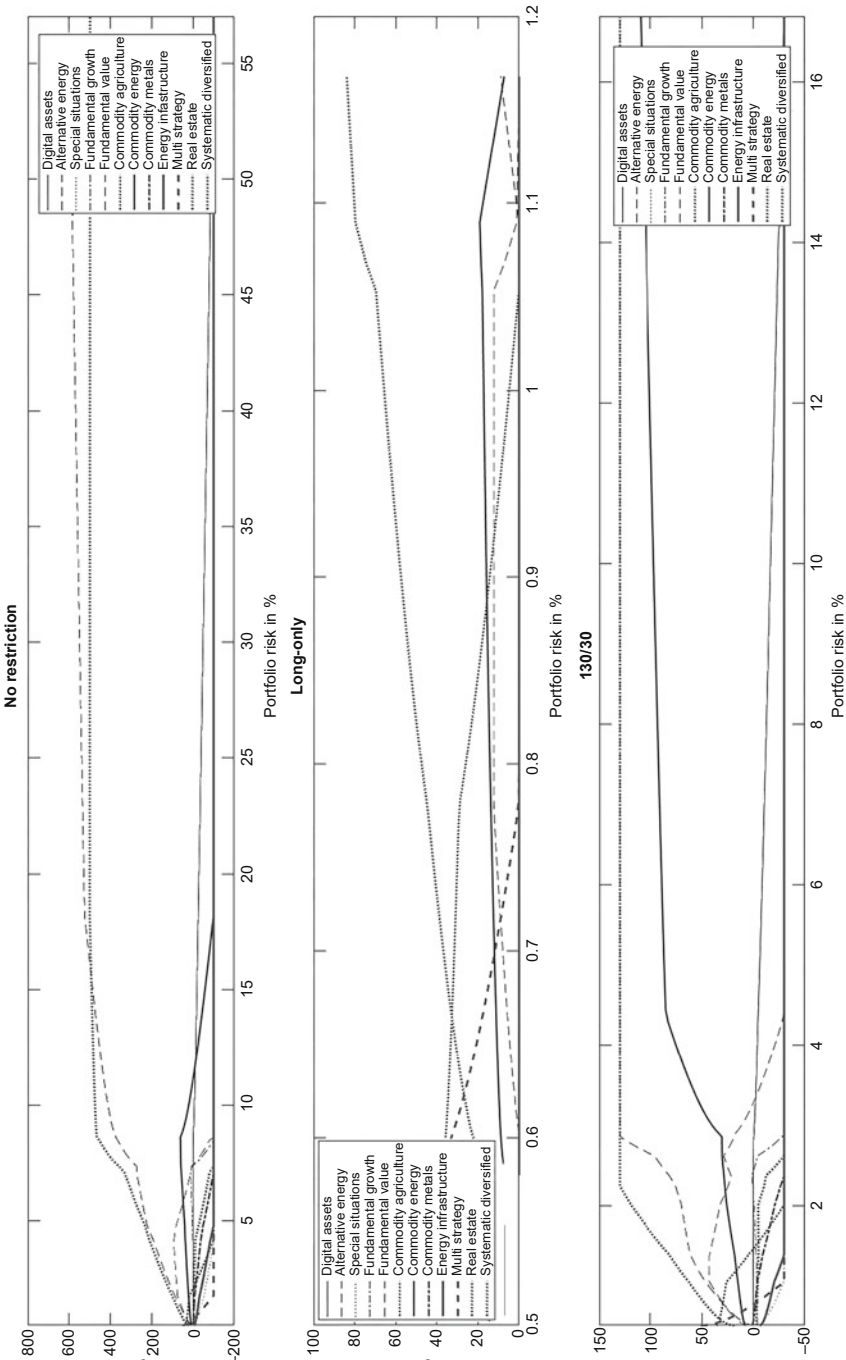


Fig. D.24 Transition map hedge funds with digital assets

Appendix E
Risk Parity Weights

See Figs. E.1, E.2, E.3, E.4, E.5 and E.6.

See Chap. 6.

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T. Glas, *Asset Pricing and Investment Styles in Digital Assets*, Advanced Studies in Diginomics and Digitalization, https://doi.org/10.1007/978-3-030-95695-0

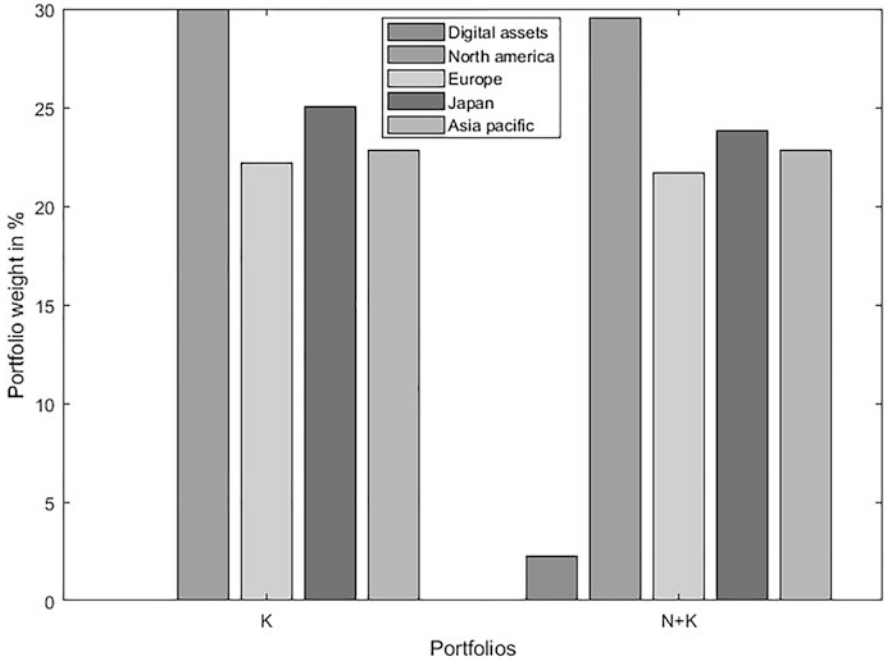


Fig. E.1 A comparison of ERC weights, equities

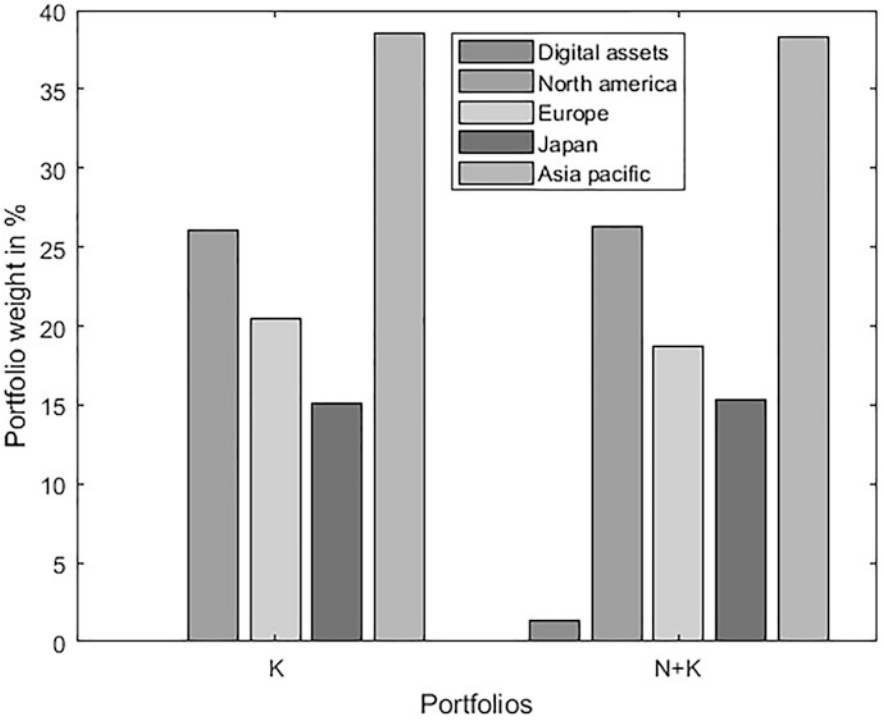


Fig. E.2 A comparison of ERC weights, bonds

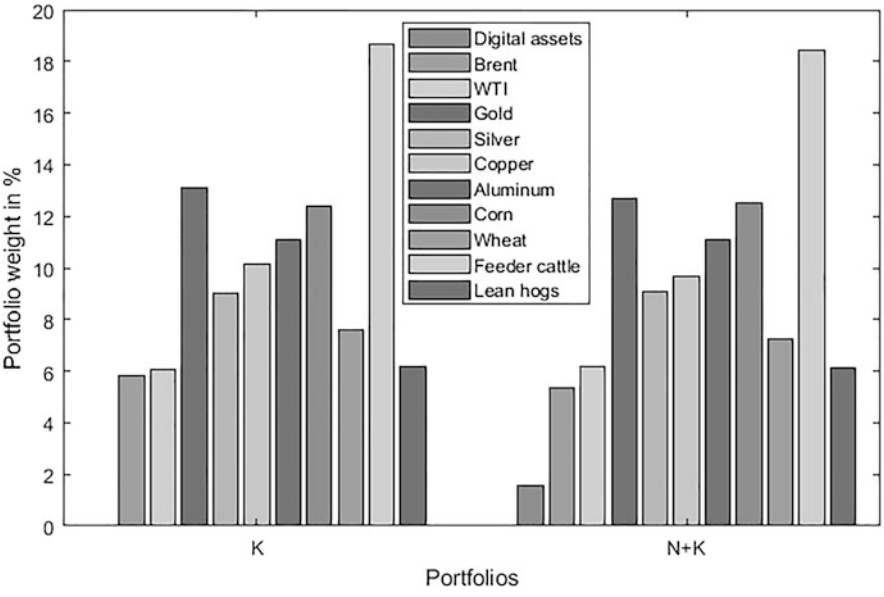


Fig. E.3 A comparison of ERC weights, commodities

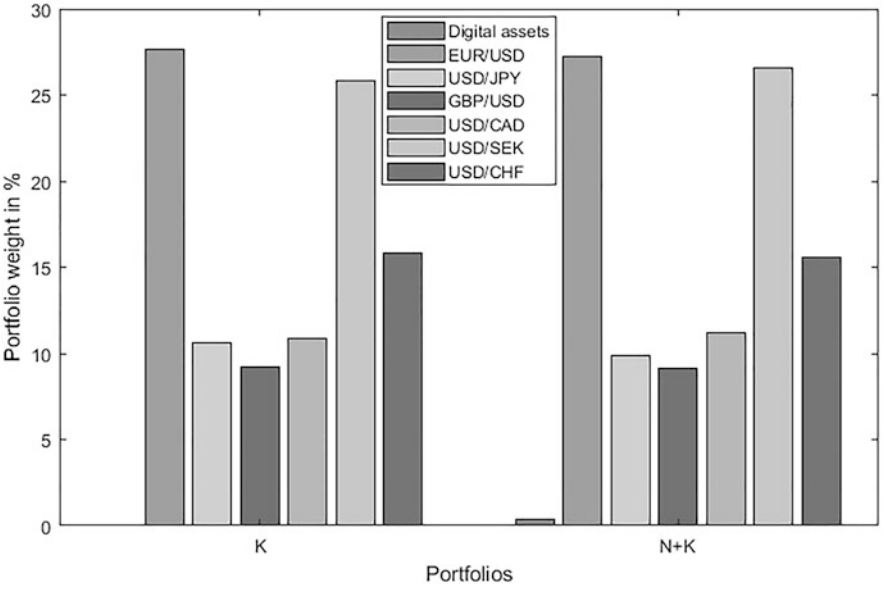


Fig. E.4 A comparison of ERC weights, FX

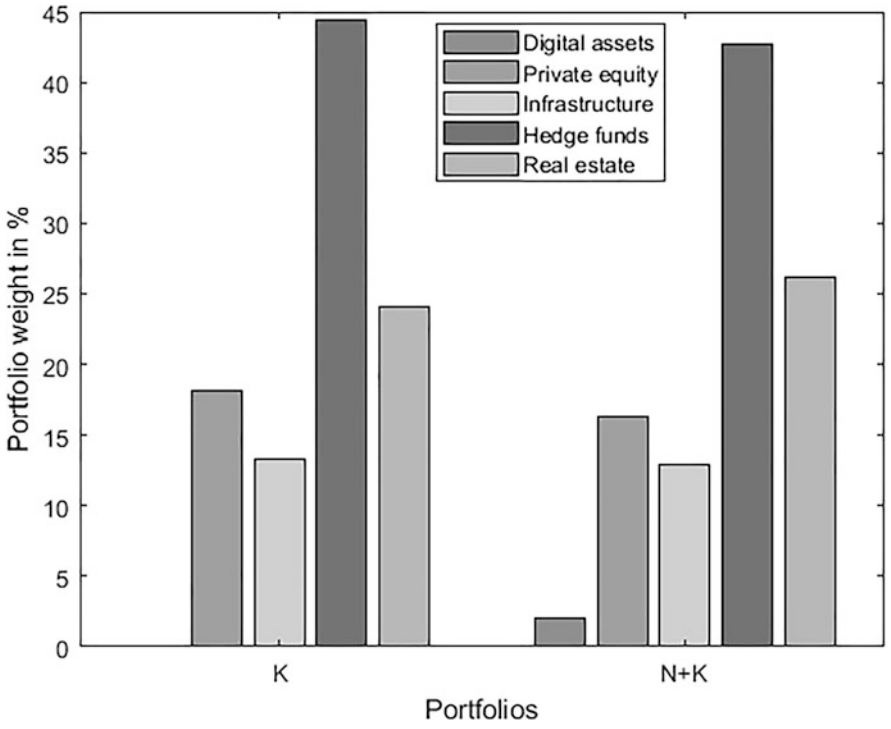


Fig. E.5 A comparison of ERC weights, alternative investments

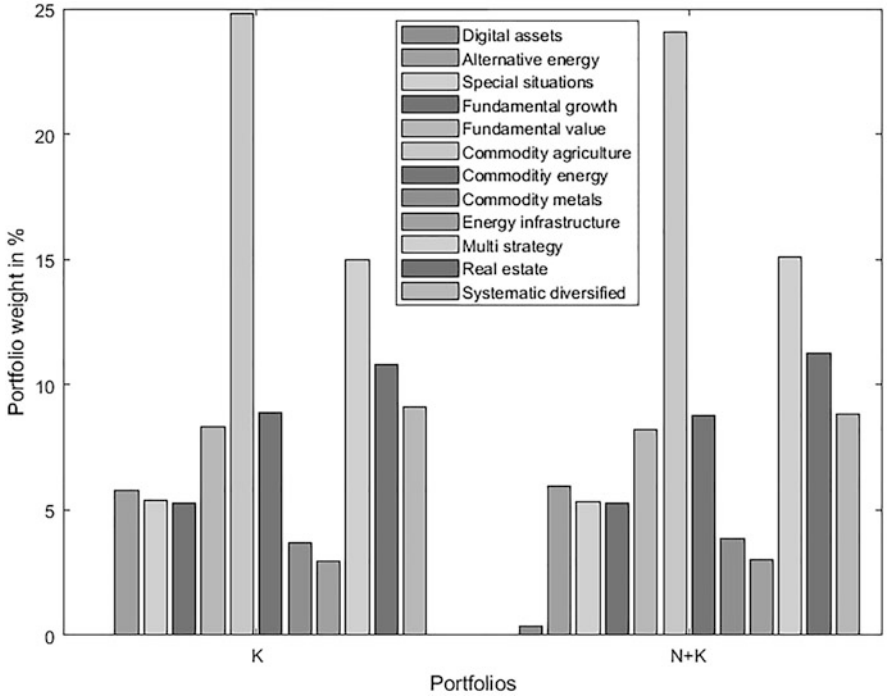


Fig. E.6 A comparison of ERC weights, hedge funds

Appendix F
Investment Style Correlations

See Tables F.1, F.2 and F.3.

See Chap. 8.

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T. Glas, *Asset Pricing and Investment Styles in Digital Assets*, Advanced Studies in Diginomics and Digitalization, https://doi.org/10.1007/978-3-030-95695-0

Table F.1 Digital asset investment style correlations

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Big														
Reversal	100.00	9.33	4.59	4.37	4.51	−2.83	−9.01	−12.24	−88.30	12.15	−13.24	15.61	17.07	−17.85
Reversal st	9.33	100.00	−0.10	−7.79	12.14	−36.17	−12.10	−8.07	0.13	39.87	−26.20	25.12	−14.43	−29.03
Amihud	4.59	−0.10	100.00	−1.68	32.28	3.73	26.75	8.74	−6.89	−6.44	0.23	−2.46	15.64	56.53
Volume	4.37	−7.79	−1.68	100.00	−3.65	12.00	3.77	−15.78	−16.85	52.19	−3.91	−12.42	−5.06	11.18
Low vol	4.51	12.14	32.28	−3.65	100.00	30.57	46.59	41.37	4.21	−6.67	−6.75	−9.97	28.90	17.80
Low vol st	−2.83	−36.17	3.73	12.00	30.57	100.00	4.53	31.02	11.21	−22.17	0.88	−11.61	23.98	29.16
Low beta	−9.01	−12.10	26.75	3.77	46.59	4.53	100.00	20.64	12.58	−10.93	−10.76	−18.90	26.58	37.82
Low-beta st	−12.24	−8.07	8.74	−15.78	41.37	31.02	20.64	100.00	6.32	−14.18	3.02	−29.74	22.98	−11.07
Momentum	−88.30	0.13	−6.89	−16.85	4.21	11.21	12.58	6.32	100.00	−23.57	−1.41	−8.20	−15.01	20.18
Size	12.15	39.87	−6.44	52.19	−6.67	−22.17	−10.93	−14.18	−23.57	100.00	−19.18	−28.30	−33.34	−21.12
Mispricing	−13.24	−26.20	0.23	−3.91	−6.75	0.88	−10.76	3.02	−1.41	−19.18	100.00	11.36	11.70	26.37
Idio risk	15.61	25.12	−2.46	−12.42	−9.97	−11.61	−18.90	−29.74	−8.20	−28.30	11.36	100.00	−11.22	1.11
Explained	17.07	−14.43	15.64	−5.06	28.90	23.98	26.58	22.98	−15.01	−33.34	11.70	−11.22	100.00	17.82
Nominal	−17.85	−29.03	56.53	11.18	17.80	29.16	37.82	−11.07	20.18	−21.12	26.37	1.11	17.82	100.00

Medium														
Reversal	100.00	−18.75	−13.74	−6.63	5.17	5.40	2.01	9.55	−56.60	23.47	−5.85	−0.52	−5.58	−2.67
Reversal st	−18.75	100.00	−9.19	−23.73	18.83	35.13	21.42	29.65	30.45	27.07	−3.95	−15.29	−8.41	−48.29
Amihud	−13.74	−9.19	100.00	11.39	56.26	6.38	17.60	−16.34	−14.77	−2.72	6.25	−5.71	9.36	43.80
Volume	−6.63	−23.73	11.39	100.00	8.69	−22.18	8.66	1.51	−17.23	10.49	−21.50	18.36	−3.69	12.88
Low vol	5.17	18.83	56.26	8.69	100.00	47.15	50.32	−21.83	−6.75	20.96	17.87	−4.45	21.06	47.76
Low vol st	5.40	35.13	6.38	−22.18	47.15	100.00	20.33	69.30	6.61	55.34	21.27	−29.90	32.95	−42.07
Low beta	2.01	21.42	17.60	8.66	50.32	20.33	100.00	18.31	11.32	33.95	−13.26	11.62	8.79	−9.34
Low-beta st	9.55	29.65	−16.34	1.51	−21.83	69.30	18.31	100.00	−5.43	61.40	24.55	8.41	35.85	−65.52
Momentum	−56.60	30.45	−14.77	−17.23	−6.75	6.61	11.32	−5.43	100.00	−5.59	−3.83	12.15	−14.97	8.89
Size	23.47	27.07	−2.72	10.49	20.96	55.34	33.95	61.40	−5.59	100.00	−23.96	−11.73	−4.71	−32.31
Mispricing	−5.85	−3.95	6.25	−21.50	17.87	21.27	−13.26	24.55	−3.83	−23.96	100.00	10.13	24.38	20.34
Idio risk	−0.52	−15.29	−5.71	18.36	−4.45	−29.90	11.62	8.41	12.15	−11.73	10.13	100.00	−41.08	−11.00
Explained	−5.58	−8.41	9.36	−3.69	21.06	32.95	8.79	35.85	−14.97	−4.71	24.38	−41.08	100.00	−25.06
Nominal	−2.67	−48.29	43.80	12.88	47.76	−42.07	−9.34	−65.52	8.89	−32.31	20.34	−11.00	−25.06	100.00

(continued)

Table F.1 (continued)

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Small														
Reversal	100.00	−9.68	−16.67	9.30	−4.06	−8.20	−16.90	−23.13	−84.76	−6.03	6.97	14.90	1.46	−15.15
Reversal st	−9.68	100.00	17.07	29.02	2.39	16.47	15.24	21.32	22.75	−0.93	−27.76	3.33	28.70	8.15
Amihud	−16.67	17.07	100.00	15.99	2.09	−3.28	8.46	−19.17	20.16	25.18	−27.56	−20.88	−14.00	22.47
Volume	9.30	29.02	15.99	100.00	1.21	−8.65	1.05	−23.27	−4.33	38.49	−36.09	−13.78	4.97	−0.81
Low vol	−4.06	2.39	2.09	1.21	100.00	56.27	53.77	21.89	1.48	−18.67	38.75	−19.28	−4.53	44.85
Low vol st	−8.20	16.47	−3.28	−8.65	56.27	100.00	21.09	69.80	6.25	−18.40	17.42	−25.76	−2.75	28.08
Low beta	−16.90	15.24	8.46	1.05	53.77	21.09	100.00	26.22	22.10	−21.51	16.30	12.29	−9.13	20.73
Low-beta st	−23.13	21.32	−19.17	−23.27	21.89	69.80	26.22	100.00	27.18	−34.61	20.02	7.05	11.12	26.61
Momentum	−84.76	22.75	20.16	−4.33	1.48	6.25	22.10	27.18	100.00	13.09	2.03	−7.85	16.55	3.62
Size	−6.03	−0.93	25.18	38.49	−18.67	−18.40	−21.51	−34.61	13.09	100.00	−14.96	−29.71	25.68	10.66
Mispricing	6.97	−27.76	−27.56	−36.09	38.75	17.42	16.30	20.02	2.03	−14.96	100.00	7.01	1.20	−7.39
Idio risk	14.90	3.33	−20.88	−13.78	−19.28	−25.76	12.29	7.05	−7.85	−29.71	7.01	100.00	−26.63	−42.99
Explained	1.46	28.70	−14.00	4.97	−4.53	−2.75	−9.13	11.12	16.55	25.68	1.20	−26.63	100.00	−6.85
Nominal	−15.15	8.15	22.47	−0.81	44.85	28.08	20.73	26.61	3.62	10.66	−7.39	−42.99	−6.85	100.00

The columns and lines represent the investment styles under consideration, with "st" denoting the short-term variants. The numbers are the correlation coefficients in %. The table reports the outcomes for all available assets

Table F.2 Equity investment style correlations

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Big														
Reversal	100.00	−23.17	−18.90	−19.16	−30.30	−24.26	−30.25	−28.83	−96.55	−8.28	−87.89	7.80	−29.79	−15.27
Reversal st	−23.17	100.00	−17.88	−19.16	−34.69	−34.78	−25.78	−27.56	20.16	−20.51	−2.29	3.96	5.58	−0.33
Amihud	−18.90	−17.88	100.00	−7.87	−7.03	−14.28	−6.14	−24.72	21.48	−28.88	27.24	−23.47	−3.87	80.92
Volume	−19.16	−19.16	−7.87	100.00	70.26	69.98	72.09	77.82	22.70	90.61	32.29	21.06	44.23	−63.54
Low vol	−30.30	−34.69	−7.03	70.26	100.00	92.93	88.68	87.15	34.56	65.83	42.73	15.20	43.51	−30.62
Low vol st	−24.26	−34.78	−14.28	69.98	92.93	100.00	79.80	89.55	28.34	66.72	34.49	17.45	48.86	−42.61
Low beta	−30.25	−25.78	−6.14	72.09	88.68	79.80	100.00	83.36	35.88	63.72	45.72	17.95	51.67	−31.81
Low-beta st	−28.83	−27.56	−24.72	77.82	87.15	89.55	83.36	100.00	29.15	74.04	38.97	12.13	55.58	−54.76
Momentum	−96.55	20.16	21.48	22.70	34.56	28.34	35.88	29.15	100.00	10.01	91.03	1.24	31.36	15.45
Size	−8.28	−20.51	−28.88	90.61	65.83	66.72	63.72	74.04	10.01	100.00	23.52	18.94	42.41	−74.83
Mispricing	−87.89	−2.29	27.24	32.29	42.73	34.49	45.72	38.97	91.03	23.52	100.00	−6.82	32.98	14.37
Idio risk	7.80	3.96	−23.47	21.06	15.20	17.45	17.95	12.13	1.24	18.94	−6.82	100.00	20.36	−28.52
Explained	−29.79	5.58	−3.87	44.23	43.51	48.86	51.67	55.58	31.36	42.41	32.98	20.36	100.00	−23.22
Nominal	−15.27	−0.33	80.92	−63.54	−30.62	−42.61	−31.81	−54.76	15.45	−74.83	14.37	−28.52	−23.22	100.00

(continued)

Table F.2 (continued)

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Medium														
Reversal	100.00	9.39	−39.66	−36.26	−55.93	−53.82	−36.43	−31.58	−98.37	8.88	−90.56	−27.91	−0.23	−50.52
Reversal st	9.39	100.00	−16.96	−21.13	−38.56	−32.86	−28.61	−22.43	−8.84	−14.74	−35.47	−13.97	−2.93	−26.10
Amihud	−39.66	−16.96	100.00	66.00	53.40	53.71	13.40	25.71	42.83	9.10	49.63	22.23	−2.83	71.28
Volume	−36.26	−21.13	66.00	100.00	47.21	46.55	26.91	39.97	34.78	−10.21	38.18	17.51	−13.37	17.75
Low vol	−55.93	−38.56	53.40	47.21	100.00	90.94	64.76	72.57	55.74	−8.19	66.34	40.15	2.42	65.23
Low vol st	−53.82	−32.86	53.71	46.55	90.94	100.00	51.51	74.67	55.62	−15.21	61.57	40.35	−4.73	68.34
Low beta	−36.43	−28.61	13.40	26.91	64.76	51.51	100.00	52.16	33.22	−4.65	44.50	21.06	8.54	22.08
Low-beta st	−31.58	−22.43	25.71	39.97	72.57	74.67	52.16	100.00	31.40	−9.33	40.20	32.29	−0.88	40.11
Momentum	−98.37	−8.84	42.83	34.78	55.74	55.62	33.22	31.40	100.00	−3.94	90.59	25.18	1.78	55.20
Size	8.88	−14.74	9.10	−10.21	−8.19	−15.21	−4.65	−9.33	−3.94	100.00	0.79	−8.83	21.86	12.46
Mispricing	−90.56	−35.47	49.63	38.18	66.34	61.57	44.50	40.20	90.59	0.79	100.00	22.76	8.93	64.38
Idio risk	−27.91	−13.97	22.23	17.51	40.15	40.35	21.06	32.29	25.18	−8.83	22.76	100.00	26.36	34.75
Explained	−0.23	−2.93	−2.83	−13.37	2.42	−4.73	8.54	−0.88	1.78	21.86	8.93	26.36	100.00	15.97
Nominal	−50.52	−26.10	71.28	17.75	65.23	68.34	22.08	40.11	55.20	12.46	64.38	34.75	15.97	100.00

Small														
Reversal	100.00	15.73	−58.70	1.33	−65.47	−67.23	−27.96	−25.62	−96.66	37.56	−89.11	−57.90	−1.76	−54.08
Reversal st	15.73	100.00	−4.89	6.22	−38.72	−31.82	−23.99	−18.67	−19.48	4.50	−42.99	−20.97	−13.81	−26.33
Amihud	−58.70	−4.89	100.00	13.91	44.14	47.53	7.27	23.93	54.24	−8.41	51.04	19.15	−2.79	47.88
Volume	1.33	6.22	13.91	100.00	−1.49	−1.86	6.83	1.59	−2.42	4.22	0.33	−9.56	30.44	−64.56
Low vol	−65.47	−38.72	44.14	−1.49	100.00	91.38	68.43	67.45	61.99	−34.48	72.93	42.05	12.10	61.54
Low vol st	−67.23	−31.82	47.53	−1.86	91.38	100.00	61.82	69.93	61.86	−33.06	74.08	39.69	20.10	61.09
Low beta	−27.96	−23.99	7.27	6.83	68.43	61.82	100.00	57.54	30.45	−17.89	40.37	24.93	24.72	26.46
Low-beta st	−25.62	−18.67	23.93	1.59	67.45	69.93	57.54	100.00	19.51	−11.07	33.21	16.82	42.34	38.03
Momentum	−96.66	−19.48	54.24	−2.42	61.99	61.86	30.45	19.51	100.00	−43.24	90.84	61.57	−0.55	51.90
Size	37.56	4.50	−8.41	4.22	−34.48	−33.06	−17.89	−11.07	−43.24	100.00	−34.09	−27.58	29.57	−22.40
Mispricing	−89.11	−42.99	51.04	0.33	72.93	74.08	40.37	33.21	90.84	−34.09	100.00	58.82	5.87	57.65
Idio risk	−57.90	−20.97	19.15	−9.56	42.05	39.69	24.93	16.82	61.57	−27.58	58.82	100.00	−1.15	31.84
Explained	−1.76	−13.81	−2.79	30.44	12.10	20.10	24.72	42.34	−0.55	29.57	5.87	−1.15	100.00	−16.55
Nominal	−54.08	−26.33	47.88	−64.56	61.54	61.09	26.46	38.03	51.90	−22.40	57.65	31.84	−16.55	100.00

The columns and lines represent the investment styles under consideration, with “st” denoting the short-term variants. The numbers are the correlation coefficients in %. The table reports the outcomes for all available assets

Table F.3 Bond investment style correlations

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Big														
Reversal	100.00	47.51	–	–	−62.55	−57.59	−46.96	−25.77	−96.30	8.65	−90.23	6.44	−26.66	−84.21
Reversal st	47.51	100.00	–	–	−46.56	−43.23	−41.70	−20.68	−50.35	−16.79	−65.75	2.27	−28.62	−63.11
Amihud	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Volume	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Low vol	−62.55	−46.56	–	–	100.00	93.47	91.21	63.64	64.77	−19.57	70.71	−16.48	16.24	75.22
Low vol st	−57.59	−43.23	–	–	93.47	100.00	80.61	82.55	56.65	0.63	65.23	−16.91	4.60	72.34
Low beta	−46.96	−41.70	–	–	91.21	80.61	100.00	62.29	51.34	−14.43	56.85	−10.40	15.68	60.09
Low−beta st	−25.77	−20.68	–	–	63.64	82.55	62.29	100.00	21.25	−8.70	27.45	−12.80	−21.22	39.83
Momentum	−96.30	−50.35	–	–	64.77	56.65	51.34	21.25	100.00	−10.25	91.83	−10.61	31.37	87.21
Size	8.65	−16.79	–	–	−19.57	0.63	−14.43	−8.70	−10.25	100.00	−1.21	43.46	29.15	12.91
Mispricing	−90.23	−65.75	–	–	70.71	65.23	56.85	27.45	91.83	−1.21	100.00	1.94	22.51	95.84
Idio risk	6.44	2.27	–	–	−16.48	−16.91	−10.40	−12.80	−10.61	43.46	1.94	100.00	6.39	−0.63
Explained	−26.66	−28.62	–	–	16.24	4.60	15.68	−21.22	31.37	29.15	22.51	6.39	100.00	20.13
Nominal	−84.21	−63.11	–	–	75.22	72.34	60.09	39.83	87.21	12.91	95.84	−0.63	20.13	100.00

Medium														
Reversal	100.00	30.60	–	–	−68.79	−69.00	−56.88	4.03	−97.69	11.33	−78.77	28.15	64.86	−75.48
Reversal st	30.60	100.00	–	–	−50.54	−56.51	−57.77	10.66	−35.80	15.27	−66.12	10.26	23.38	−62.05
Amihud	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Volume	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Low vol	−68.79	−50.54	–	–	100.00	93.21	86.94	9.35	68.22	−47.01	80.82	−20.61	−49.53	83.60
Low vol st	−69.00	−56.51	–	–	93.21	100.00	81.21	23.04	68.66	−29.44	76.99	−36.63	−52.58	79.50
Low beta	−56.88	−57.77	–	–	86.94	81.21	100.00	13.59	60.11	−45.22	77.25	−15.62	−33.08	74.87
Low−beta st	4.03	10.66	–	–	9.35	23.04	13.59	100.00	−7.94	−7.70	−22.97	−33.05	3.31	−23.82
Momentum	−97.69	−35.80	–	–	68.22	68.66	60.11	−7.94	100.00	−11.31	80.30	−21.25	−54.80	78.76
Size	11.33	15.27	–	–	−47.01	−29.44	−45.22	−7.70	−11.31	100.00	−34.55	−19.84	18.98	−31.95
Mispricing	−78.77	−66.12	–	–	80.82	76.99	77.25	−22.97	80.30	−34.55	100.00	−5.94	−52.83	91.83
Idio risk	28.15	10.26	–	–	−20.61	−36.63	−15.62	−33.05	−21.25	−19.84	−5.94	100.00	42.58	−10.27
Explained	64.86	23.38	–	–	−49.53	−52.58	−33.08	3.31	−54.80	18.98	−52.83	42.58	100.00	−49.98
Nominal	−75.48	−62.05	–	–	83.60	79.50	74.87	−23.82	78.76	−31.95	91.83	−10.27	−49.98	100.00

(continued)

Table F.3 (continued)

	Reversal	Reversal st	Amihud	Volume	Low vol	Low vol st	Low beta	Low beta st	Momentum	Size	Mispricing	Idio risk	Explained	Nominal
Small														
Reversal	100.00	29.20	–	–	−70.25	−69.82	−35.53	−21.92	−97.42	29.35	−83.85	3.26	31.73	−80.89
Reversal st	29.20	100.00	–	–	−31.01	−33.40	−24.30	−30.32	−27.95	40.34	−44.01	−1.42	32.33	−30.11
Amihud	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Volume	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Low vol	−70.25	−31.01	–	–	100.00	92.48	54.05	39.61	67.75	−23.62	72.75	−11.28	−19.33	79.74
Low vol st	−69.82	−33.40	–	–	92.48	100.00	62.38	59.34	68.44	−25.43	75.70	−18.29	−26.21	77.52
Low beta	−35.53	−24.30	–	–	54.05	62.38	100.00	54.58	32.53	−38.61	48.86	−21.89	−26.47	38.22
Low−beta st	−21.92	−30.32	–	–	39.61	59.34	54.58	100.00	19.82	−24.61	33.26	−14.78	−24.43	19.89
Momentum	−97.42	−27.95	–	–	67.75	68.44	32.53	19.82	100.00	−32.22	86.44	5.95	−26.12	81.35
Size	29.35	40.34	–	–	−23.62	−25.43	−38.61	−24.61	−32.22	100.00	−37.81	14.11	42.89	−13.40
Mispricing	−83.85	−44.01	–	–	72.75	75.70	48.86	33.26	86.44	−37.81	100.00	7.20	−23.56	88.90
Idio risk	3.26	−1.42	–	–	−11.28	−18.29	−21.89	−14.78	5.95	14.11	7.20	100.00	36.09	−5.56
Explained	31.73	32.33	–	–	−19.33	−26.21	−26.47	−24.43	−26.12	42.89	−23.56	36.09	100.00	−18.79
Nominal	−80.89	−30.11	–	–	79.74	77.52	38.22	19.89	81.35	−13.40	88.90	−5.56	−18.79	100.00

The columns and lines represent the investment styles under consideration, with "st" denoting the short-term variants. The numbers are the correlation coefficients in %. The table reports the outcomes for all available assets

Appendix G
Independence of Investment Styles

See Tables G.1, G.2 and G.3.

See Chap. 8.

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T. Glas, *Asset Pricing and Investment Styles in Digital Assets*, Advanced Studies in Diginomics and Digitalization, https://doi.org/10.1007/978-3-030-95695-0

Table G.1 Digital asset investment style independence

Style	Intercept	Market	Size	Volume	Short-term reversal	Adjusted R^2 (%)
Big						
Market	0.03	–	0.18	−0.18	0.36	1.85
	(0.54)	–	(0.67)	(−0.61)	(1.33)	
Size	−0.02	0.05	–	0.62	0.42	39.18
	(−0.84)	(0.67)	–	(5.16)[c]	(3.32)[c]	
Volume	0.00	−0.04	0.56	–	−0.28	31.34
	(0.04)	(−0.61)	(5.16)[c]	–	(−2.21)[b]	
Short-term reversal	−0.05	0.10	0.43	−0.32	–	19.04
	(−1.91)[a]	(1.33)	(3.32)[c]	(−2.21)[b]	–	
Medium						
Market	−0.04	–	−0.21	−0.33	0.13	3.57
	(−0.69)	–	(−0.98)	(−1.58)	(0.73)	
Size	−0.01	−0.09	–	0.13	0.23	4.89
	(−0.39)	(−0.98)	–	(0.97)	(2.10)[b]	
Volume	0.08	−0.14	0.14	–	−0.20	7.54
	(2.49)[b]	(−1.58)	(0.97)	–	(−1.76)[a]	
Short-term reversal	0.08	0.08	0.36	−0.29	–	8.47
	(1.90)[a]	(0.73)	(2.10)[b]	(−1.76)[a]	–	
Small						
Market	−0.15	–	0.55	−0.23	0.36	13.88
	(−2.50)[b]	–	(2.97)[c]	(−1.33)	(1.82)[a]	
Size	0.07	0.27	–	0.37	−0.16	23.63
	(1.56)	(2.97)[c]	–	(3.30)[c]	(−1.13)	
Volume	0.07	−0.15	0.49	–	0.38	20.30
	(1.47)	(−1.33)	(3.30)[c]	–	(2.43)[b]	
Short-term reversal	0.13	0.17	−0.16	0.28	–	9.38
	(3.13)[c]	(1.82)[a]	(−1.13)	(2.43)[b]	–	

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
This table shows the regression coefficients of a multivariate regression against a specific investment style. Below the coefficient, the *t*-statistics is reported in parentheses. The investment styles are based on a dataset comprised of all available assets of the respective asset class

Table G.2 Equity investment style independence

Style	Intercept	Market	Size	Volume	Short-term reversal	Adjusted R^2 (%)
Big						
Market	0.01	–	0.12	−0.56	0.36	21.67
	(1.97)[a]	–	(0.40)	(−1.72)[a]	(2.36)[b]	
Size	0.00	0.02	–	0.95	−0.05	82.02
	(0.17)	(0.40)	–	(15.07)[c]	(−0.67)	
Volume	0.00	−0.09	0.84	–	0.03	82.77
	(0.27)	(−1.72)[a]	(15.07)[c]	–	(0.42)	
Short-term reversal	−0.00	0.25	−0.17	0.11	–	8.18
	(−0.09)	(2.36)[b]	(−0.67)	(0.42)	–	
Medium						
Market	0.01	–	0.22	−0.23	0.12	1.60
	(2.15)[b]	–	(0.83)	(−1.24)	(1.02)	
Size	0.00	0.05	–	−0.08	−0.08	−0.11
	(0.14)	(0.83)	–	(−0.81)	(−1.41)	
Volume	0.00	−0.11	−0.15	–	−0.13	3.65
	(0.54)	(−1.24)	(−0.81)	–	(−1.55)	
Short-term reversal	−0.00	0.15	−0.41	−0.32	–	4.14
	(−0.53)	(1.02)	(−1.41)	(−1.55)	–	
Small						
Market	0.00	–	0.35	−0.05	0.22	2.67
	(1.05)	–	(1.47)	(−0.26)	(1.52)	
Size	0.01	0.10	–	0.07	0.00	−0.47
	(4.78)[c]	(1.47)	–	(0.70)	(0.00)	
Volume	0.00	−0.02	0.12	–	0.05	−3.94
	(1.01)	(−0.26)	(0.70)	–	(0.49)	
Short-term reversal	−0.00	0.18	0.00	0.08	–	−0.61
	(−0.52)	(1.52)	(0.00)	(0.49)	–	

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
This table shows the regression coefficients of a multivariate regression against a specific investment style. Below the coefficient, the *t*-statistics is reported in parentheses. The investment styles are based on a dataset comprised of all available assets of the respective asset class

Table G.3 Bond investment style independence

Style	Intercept	Market	Size	Volume	Short-term reversal	Adjusted R^2 (%)
Big						
Market	−0.00	–	−0.13	–	0.16	3.39
	(−0.56)	–	(−0.60)	–	(1.81)[a]	
Size	−0.00	−0.05	–	–	−0.06	0.09
	(−1.29)	(−0.60)	–	–	(−1.11)	
Short-term reversal	−0.00	0.33	−0.35	–	–	4.82
	(−0.25)	(1.81)[a]	(−1.11)	–	–	
Medium						
Market	−0.00	–	0.07	–	0.16	9.67
	(−0.43)	–	(0.66)	–	(2.69)[c]	
Size	−0.00	0.11	–	–	0.07	−0.28
	(−1.23)	(0.66)	–	–	(0.88)	
Short-term reversal	−0.00	0.68	0.18	–	–	10.18
	(−1.27)	(2.69)[c]	(0.88)	–	–	
Small						
Market	−0.00	–	0.06	–	0.13	7.05
	(−1.10)	–	(0.77)	–	(1.92)[a]	
Size	0.00	0.18	–	–	0.33	14.26
	(1.93)[a]	(0.77)	–	–	(2.98)[c]	
Short-term reversal	−0.01	0.48	0.41	–	–	18.58
	(−2.29)[b]	(1.92)[a]	(2.98)[c]	–	–	

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level
This table shows the regression coefficients of a multivariate regression against a specific investment style. Below the coefficient, the *t*-statistics is reported in parentheses. The investment styles are based on a dataset comprised of all available assets of the respective asset class

Appendix H
Results Based on USD Prices

See Tables H.1, H.2, H.3, H.4 and H.5.

See Chap. 9.

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T. Glas, *Asset Pricing and Investment Styles in Digital Assets*, Advanced Studies in Diginomics and Digitalization, https://doi.org/10.1007/978-3-030-95695-0

Table H.1 Regression results for the existing models (USD)

	All			Mineable			Non-mineable		
	Eq. (9.1)	Eq. (9.2)	Eq. (9.3)	Eq. (9.1)	Eq. (9.2)	Eq. (9.3)	Eq. (9.1)	Eq. (9.2)	Eq. (9.3)
alpha	−34.83	−6.36	−7.30	−20.27	−4.22	−8.51	−44.66	−7.55	−0.528
SA (%)	18.21	7.60	9.90	11.78	5.59	12.57	22.85	8.76	7.80
$\bar{\beta}_1$	1.7839	0.5420	1.9753	0.9717	0.3941	1.8323	2.3413	0.6331	2.1237
$\bar{\beta}_2$	0.0918		0.5696	0.1919		1.0439		0.0247	0.3313
$\bar{\beta}_3$			−0.1804			−0.5180			−0.0160
$\bar{\beta}_4$			−0.1030			0.0038			−0.2464
R^2	34.13	15.59	58.71	20.17	8.24	43.67	43.86	20.45	69.47

This table shows the regression results for the three asset pricing models above. I ran these models against each available digital asset in the three datasets all, mineable, and non-mineable. Thus, the values in the tables are averaged over all assets. Alpha, significant alphas (SA), and R^2 are percentage values. The value of significant alphas is the number of all alphas significant at the 10% level divided by the number of all digital assets. β_1 to β_4 are the averaged regression coefficients of all digital assets

Table H.2 Explained variance by principal component (USD)

	All			Mineable			Non-mineable		
PC	PLS	Random	Missing	PLS	Random	Missing	PLS	Random	Missing
PC 1	29.54	76.17	30.71	30.38	86.59	64.83	26.14	51.41	54.98
PC 2	16.38	4.82	6.48	16.67	3.13	5.36	16.49	9.56	8.13
PC 3	9.01	3.37	5.68	8.29	2.05	2.18	11.81	6.90	4.45
PC 4	6.22	2.61	4.60	6.71	1.39	1.89	5.98	5.42	3.70
PC 5	5.98	2.10	4.08	5.66	1.08	1.63	4.74	4.39	3.13
PC 6	5.33	1.74	3.53	4.94	0.88	1.42	4.26	3.63	2.84
PC 7	3.46	1.46	3.00	3.23	0.73	1.40	4.22	3.04	2.49
PC 8	3.30	1.24	2.76	3.07	0.61	1.30	3.88	2.56	2.01
PC 9	2.45	1.05	2.44	2.89	0.52	1.11	3.50	2.17	1.82
PC 10	2.25	0.90	2.13	2.65	0.45	1.07	2.48	1.85	1.47
Sum	83.92	95.46	65.40	84.48	97.43	82.19	83.49	90.93	85.02

All values are percentage values. PLS denotes the principal components based on a PLS regression, whereas the random and missing PCs both stem from a PCA

Table H.3 Regression results and Fama–MacBeth coefficients (USD)

	All			Mineable			Non-mineable		
PC	PLS	Random	Missing	PLS	Random	Missing	PLS	Random	Missing
Regression results									
alpha	−2.84	−0.94	−8.78	−2.77	−0.16	−7.02	−2.97	3.73	−6.39
SA	6.73	0.00	9.03	7.58	0.00	6.39	6.84	0.00	12.04
$\bar{\beta}_1$	−0.0487	0.1909	0.0267	0.0659	0.1859	0.2972	−0.0869	0.1905	−0.1052
$\bar{\beta}_2$	−1.2992	0.0002	−0.0045	−1.4466	−0.0001	0.0715	−1.2017	0.0001	0.0336
$\bar{\beta}_3$	−1.2154	−0.0001	−0.0039	−1.1272	−0.0001	−0.0117	−1.3357	0.0003	−0.0113
R^2	47.65	81.89	53.64	33.00	90.64	40.83	58.13	65.73	64.02
Fama–MacBeth									
Univariate									
PC 1	0.0024	0.0841	0.6609	−0.0078	0.3911	0.2149	−0.0064	0.0358	−0.1824
	(0.12)	(0.21)	(1.20)	(−0.40)	(0.70)	(1.37)	(−0.33)	(0.17)	(−0.77)
PC 2	−0.0033	−0.0002	0.4043	−0.0048	0.0002	−0.4495	0.0002	0.0006	−0.0632
	(−0.20)	(−0.00)	(0.90)	(−0.29)	(0.00)	(−1.39)	(0.01)	(0.00)	(−0.25)
PC 3	−0.0070	−0.0002	−0.0298	−0.0030	−0.0004	0.1890	−0.0057	0.0008	−0.0967
	(−0.40)	(−0.00)	(−0.15)	(−0.18)	(−0.01)	(0.78)	(−0.29)	(0.01)	(−0.36)

Multivariate									
PC 1	0.0145	−0.0107	0.8114	−0.0071	0.1280	0.6264	0.0102	0.0642	−0.1564
	(0.50)	(−0.01)	(0.80)	(−0.25)	(0.12)	(0.94)	(0.30)	(0.24)	(−0.21)
PC 2	−0.0072	0.0008	0.1780	−0.0196	−0.0004	1.0230	0.0010	0.0019	0.1416
	(−0.36)	(0.00)	(0.31)	(−0.89)	(−0.00)	(0.78)	(0.05)	(0.01)	(0.11)
PC 3	−0.0067	−0.0002	0.0289	−0.0033	−0.0002	0.2769	−0.0000	0.0013	−0.2841
	(−0.34)	(−0.00)	(0.11)	(−0.18)	(−0.00)	(0.96)	(−0.00)	(0.01)	(−0.68)
GRS									
F-stat		159.39			563.75			89.65	
p-value		(0.00)			(0.00)			(0.00)	

This table shows the regression results of the identified principal components. I ran these regressions against each available digital asset in the dataset. Thus, the values in the tables are averaged over all assets. Alpha, significant alphas (SA), and R^2 are percentage values. The SA value is the number of all alphas significant at the 10% level divided by the number of all digital assets. β_1 to β_3 are the averaged regression coefficients of all digital assets. Further, I show the Fama–MacBeth regression coefficients for the first three PCs as described in Table H.2. Below the coefficients I report the respective t-statistics in parentheses. Moreover, I report the GRS test statistics for the Monte Carlo simulation approach: F-stat denotes the averaged GRS test statistics, and their corresponding p-values are given in parentheses below. Data is given in USD terms, since that analysis focuses on an "external" perspective

Table H.4 VIFs and multivariate FM regression results (USD)

		All			Mineable		
Var	VIF	All	All but tiny	Tiny	All	All but tiny	Tiny
2	8.57	–	–	–	–	–	–
		–	–	–	–	–	–
3	2.71	−0.0139	−0.0206	−0.1487	−0.0069	−0.0150	−0.1694
		(−0.94)	(−1.31)	(−1.51)	(−0.30)	(−0.52)	(−1.53)
4	1.47	−0.0099	−0.0056	−0.0048	−0.0073	−0.0135	−0.0109
		(−0.69)	(−0.31)	(−0.23)	(−0.32)	(−0.47)	(−0.80)
5	1.62	**−0.0105**	−0.0056	**−0.0547**	−0.0011	−0.0012	−0.0656
		(−2.64)	(−1.32)	**(−5.33)**	(−0.19)	(−0.17)	(−1.31)
6	2.64	−0.0023	−0.0099	**−0.4479**	0.0958	−0.0406	−0.2161
		(−0.04)	(−0.27)	**(−3.04)**	(1.09)	(−0.72)	(−1.04)
7	2.43	0.0397	0.1665	−0.2628	0.3315	0.1856	−0.2618
		(0.28)	(1.18)	(−0.76)	(1.84)	(0.96)	(−0.33)
8	1.96	0.0691	0.0718	0.0101	0.0316	0.0786	**0.2246**
		(1.44)	(1.55)	(0.11)	(0.65)	(1.42)	**(2.08)**
9	1.40	**0.0242**	0.0146	0.0449	0.0107	0.0083	0.0065
		(2.69)	(1.62)	(1.90)	(0.70)	(0.51)	(0.23)
10	8.69	–	–	–	–	–	–
		–	–	–	–	–	–
11	5.41	0.0118	0.0033	**−0.0872**	−0.0019	−0.0074	**−0.1093**
		(1.75)	(0.49)	**(−2.22)**	(−0.22)	(−0.85)	**(−2.31)**
12	3.19	0.3250	0.4979	−141,697.0501	0.5800	−0.2971	794,968.0340
		(0.47)	(0.72)	(−1.08)	(0.48)	(−0.56)	(1.64)
13	3.72	0.0137	0.0080	0.2113	0.0120	0.0243	0.0823
		(0.43)	(0.25)	(1.01)	(0.17)	(0.29)	(0.20)
17	3.36	**−0.0067**	−0.0056	**−0.0270**	−0.0057	−0.0009	−0.0414
		(−2.15)	(−1.85)	**(−2.23)**	(−1.95)	(−0.31)	(−1.04)
Dummy	2.73	−0.0128	−0.0185	0.0527	–	–	–
		(−0.87)	(−1.24)	(0.97)	–	–	–

This table presents the VIFs and the FM regression results for the identified characteristics. The multivariate FM coefficients are reported for every characteristic with the respective t-value in parentheses below. The variables 2–17 are in line with the numbering in Table 9.2. "Dummy" is the blockchain dummy that takes values of 1 for digital assets connected to their own blockchain and 0 otherwise. VIF is the variance inflation factor for every variable over the full data history. To calculate the VIFs, dummy is the return spread between an equally weighted index of mineable digital assets and non-mineable assets. I exclude characteristics with VIF > 7 from the rolling multivariate FM regression analysis. The data sets "All" and "Mineable" include all digital assets and only digital assets with their own blockchain, respectively. I do not consider non-mineable digital assets here since these emerged mainly since 2016, which leaves too short a data history. However, I split the dataset in line with Green et al. (2017). "All" again includes all digital assets within the former mentioned categories. "All but tiny" excludes all digital assets with market capitalizations smaller than the 20th percentile of the respective dataset. "Tiny," on the other hand, contains only digital assets with market capitalizations equal to or less than the 20th percentile of the respective dataset. T-values equal or larger than |2| are highlighted in bold

Table H.5 Portfolio sorts based on predicted returns (USD)

Long–short quintile portfolios	Mean (%)	*t*-stat	SD (%)	Return/risk	Skewness	Kurtosis
All						
12 characteristics	0.79	(0.17)	32.17	0.02	0.51	2.91
5, 6, 8, 9, 11 and 17	4.37	(0.76)	39.83	0.11	1.05	4.58
5, 9 and 17	10.92	(3.75)[c]	19.53	0.56	0.62	3.25
5, 6, 11 and 17	5.87	(2.13)[b]	18.93	0.31	−0.49	4.08
8 and 11	3.87	(0.52)	51.41	0.08	0.84	4.04
5 only	6.23	(2.02)[b]	21.18	0.29	−0.17	3.10
6 only	−1.41	(−0.33)	29.85	−0.05	−0.74	3.39
8 only	1.68	(0.16)	71.28	0.02	0.75	2.93
9 only	−0.91	(−0.38)	16.69	−0.05	0.27	2.78
11 only	6.66	(2.22)[b]	20.55	0.32	0.21	3.62
17 only	14.56	(4.76)[c]	20.04	0.73	0.77	3.49
5 and 17	10.02	(3.23)[c]	21.03	0.48	0.38	3.44
11 and 17	12.09	(4.30)[c]	18.62	0.65	0.68	3.59
5 and 11	6.04	(2.35)[b]	17.63	0.34	−0.26	4.46
All but tiny						
12 characteristics	4.13	(0.93)	29.91	0.13	0.83	3.25
5, 6, 8, 9, 11 and 17	4.09	(0.80)	35.58	0.12	0.99	4.06
5, 9 and 17	7.27	(2.55)[b]	19.52	0.37	0.50	3.69
5, 6, 11 and 17	9.55	(3.37)[c]	19.14	0.50	0.34	3.52
8 and 11	2.35	(0.34)	47.69	0.05	1.11	3.60
5 only	9.52	(3.16)[c]	20.44	0.47	−0.26	2.75
6 only	0.53	(0.12)	31.00	0.02	0.56	4.17
8 only	−3.86	(−0.42)	63.45	−0.06	0.27	3.63
9 only	2.00	(0.91)	15.35	0.13	−0.19	2.94
11 only	19.28	(3.89)[c]	33.14	0.58	0.01	2.90
17 only	11.70	(3.82)[c]	20.51	0.57	1.15	4.84
5 and 17	7.57	(2.39)[b]	21.74	0.35	0.35	3.44
11 and 17	10.10	(3.29)[c]	20.74	0.49	0.49	3.68
5 and 11	10.24	(3.49)[c]	19.81	0.52	−0.47	2.31

[a] Denotes significance at the 10% level
[b] Denotes significance at the 5% level
[c] Denotes significance at the 1% level

This table presents the return moments of a dollar-neutral long–short portfolio. The hedge portfolio is formed by going long for the top quintile and short for the bottom quintile from a portfolio sort based on predicted returns. The returns are predicted in accordance with Lewellen (2015). I sort for high predicted returns in all cases. Quintile portfolios are equally weighted and based on the dataset containing all available digital assets or all but tiny digital assets. For the FM coefficients I use the average FM coefficient over a rolling 12-month lookback period to predict the returns. "Mean" is the monthly percentage hedge portfolio return. "*t*-stat" is the *t*-value of the mean monthly returns in parentheses. "SD" is the standard deviation of the mean monthly hedge portfolio returns in %. I calculated the risk-adjusted return "return/risk" by dividing the return by the standard deviation. "Skewness" and "kurtosis" are the third and fourth return moments

Appendix I
Full Enumeration

Table I.1 presents an excerpt of the full enumeration results of possible risk factors driving digital asset returns, as mentioned in Chap. 9. The full enumeration aims to identify risk factors in digital asset returns by investigating every possible combination of variables that are assumed to explain the returns of digital assets. Therefore, I considered two approaches, a pooled OLS panel regression and rolling FM regressions. Both methods are explained in more detail in the methodology section of this book.

For better comprehensibility, the results reported in Table I.1 can be thought of as some sort of forward search algorithm. A full enumeration would yield too many instances to show here. In the table I report only significant outcomes of combinations involving six characteristics at once. A forward search algorithm would, for instance, identify single variables that significantly explain the cross-section of digital asset returns to some extent in a first step. From that starting point, each of the single statistically significant variables is combined with every other variable. Therefore, the second step is a regression model containing two independent variables. Models with two statistically significant variables are kept, whereas models with less than two significant variables are dismissed. This procedure is repeated as long as no further significant variable can be added. Table I.1 is based on a full enumeration. However, it only reports the outcomes that would have been arrived at with a forward search algorithm.

In the case here, the enumeration stops after combinations of five simultaneously significant variables. There is no possible combination of variables that contains six or more statistically significant ones. All of the pooled OLS panel regression models contain one insignificant variable, which is also the final step at which the enumeration stops. Thus, 23 combinations seem to exist, each comprised of five variables, which explain the cross-section of digital asset returns with statistical

See Chap. 9.

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Table I.1 Results Sect. 9.6

Var 1	Var 2	Var 3	Var 4	Var 5	Var 6	Lambda 1	Lambda 2	Lambda 3	Lambda 4	Lambda 5	Lambda 6	SE(L1)	SE(L2)	SE(L3)	SE(L4)	SE(L5)	SE(L6)	T(L1)	T(L2)	T(L3)	T(L4)	T(L5)	T(L6)	Adj. R^2	Sum abs(t)
Pooled OLS panel regression																									
2	3	4	10	11	13	0	0.0009	−0.0026	0	−0.0066	−0.1005	0	0.0004	0.0006	0	0.0049	0.0341	7.68	2.51	4.24	16.79	1.34	2.95	0.26	35.51
2	4	5	10	11	13	0	−0.0017	−0.0138	0	0.0097	−0.0893	0	0.0006	0.0058	0	0.0058	0.035	7.51	2.65	2.4	16.72	1.68	2.55	0.26	33.51
2	3	7	10	13	18	0	0.0005	0.2955	0	−0.1631	0.0118	0	0	0.1264	0	0.032	0.015	2.45	16.22	2.34	2.34	5.1	0.79	0.21	29.24
2	3	7	10	12	13	0	0.0005	0.3002	0	0.1059	−0.1653	0	0	0.1258	0	0.1764	0.0314	2.38	16.23	2.39	2.32	0.6	5.27	0.21	29.18
2	3	7	10	13	17	0	0.0005	0.3004	0	−0.1652	0	0	0	0.1254	0	0.0314	0	2.38	16.24	2.4	2.32	5.26	0.53	0.21	29.13
2	3	7	9	10	13	0	0.0005	0.3077	−0.0019	0	−0.1651	0	0	0.1264	0.006	0	0.0314	2.38	16.05	2.43	0.32	2.27	5.26	0.21	28.71
2	3	5	7	10	13	0	0.0005	−0.0077	0.2254	0	−0.153	0	0	0.0036	0.1342	0	0.0309	2.24	14.37	2.12	1.68	2.49	4.96	0.21	27.85
2	4	5	8	10	11	0	−0.002	−0.016	0.0408	0	0.0122	0	0.0008	0.0057	0.0372	0	0.0054	6.71	2.49	2.81	1.1	12.17	2.27	0.26	27.54
2	3	5	10	13	17	0	0.0005	−0.011	0	−0.142	0	0	0.0001	0.0039	0	0.0302	0	2.5	9.86	2.85	2.58	4.7	1.42	0.21	23.91
2	3	5	9	10	13	0	0.0005	−0.0108	0.0042	0	−0.1422	0	0.0001	0.0039	0.0069	0	0.0303	2.53	10.48	2.8	0.61	2.68	4.69	0.21	23.77
2	3	5	10	12	13	0	0.0005	−0.0108	0	0.175	−0.1422	0	0.0001	0.0039	0	0.2027	0.0303	2.49	9.87	2.79	2.58	0.86	4.7	0.21	23.3
2	3	5	10	13	18	0	0.0005	−0.0106	0	−0.1403	−0.0005	0	0.0001	0.0038	0	0.0306	0.0143	2.51	9.79	2.76	2.58	4.58	0.04	0.21	22.26
2	3	4	5	10	13	0	0.001	−0.0024	−0.0073	0	−0.0928	0	0.0003	0.0004	0.0041	0	0.0308	2.86	3.02	5.64	1.78	2.39	3.01	0.25	18.71
2	8	10	11	13	17	0	0.0401	0	−0.0142	−0.1574	0	0	0.0396	0	0.0049	0.0342	0	3.7	1.01	3.68	2.91	4.6	2.22	0.21	18.12
2	4	5	6	10	11	0	−0.0016	−0.0151	0.0819	0	0.0122	0	0.0007	0.0057	0.0849	0	0.0056	6.29	2.49	2.65	0.96	3.23	2.18	0.26	17.82
2	3	4	10	13	17	0	0.001	−0.0019	0	−0.1039	0	0	0.0003	0.0006	0	0.0317	0	2.91	3	3.39	2.51	3.28	0.42	0.24	15.51
2	3	4	10	12	13	0	0.001	−0.0019	0	0.0767	−0.104	0	0.0003	0.0006	0	0.1764	0.0317	2.91	3	3.33	2.51	0.43	3.28	0.24	15.47
2	3	4	10	13	18	0	0.001	−0.0018	0	−0.1016	0.0021	0	0.0003	0.0006	0	0.033	0.0126	2.93	2.95	3.06	2.52	3.08	0.17	0.25	14.7
2	3	4	9	10	13	0	0.0009	−0.0019	0.001	0	−0.1036	0	0.0004	0.0007	0.0094	0	0.0317	2.94	2.63	2.68	0.11	2.54	3.27	0.24	14.16
2	3	4	8	10	13	0	0.0008	−0.0023	0.037	0	−0.1076	0	0.0004	0.0008	0.0389	0	0.0318	2.28	2.09	3.09	0.95	2.08	3.39	0.25	13.88
Rolling FM regression																									
3	7	11	13	17	18	−0.0236	0.1571	−0.0104	−0.0705	0.0003	−0.0144	0.0062	0.0767	0.0036	0.0162	0.0001	0.0106	3.8	2.05	2.91	4.35	2.81	1.36	–	17.28
3	5	7	11	13	17	−0.0236	−0.0082	0.1432	0.0003	−0.0735	0.0002	0.0069	0.0035	0.0702	0.0051	0.0143	0.0001	3.41	2.38	2.04	0.05	5.14	2.58	–	15.6
4	5	6	7	8	17	−0.1648	−0.0092	−0.1355	−0.1587	0.0804	0.0002	8.8479	0.0028	0.0567	0.0729	0.0402	0.0001	0.02	3.24	2.39	2.18	2	2.67	–	12.5

Here I report the results of the full enumeration of 17 different characteristics. The first part of the table includes the results of the pooled OLS panel regression, where I report "Var" the number of the variables (analogous to Table 9.2, except for variable 18 which is the blockchain dummy), the lambda of the variables, the standard error (SE) of the lambdas, and the respective t-value. Additionally, I outline the R^2 as well as the sum of the absolute t-values. Values of "0" are too small to be shown with 4 digits. The second part of the table reports the same numbers for the rolling FM regressions

significance. “Var 1” to “Var 6” in the table describe the respective independent variables (see Table 9.2 for details). The table also contains the lambdas, standard errors, and t-statistics for every model and also reports the adjusted R^2 and the sum of the absolute t-values.

Interestingly, the pooled OLS panel regression clearly yields more possible combinations than the rolling FM regression approach. The limit of a rolling FM regression, however, is commonly considered to be five variables at once, which means the panel regression is more powerful for a larger number of possible variables. Since I did not know which of the above reported models is the “correct” one, I used this analysis as the basis for the remaining tests as described in Chap. 9.

Appendix J
Digital Asset Data Details II

Below is a list of all the digital assets used in the analyses from www.coingecko.com:

(1) 01coin (2) 02-token (3) 0cash (4) 0chain (5) 0x (6) 0xcert (7) 0xdark
(8) 0xeth-cash (9) 0xeth-classic (10) 0xeth-sv (11) 0xethereum-token (12) 0xkojo
(13) 0xmaker-token (14) 0xmonero (15) 0xzar (16) 100-waves-eth-btc-set
(17) 100-waves-eth-usd-yield-set (18) 100-waves-ethusd-ether-hoard-set
(19) 10m-token (20) 10x-long-algorand-token (21) 10x-long-altcoin-index-token
(22) 10x-long-bitcoin-cash-token (23) 10x-long-bitcoin-sv-token
(24) 10x-long-bitcoin-token (25) 10x-long-bnb-token (26) 10x-long-eos-token
(27) 10x-long-ethereum-classic-token (28) 10x-long-ethereum-token
(29) 10x-long-exchange-token-index-token (30) 10x-long-huobi-token-token
(31) 10x-long-leo-token (32) 10x-long-litecoin-token
(33) 10x-long-midcap-index-token (34) 10x-long-okb-token
(35) 10x-long-shitcoin-index-token (36) 10x-long-trx-token
(37) 10x-long-xrp-token (38) 10x-short-algorand-token
(39) 10x-short-altcoin-index-token (40) 10x-short-bitcoin-cash-token
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See Chap. 11.

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